CLF-C02

CLOUD PRACTITIONER

LEARNER GUIDE SLIDES · WSQ

AWS Certified Cloud Practitioner (CLF-C02)

WSQ Course Code: TGS-2023039183

Conducted by Tertiary Infotech Academy Pte Ltd · UEN 201200696W

Version v13 · 31 July 2026

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It is mandatory for you to take both AM, PM and Assessment digital attendance for WSQ funded courses.

- The trainer or administrator will show you the digital attendance QR code generated from SSG portal.
- Please scan the QR code from your mobile phone camera and submit your attendance.
- Digital Attendance (Mandatory)

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ICE-BREAKER

Let's Know Each Other

- Your name and organisation / role.
- Your experience with AWS or cloud services (if any).
- What you want to understand or achieve with AWS after this course.

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Ground Rules

- Set your mobile phone to silent mode
- Participate actively in the class. No question is stupid.
- Mutual respect. Agree to disagree.
- One conversation at one time.
- Be punctual. Back from breaks on time.
- Exit the class silently if you need to step out for phone call, toilet break etc.
- 75% attendance is required

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Lesson Plan

- Day 1
- Digital Attendance (AM)
- Trainer and Learners Introduction
- Learning Outcomes
- Course Outline
- Topic 1: Cloud Concepts
- Topic 2: Security and Compliance
- Lunch Break
- Digital Attendance (PM)
- Topic 3: AWS Compute Services
- Topic 4: AWS Database and Storage Services
- End of Day 1

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Lesson Plan

- Day 2
- Digital Attendance (AM)
- Topic 5: AWS Networking and Other Services
- Topic 6: AWS Analytics Services
- Lunch Break
- Digital Attendance (PM)
- Topic 7: Support
- Revision
- Course Feedback and TRAQOM Survey
- Digital Attendance (Assessment)
- Final Assessment
- End of Class

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TSC Title: Cloud Computing

1 TSC Code: ICT-DIT-3020-1.1

2 Skills Framework

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TSC Abilities:

- A1: Set up environment for the implementation of cloud solutions in accordance to user requirements
- A2: Run tests for cloud solutions to minimise integration-related issues
- A3: Detect issues with cloud solutions to identify solutions
- A4: Resolve cloud integration and implementation issues
- A5: Conduct implementation tests on cloud solutions to measure its effectiveness against defined metrics.
- A6: Escalate unresolved issues related to implementation of cloud solutions
- TSC Knowledge:
- K1: Security and quality standards
- K2: Diagnostic tools for cloud-related problems
- K3: Types of installation tests for cloud solutions
- K4: Technical solutions or techniques to resolve cloud integration-related issues
- K5: Database administration and maintenance tools and techniques

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Learning Outcomes

- LO1: Set up and test AWS cloud solutions in accordance to security standards.
- LO2: Detect and resolve AWS cloud integration issues related to AWS services.
- LO3: Conduct implementation tests and escalate unresolved AWS cloud issues.

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TOPIC 1

Introduction to Cloud Concepts

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TOPIC 3

Define methods of deploying and operating in the AWS Cloud.

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TOPIC 6

AWS artificial intelligence and machine learning (AI/ML) services and analytics services.

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TOPIC 6

Access AWS Skills Builder

- https://skillbuilder.aws/
- Builder ID Login:
- training@tertiaryinfotech.com
- training2@tertiaryinfotech.com
- training3@tertiaryinfotech.com
- training4@tertiaryinfotech.com
- training5@tertiaryinfotech.com
- PW: Tertiary@888
- https://skillbuilder.aws/
- Builder ID Login:
- training6@tertiaryinfotech.com
- training7@tertiaryinfotech.com
- training8@tertiaryinfotech.com

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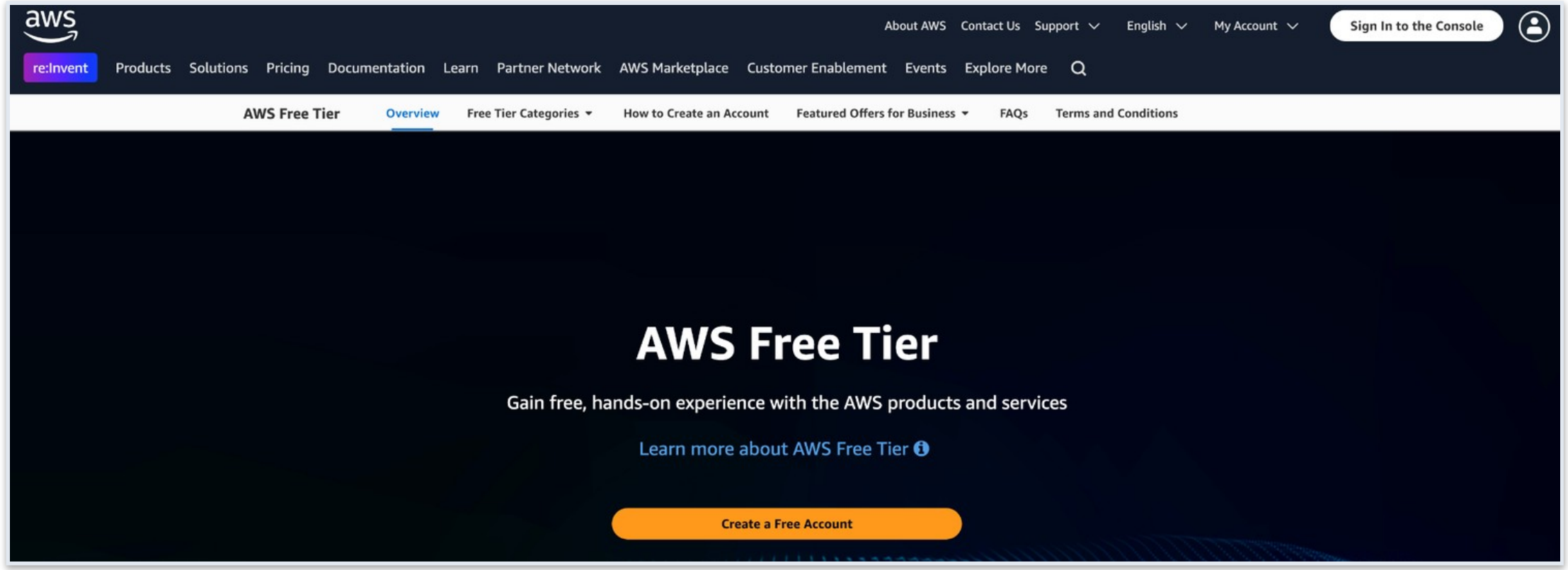
TOPIC 6

Sign Up Free Trial AWS Account



https://aws.amazon.com/free/

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Download Sample Codes

- Download the sample codes use for this course below
- https://drive.google.com/drive/folders/1KKpRQ1HWw380u4uwp2xPVnv8vNoDdTVN

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Written Assessment (Short Answer Questions) - 1 hr

- Practical Performance (PP) - 1 hr
- Assessment format: Open Book
- Open book assessment ONLY includes Slides, Learner Guide or any approved materials.
- Appeal process
- Final Assessment

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Briefing for Assessment

1. Place phones & other materials under the table or on the floor
2. No photos or recording of assessment scripts
3. No discussion during assessment
4. Use black/blue pen for assessment [hard copies]
5. No usage of liquid paper or correction tape
6. Assessment scripts will be collected when the time is up

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Criteria for Funding

1. Minimum attendance rate of 75% based on SSG Digital Attendance record.

2. Complete the assessment and be assessed as ‘Competent’

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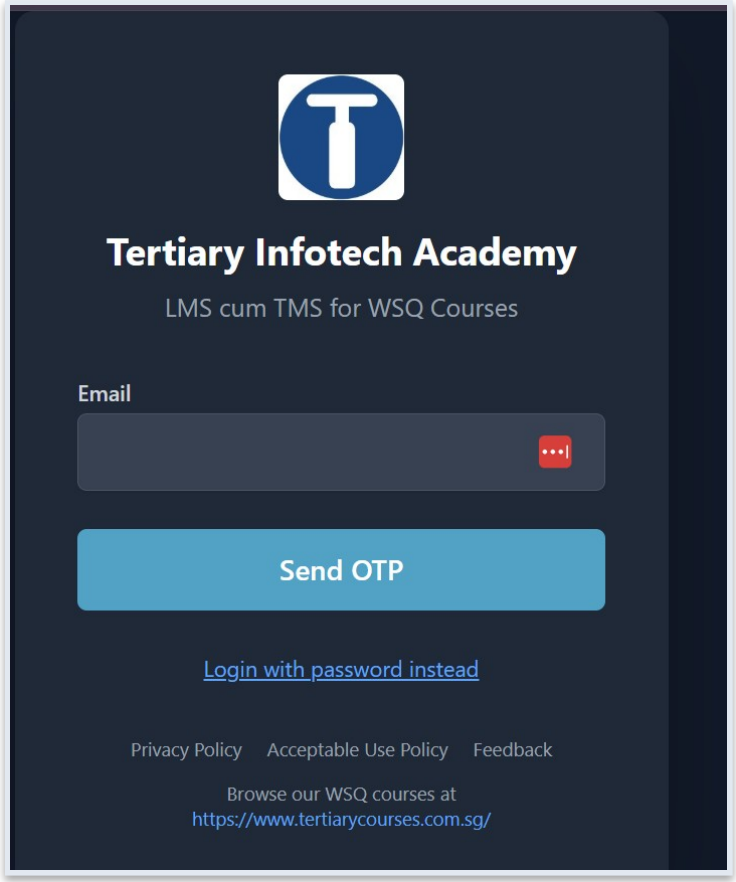
TOPIC 6

LMS/TMS



https://lms-tms.tertiaryinfotech.com

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TOPIC 1

Cloud Concepts

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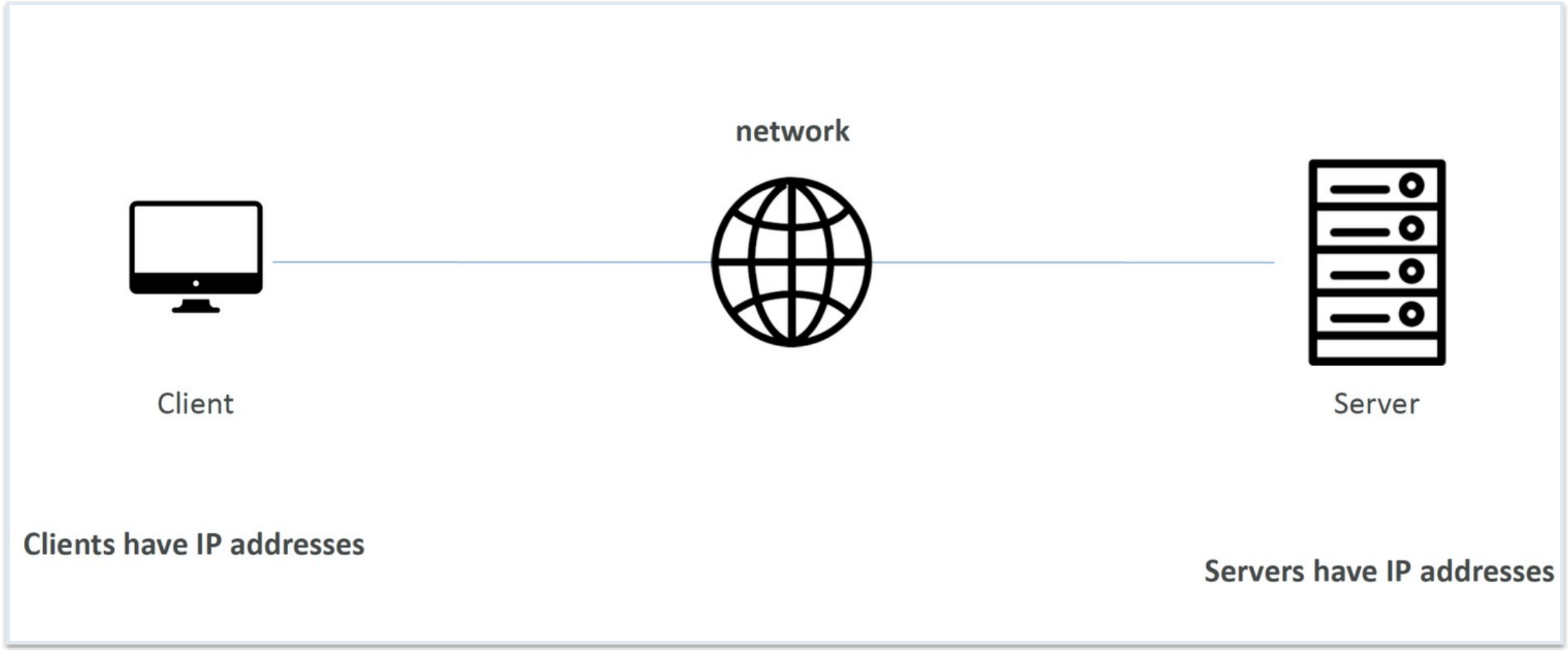
TOPIC 1

How websites work



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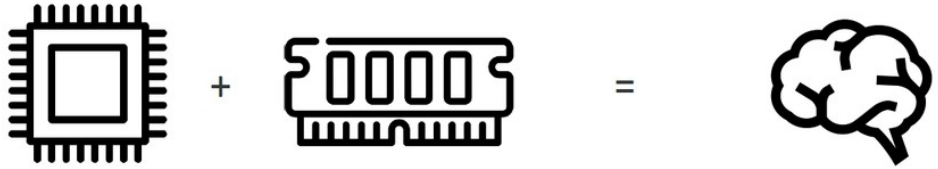
TOPIC 1

What is a server composed of?

- Compute: CPU
- Memory: RAM



- Storage: Data

- Database: Store data in a structured way

- Network: Routers, switch, DNS server

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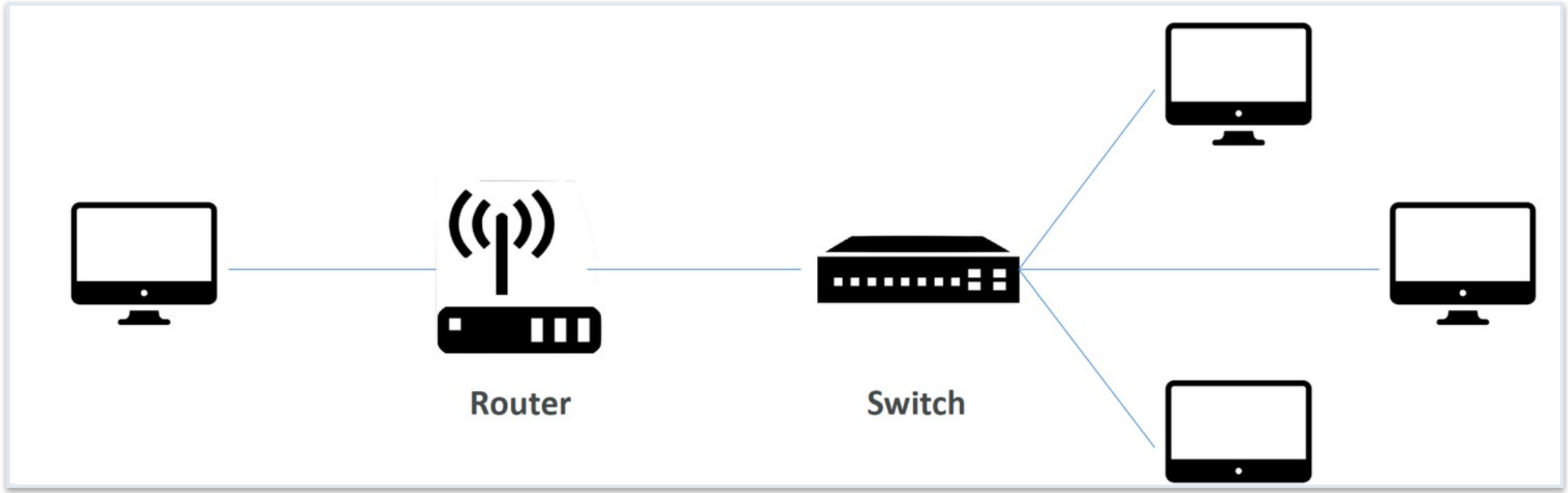
TOPIC 1

IT Terminology



Network: cables, routers and servers connected with each other

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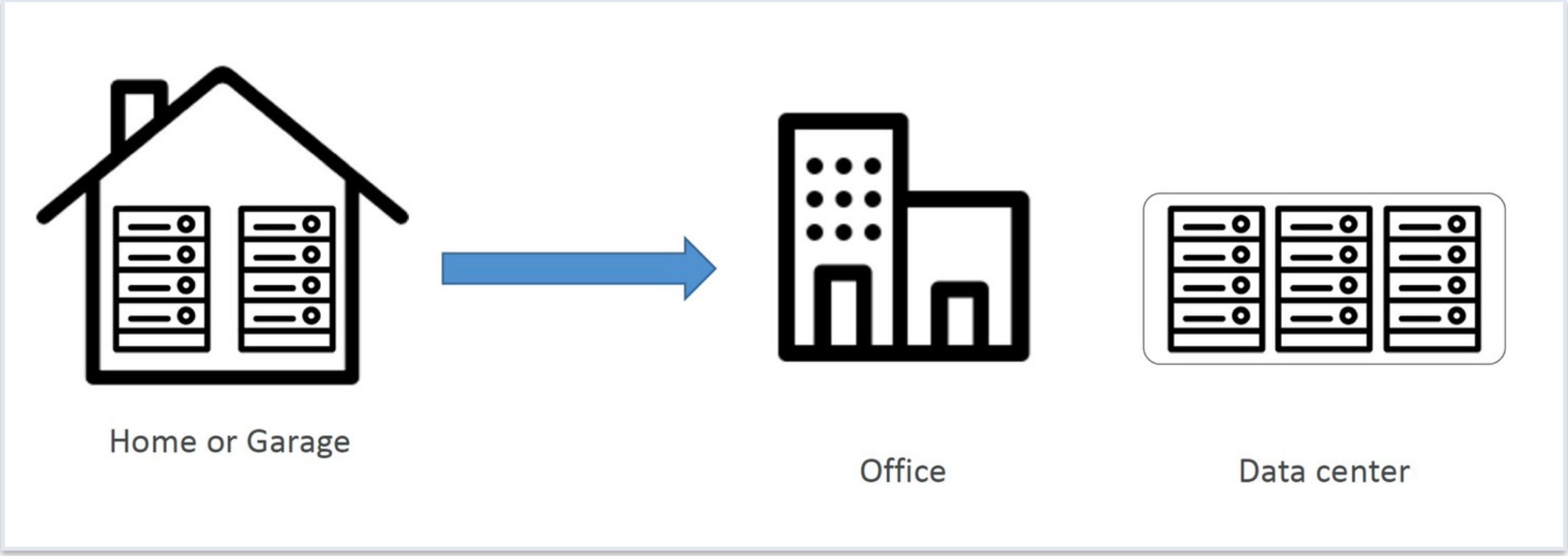
TOPIC 1

Traditionally, how to build infrastructure



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TOPIC 1

Problems with traditional IT approach

- Pay for the rent for the data center
- Pay for power supply, cooling, maintenance
- Adding and replacing hardware takes time
- Scaling is limited
- Hire 24/7 team to monitor the infrastructure
- How to deal with disasters? (earthquake, power shutdown, fire…)
- Can we externalize all this?

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What is Cloud Computing?

- Cloud computing is the on-demand delivery of compute power, database storage, applications, and other IT resources
- Through a cloud services platform with pay-as-you-go pricing
- You can provision exactly the right type and size of computing resources you need
- You can access as many resources as you need, almost instantly
- Simple way to access servers, storage, databases and a set of application services
- Amazon Web Services owns and maintains the network-connected hardware
- required for these application services, while you provision and use what you need via a web application.

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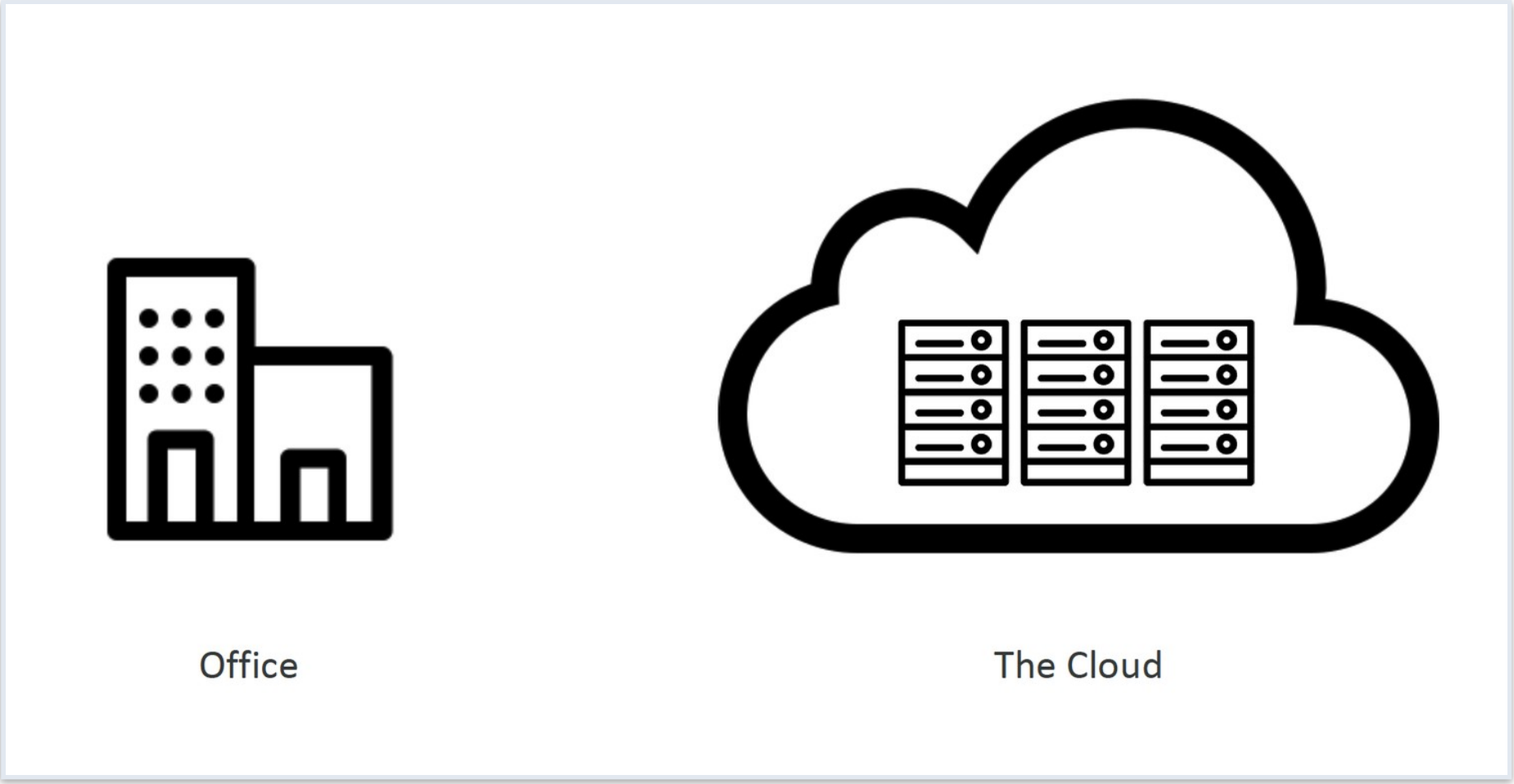
What is Cloud Computing?



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TOPIC 1

You've been using some Cloud services

Gmail

- E-mail cloud service
- Pay for ONLY your emails stored (no infrastructure, etc.)

Dropbox

- Cloud Storage Service
- Originally built on AWS

Netflix

- Built on AWS
- Video on Demand

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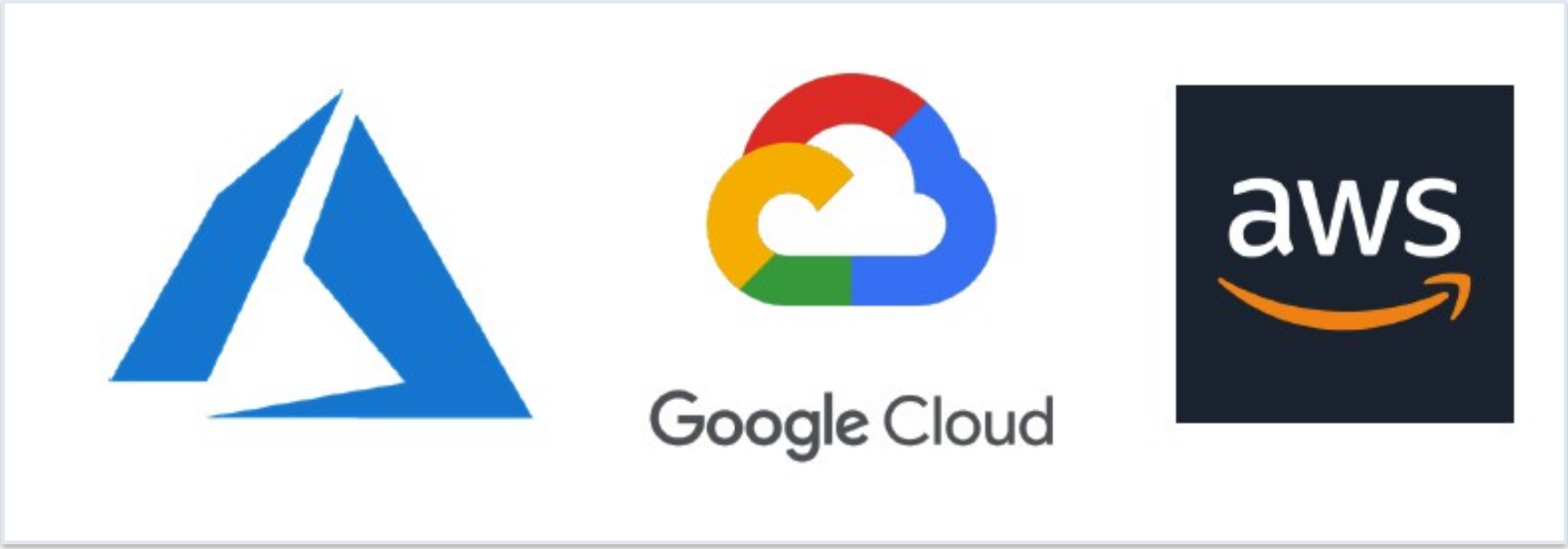
TOPIC 1

The Deployment Models of the Cloud



Private Cloud:

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The Five Characteristics of Cloud Computing

- On-demand self service:
- Users can provision resources and use them without human interaction from the service provider
- Broad network access:
- Resources available over the network, and can be accessed by diverse client platforms
- Multi-tenancy and resource pooling:
- Multiple customers can share the same infrastructure and applications with security and privacy
- Multiple customers are serviced from the same physical resources
- Rapid elasticity and scalability:
- Automatically and quickly acquire and dispose resources when needed
- Quickly and easily scale based on demand
- Measured service:
- Usage is measured, users pay correctly for what they have used

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Six Advantages of Cloud Computing

- Trade capital expense (CAPEX) for operational expense (OPEX)
- Pay On-Demand: don’t own hardware
- Reduced Total Cost of Ownership (TCO) & Operational Expense (OPEX)
- Benefit from massive economies of scale
- Prices are reduced as AWS is more efficient due to large scale
- Stop guessing capacity
- Scale based on actual measured usage
- Increase speed and agility
- Stop spending money running and maintaining data centers
- Go global in minutes: leverage the AWS global infrastructure

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Problems solved by the Cloud

- Flexibility: change resource types when needed
- Cost-Effectiveness: pay as you go, for what you us
- Scalability: accommodate larger loads by making hardware stronger or adding additional nodes
- Elasticity: ability to scale out and scale-in when needed
- High-availability and fault-tolerance: build across data centers
- Agility: rapidly develop, test and launch software applications

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Types of Cloud Computing

- Infrastructure as a Service (IaaS)
- Provide building blocks for cloud IT
- Provides networking, computers, data storage space
- Highest level of flexibility
- Easy parallel with traditional on-premises IT
- Platform as a Service (PaaS)
- Removes the need for your organization to manage the underlying infrastructure
- Focus on the deployment and management of your applications
- Software as a Service (SaaS)
- Completed product that is run and managed by the service provider

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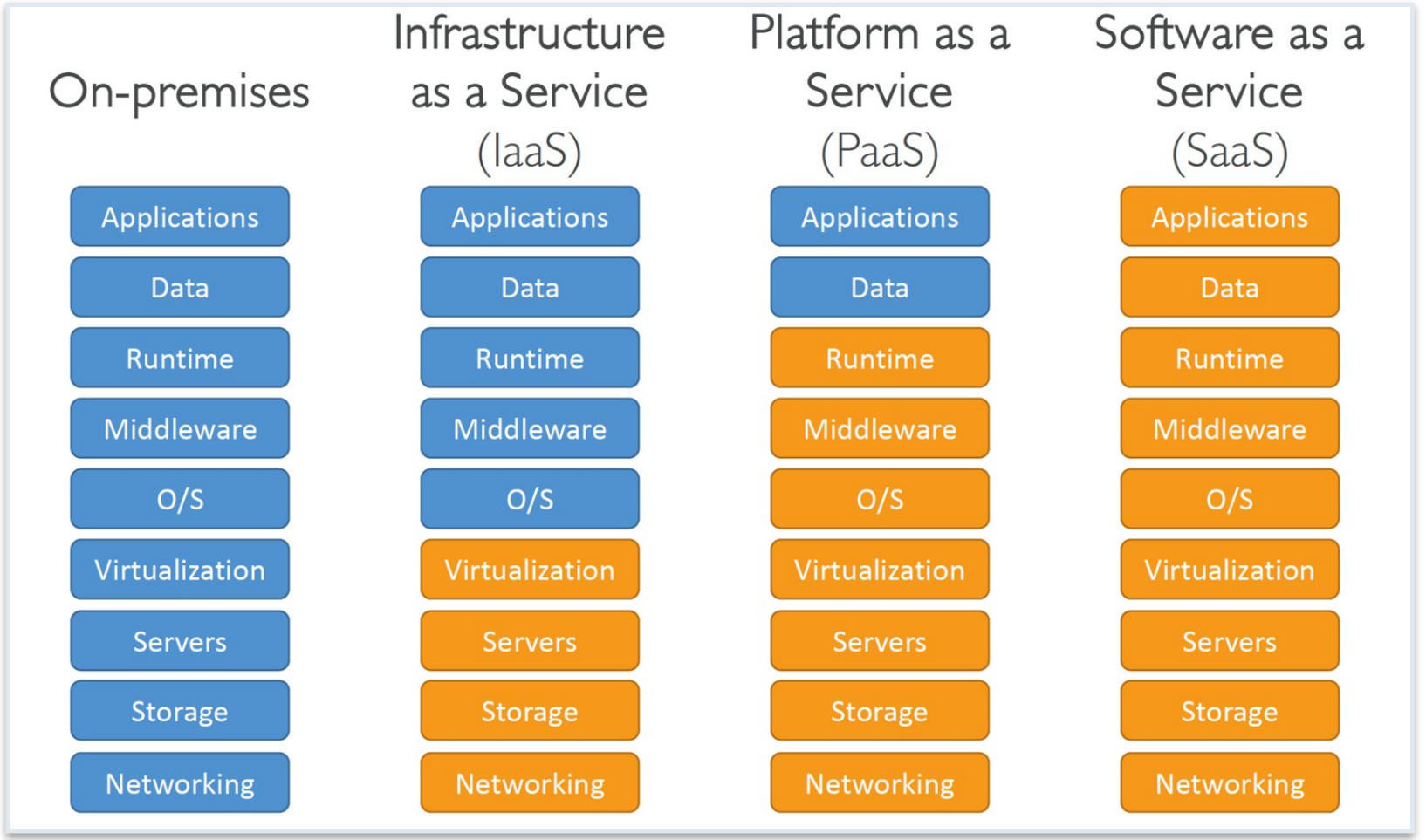
Reference



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Example of Cloud Computing Types

- Infrastructure as a Service:
- Amazon EC2 (on AWS)
- GCP, Azure, Rackspace, Digital Ocean, Linode
- Platform as a Service:
- Elastic Beanstalk (on AWS)
- Heroku, Google App Engine (GCP), Windows Azure (Microsoft)
- Software as a Service:
- Many AWS services (ex: Rekognition for Machine Learning)
- Google Apps (Gmail), Dropbox, Zoom

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TOPIC 1

Pricing of the Cloud – Quick Overview

AWS has 3 pricing fundamentals, following the pay-as-you-go pricing model

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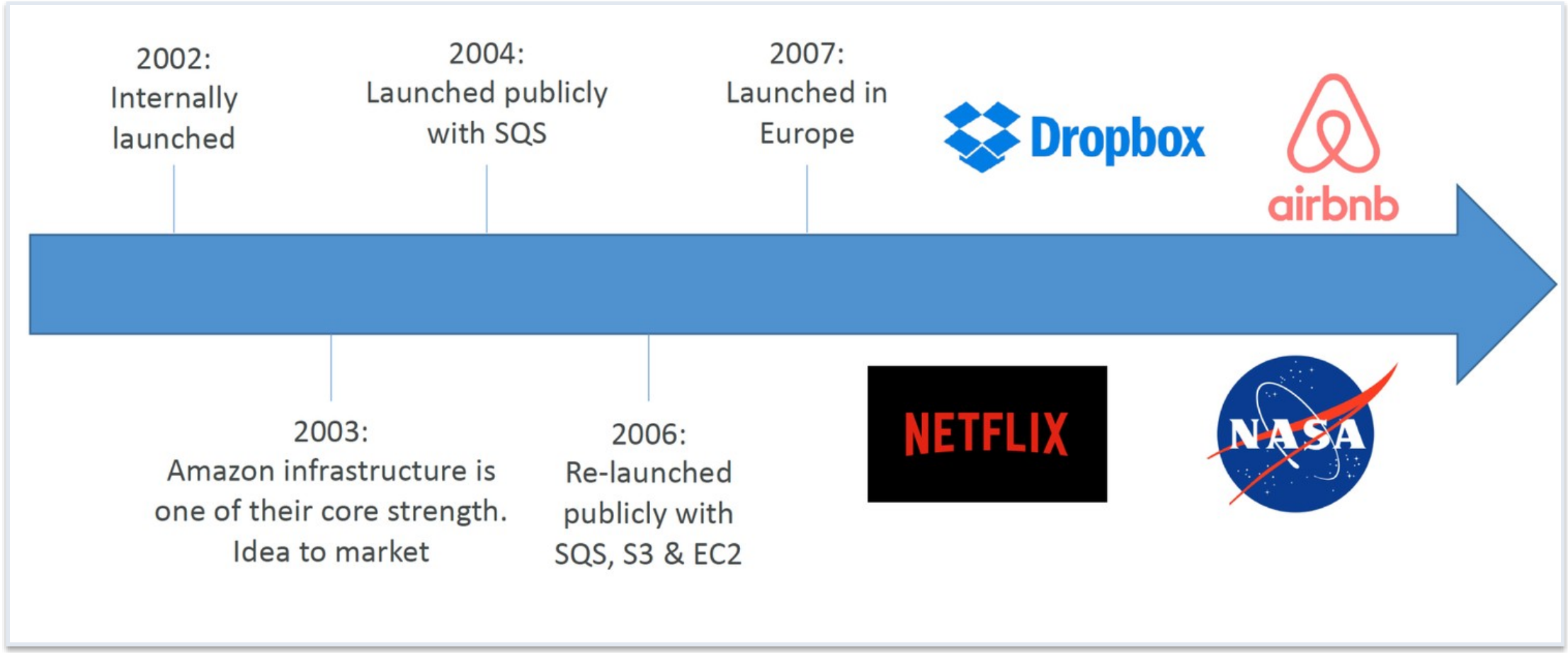
AWS Cloud History



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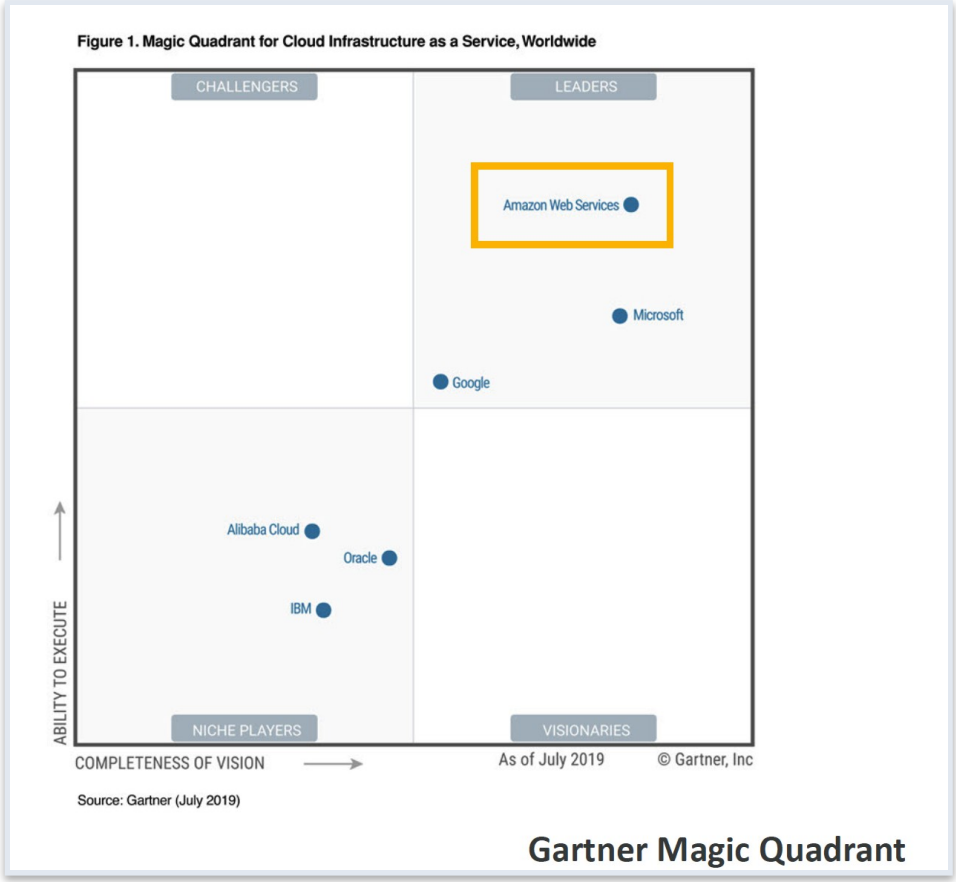
AWS Cloud Number Facts



In 2019, AWS had $35.02 billion in annual revenue

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AWS Cloud Use Cases

AWS enables you to build sophisticated, scalable applications

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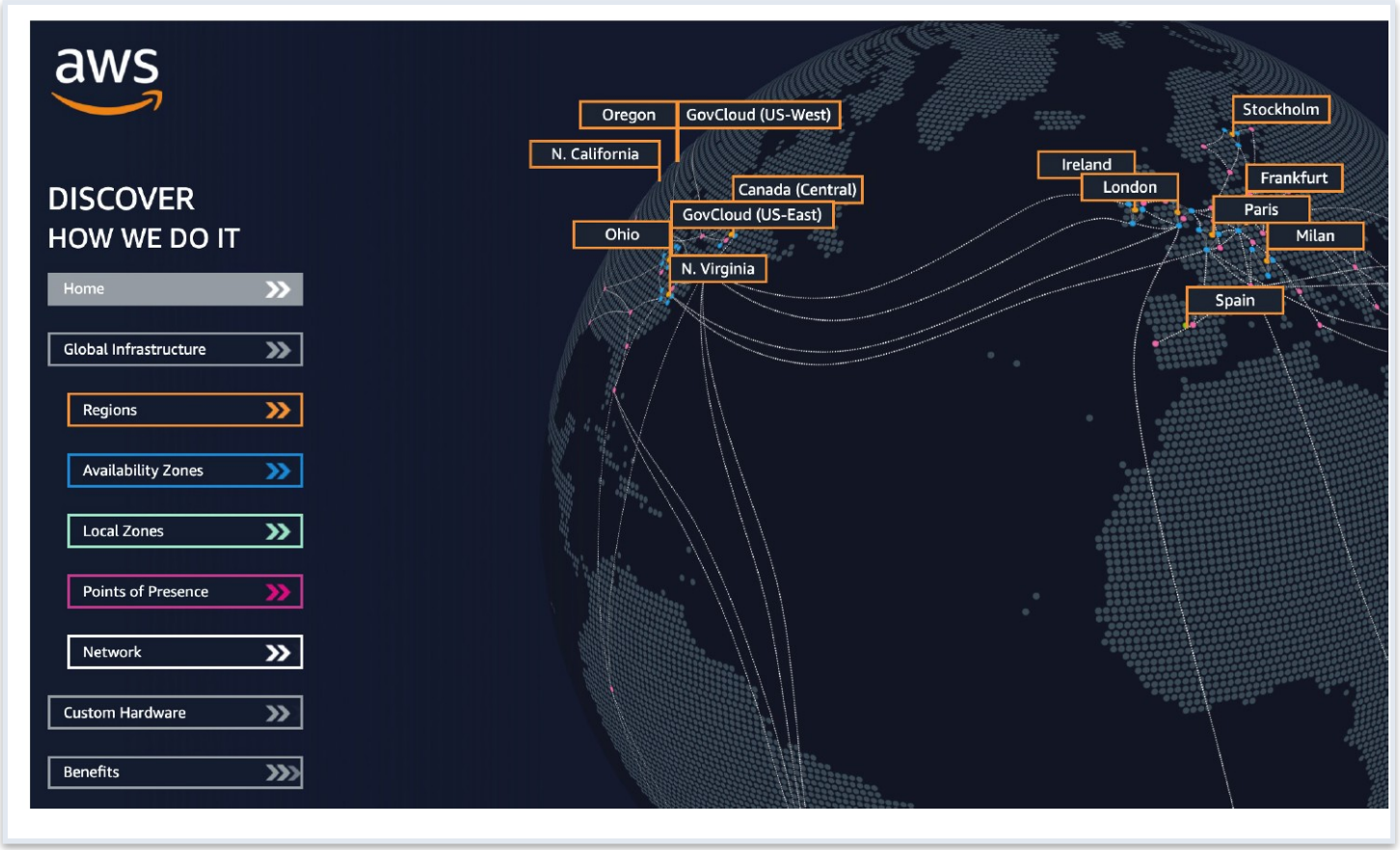
AWS Global Infrastructure



AWS Regions

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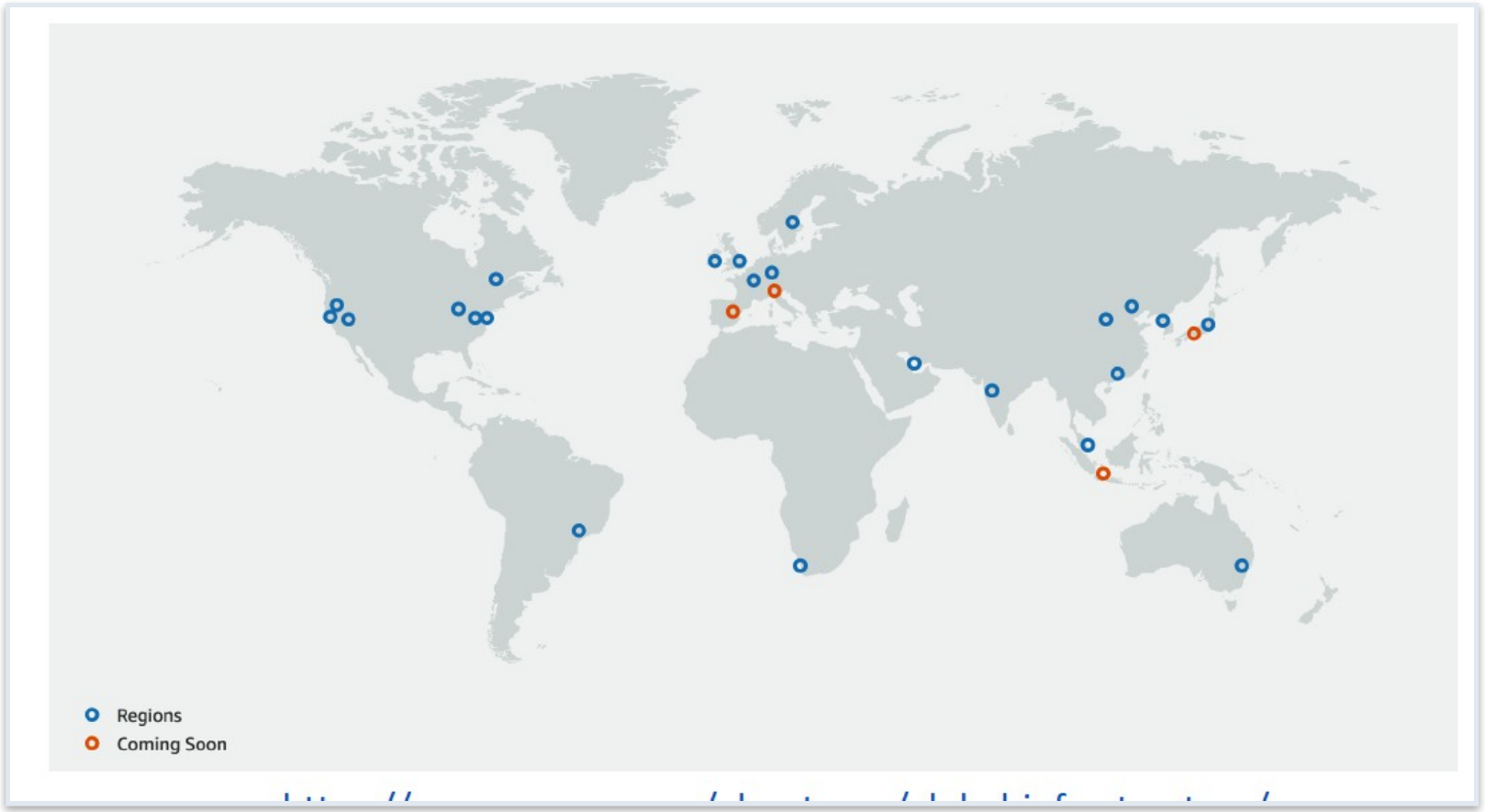
AWS Regions



AWS has Regions all around the world

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How to choose an AWS Region?

Compliance with data governance and legal requirements: data never leaves a region without your explicit permission

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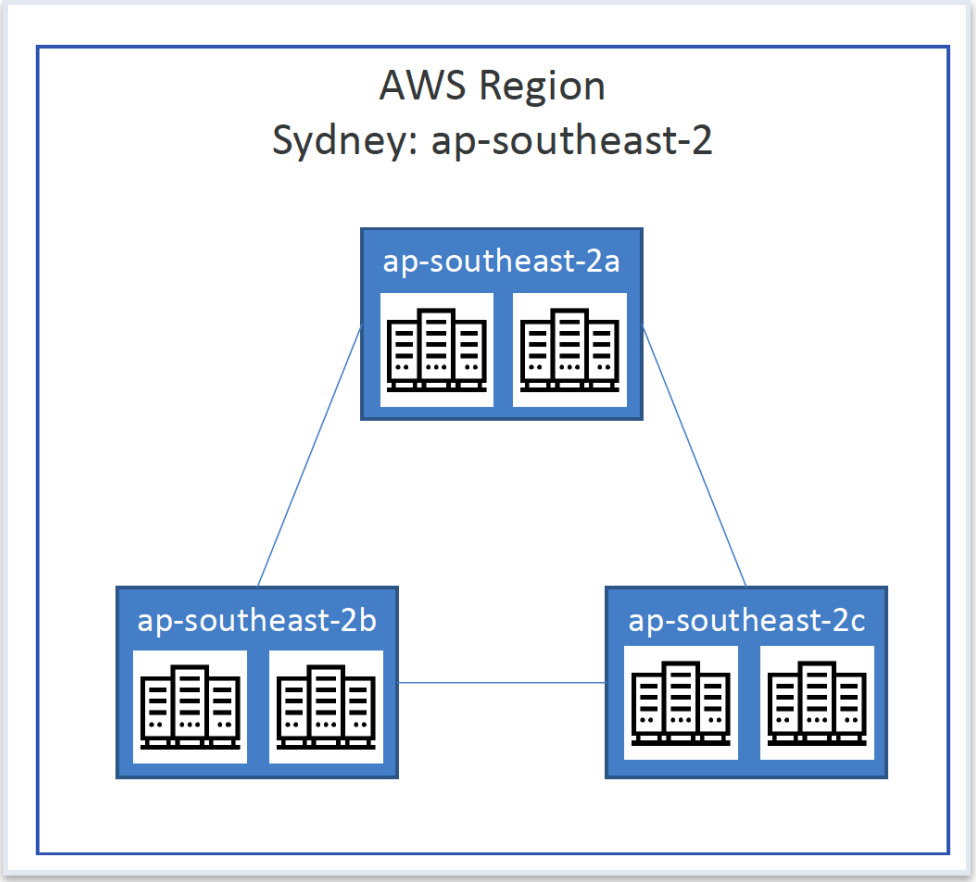
AWS Availability Zones



Each region has many availability zones (usually 3, min is 3, max is 6). Example:

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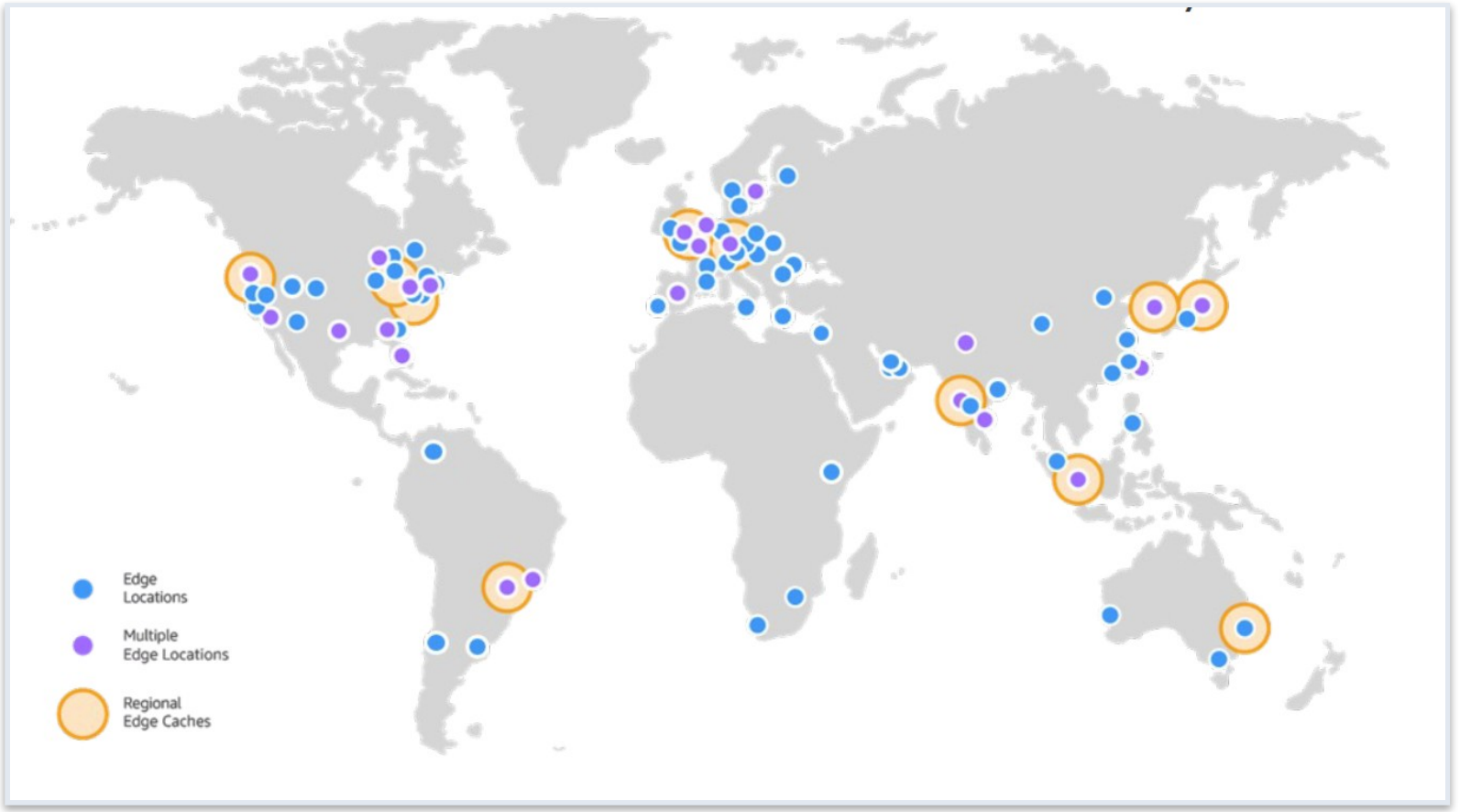
AWS Points of Presence (Edge Locations)



Amazon has 400+ Points of Presence (400+ Edge Locations & 10+

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Tour of the AWS Console

- AWS has Global Services:
- Identity and Access Management (IAM)
- Route 53 (DNS service)
- CloudFront (Content Delivery Network)
- WAF (Web Application Firewall)
- Most AWS services are Region-scoped:
- Amazon EC2 (Infrastructure as a Service)
- Elastic Beanstalk (Platform as a Service)
- Lambda (Function as a Service)
- Rekognition (Software as a Service)
- Region Table: https://aws.amazon.com/about-aws/global-infrastructure/regional-product-services

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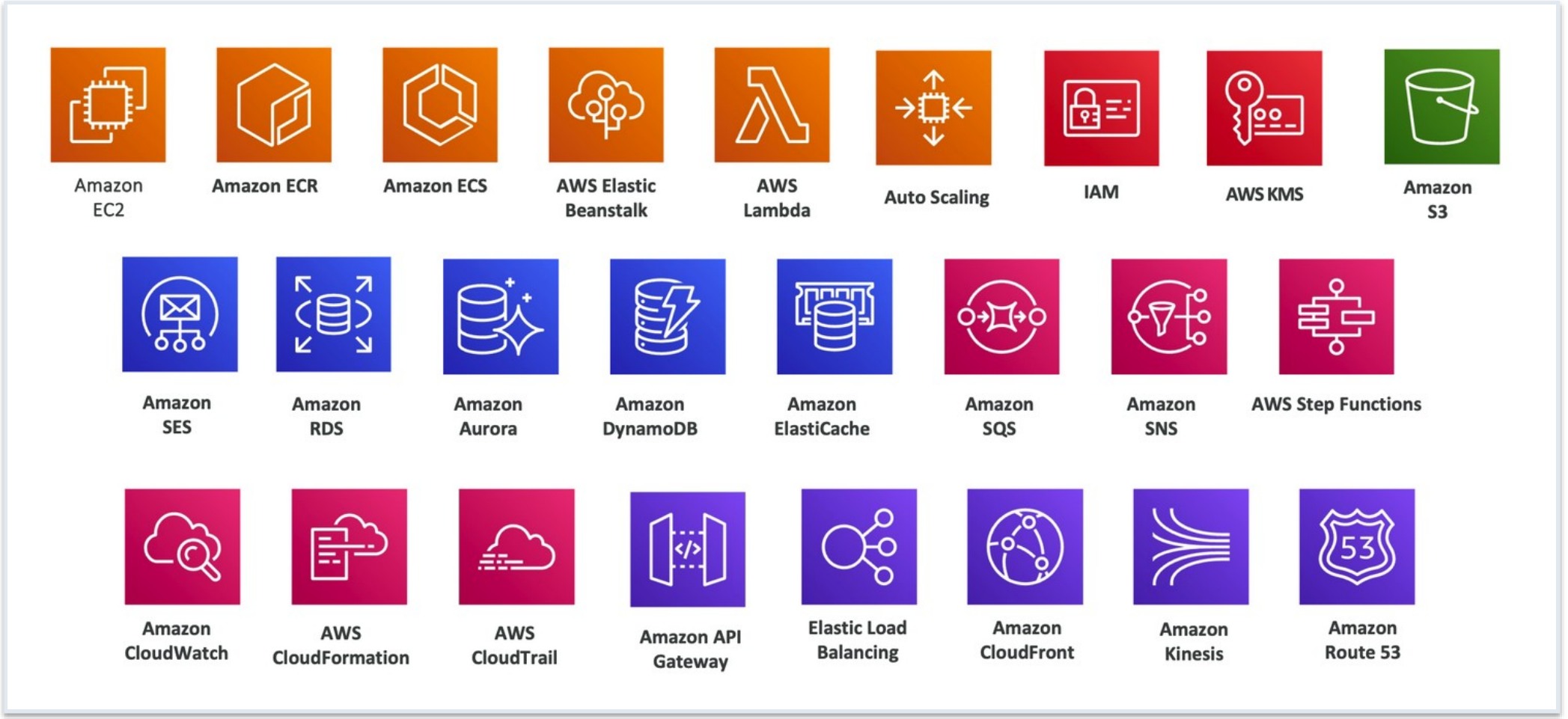
AWS Services (>200)



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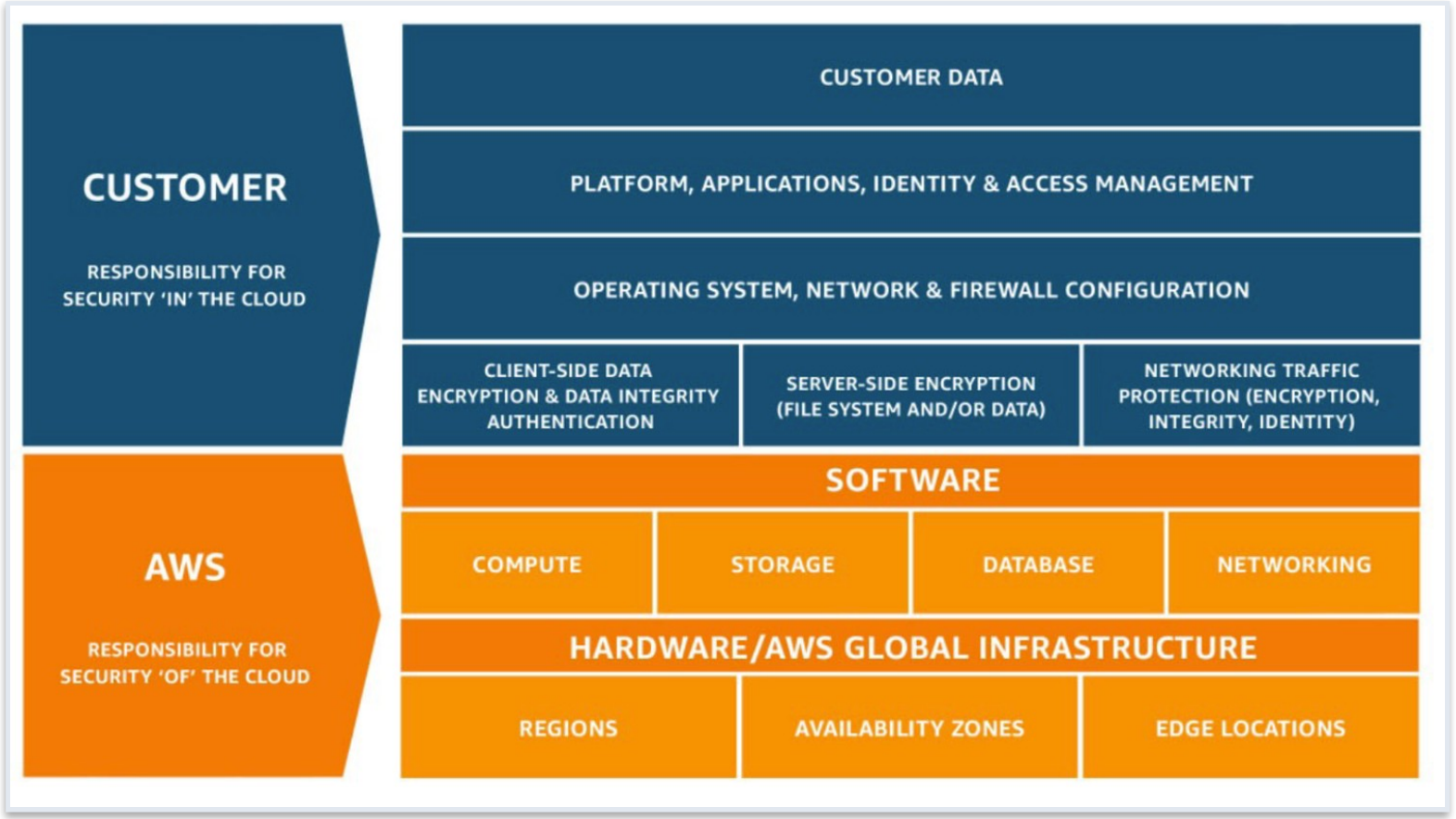
Shared Responsibility Model diagram



https://aws.amazon.com/compliance/shared-responsibility-model/

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AWS Acceptable Use Policy

1. https://aws.amazon.com/aup/
2. No Illegal, Harmful, or Offensive Use or Content
3. No Security Violations
4. No Network Abuse
5. No E-Mail or Other Message Abuse

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TOPIC 2

Security and Compliance

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IAM

IAM

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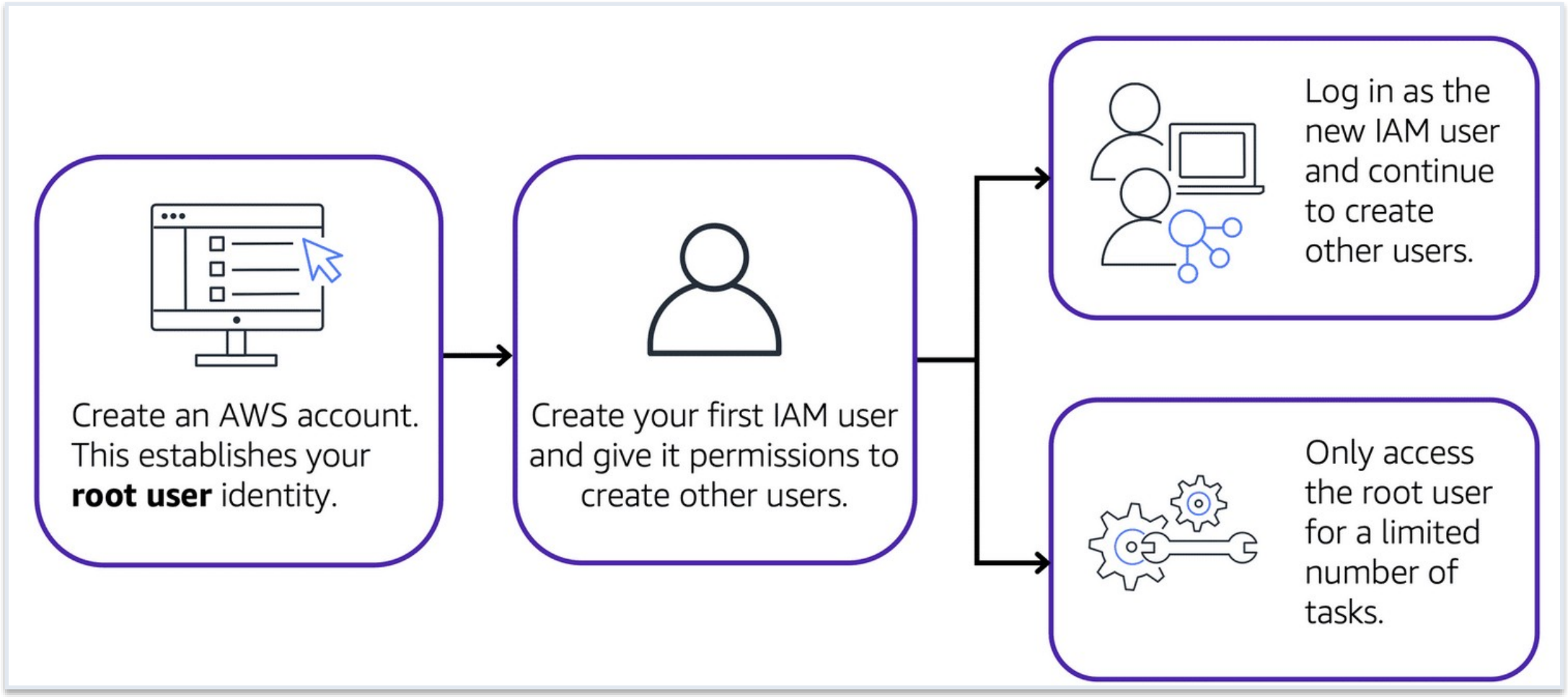
AWS account root user



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IAM: Users & Groups

1. https://aws.amazon.com/aup/
2. No Illegal, Harmful, or Offensive Use or Content
3. No Security Violations
4. No Network Abuse
5. No E-Mail or Other Message Abuse

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IAM: Users & Groups

IAM = Identity and Access Management, Global service

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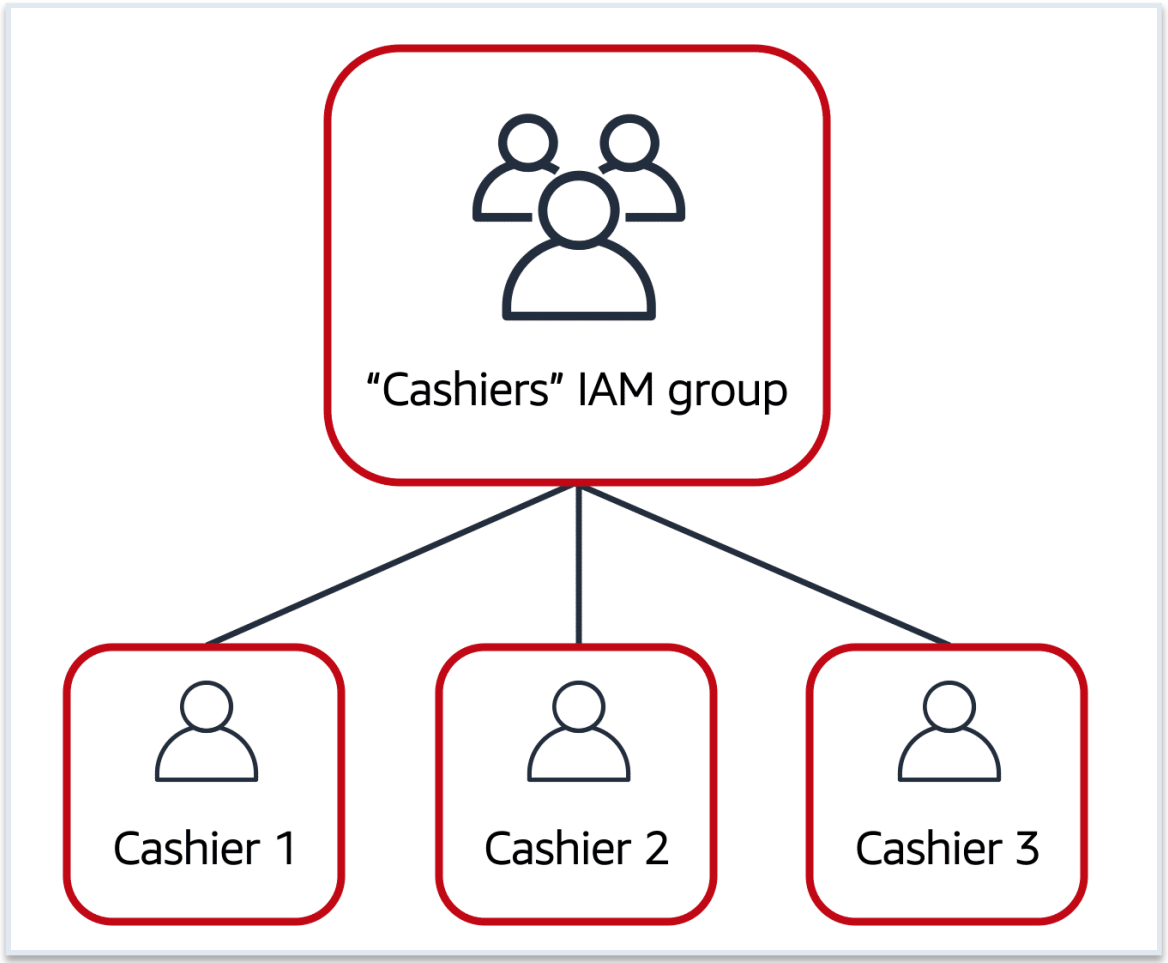
IAM groups



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IAM roles

An IAM role is an identity that you can assume to gain temporary access to permissions.

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IAM: Permissions

```
{
    "Version": "2012-10-17",
    "Statement": [
        {
            "Effect": "Allow",
            "Action": "ec2:Describe*",
            "Resource": "*"
        },
        {
            "Effect": "Allow",
            "Action": "elasticloadbalancing:Describe*",
            "Resource": "*"
        },
        {
            "Effect": "Allow",
            "Action": [
                "cloudwatch:ListMetrics",
                "cloudwatch:GetMetricStatistics",
                "cloudwatch:Describe*"
            ],
            "Resource": "*"
        }
    ]
}
```

Users or Groups can be assigned JSON documents called policies

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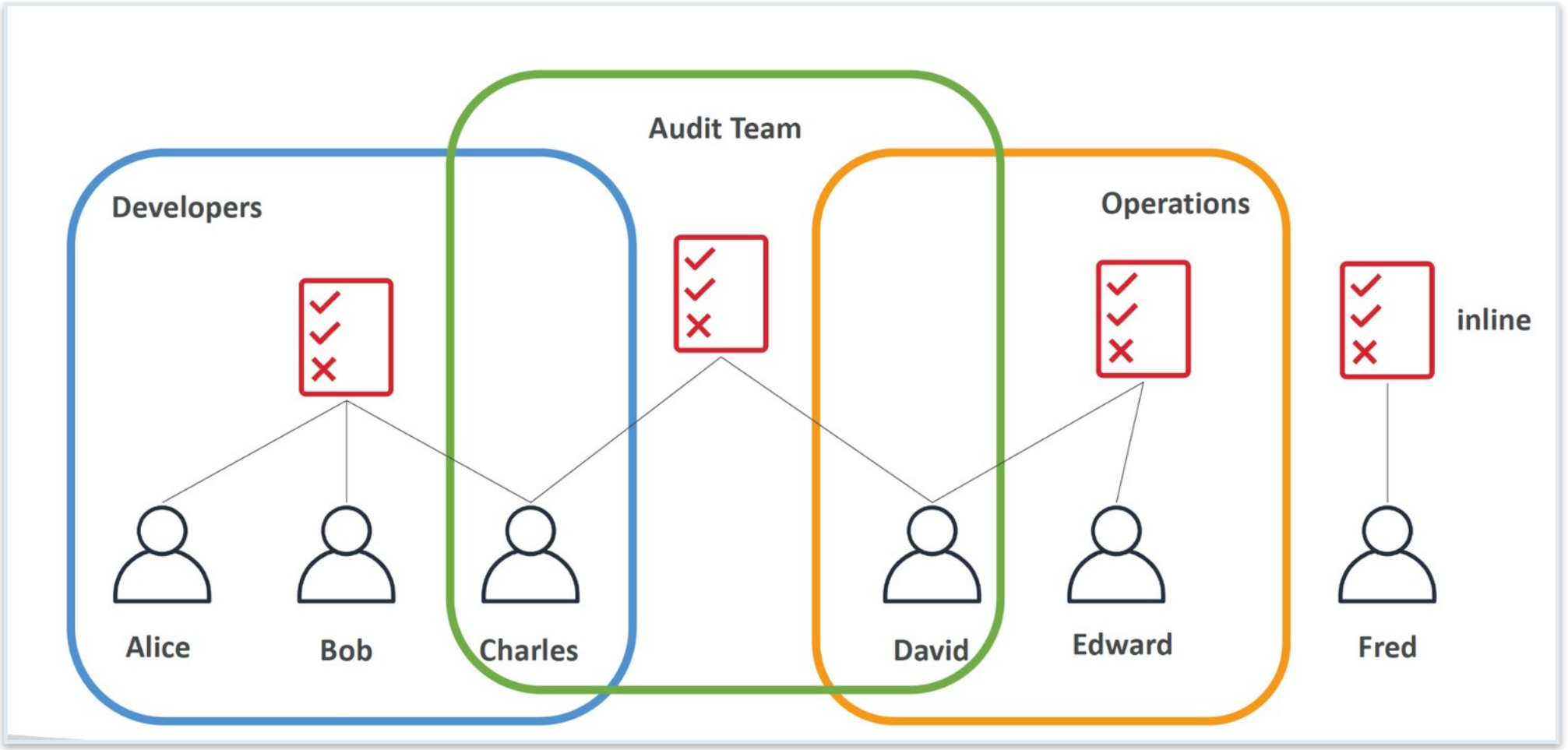
IAM Policies inheritance



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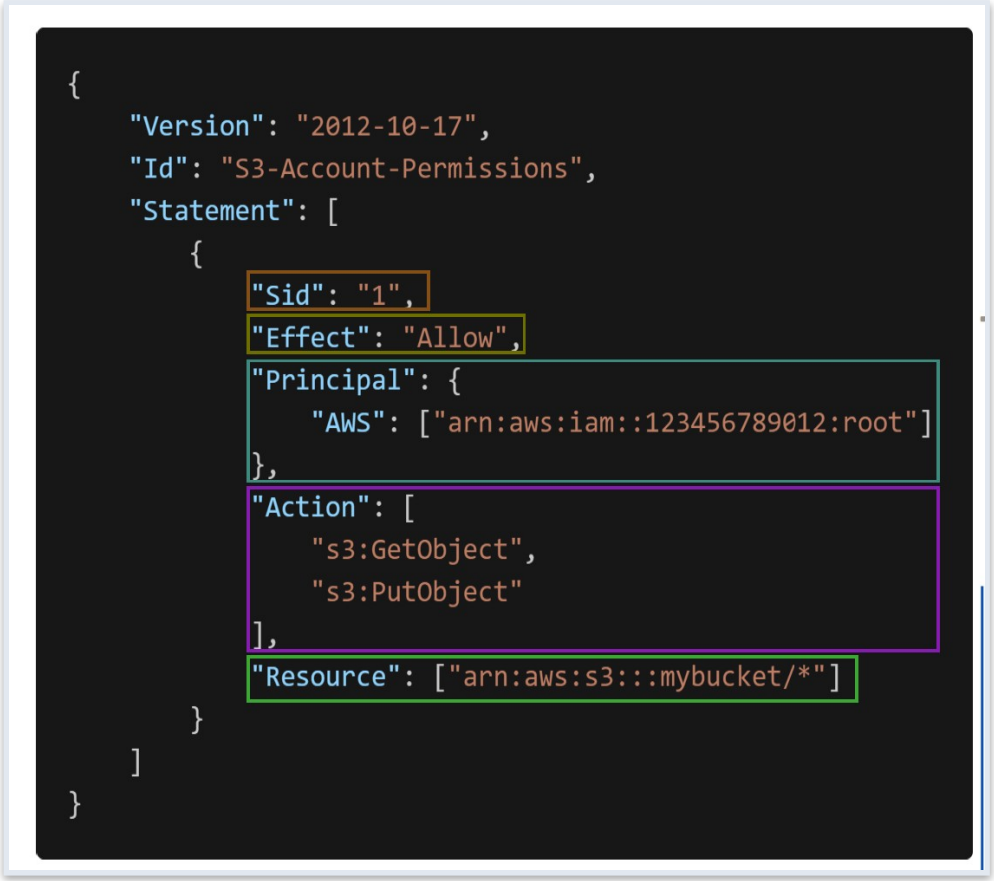
IAM Policies Structure



Consists of

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Example: IAM policy

```
{
   "Version": "2012-10-17",
   "Statement": {
      "Effect": "Allow",
      "Action": "s3:ListObject",
      "Resource": "arn:aws:s3:::
AWSDOC-EXAMPLE-BUCKET"
   }
}
```


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IAM – Password Policy

- Strong passwords = higher security for your account
- In AWS, you can setup a password policy:
- Set a minimum password length
- Require specific character types:
- including uppercase letters
- lowercase letters
- numbers
- non-alphanumeric characters
- Allow all IAM users to change their own passwords
- Require users to change their password after some time (password expiration)
- Prevent password re-use

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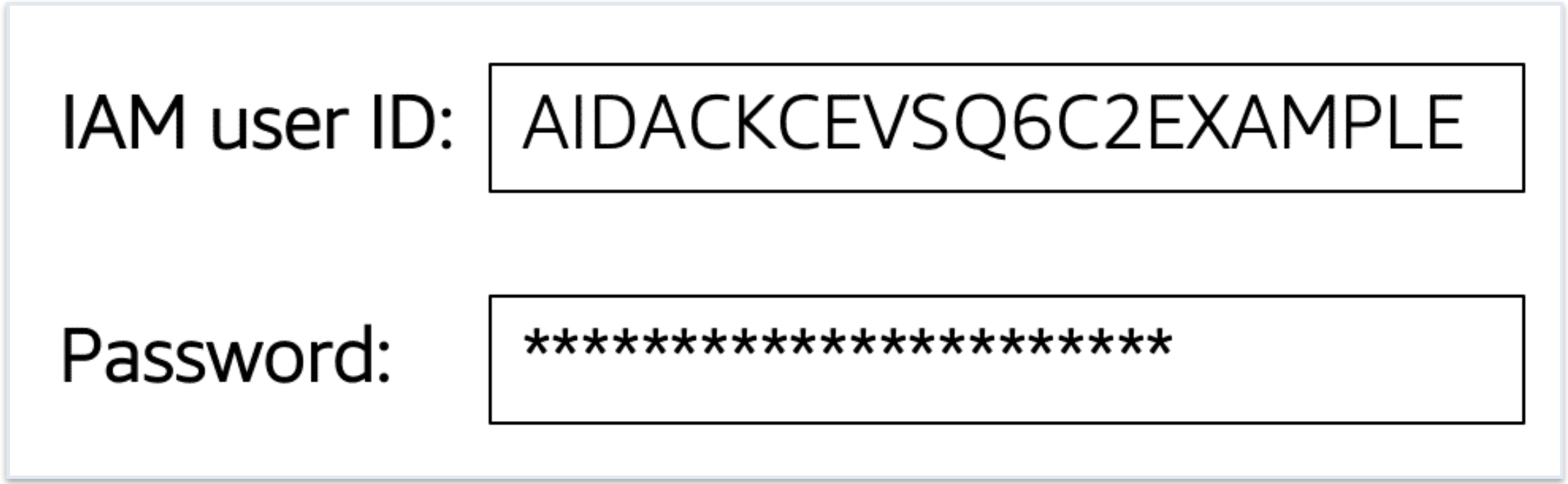
Multi-factor authentication



In IAM, multi-factor authentication (MFA) provides an extra layer of security for your AWS account.

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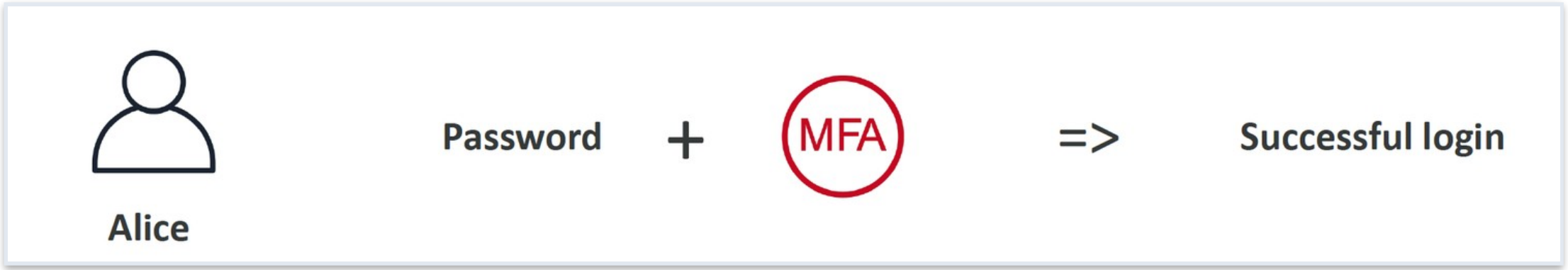
Multi Factor Authentication - MFA



Users have access to your account and can possibly change configurations or delete resources in your AWS account

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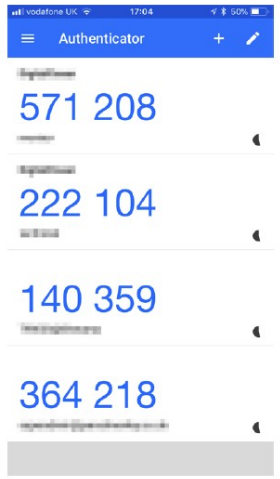
MFA devices options in AWS

Virtual MFA device



Google Authenticator
(phone only)

Authy
(phone only)

Support for multiple tokens on a single device.

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MFA devices options in AWS

Hardware Key Fob MFA Device

Provided by Gemalto (3rd party)

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How can users access AWS ?

- To access AWS, you have three options:
- AWS Management Console (protected by password + MFA)
- AWS Command Line Interface (CLI): protected by access keys
- AWS Software Developer Kit (SDK) - for code: protected by access keys
- Access Keys are generated through the AWS Console
- Users manage their own access keys
- Access Keys are secret, just like a password. Don’t share them
- Access Key ID ~= username
- Secret Access Key ~= password

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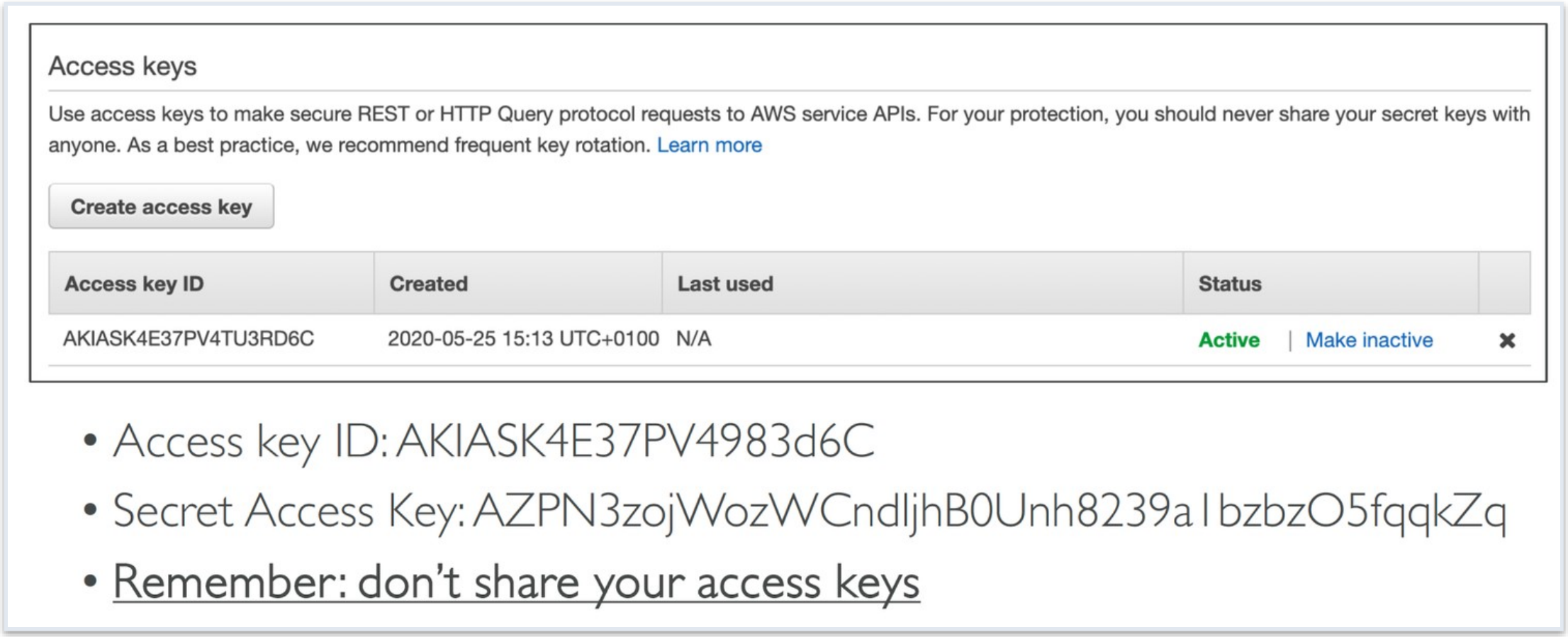
Example (Fake) Access Keys



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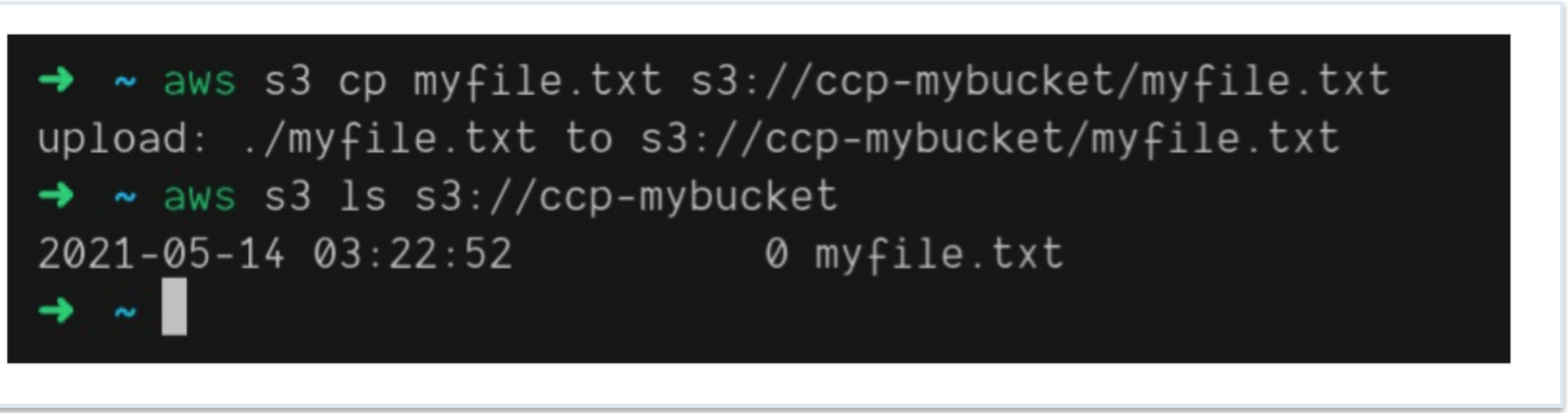
What's the AWS CLI?



A tool that enables you to interact with AWS services using commands in your command-line shell

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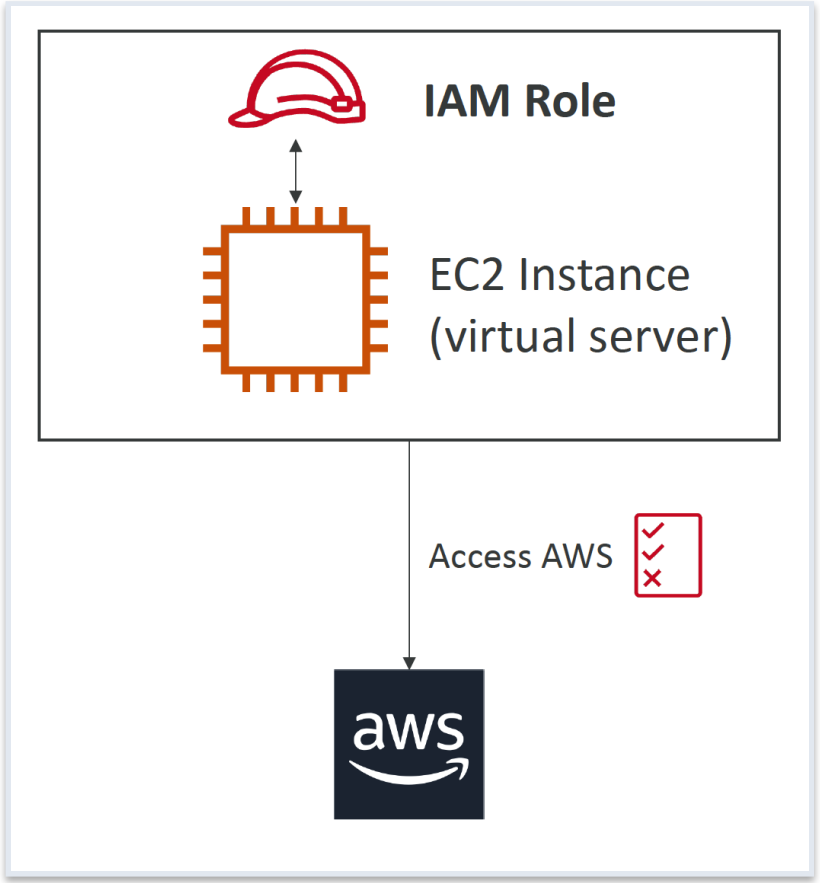
IAM Roles for Services



Some AWS service will need to perform actions on your behalf

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IAM Security Tools

- IAM Credentials Report (account-level)
- a report that lists all your account's users and the status of their various credentials
- IAM Access Advisor (user-level)
- Access advisor shows the service permissions granted to a user and when thoseservices were last accessed.
- You can use this information to revise your policies

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IAM Guidelines & Best Practices

- Don’t use the root account except for AWS account setup
- One physical user = One AWS user
- Assign users to groups and assign permissions to groups
- Create a strong password policy
- Use and enforce the use of Multi Factor Authentication (MFA)
- Create and use Roles for giving permissions to AWS services
- Use Access Keys for Programmatic Access (CLI / SDK)
- Audit permissions of your account using IAM Credentials Report & IAM Access Advisor
- Never share IAM users & Access Keys

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Shared Responsibility Model for IAM

- Infrastructure (global network security)
- Configuration and vulnerability analysis
- Compliance validation

You

- Users, Groups, Roles, Policies management and monitoring
- Enable MFA on all accounts
- Rotate all your keys often
- Use IAM tools to apply appropriate permissions
- Analyze access patterns & review permissions

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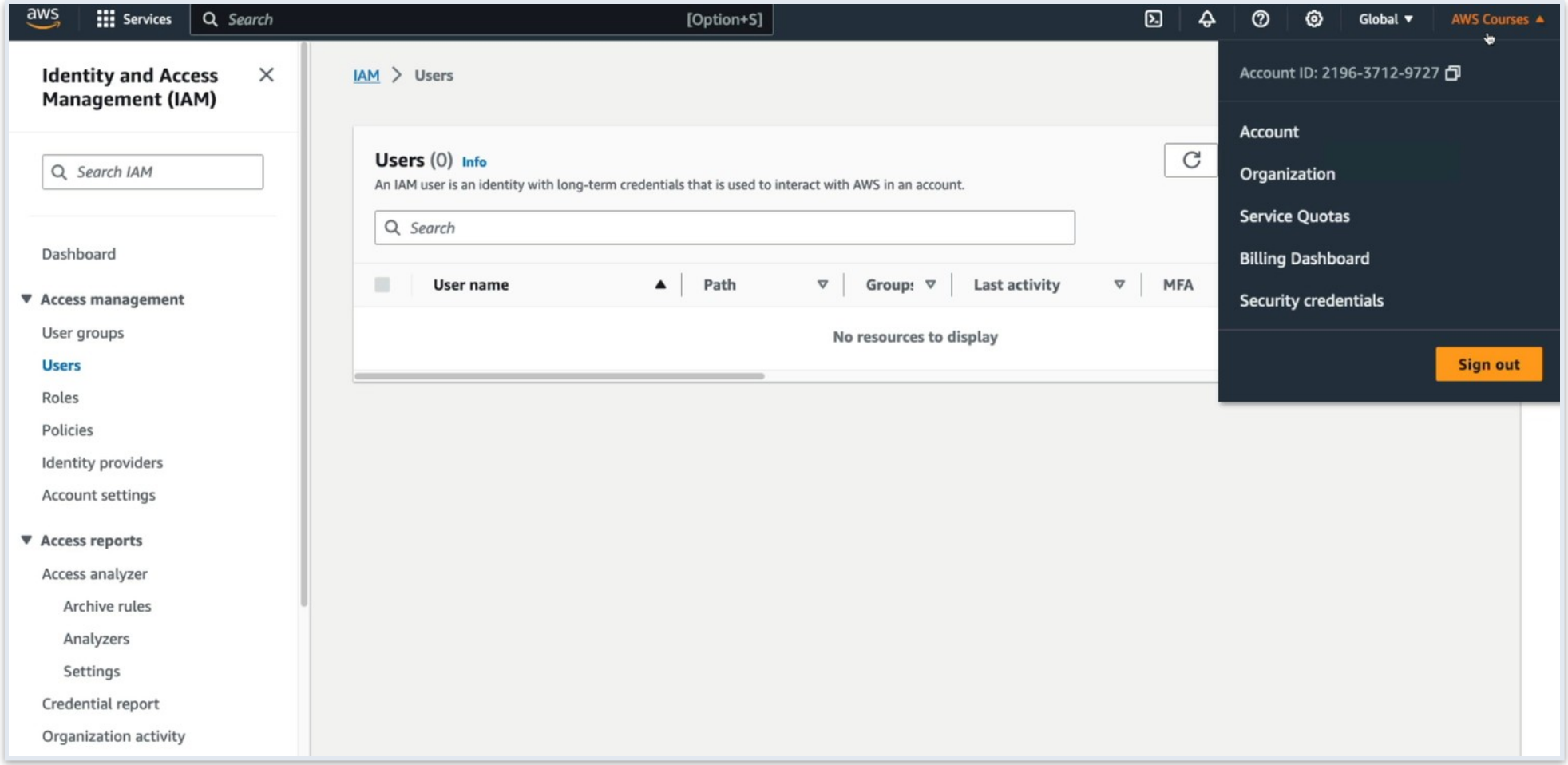
Activity: IAM Hand On



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IAM Section – Summary

- Users: mapped to a physical user, has a password for AWS Console
- Groups: contains users only
- Policies: JSON document that outlines permissions for users or groups
- Roles: for EC2 instances or AWS services
- Security: MFA + Password Policy
- AWS CLI: manage your AWS services using the command-line
- AWS SDK: manage your AWS services using a programming language
- Access Keys: access AWS using the CLI or SDK
- Audit: IAM Credential Reports & IAM Access Advisor

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TOPIC 3

AWS Compute Services

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EC2

EC2

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Amazon EC2

- EC2 is one of the most popular of AWS' offering
- EC2 = Elastic Compute Cloud = Infrastructure as a Service
- It mainly consists in the capability of :
- Renting virtual machines (EC2)
- Storing data on virtual drives (EBS)
- Distributing load across machines (ELB)
- Scaling the services using an auto-scaling group (ASG)
- Knowing EC2 is fundamental to understand how the Cloud works

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EC2 sizing & configuration options

- Operating System (OS): Linux, Windows or Mac OS
- How much compute power & cores (CPU)
- How much random-access memory (RAM)
- How much storage space:
- Network-attached (EBS & EFS)
- hardware (EC2 Instance Store)
- Network card: speed of the card, Public IP address
- Firewall rules: security group
- Bootstrap script (configure at first launch): EC2 User Data

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EC2 User Data

- It is possible to bootstrap our instances using an EC2 User data script.
- bootstrapping means launching commands when a machine starts
- That script is only run once at the instance first start
- EC2 user data is used to automate boot tasks such as:
- Installing updates
- Installing software
- Downloading common files from the internet
- Anything you can think of
- The EC2 User Data Script runs with the root user

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Activity: Launching an EC2 Instance running Linux

1. We'll be launching our first virtual server using the AWS Console
2. We'll get a first high-level approach to the various parameters
3. We'll see that our web server is launched using EC2 user data
4. We'll learn how to start / stop / terminate our instance.

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EC2 Instance Types - Overview

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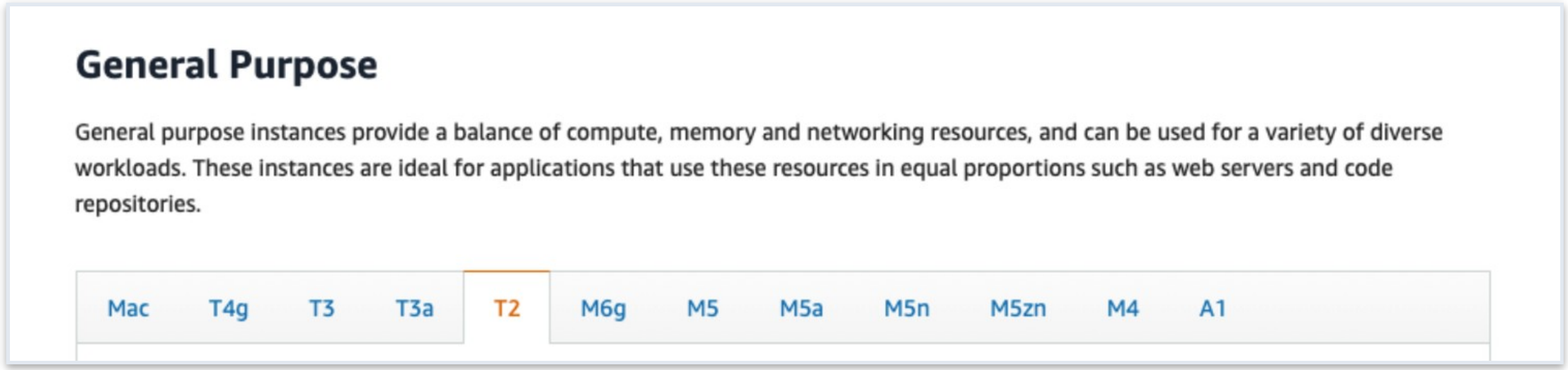
EC2 Instance Types – General Purpose



Great for a diversity of workloads such as web servers or code repositories

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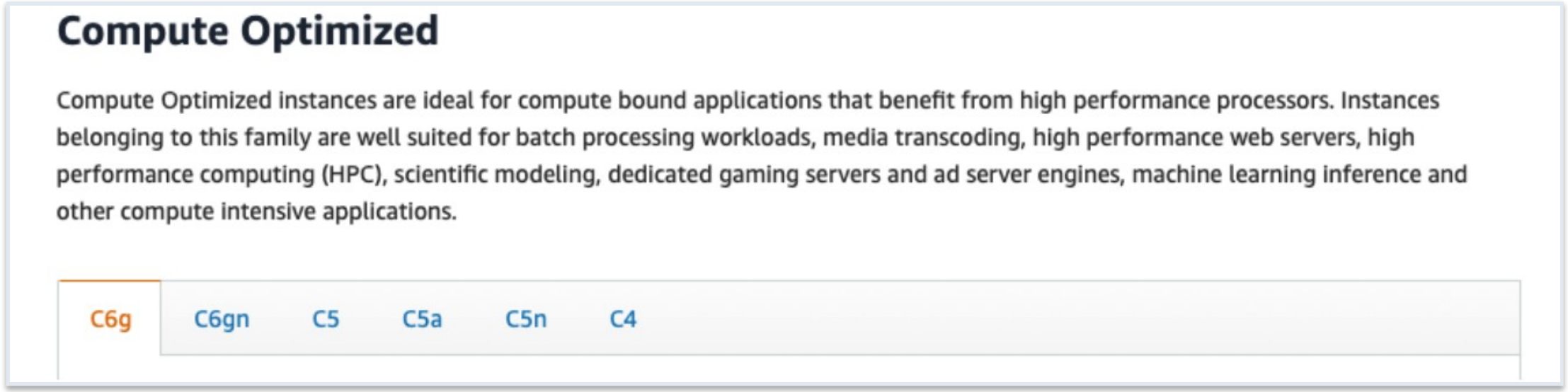
EC2 Instance Types – Compute Optimized



Great for compute-intensive tasks that require high performance processors:

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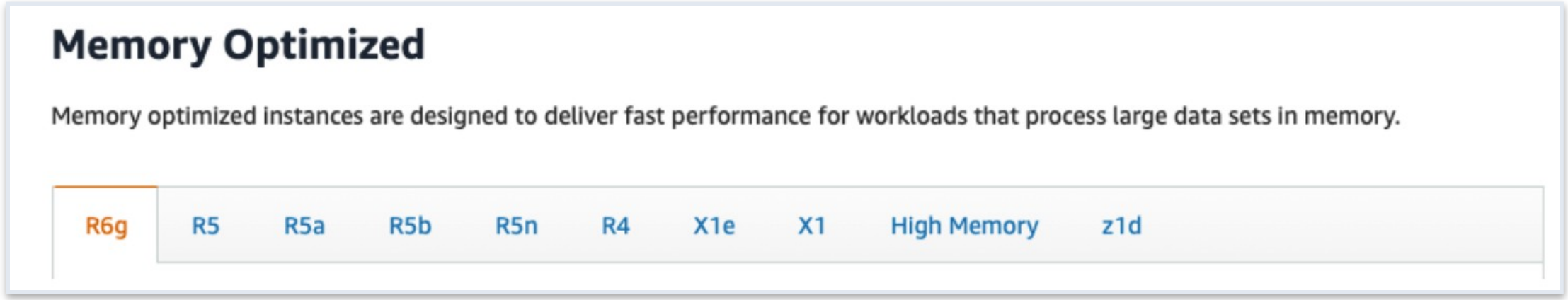
EC2 Instance Types – Memory Optimized



Fast performance for workloads that process large data sets in memory

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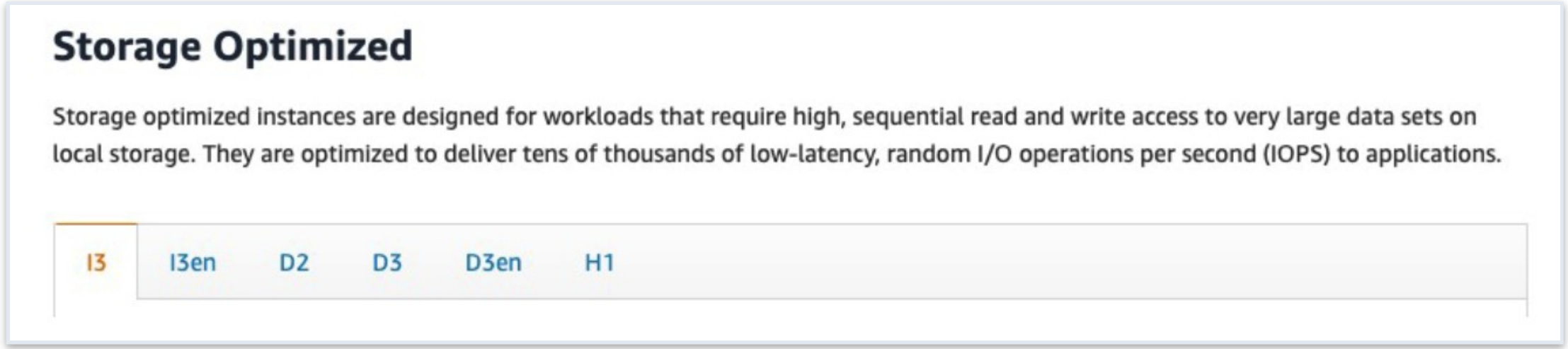
EC2 Instance Types – Storage Optimized



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EC2 Instance Types: example

Instance	vCPU	Mem (GiB)	Storage	Network Performance	EBS Bandwidth (Mbps)
t2.micro	1	1	EBS-Only	Low to Moderate	
t2.xlarge	4	16	EBS-Only	Moderate	
c5d.4xlarge	16	32	1 x 400 NVMe SSD	Up to 10 Gbps	4,750
r5.16xlarge	64	512	EBS Only	20 Gbps	13,600
m5.8xlarge	32	128	EBS Only	10 Gbps	6,800

t2.micro is part of the AWS free tier (up to 750 hours per month)

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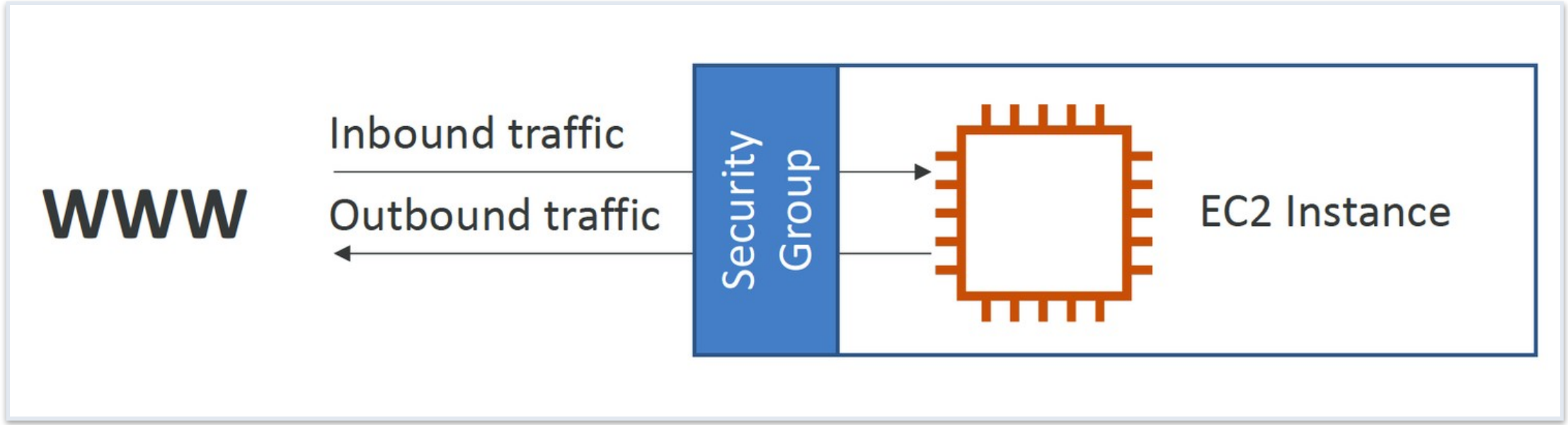
Introduction to Security Groups



Security Groups are the fundamental of network security in AWS

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Security Groups Deeper Dive

Type	Protocol	Port Range	Source	Description
HTTP	TCP	80	0.0.0.0/0	test http page
SSH	TCP	22	122.149.196.85/32	
Custom TCP Rule	TCP	4567	0.0.0.0/0	java app

Security groups are acting as a “firewall” on EC2 instances

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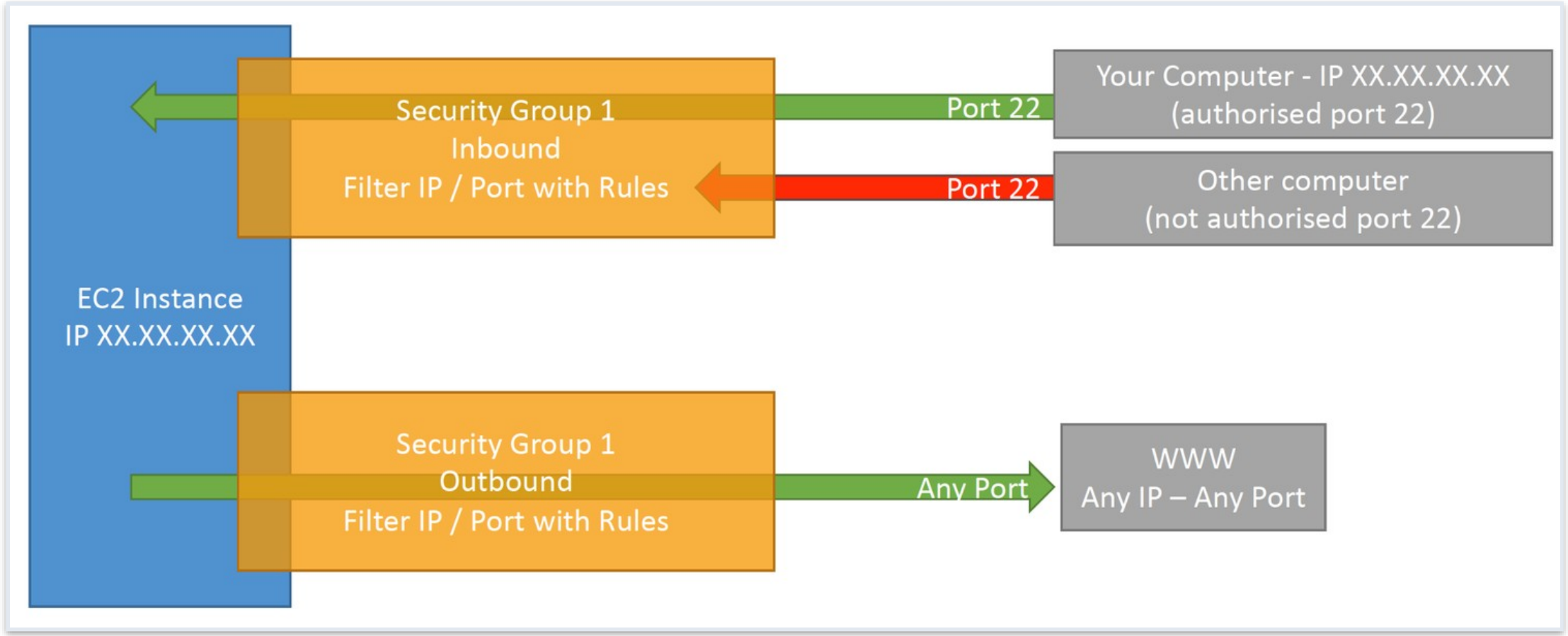
Security Groups Diagram



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Security Groups Good to know

- Can be attached to multiple instances
- Locked down to a region / VPC combination
- Does live “outside” the EC2 – if traffic is blocked the EC2 instance won’t see it
- It’s good to maintain one separate security group for SSH access
- If your application is not accessible (time out), then it’s a security group issue
- If your application gives a “connection refused“ error, then it’s an application error or it’s not launched
- All inbound traffic is blocked by default
- All outbound traffic is authorised by default

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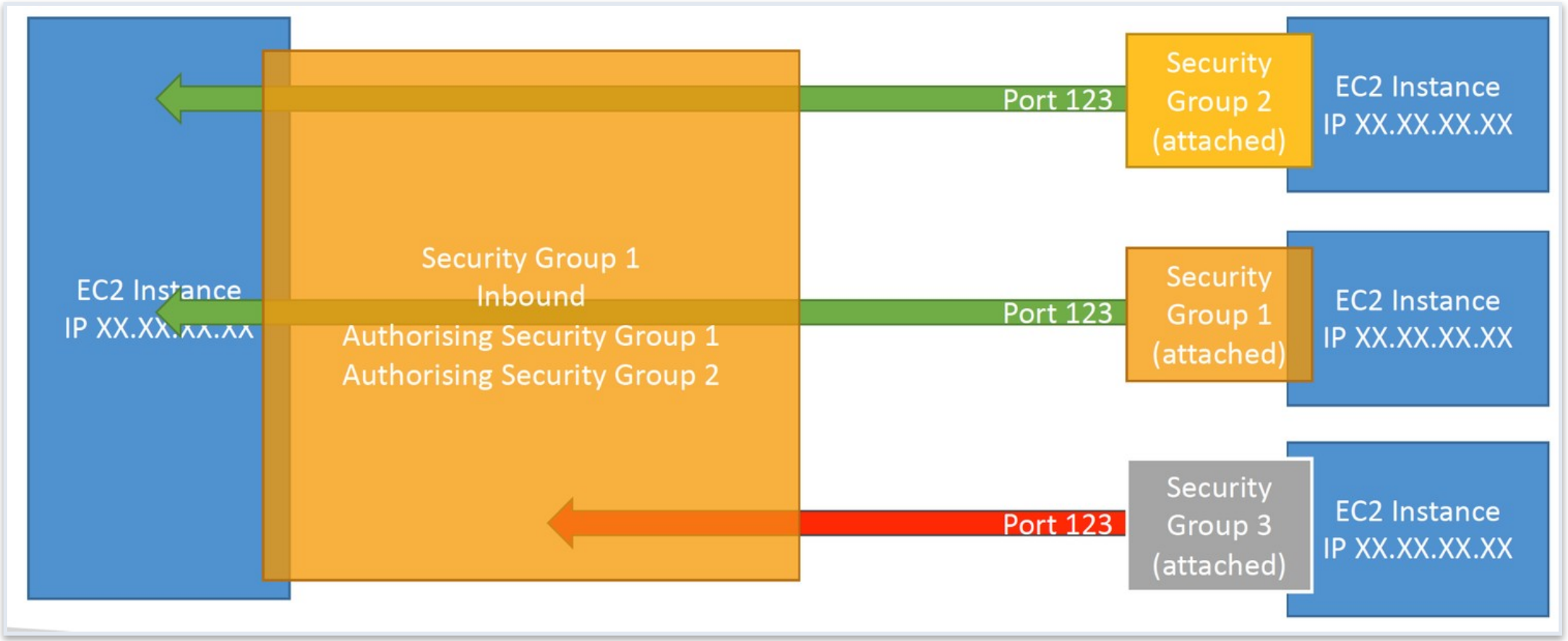
Referencing other security groups



Diagram

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Classic Ports to know

1. 22 = SSH (Secure Shell) - log into a Linux instance
2. 21 = FTP (File Transfer Protocol) – upload files into a file share
3. 22 = SFTP (Secure File Transfer Protocol) – upload files using SSH
4. 80 = HTTP – access unsecured websites
5. 443 = HTTPS – access secured websites
6. 3389 = RDP (Remote Desktop Protocol) – log into a Windows instance

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SSH Summary Table

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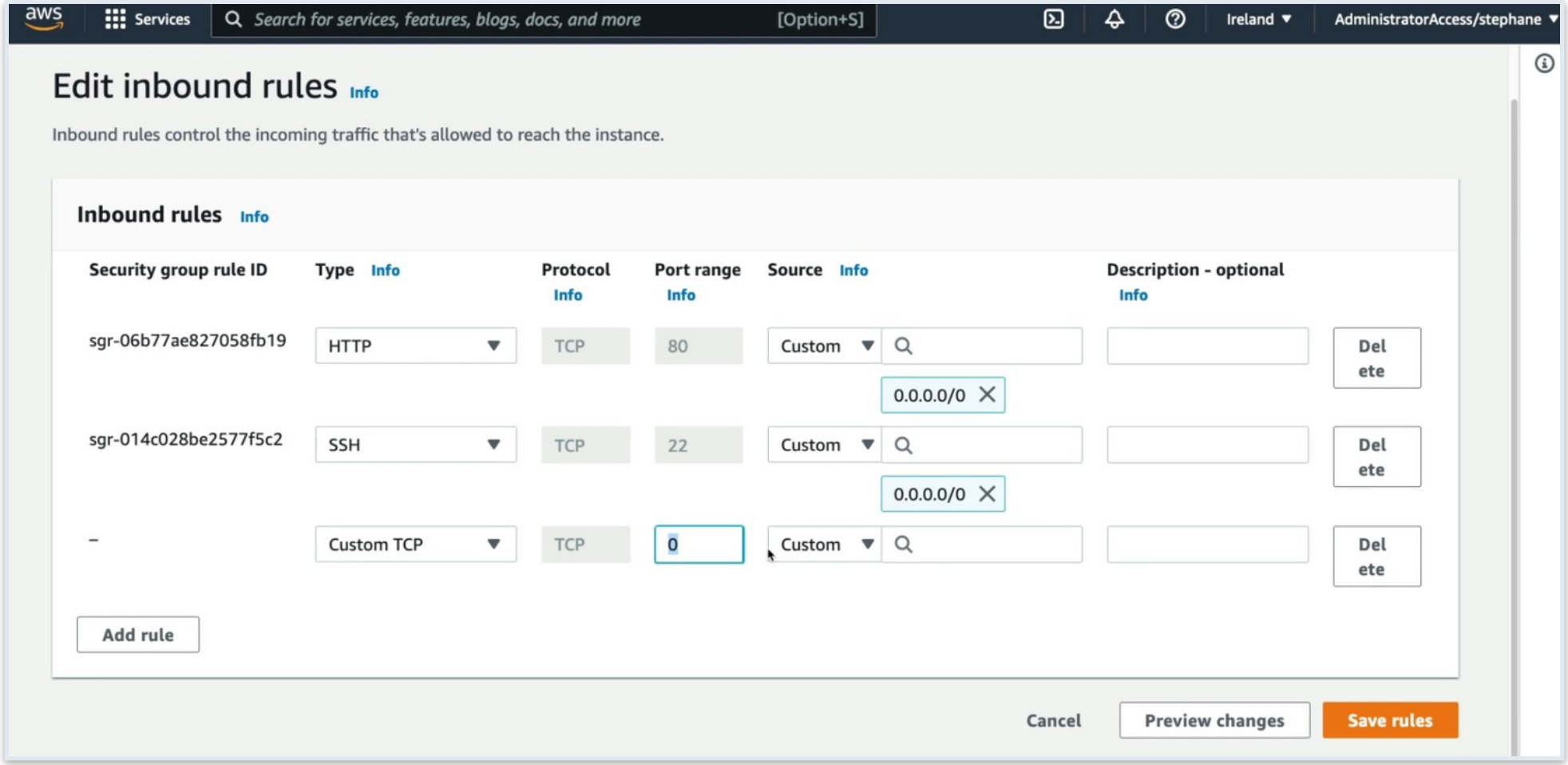
Activity: Security Tools Hand On



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EC2 Instance Connect

1. Connect to your EC2 instance within your browser
2. No need to use your key file that was downloaded
3. The “magic” is that a temporary key is uploaded onto EC2 by AWS
4. Works only out-of-the-box with Amazon Linux 2
5. Need to make sure the port 22 is still opened!

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EC2 Instances Purchasing Options

- On-Demand Instances – short workload, predictable pricing, pay by second
- Reserved (1 & 3 years)
- Reserved Instances – long workloads
- Convertible Reserved Instances – long workloads with flexible instances
- Savings Plans (1 & 3 years) –commitment to an amount of usage, long workload
- Spot Instances – short workloads, cheap, can lose instances (less reliable)
- Dedicated Hosts – book an entire physical server, control instance placement
- Dedicated Instances – no other customers will share your hardware
- Capacity Reservations – reserve capacity in a specific AZ for any duration

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EC2 On Demand

- Pay for what you use:
- Linux or Windows - billing per second, after the first minute
- All other operating systems - billing per hour
- Has the highest cost but no upfront payment
- No long-term commitment
- Recommended for short-term and un-interrupted workloads, where you can't predict how the application will behave

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EC2 Reserved Instances

- Up to 72% discount compared to On-demand
- You reserve a specific instance attributes (Instance Type, Region, Tenancy, OS)
- Reservation Period – 1 year (+discount) or 3 years (+++discount)
- Payment Options – No Upfront (+), Partial Upfront (++), All Upfront (+++)
- Reserved Instance's Scope – Regional or Zonal (reserve capacity in an AZ)
- Recommended for steady-state usage applications (think database)
- You can buy and sell in the Reserved Instance Marketplace
- Convertible Reserved Instance
- Can change the EC2 instance

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EC2 Savings Plans

- Get a discount based on long-term usage (up to 72% - same as RIs)
- Commit to a certain type of usage ($10/hour for 1 or 3 years)
- Usage beyond EC2 Savings Plans is billed at the On-Demand price
- Locked to a specific instance family & AWS region (e.g., M5 in us-east-1)
- Flexible across:
- Instance Size (e.g., m5.xlarge, m5.2xlarge)
- OS (e.g., Linux, Windows)
- Tenancy (Host, Dedicated, Default)

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EC2 Spot Instances

- Can get a discount of up to 90% compared to On-demand
- Instances that you can “lose” at any point of time if your max price is less than the
- current spot price
- The MOST cost-efficient instances in AWS
- Useful for workloads that are resilient to failure
- Batch jobs
- Data analysis
- Image processing
- Any distributed workloads
- Workloads with a flexible start and end time
- Not suitable for critical jobs or databases

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EC2 Dedicated Hosts

- A physical server with EC2 instance capacity fully dedicated to your use
- Allows you address compliance requirements and use your existing server bound software licenses (per-socket, per-core, pe—VM software licenses)
- Purchasing Options:
- On-demand – pay per second for active Dedicated Host
- Reserved - 1 or 3 years (No Upfront, Partial Upfront, All Upfront)
- The most expensive option
- Useful for software that have complicated licensing model (BYOL – Bring Your Own License)
- Or for companies that have strong regulatory or compliance needs

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EC2 Dedicated Instances

Characteristic	Dedicated Instances	Dedicated Hosts
Enables the use of dedicated physical servers	X	X
Per instance billing (subject to a $2 per region fee)	X	
Per host billing		X
Visibility of sockets, cores, host ID		X
Affinity between a host and instance		X
Targeted instance placement		X
Automatic instance placement	X	X
Add capacity using an allocation request		X

Instances run on hardware that’s dedicated to you

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EC2 Capacity Reservations

- Reserve On-Demand instances capacity in a specific AZ for any duration
- You always have access to EC2 capacity when you need it
- No time commitment (create/cancel anytime), no billing discounts
- Combine with Regional Reserved Instances and Savings Plans to benefit from billing discounts
- You’re charged at On-Demand rate whether you run instances or not
- Suitable for short-term, uninterrupted workloads that needs to be in a specific AZ

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Which purchasing option is right for me?

On demand: coming and staying in resort whenever we like, we pay the full price

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Price Comparison

Price Type	Price (per hour)
On-Demand	$0.10
Spot Instance (Spot Price)	$0.038 - $0.039 (up to 61% off)
Reserved Instance (1 year)	$0.062 (No Upfront) - $0.058 (All Upfront)
Reserved Instance (3 years)	$0.043 (No Upfront) - $0.037 (All Upfront)
EC2 Savings Plan (1 year)	$0.062 (No Upfront) - $0.058 (All Upfront)
Reserved **Convertible** Instance (1 year)	$0.071 (No Upfront) - $0.066 (All Upfront)
Dedicated Host	On-Demand Price
Dedicated Host Reservation	Up to 70% off
Capacity Reservations	On-Demand Price

Example – m4.large – us-east-1

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Shared Responsibility Model for EC2

- Infrastructure (global network security)
- Isolation on physical hosts
- Replacing faulty hardware
- Compliance validation

- Security Groups rules
- Operating-system patches and updates
- Software and utilities installed on the EC2 instance
- IAM Roles assigned to EC2 & IAM user access management
- Data security on your instance

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EC2 Summary

- EC2 Instance: AMI (OS) + Instance Size (CPU + RAM) + Storage +security groups + EC2 User Data
- Security Groups: Firewall attached to the EC2 instance
- EC2 User Data: Script launched at the first start of an instance
- SSH: start a terminal into our EC2 Instances (port 22)
- EC2 Instance Role: link to IAM roles
- Purchasing Options: On-Demand, Spot, Reserved (Standard + Convertible), Dedicated Host, Dedicated Instance

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EC2 INSTANCE STORAGE

EC2 Instance Storage

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What’s an EBS Volume?

- An EBS (Elastic Block Store) Volume is a network drive you can attach to your instances while they run
- It allows your instances to persist data, even after their termination
- They can only be mounted to one instance at a time (at the CCP level)
- They are bound to a specific availability zone
- Analogy: Think of them as a “network USB stick”
- Free tier: 30 GB of free EBS storage of type General Purpose (SSD) or Magnetic per month

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EBS Volume

- It’s a network drive (i.e. not a physical drive)
- It uses the network to communicate the instance, which means there might be a bit of latency
- It can be detached from an EC2 instance and attached to another one quickly
- It’s locked to an Availability Zone (AZ)
- An EBS Volume in us-east-1a cannot be attached to us-east-1b
- To move a volume across, you first need to snapshot it
- Have a provisioned capacity (size in GBs, and IOPS)
- You get billed for all the provisioned capacity
- You can increase the capacity of the drive over time

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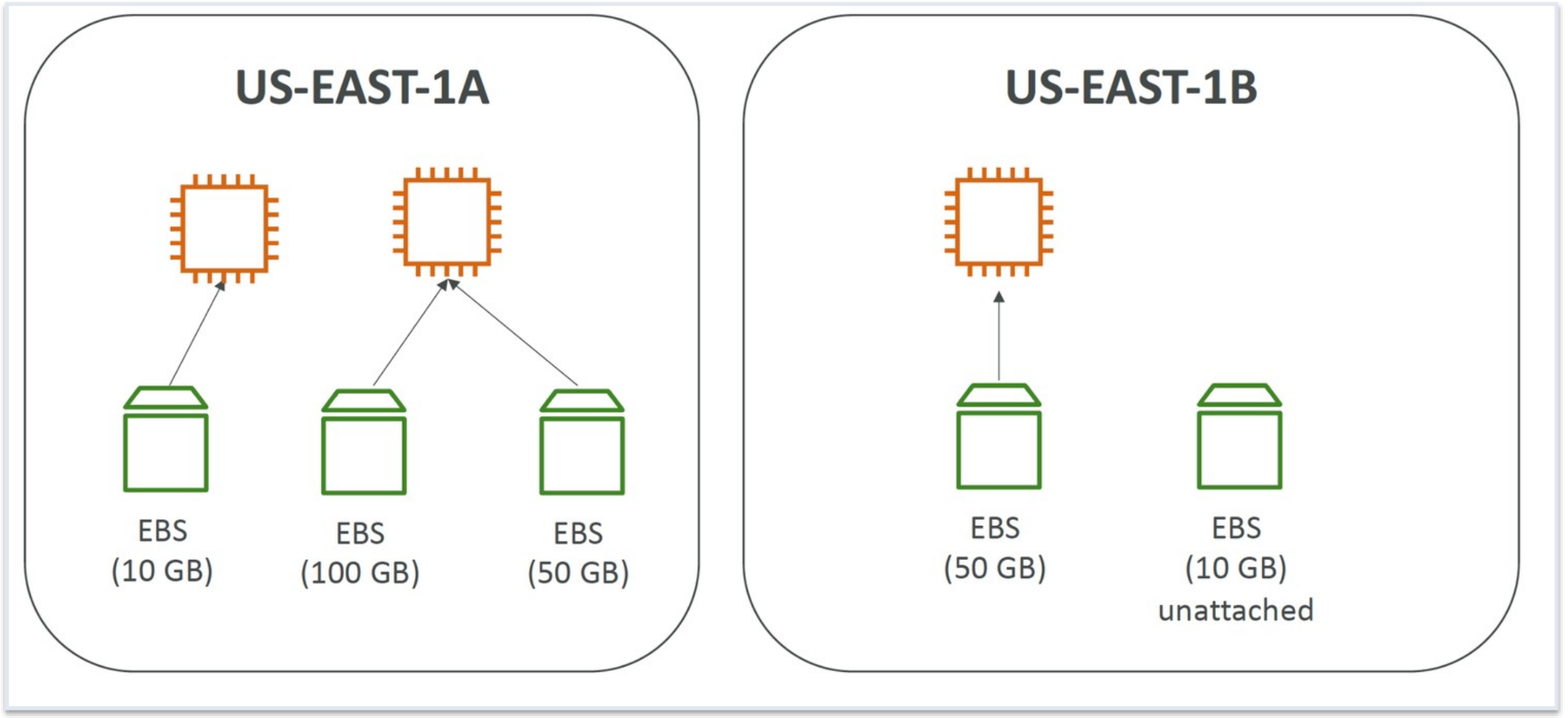
EBS Volume - Example



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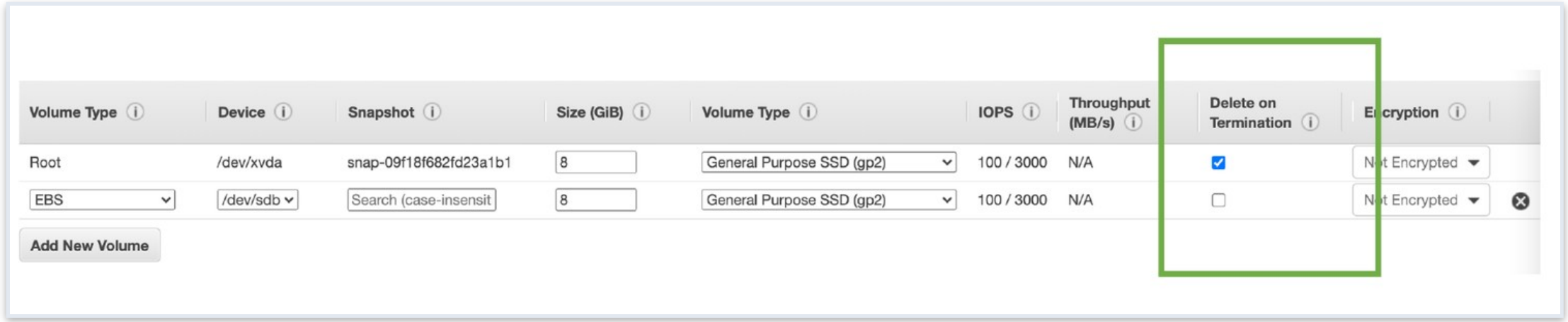
EBS – Delete on Termination attribute



Controls the EBS behaviour when an EC2 instance terminates

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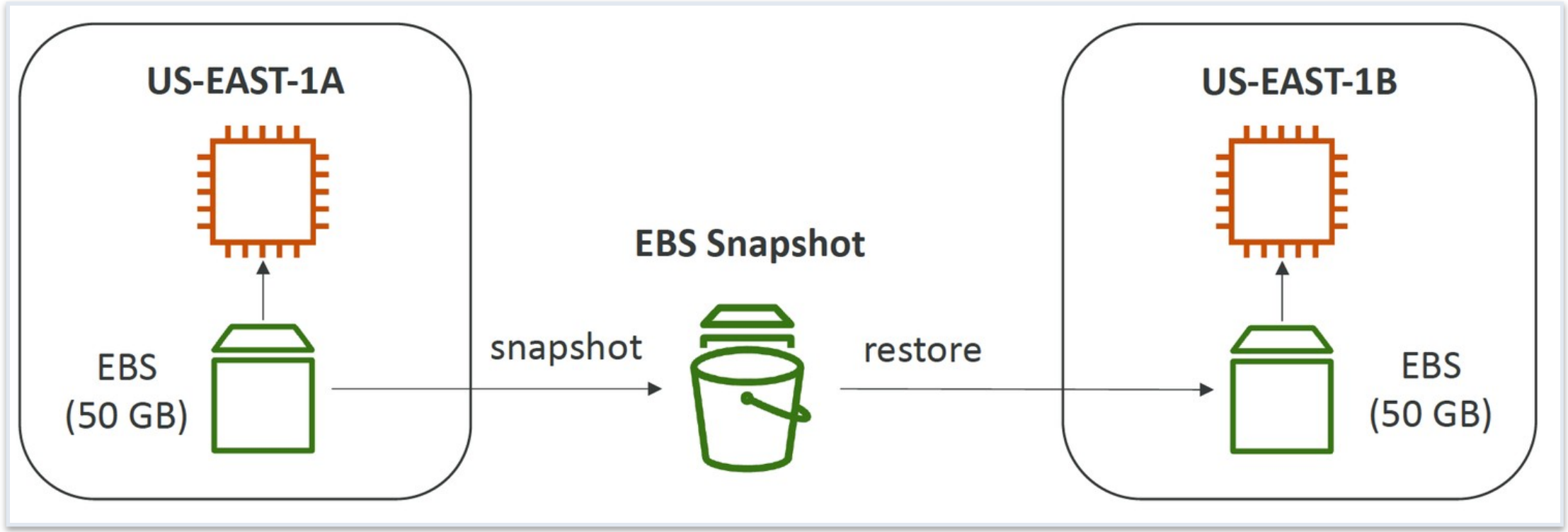
EBS Snapshots



Make a backup (snapshot) of your EBS volume at a point in time

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EBS Snapshots Features

EBS Snapshot Archive

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AMI Overview

- AMI = Amazon Machine Image
- AMI are a customization of an EC2 instance
- You add your own software, configuration, operating system, monitoring…
- Faster boot / configuration time because all your software is pre-packaged
- AMI are built for a specific region (and can be copied across regions)
- You can launch EC2 instances from:
- A Public AMI: AWS provided
- Your own AMI: you make and maintain them yourself
- An AWS Marketplace AMI: an AMI someone else made (and potentially sells)

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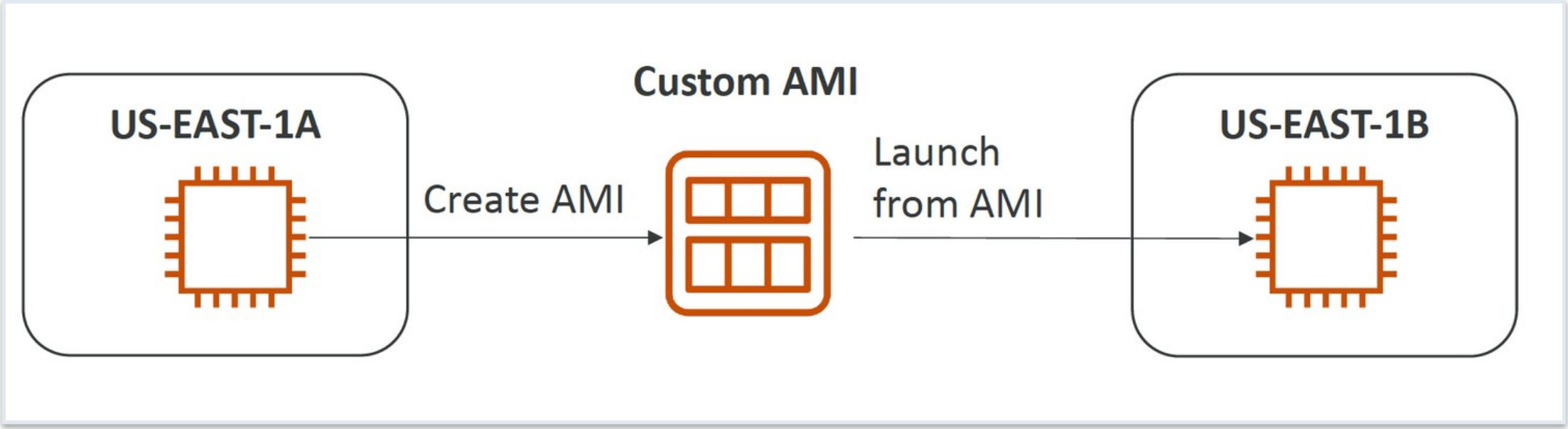
AMI Process (from an EC2 instance)



Start an EC2 instance and customize it

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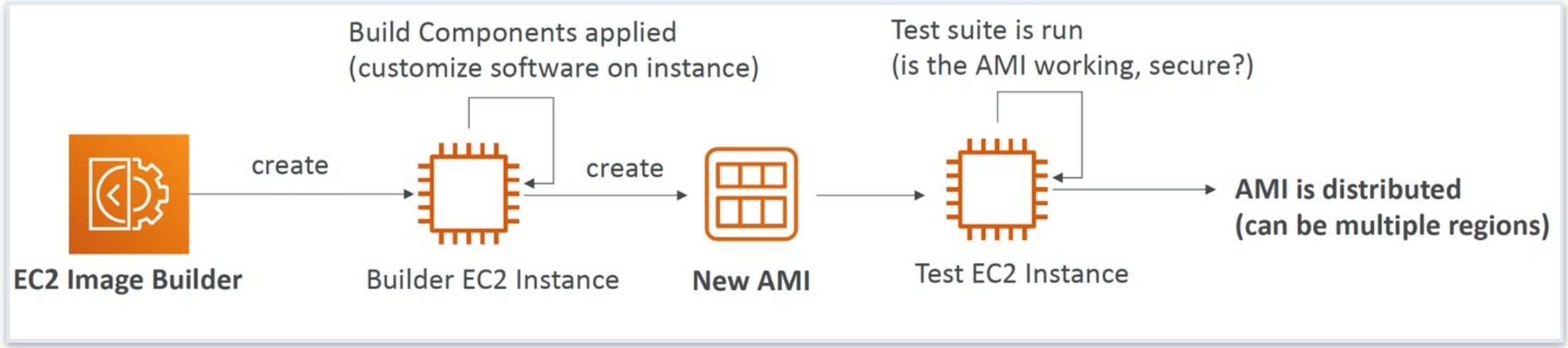
EC2 Image Builder



Used to automate the creation of Virtual Machines or container images

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EC2 Instance Store

- EBS volumes are network drives with good but “limited” performance
- If you need a high-performance hardware disk, use EC2 Instance Store
- Better I/O performance
- EC2 Instance Store lose their storage if they’re stopped (ephemeral)
- Good for buffer / cache / scratch data / temporary content
- Risk of data loss if hardware fails
- Backups and Replication are your responsibility

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Local EC2 Instance Store

Instance Size	100% Random Read IOPS	Write IOPS
i3.large *	100,125	35,000
i3.xlarge *	206,250	70,000
i3.2xlarge	412,500	180,000
i3.4xlarge	825,000	360,000
i3.8xlarge	1.65 million	720,000
i3.16xlarge	3.3 million	1.4 million
i3.metal	3.3 million	1.4 million
i3en.large *	42,500	32,500
i3en.xlarge *	85,000	65,000
i3en.2xlarge *	170,000	130,000
i3en.3xlarge	250,000	200,000
i3en.6xlarge	500,000	400,000
i3en.12xlarge	1 million	800,000
i3en.24xlarge	2 million	1.6 million
i3en.metal	2 million	1.6 million

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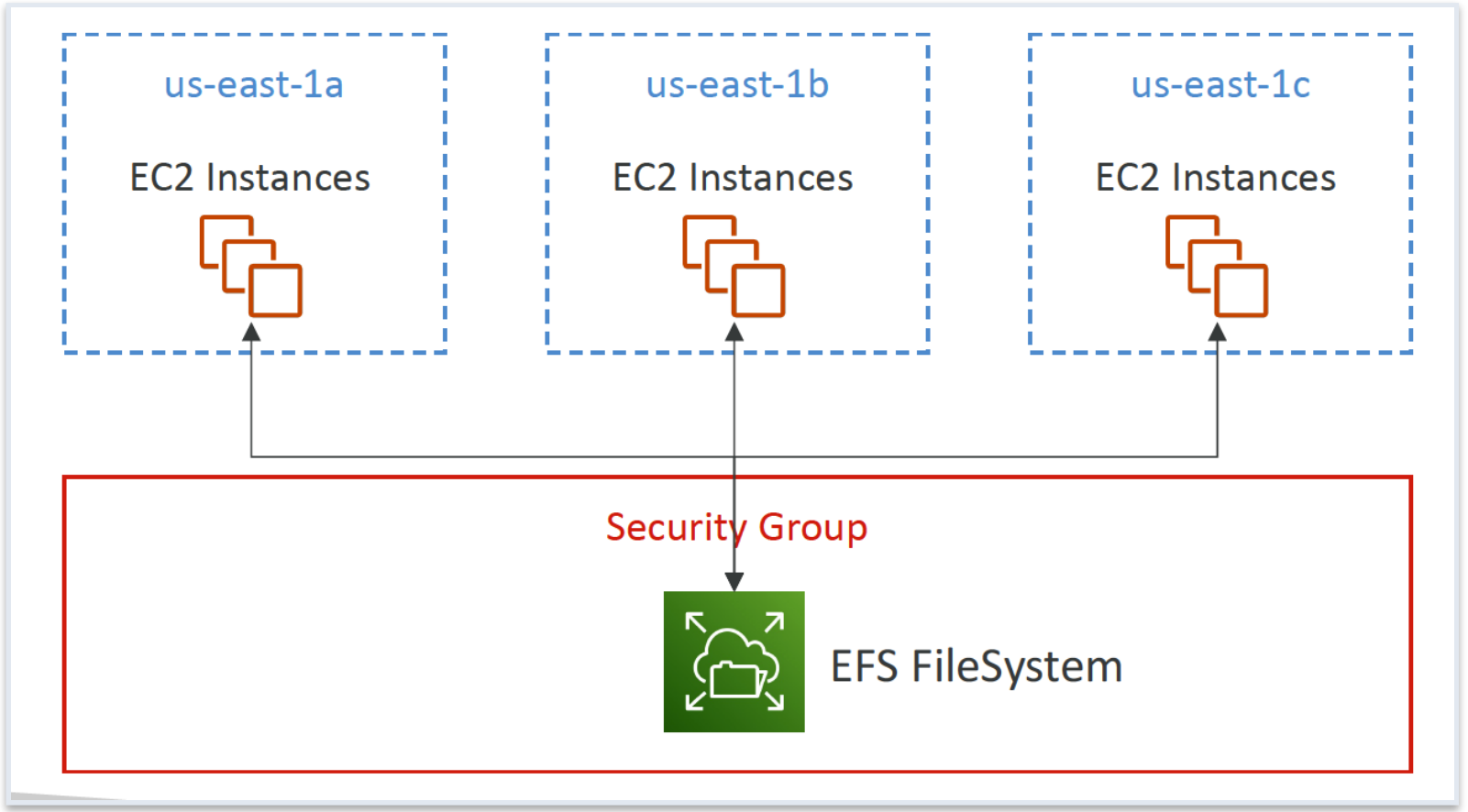
EFS – Elastic File System



Managed NFS (network file system) that can be mounted on 100s of EC2

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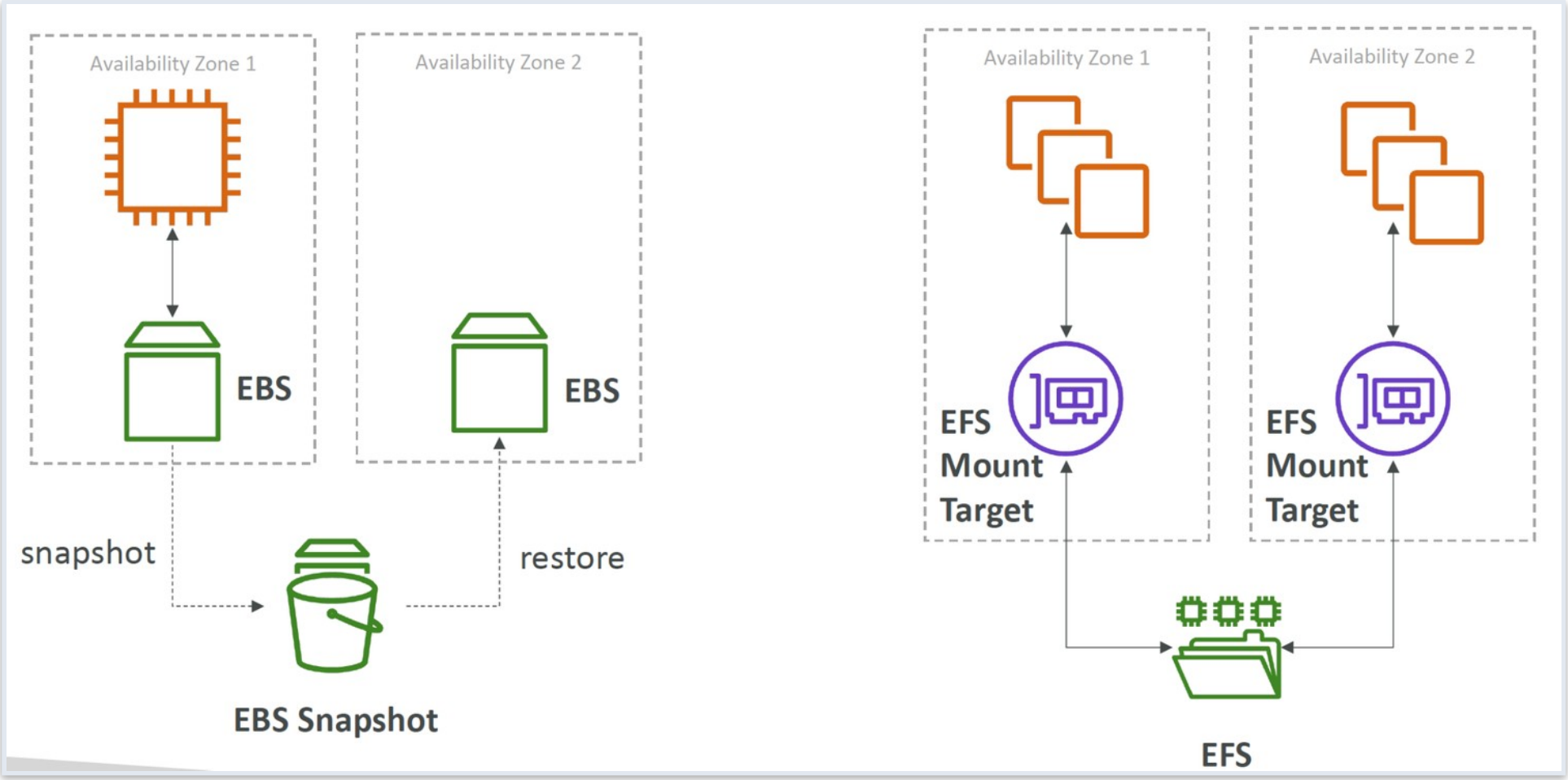
EBS vs EFS



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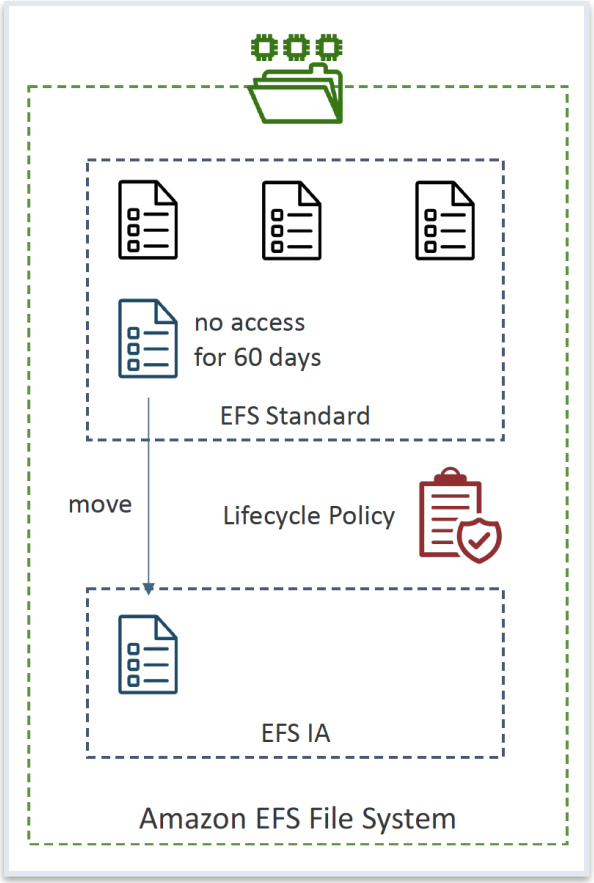
EFS Infrequent Access (EFS-IA)



Storage class that is cost-optimized for files not accessed every day

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Shared Responsibility Model for EC2 Storage

- Infrastructure
- Replication for data for EBS volumes & EFS drives
- Replacing faulty hardware
- Ensuring their employees cannot access your data

- Setting up backup / snapshot procedures
- Setting up data encryption
- Responsibility of any data on the drives
- Understanding the risk of using EC2 Instance Store

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Amazon FSx – Overview

Launch 3rd party high-performance file systems on AWS

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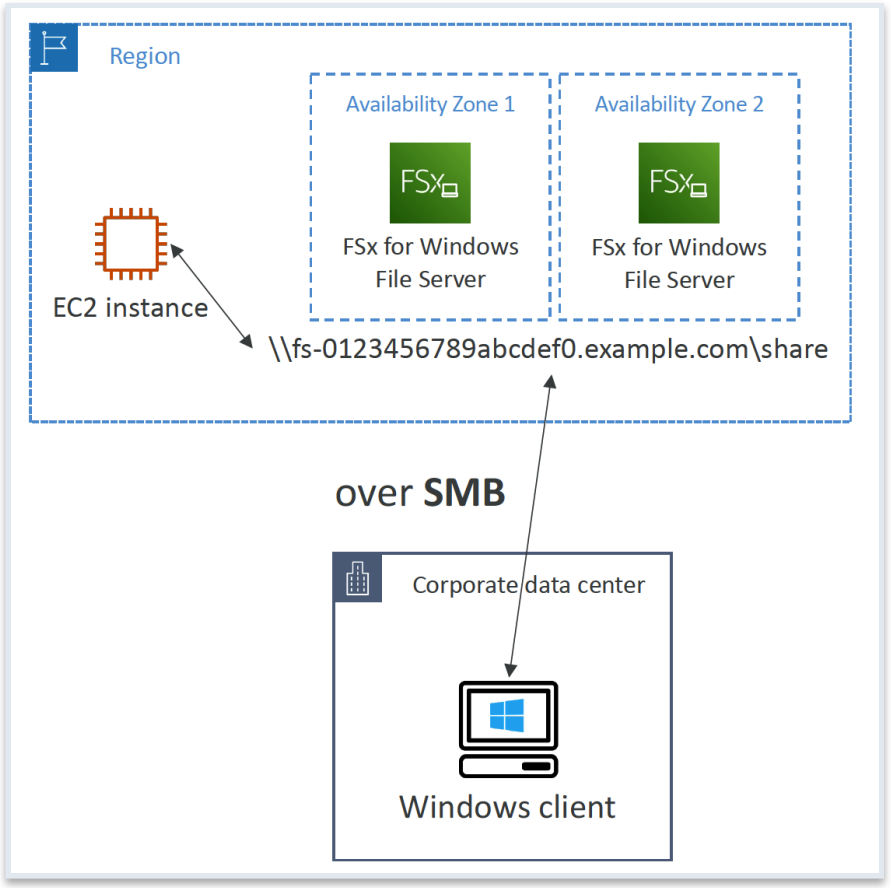
Amazon FSx for Windows File Server



A fully managed, highly reliable, and scalable Windows native shared file system

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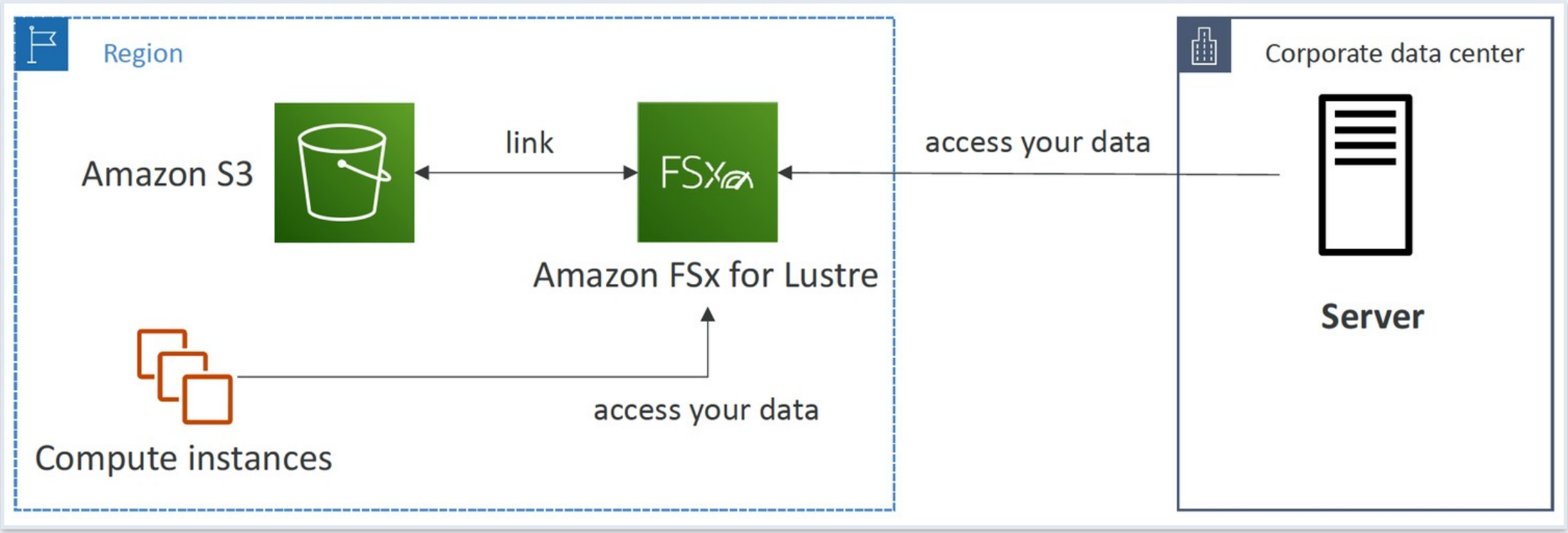
Amazon FSx for Lustre



A fully managed, high-performance, scalable file storage for High Performance Computing (HPC)

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EC2 Instance Storage - Summary

- EBS volumes:
- network drives attached to one EC2 instance at a time
- Mapped to an Availability Zones
- Can use EBS Snapshots for backups / transferring EBS volumes across AZ
- AMI: create ready-to-use EC2 instances with our customizations
- EC2 Image Builder: automatically build, test and distribute AMIs
- EC2 Instance Store:
- High performance hardware disk attached to our EC2 instance
- Lost if our instance is stopped / terminated
- EFS: network file system, can be attached to 100s of instances in a region
- EFS-IA: cost-optimized storage class for infrequent accessed files
- FSx for Windows: Network File System for Windows servers
- FSx for Lustre: High Performance Computing Linux file system

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OTHER COMPUTE

Other Compute

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What is Docker?

- Docker is a software development platform to deploy apps
- Apps are packaged in containers that can be run on any OS
- Apps run the same, regardless of where they’re run
- Any machine
- No compatibility issues
- Predictable behavior
- Less work
- Easier to maintain and deploy
- Works with any language, any OS, any technology
- Scale containers up and down very quickly (seconds)

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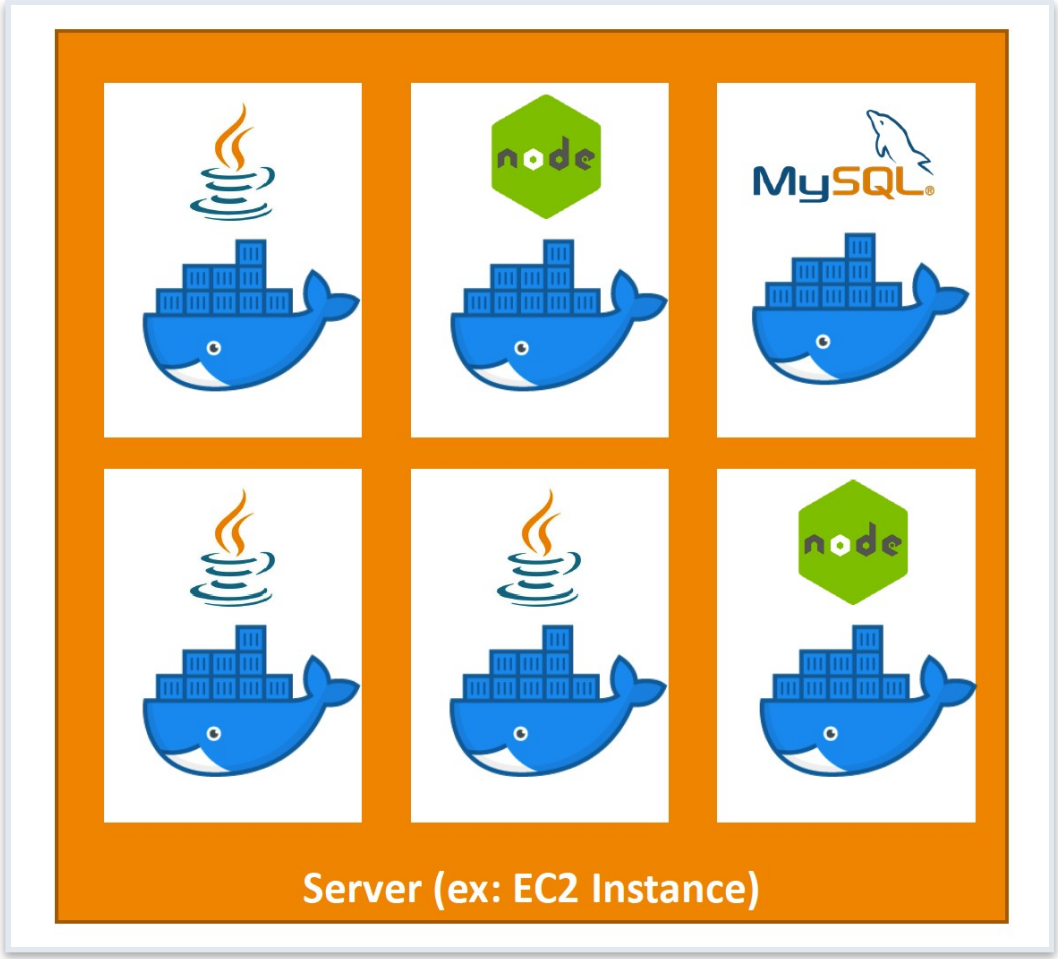
Docker on an OS



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Where Docker images are stored?

- Docker images are stored in Docker Repositories
- Public: Docker Hub https://hub.docker.com/
- Find base images for many technologies or OS:
- Ubuntu
- MySQL
- NodeJS, Java…
- Private: Amazon ECR (Elastic Container Registry)

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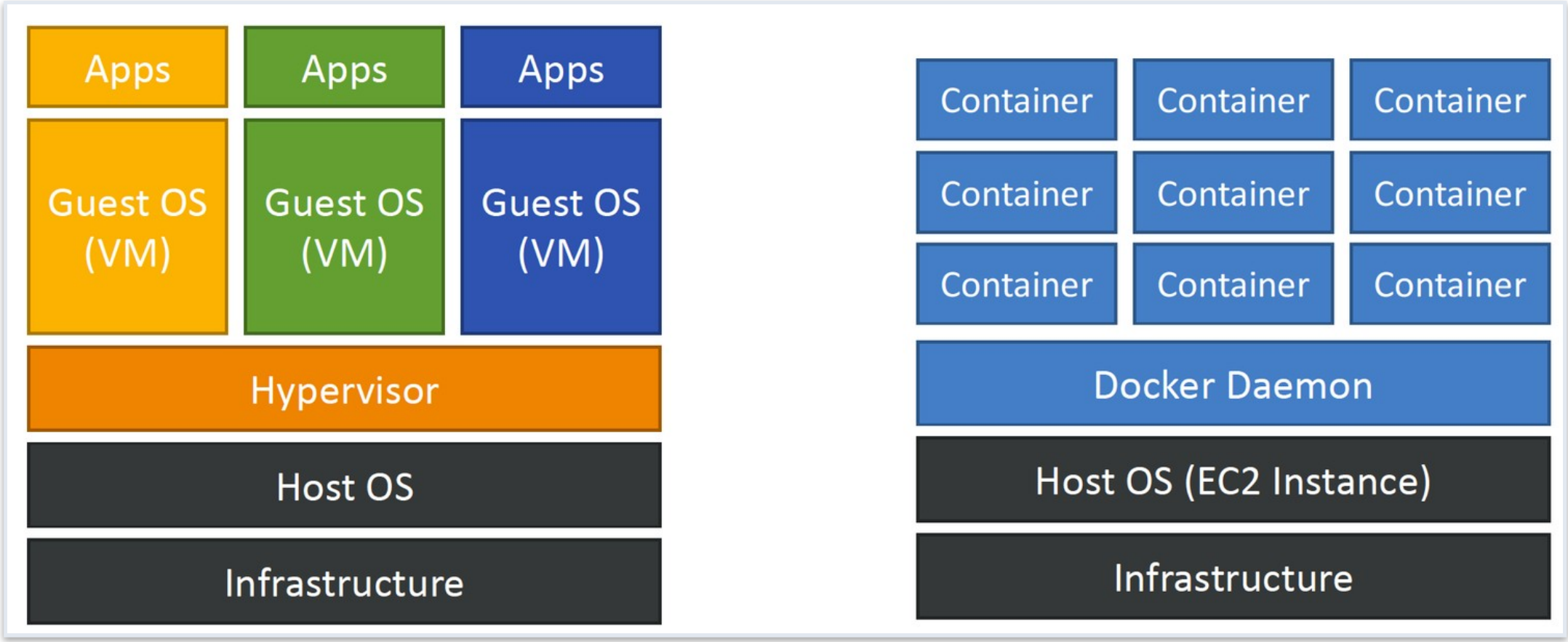
Docker versus Virtual Machines



Docker is ”sort of” a virtualization technology, but not exactly

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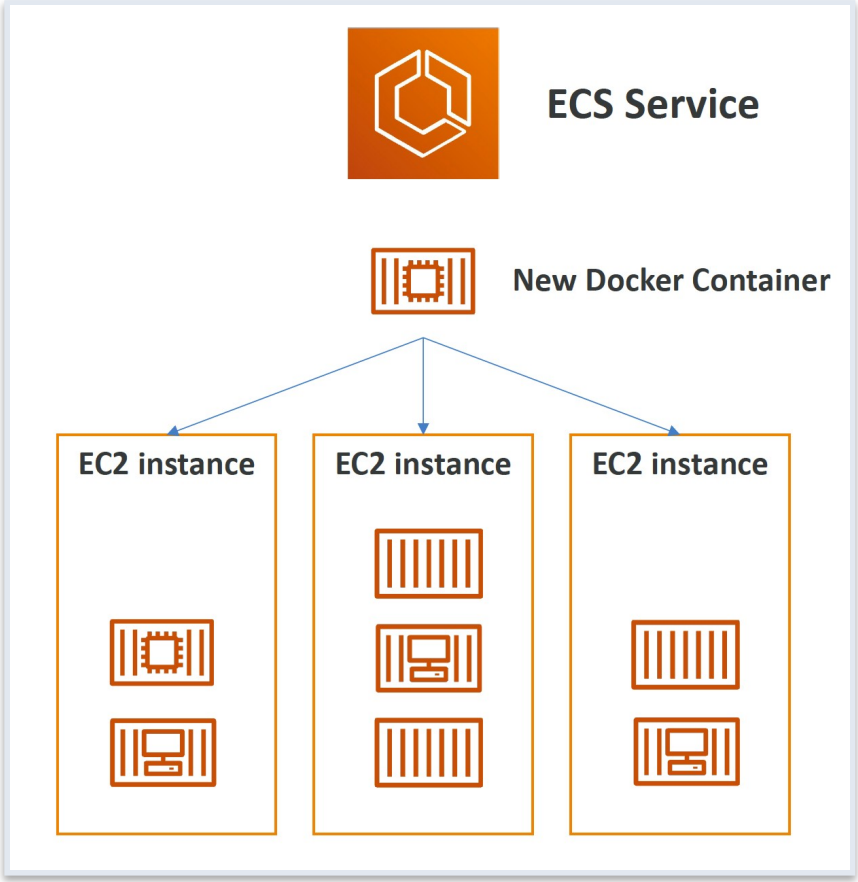
ECS



ECS = Elastic Container Service

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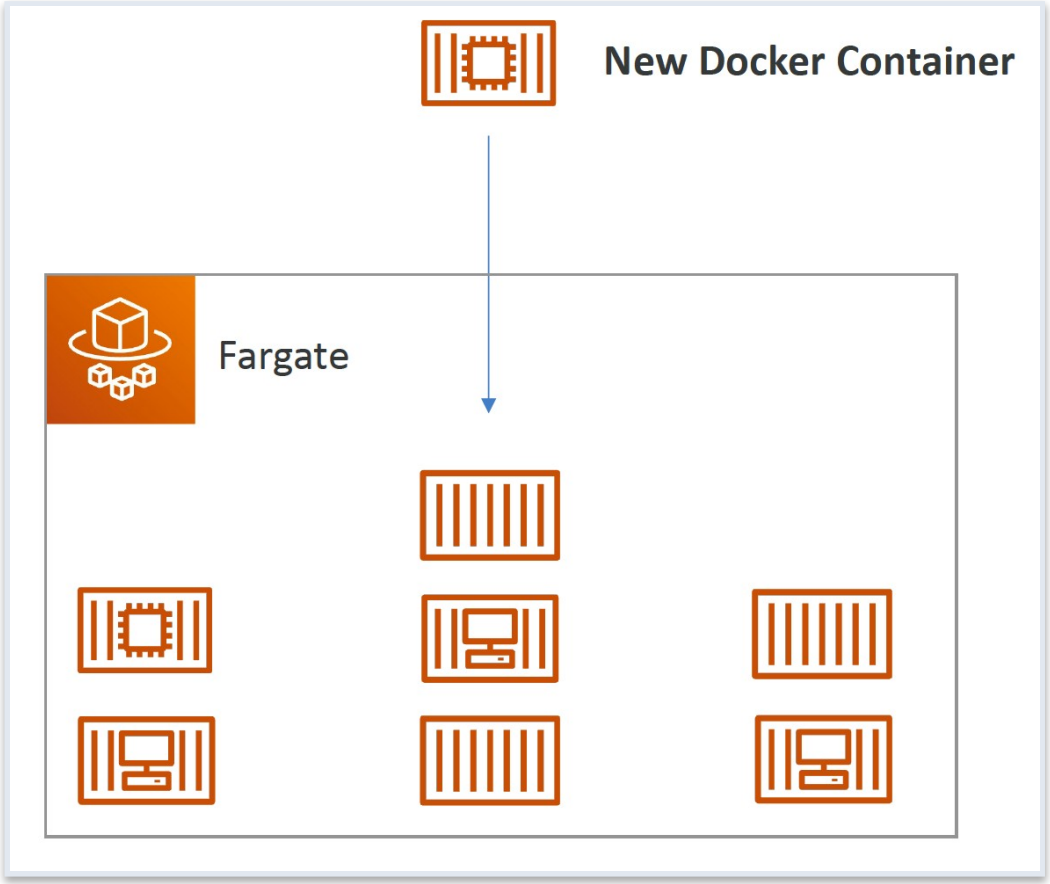
Fargate



Launch Docker containers on AWS

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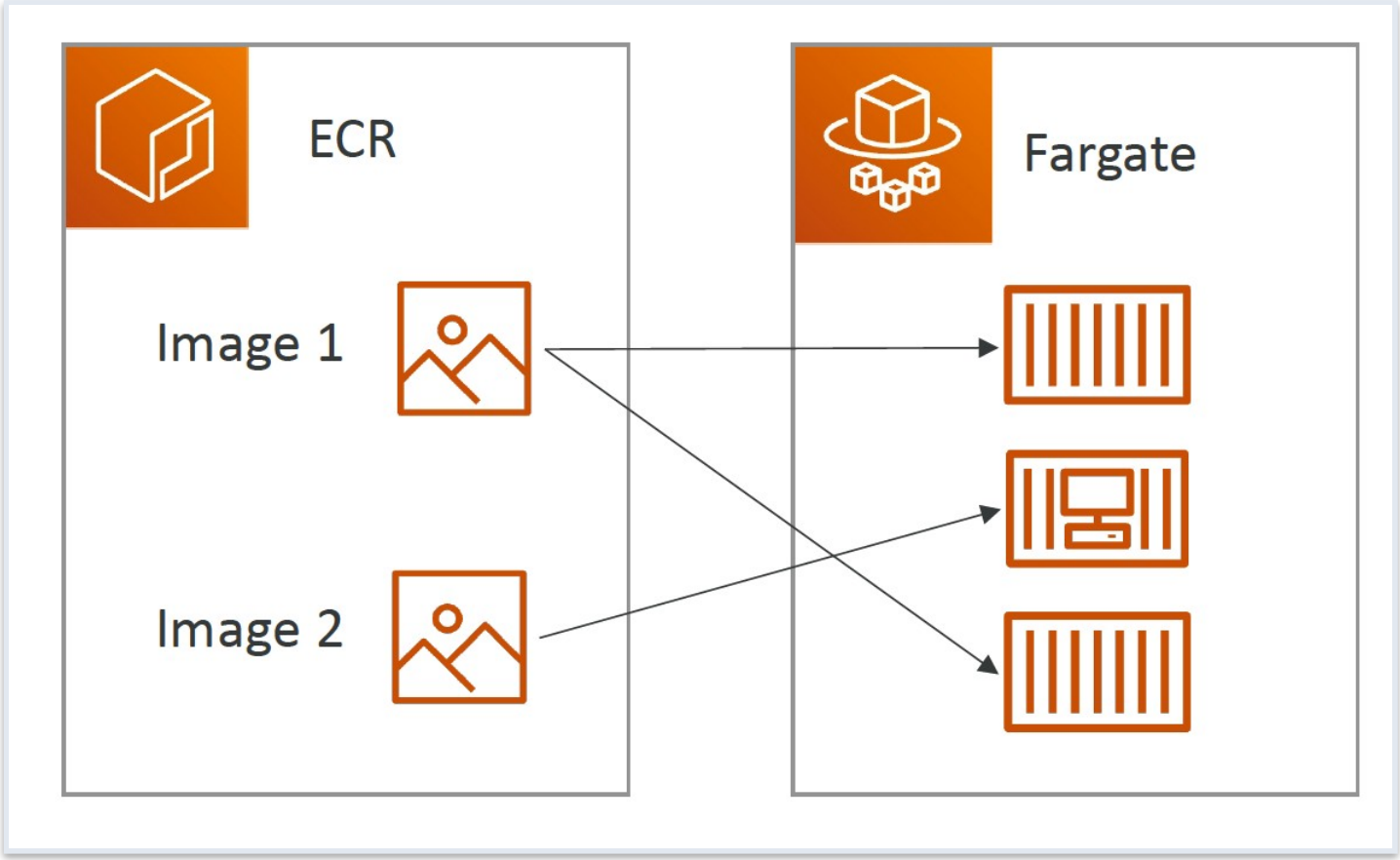
ECR



Elastic Container Registry

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What’s serverless?

- Serverless is a new paradigm in which the developers don’t have to manage servers anymore…
- They just deploy code
- They just deploy… functions !
- Initially... Serverless == FaaS (Function as a Service)
- Serverless was pioneered by AWS Lambda but now also includes anything that’s managed: “databases, messaging, storage, etc.”
- Serverless does not mean there are no servers… it means you just don’t manage / provision / see them

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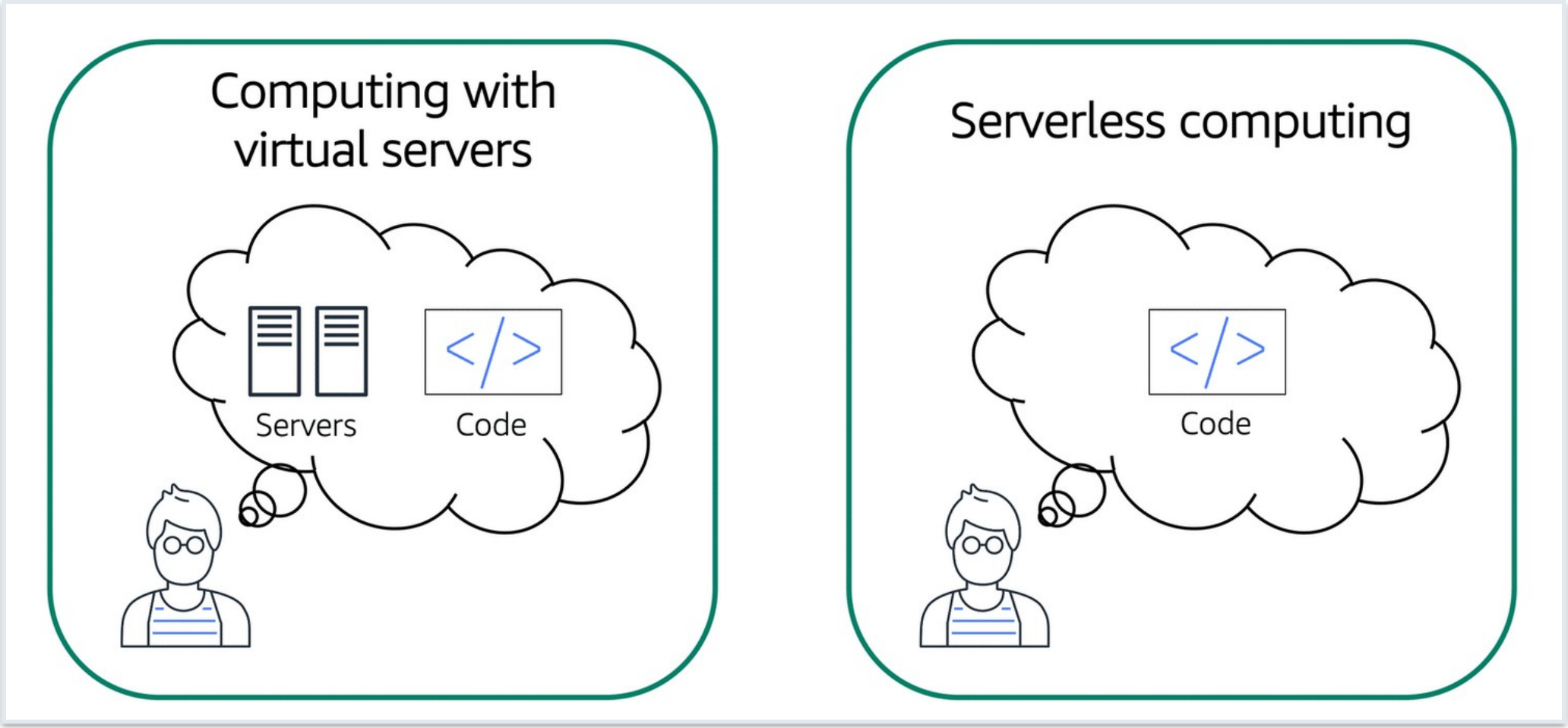
Serverless computing



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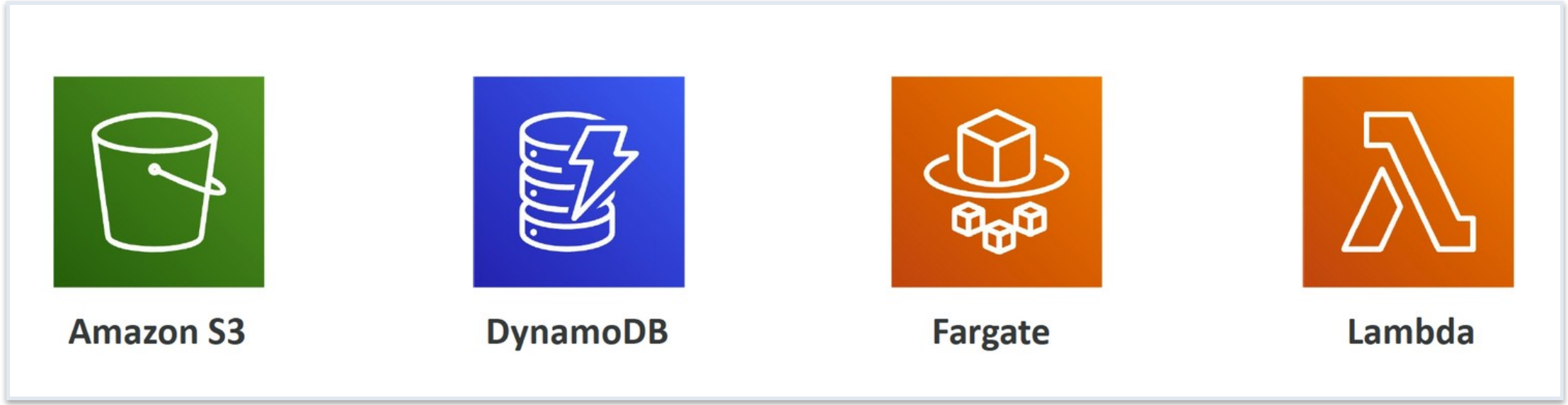
Example of Serverless



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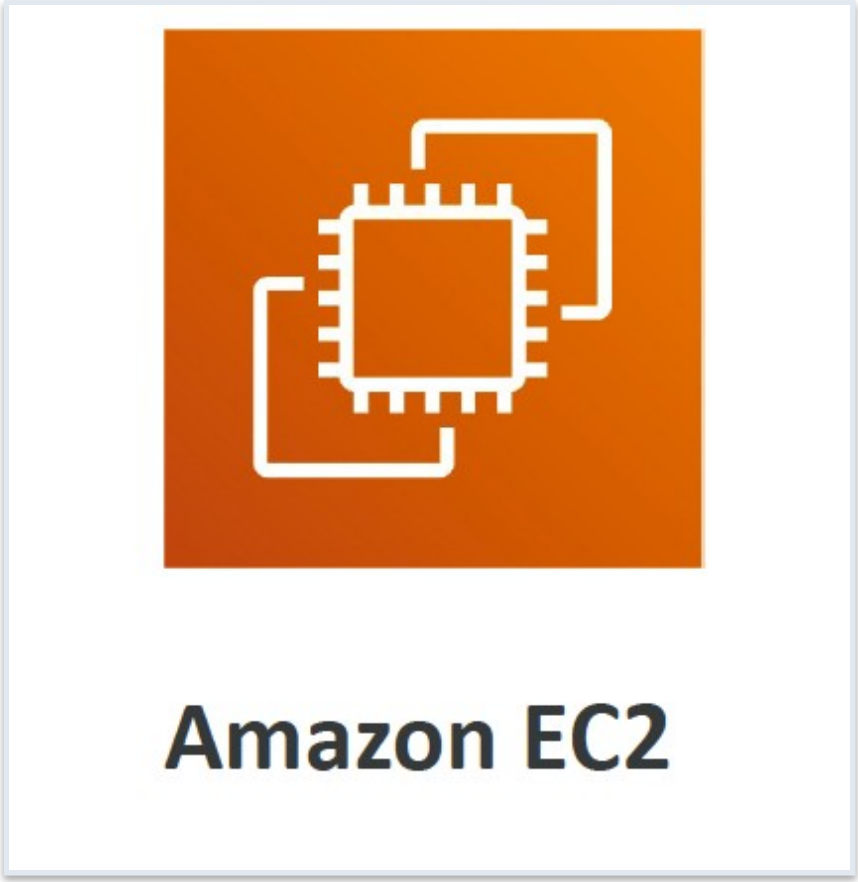
Why AWS Lambda



Virtual Servers in the Cloud

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Benefits of AWS Lambda

- Easy Pricing:
- Pay per request and compute time
- Free tier of 1,000,000 AWS Lambda requests and 400,000 GBs of compute time
- Integrated with the whole AWS suite of services
- Event-Driven: functions get invoked by AWS when needed
- Integrated with many programming languages
- Easy monitoring through AWS CloudWatch
- Easy to get more resources per functions (up to 10GB of RAM!)
- Increasing RAM will also improve CPU and network!

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AWS Lambda language support

- Node.js (JavaScript)
- Python
- Java (Java 8 compatible)
- C# (.NET Core)
- Golang
- C# / Powershell
- Ruby
- Custom Runtime API (community supported, example Rust)
- Lambda Container Image
- The container image must implement the Lambda Runtime API
- ECS / Fargate is preferred for running arbitrary Docker images

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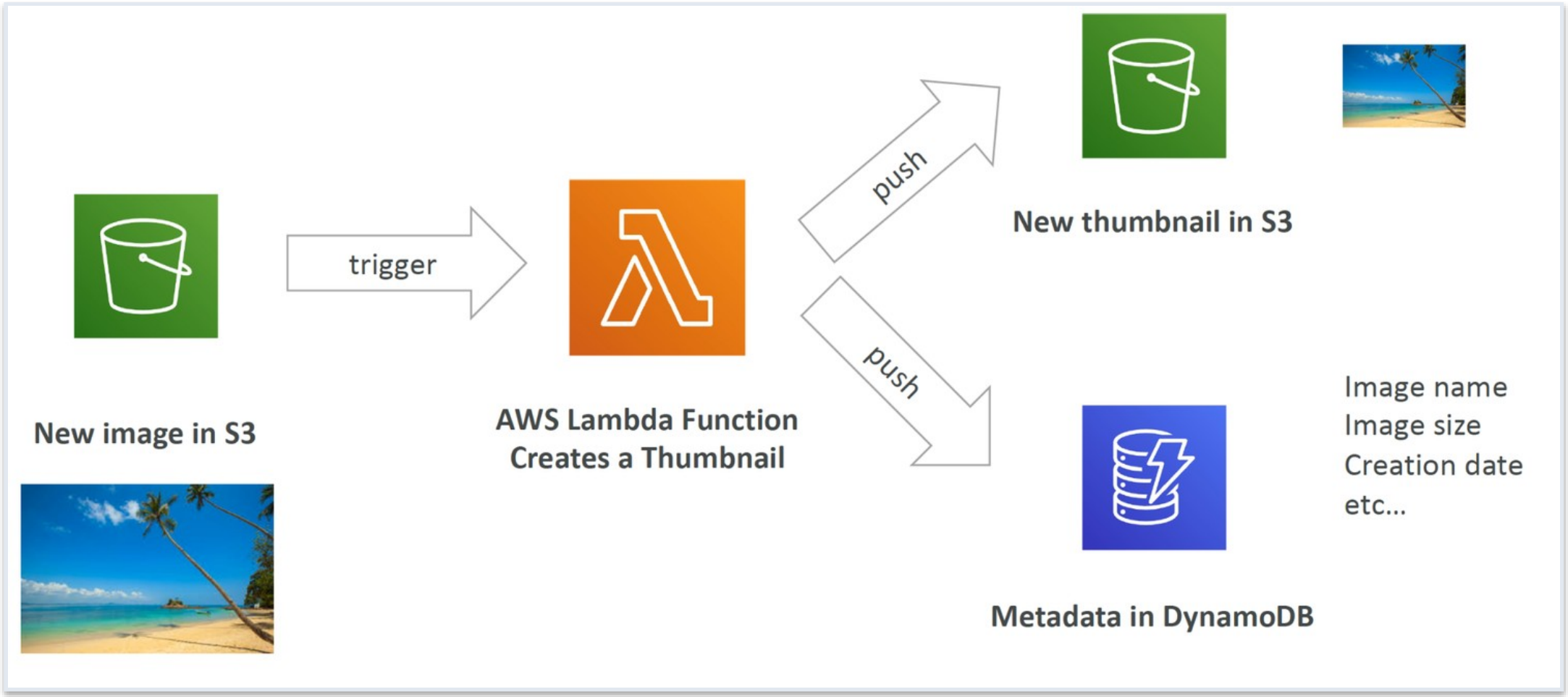
Example: Serverless Thumbnail creation



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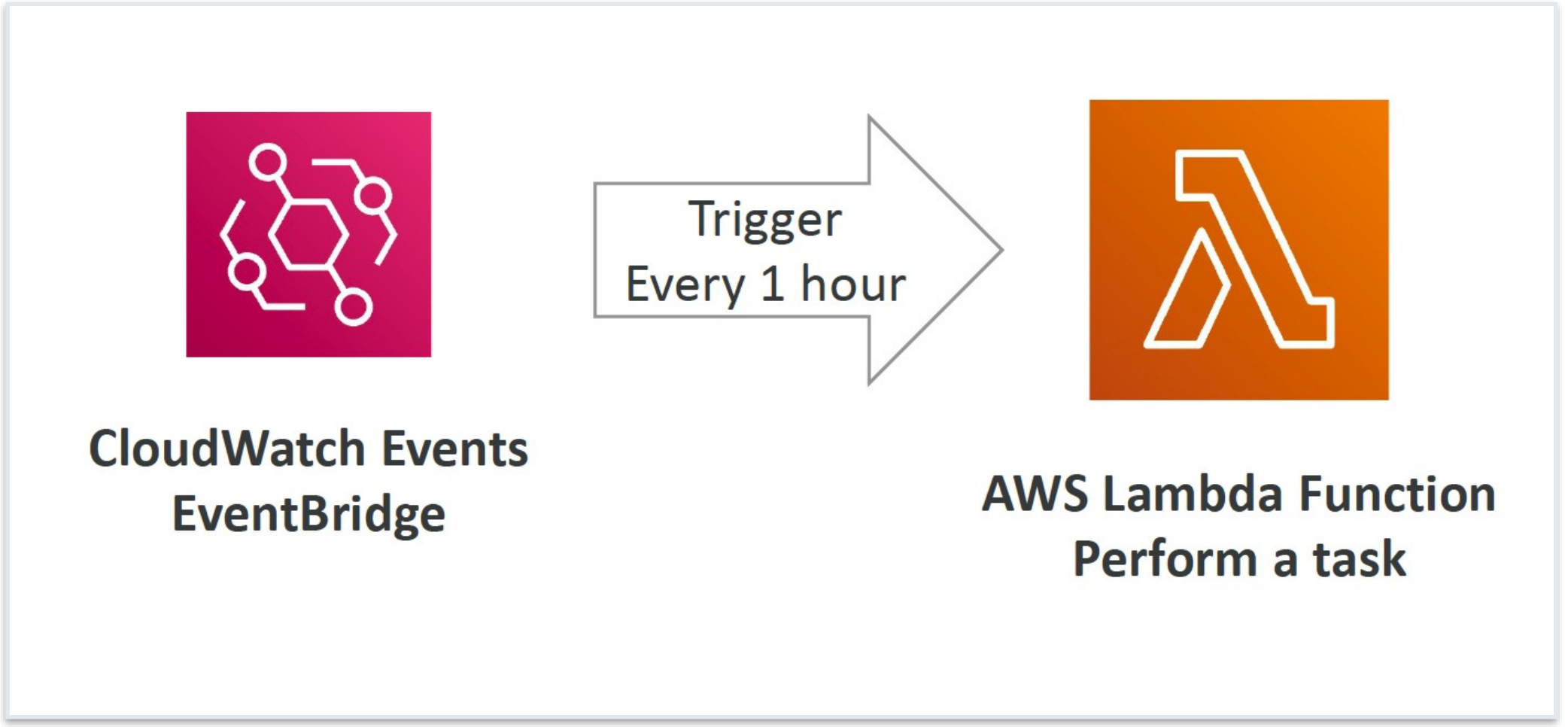
Example: Serverless CRON Job



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AWS Lambda Pricing: example

- You can find overall pricing information here: https://aws.amazon.com/lambda/pricing/
- Pay per calls:
- First 1,000,000 requests are free
- $0.20 per 1 million requests thereafter ($0.0000002 per request)
- Pay per duration: (in increment of 1 ms)
- 400,000 GB-seconds of compute time per month for FREE
- == 400,000 seconds if function is 1GB RAM
- == 3,200,000 seconds if function is 128 MB RAM
- After that $1.00 for 600,000 GB-seconds
- It is usually very cheap to run AWS Lambda so it’s very popular

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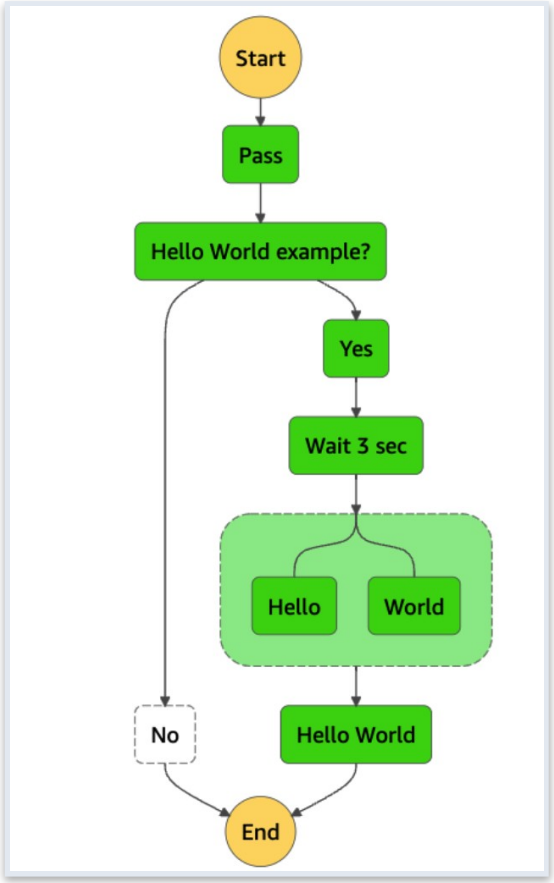
AWS Step Functions



Build serverless visual workflow to orchestrate your Lambda functions

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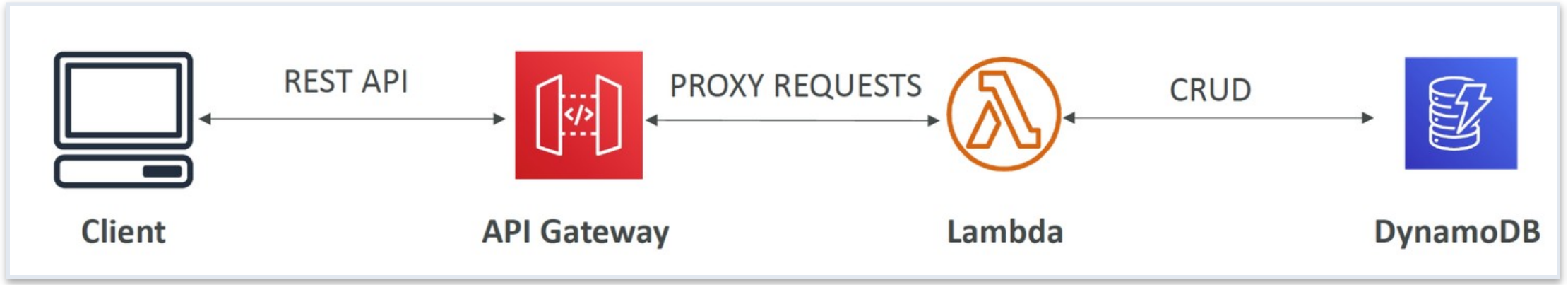
Amazon API Gateway



Example: building a serverless API

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AWS Batch

- Fully managed batch processing at any scale
- Efficiently run 100,000s of computing batch jobs on AWS
- A “batch” job is a job with a start and an end (opposed to continuous)
- Batch will dynamically launch EC2 instances or Spot Instances
- AWS Batch provisions the right amount of compute / memory
- You submit or schedule batch jobs and AWS Batch does the rest!
- Batch jobs are defined as Docker images and run on ECS
- Helpful for cost optimizations and focusing less on the infrastructure

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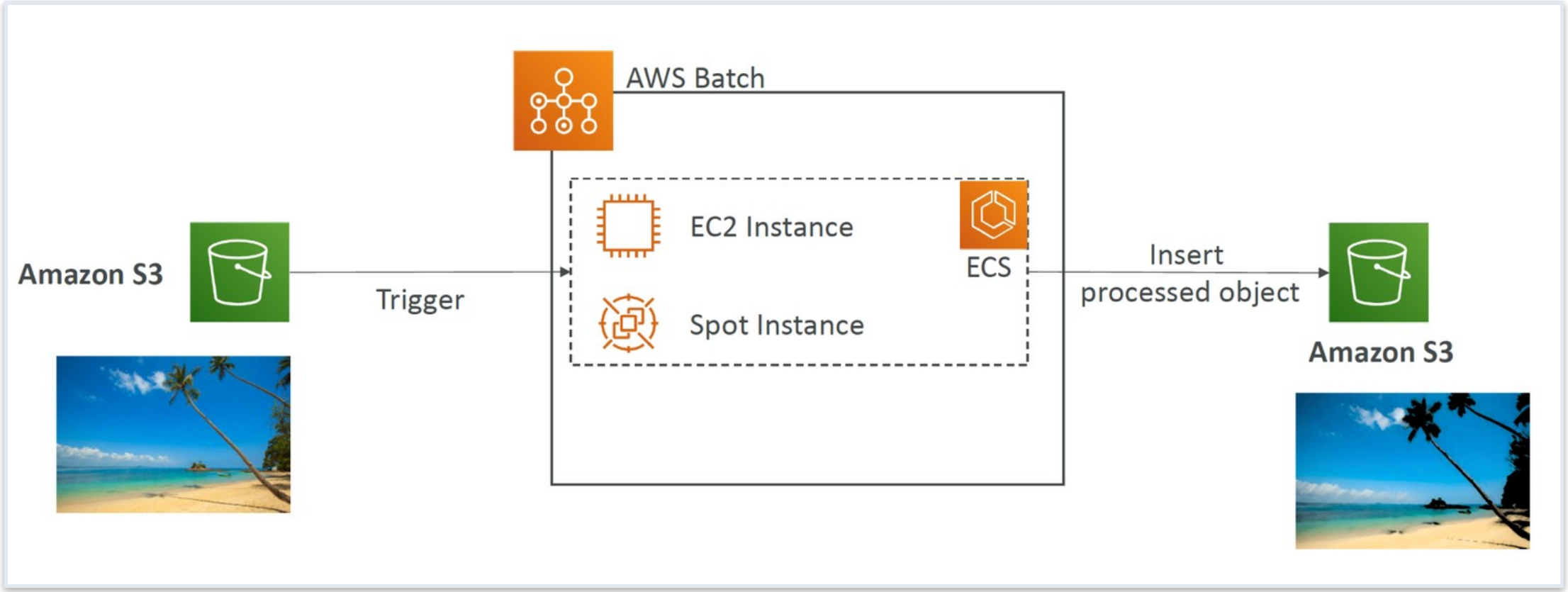
AWS Batch – Simplified Example



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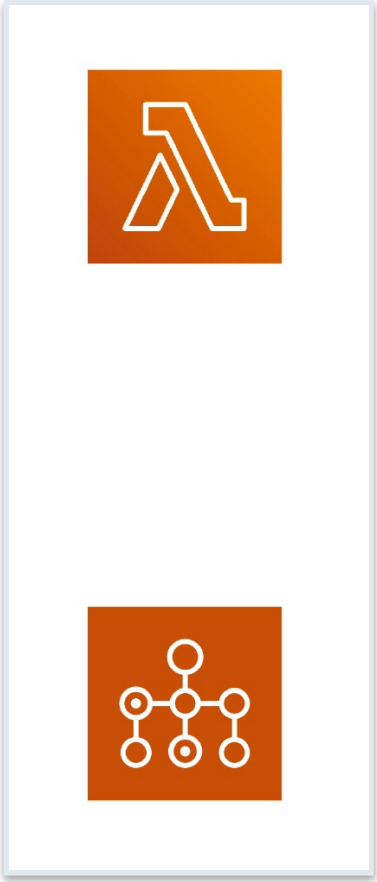
Batch vs Lambda



Lambda:

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Amazon Lightsail

- Virtual servers, storage, databases, and networking
- Low & predictable pricing
- Simpler alternative to using EC2, RDS, ELB, EBS, Route 53…
- Great for people with little cloud experience!
- Can setup notifications and monitoring of your Lightsail resources
- Use cases:
- Simple web applications (has templates for LAMP, Nginx, MEAN, Node.js…)
- Websites (templates for WordPress, Magento, Plesk, Joomla) Dev / Test environment
- Has high availability but no auto-scaling, limited AWS integrations

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Other Compute - Summary

- Docker: container technology to run applications
- ECS: run Docker containers on EC2 instances
- Fargate:
- Run Docker containers without provisioning the infrastructure
- Serverless offering (no EC2 instances)
- ECR: Private Docker Images Repository
- Batch: run batch jobs on AWS across managed EC2 instances
- Lightsail: predictable & low pricing for simple application & DB stacks

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Lambda Summary

- Lambda is Serverless, Function as a Service, seamless scaling, reactive
- Lambda Billing:
- By the time run x by the RAM provisioned
- By the number of invocations
- Language Support: many programming languages except (arbitrary) Docker
- Invocation time: up to 15 minutes
- Use cases:
- Create Thumbnails for images uploaded onto S3
- Run a Serverless cron job
- API Gateway: expose Lambda functions as HTTP API

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DEPLOYING AND MANAGING

Infrastructure at Scale

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What is CloudFormation

- CloudFormation is a declarative way of outlining your AWS Infrastructure, for any resources (most of them are supported).
- For example, within a CloudFormation template, you say:
- I want a security group
- I want two EC2 instances using this security group
- I want an S3 bucket
- I want a load balancer (ELB) in front of these machines
- Then CloudFormation creates those for you, in the right order, with the exact configuration that you specify

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Benefits of AWS CloudFormation (1/2)

- Infrastructure as code
- No resources are manually created, which is excellent for control
- Changes to the infrastructure are reviewed through code
- Cost
- Each resources within the stack is tagged with an identifier so you can easily see how much a stack costs you
- You can estimate the costs of your resources using the CloudFormation template
- Savings strategy: In Dev, you could automation deletion of templates at 5 PM and recreated at 8 AM, safely

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Benefits of AWS CloudFormation (2/2)

- Productivity
- Ability to destroy and re-create an infrastructure on the cloud on the fly
- Automated generation of Diagram for your templates!
- Declarative programming (no need to figure out ordering and orchestration)
- Don’t re-invent the wheel
- Leverage existing templates on the web!
- Leverage the documentation
- Supports (almost) all AWS resources:
- Everything we’ll see in this course is supported
- You can use “custom resources” for resources that are not supported

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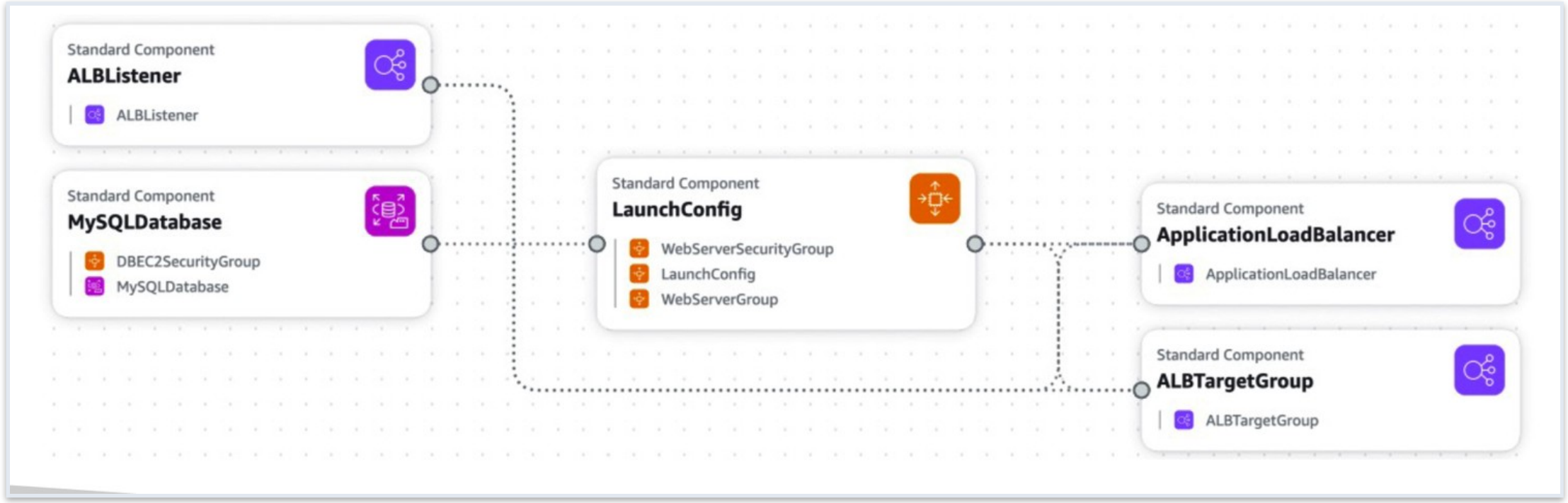
CloudFormation + Application Composer



Example: WordPress CloudFormation Stack

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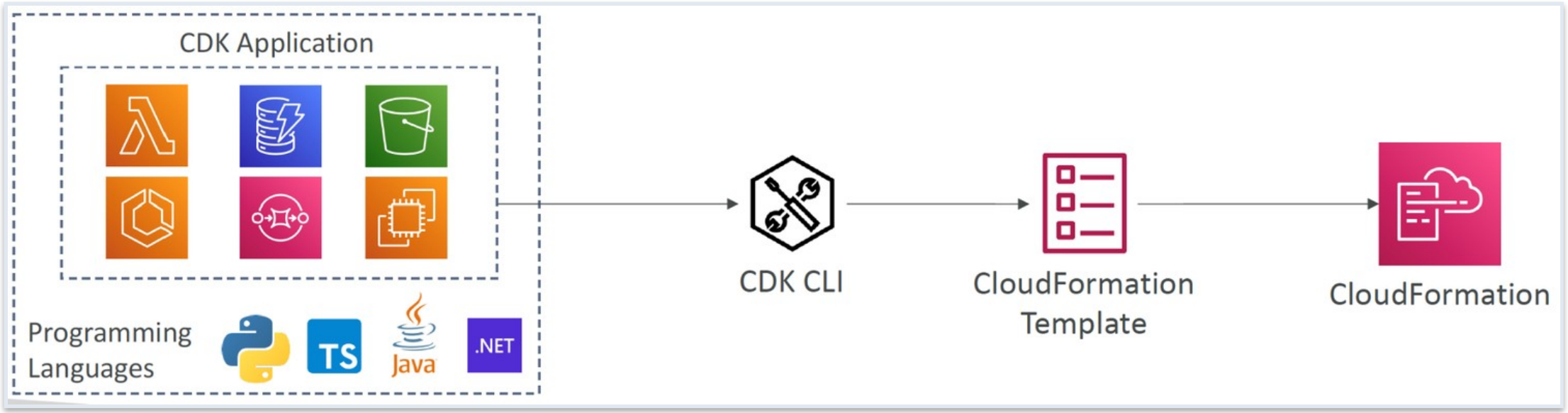
AWS Cloud Development Kit (CDK)



Define your cloud infrastructure using a familiar language:

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CDK Example

```
export class MyEcsConstructStack extends core.Stack {
  constructor(scope: core.App, id: string, props?: core.StackProps)
    super(scope, id, props);

    const vpc = new ec2.Vpc(this, "MyVpc", {
      maxAzs: 3 // Default is all AZs in region
    });

    const cluster = new ecs.Cluster(this, "MyCluster", {
      vpc: vpc
    });

    // Create a load-balanced Fargate service and make it public
    new ecs_patterns.ApplicationLoadBalancedFargateService(this, "My
      cluster: cluster, // Required
      cpu: 512, // Default is 256
      desiredCount: 6, // Default is 1
      taskImageOptions: { image: ecs.ContainerImage.fromRegistry("am
      memoryLimitMiB: 2048, // Default is 512
      publicLoadBalancer: true // Default is false
    });
  }
}
```


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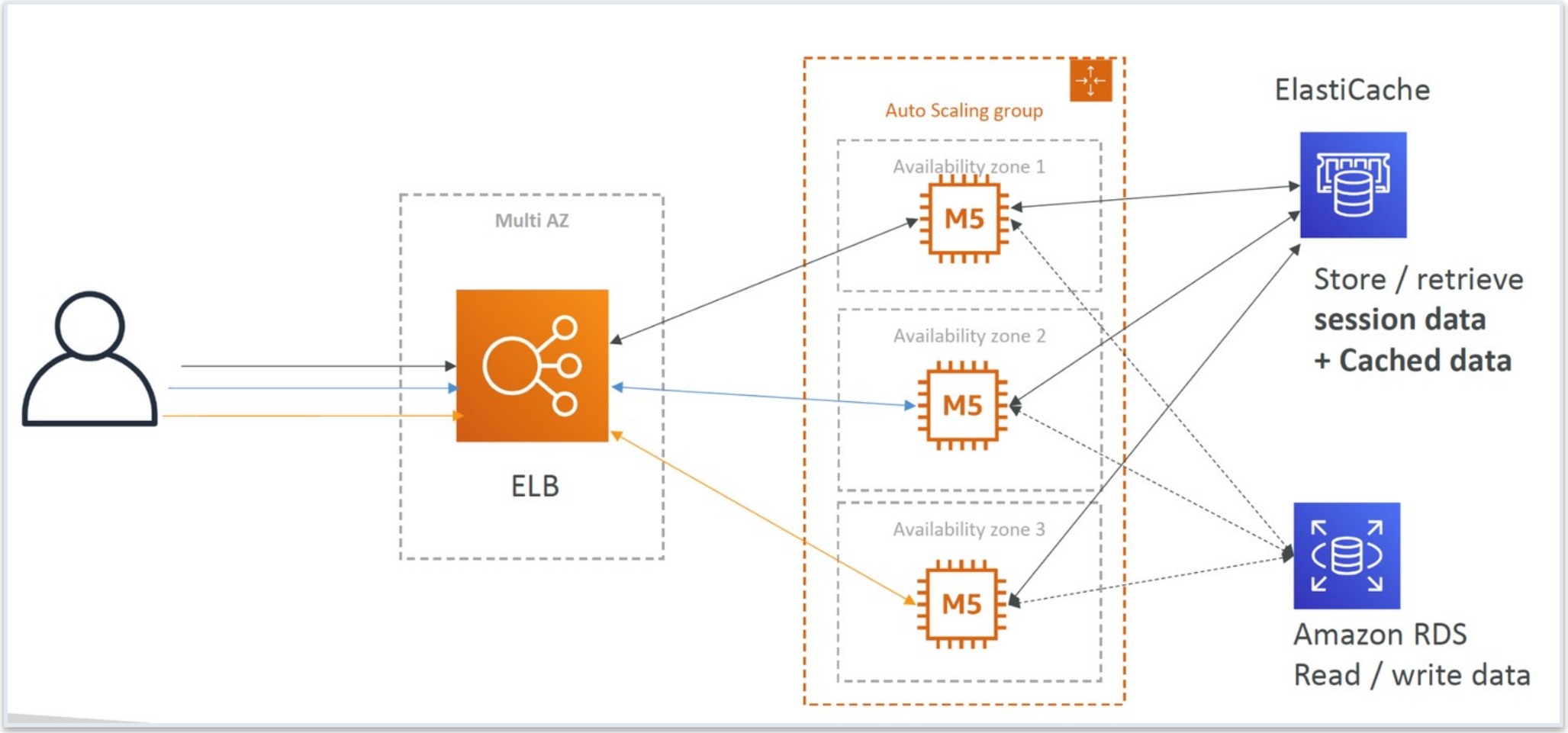
Typical architecture: Web App 3-tier



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Developer problems on AWS

- Managing infrastructure
- Deploying Code
- Configuring all the databases, load balancers, etc
- Scaling concerns
- Most web apps have the same architecture (ALB + ASG)
- All the developers want is for their code to run!
- Possibly, consistently across different applications and environments

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AWS Elastic Beanstalk Overview

- Elastic Beanstalk is a developer centric view of deploying an application on AWS
- It uses all the component's we've seen before: EC2, ASG, ELB, RDS, etc…
- But it's all in one view that's easy to make sense of!
- We still have full control over the configuration
- Beanstalk = Platform as a Service (PaaS)
- Beanstalk is free but you pay for the underlying instances

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Elastic Beanstalk

- Managed service
- Instance configuration / OS is handled by Beanstalk
- Deployment strategy is configurable but performed by Elastic Beanstalk
- Capacity provisioning
- Load balancing & auto-scaling
- Application health-monitoring & responsiveness
- Just the application code is the responsibility of the developer
- Three architecture models:
- Single Instance deployment: good for dev
- LB + ASG: great for production or pre-production web applications
- ASG only: great for non-web apps in production (workers, etc..)

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Elastic Beanstalk

- Support for many platforms:
- Go
- Java SE
- Java with Tomcat
- .NET on Windows Server with IIS
- Node.js
- PHP
- Python
- Ruby
- Packer Builder
- Single Container Docker
- Multi-Container Docker
- Preconfigured Docker

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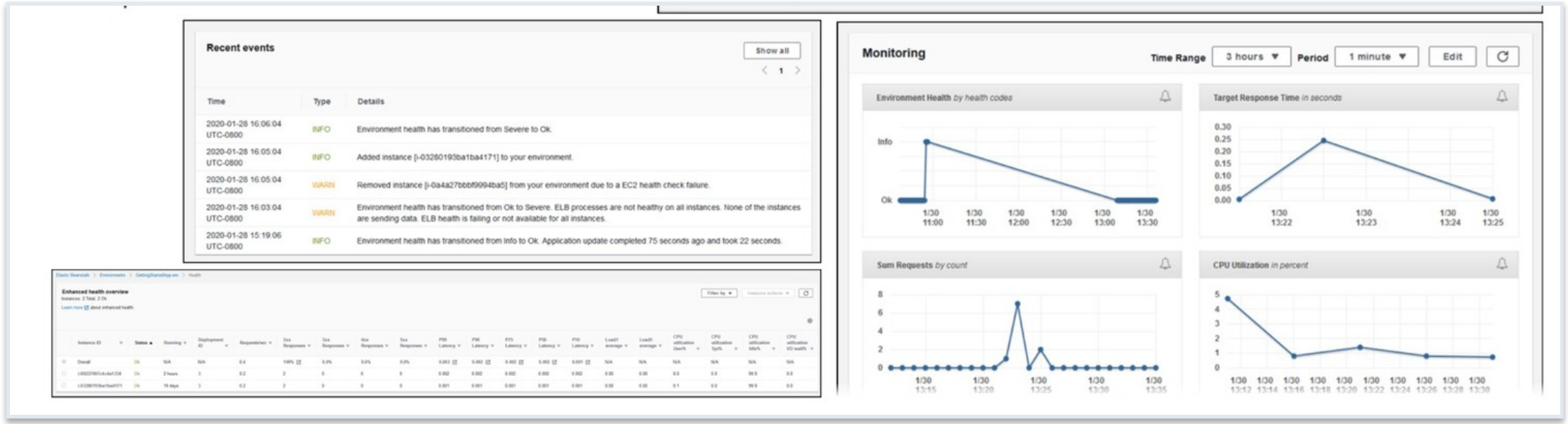
Elastic Beanstalk – Health Monitoring



Health agent pushes metrics to CloudWatch

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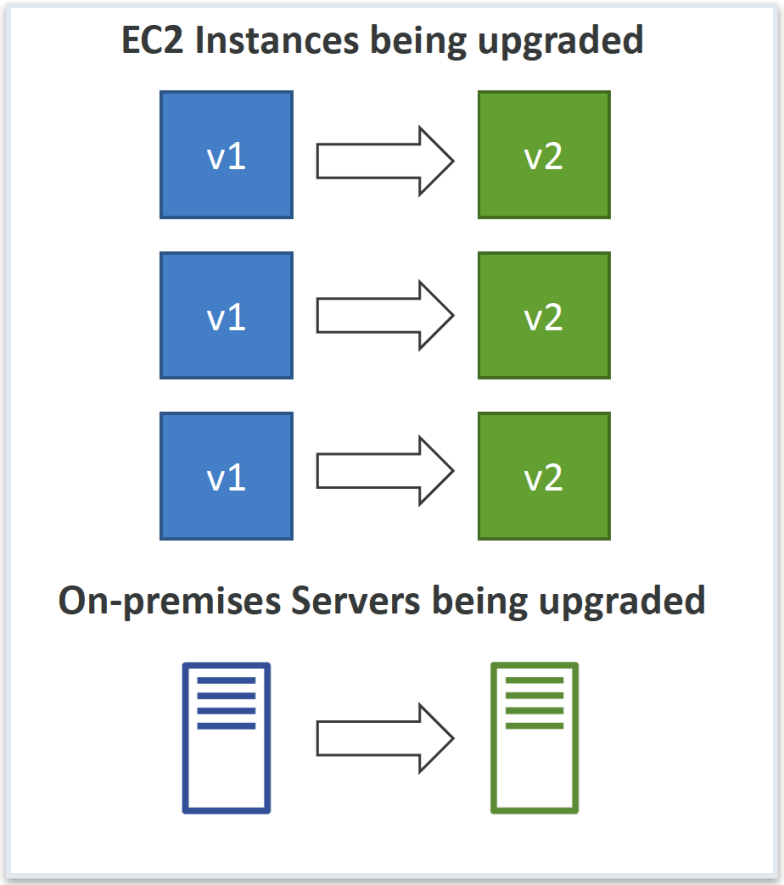
AWS CodeDeploy



We want to deploy our application automatically

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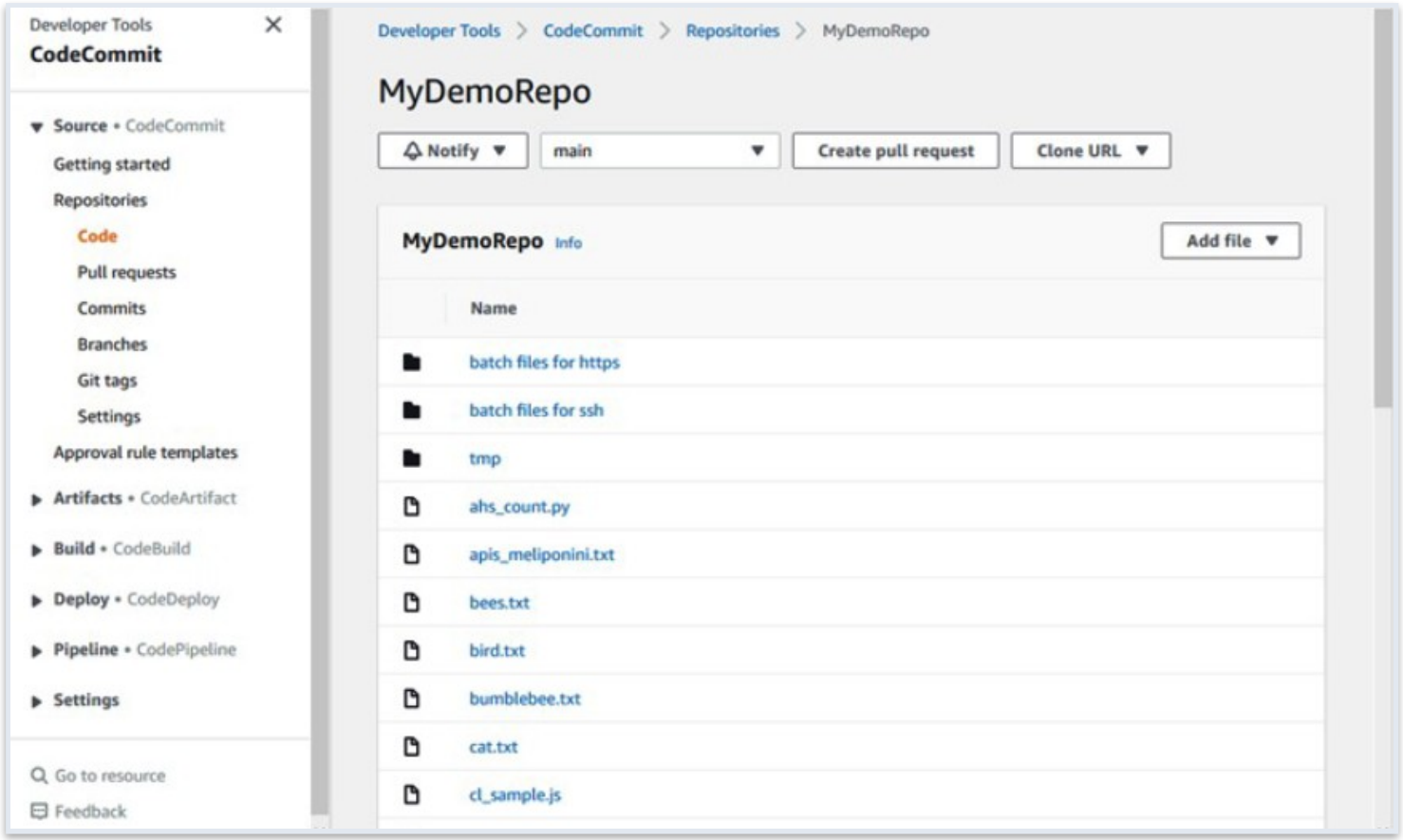
AWS CodeCommit



Before pushing the application code to servers, it needs to be stored somewhere

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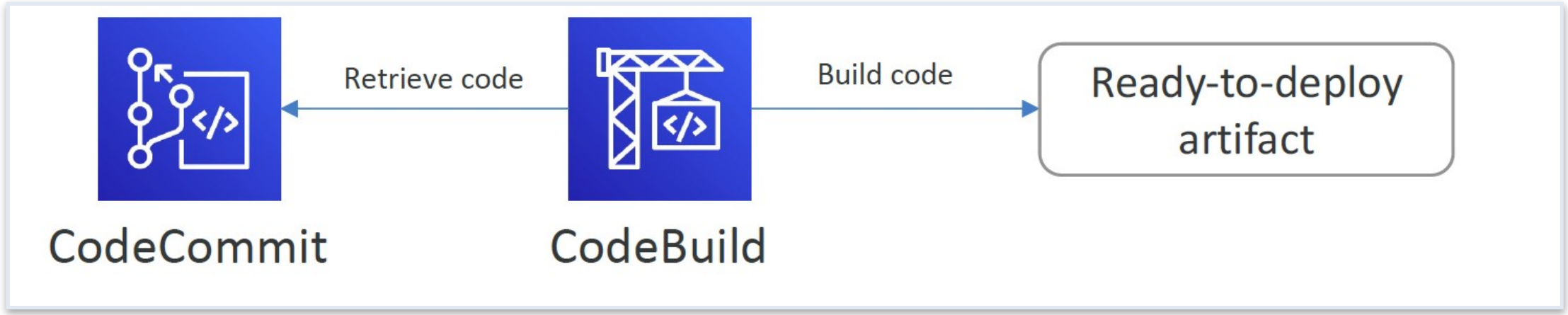
AWS CodeBuild



Code building service in the cloud (name is obvious)

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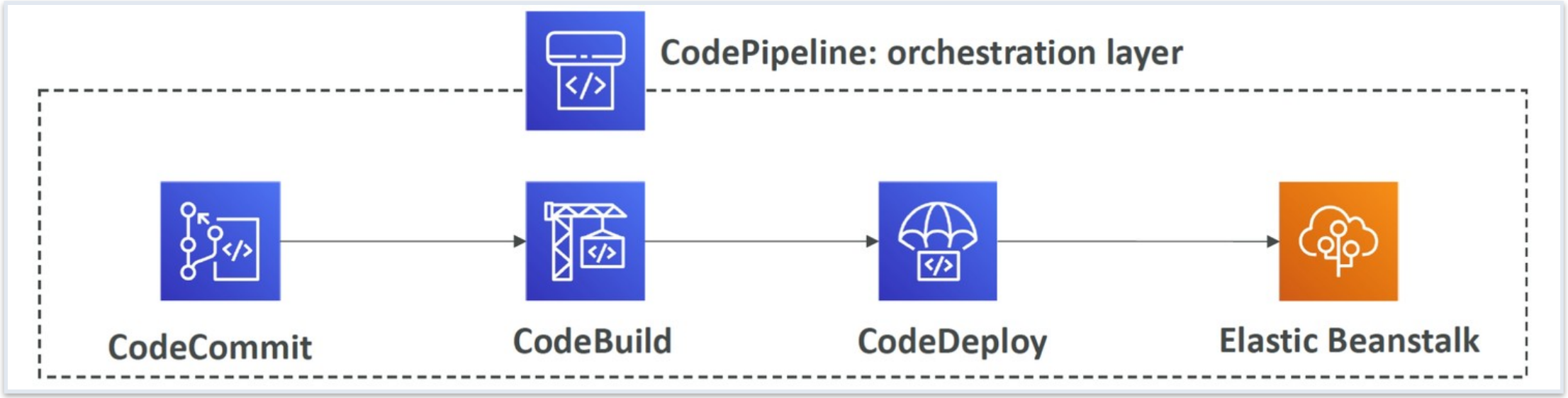
AWS CodePipeline



Orchestrate the different steps to have the code automatically pushed to production

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AWS CodeArtifact

- Software packages depend on each other to be built (also called code dependencies), and new ones are created
- Storing and retrieving these dependencies is called artifact management
- Traditionally you need to setup your own artifact management system
- CodeArtifact is a secure, scalable, and cost-effective artifact management for software development
- Works with common dependency management tools such as Maven, Gradle, npm, yarn, twine, pip, and NuGet
- Developers and CodeBuild can then retrieve dependencies straight from CodeArtifact

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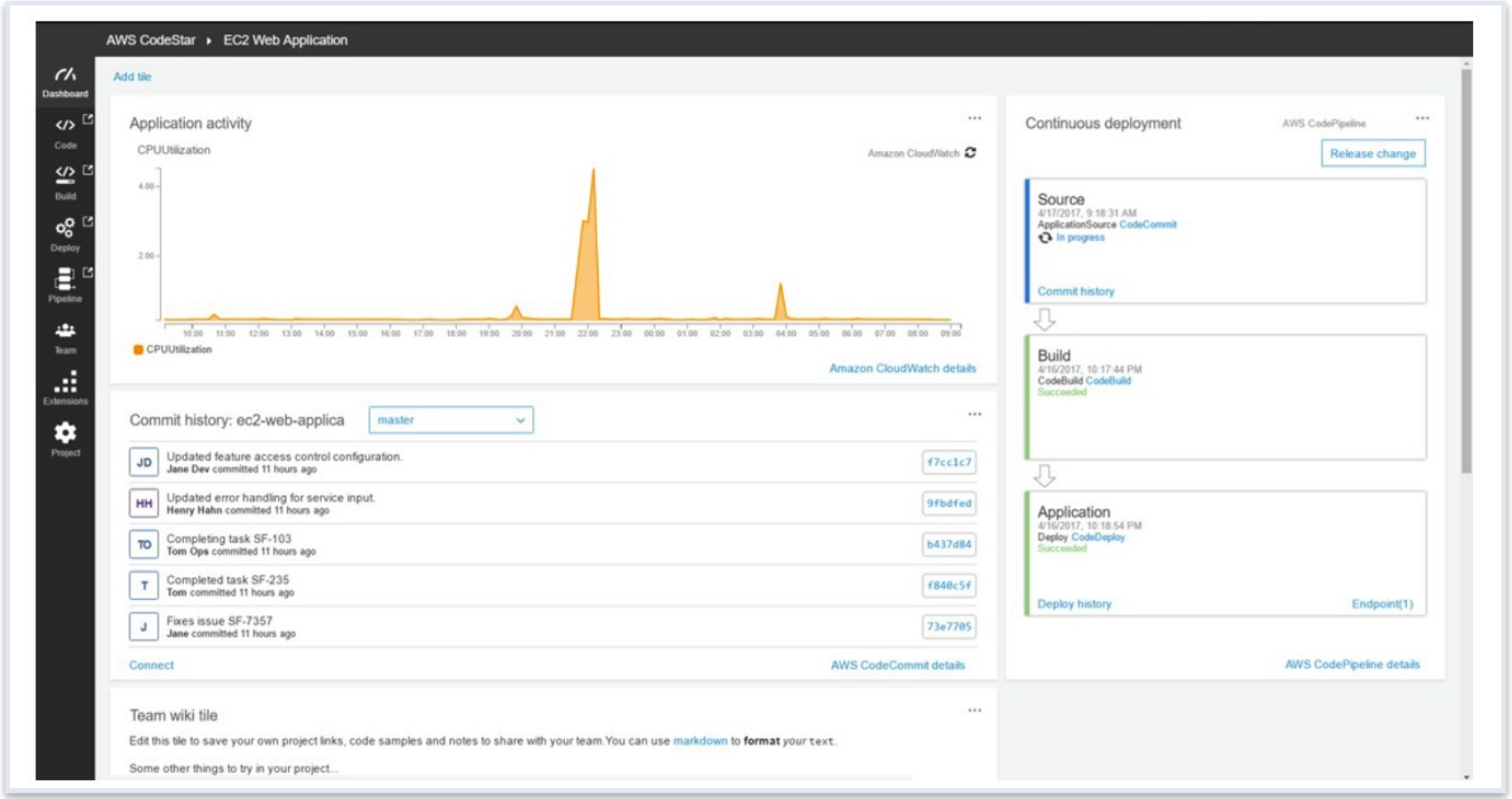
AWS CodeStar



Unified UI to easily manage software development activities in one place

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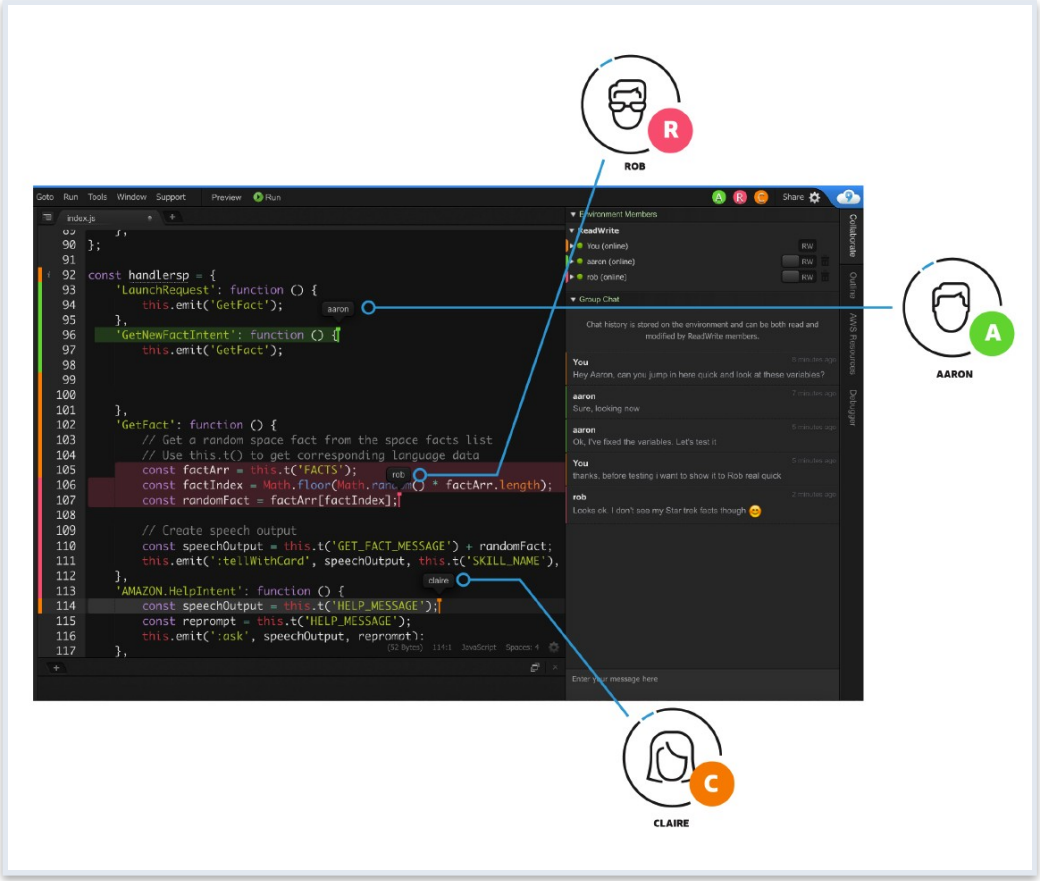
AWS Cloud9



AWS Cloud9 is a cloud IDE (Integrated Development Environment) for writing, running and debugging code

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AWS Systems Manager (SSM)

- Helps you manage your EC2 and On-Premises systems at scale
- Another Hybrid AWS service
- Get operational insights about the state of your infrastructure
- Suite of 10+ products
- Most important features are:
- Patching automation for enhanced compliance
- Run commands across an entire fleet of servers
- Store parameter configuration with the SSM Parameter Store
- Works for Linux, Windows, MacOS, and Raspberry Pi OS (Raspbian)

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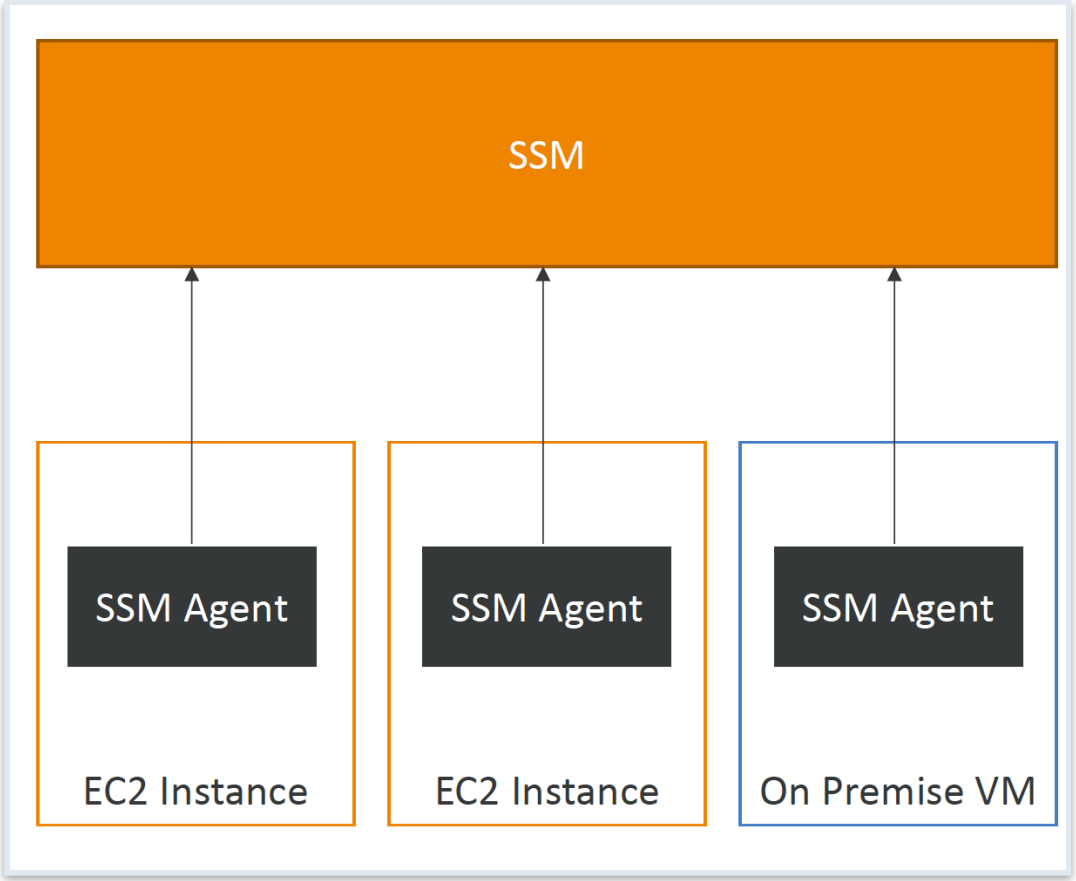
How Systems Manager works



We need to install the SSM agent onto the systems we control

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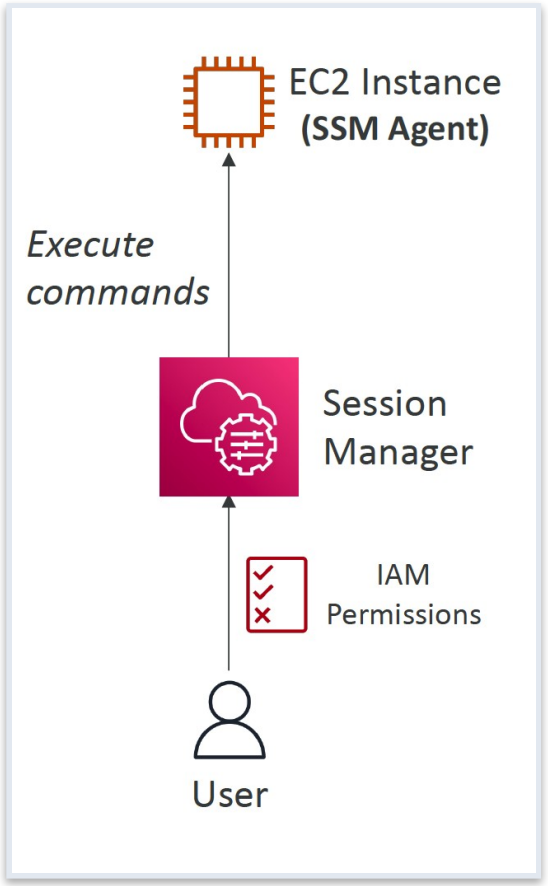
Systems Manager – SSM Session Manager



Allows you to start a secure shell on your EC2 and on-premises servers

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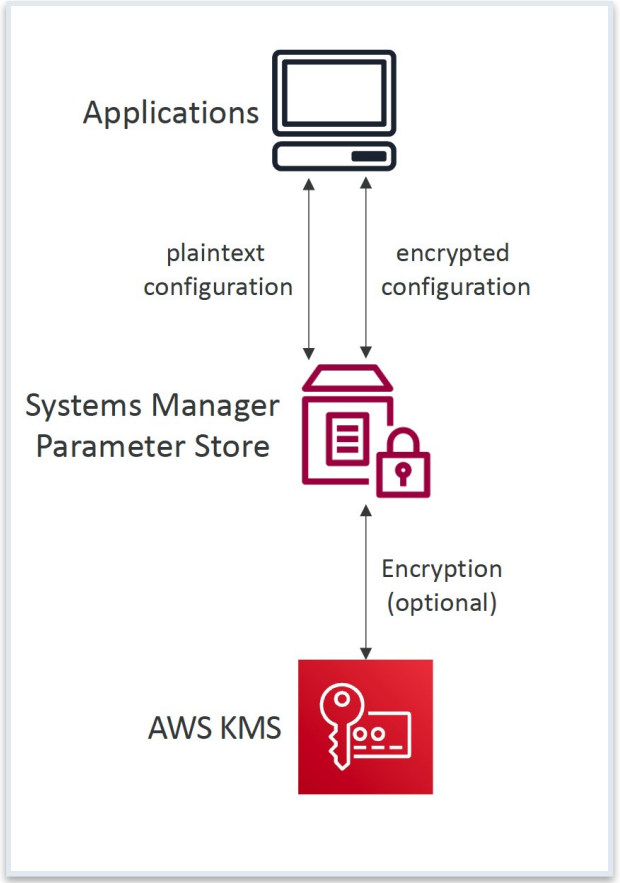
Systems Manager Parameter Store



Secure storage for configuration and secrets

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Deployment - Summary

- CloudFormation: (AWS only)
- Infrastructure as Code, works with almost all of AWS resources
- Repeat across Regions & Accounts
- Beanstalk: (AWS only)
- Platform as a Service (PaaS), limited to certain programming languages or Docker
- Deploy code consistently with a known architecture: ex, ALB + EC2 + RDS
- CodeDeploy (hybrid): deploy & upgrade any application onto servers
- Systems Manager (hybrid): patch, configure and run commands at scale

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Developer Services - Summary

- CodeCommit: Store code in private git repository (version controlled)
- CodeBuild: Build & test code in AWS
- CodeDeploy: Deploy code onto servers
- CodePipeline: Orchestration of pipeline (from code to build to deploy)
- CodeArtifact: Store software packages / dependencies on AWS
- CodeStar: Unified view for allowing developers to do CICD and code
- Cloud9: Cloud IDE (Integrated Development Environment) with collab
- AWS CDK: Define your cloud infrastructure using a programming language

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TOPIC 4

AWS Database and Storage Services

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AMAZON S3

Amazon S3

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Amazon S3

1. Amazon S3 is one of the main building blocks of AWS
2. It’s advertised as ”infinitely scaling” storage
3. Many websites use Amazon S3 as a backbone
4. Many AWS services use Amazon S3 as an integration as well

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Amazon S3 Use cases

Backup and storage

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Amazon S3 - Buckets

Amazon S3 allows people to store objects (files) in “buckets” (directories)

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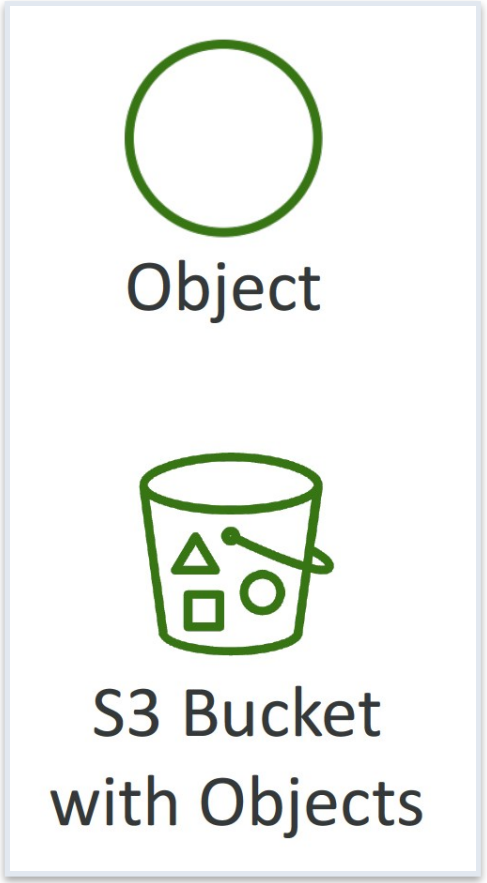
Amazon S3 - Objects



Objects (files) have a Key

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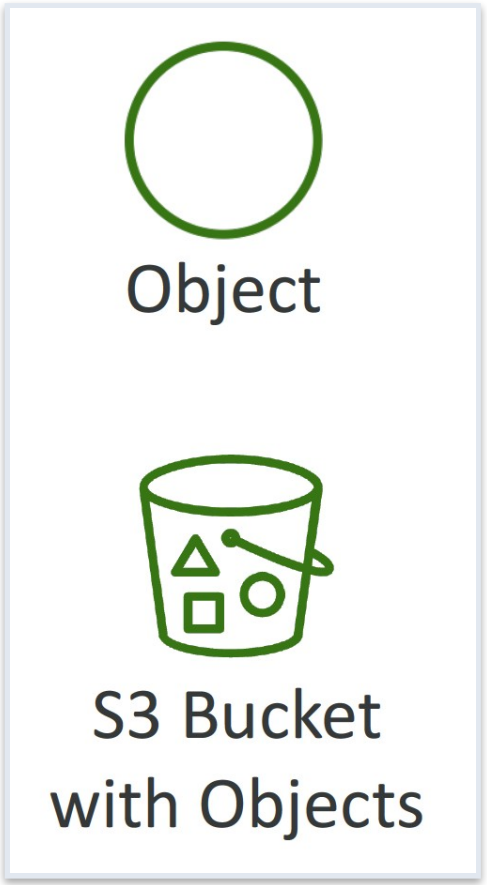
Amazon S3 – Objects (cont.)



Object values are the content of the body:

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Amazon S3 – Security

- User-Based
- IAM Policies – which API calls should be allowed for a specific user from IAM
- Resource-Based
- Bucket Policies – bucket wide rules from the S3 console - allows cross account
- Object Access Control List (ACL) – finer grain (can be disabled)
- Bucket Access Control List (ACL) – less common (can be disabled)
- Note: an IAM principal can access an S3 object if
- The user IAM permissions ALLOW it OR the resource policy ALLOWS it
- AND there’s no explicit DENY
- Encryption: encrypt objects in Amazon S3 using encryption keys

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S3 Bucket Policies

```
{
    "Version": "2012-10-17",
    "Statement": [
        {
            "Sid": "PublicRead",
            "Effect": "Allow",
            "Principal": "*",
            "Action": [
                "s3:GetObject"
            ],
            "Resource": [
                "arn:aws:s3:::examplebucket/*"
            ]
        }
    ]
}
```

JSON based policies

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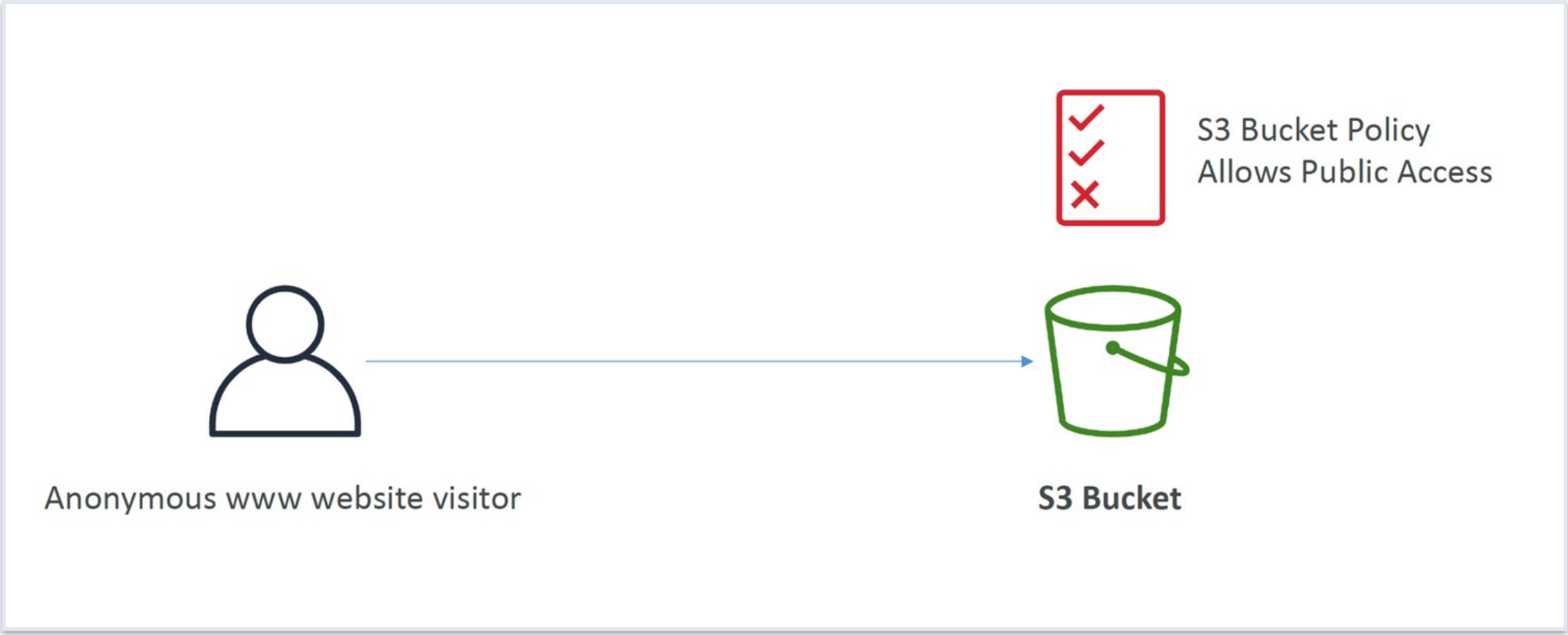
Example: Public Access - Use Bucket Policy



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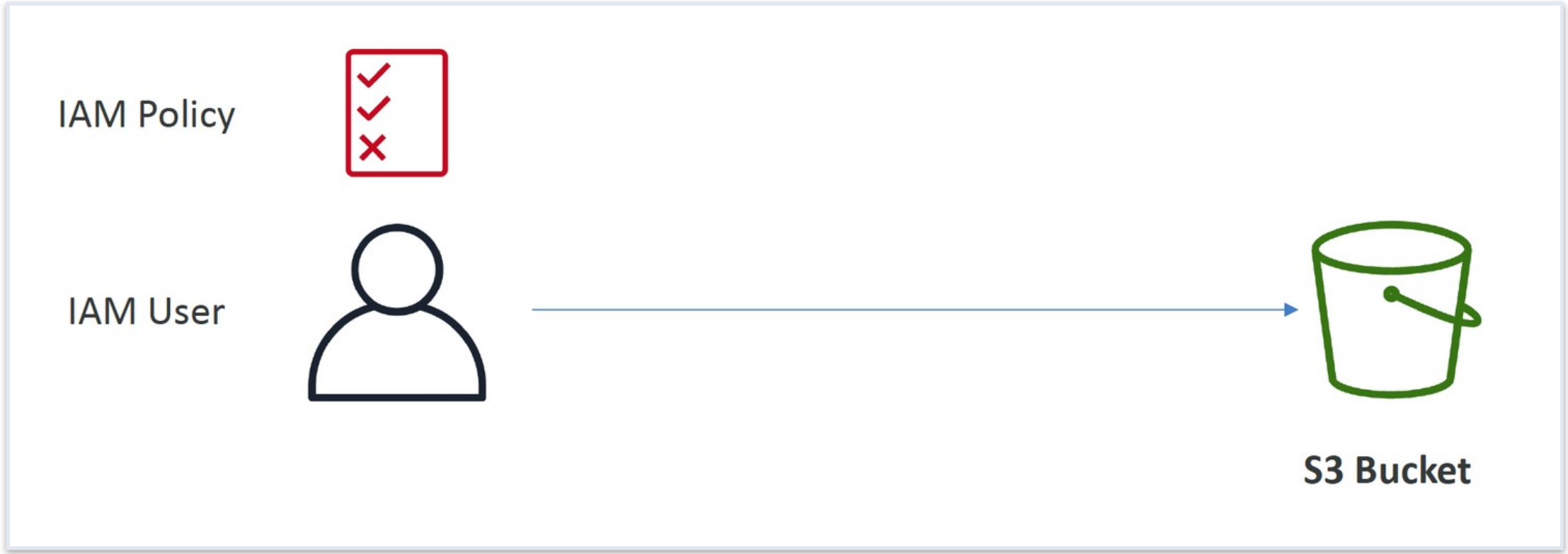
Example: User Access to S3 – IAM permissions



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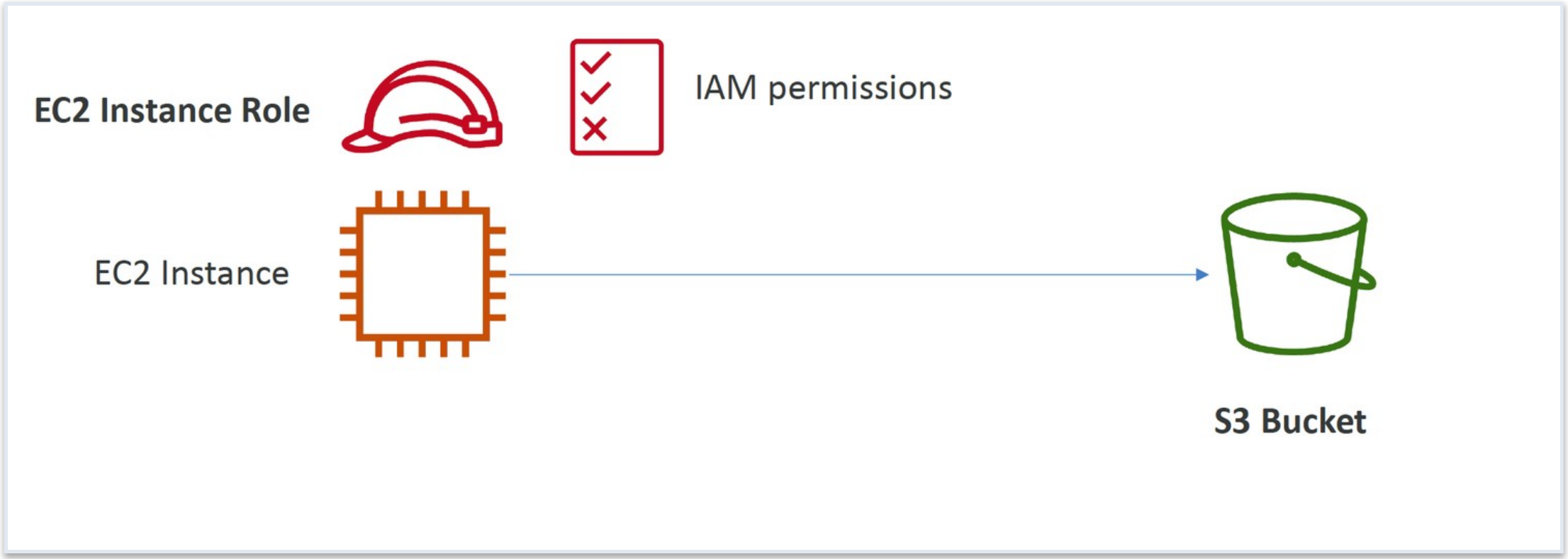
Example: EC2 instance access - Use IAM Roles



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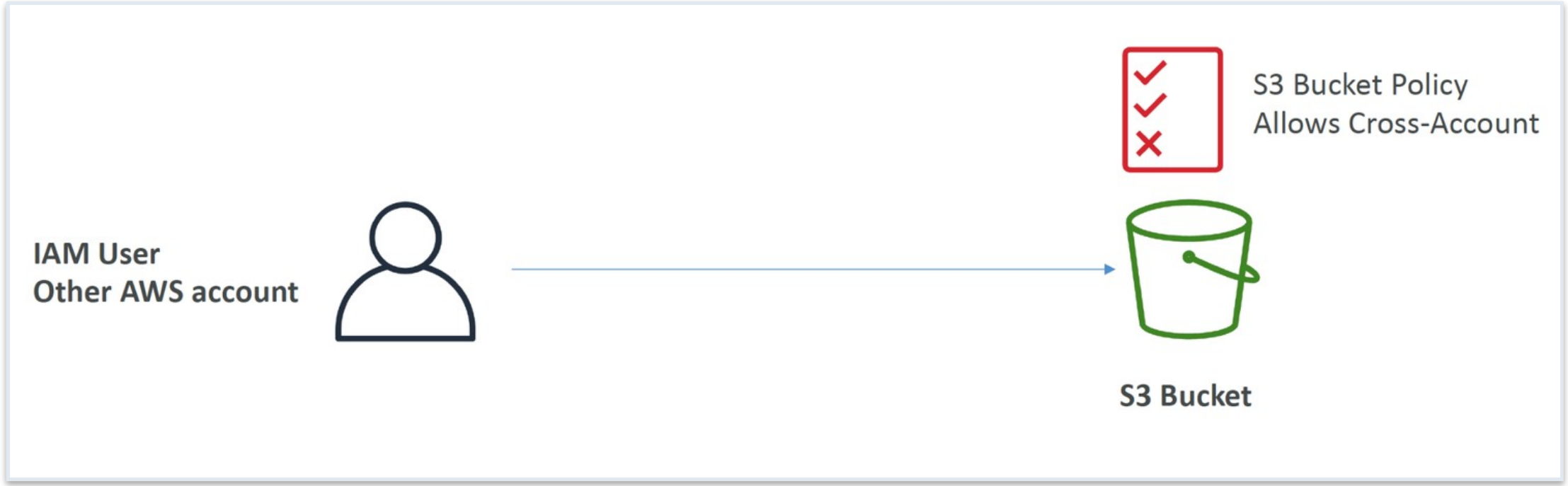
Advanced: Cross-Account Access –



Use Bucket Policy

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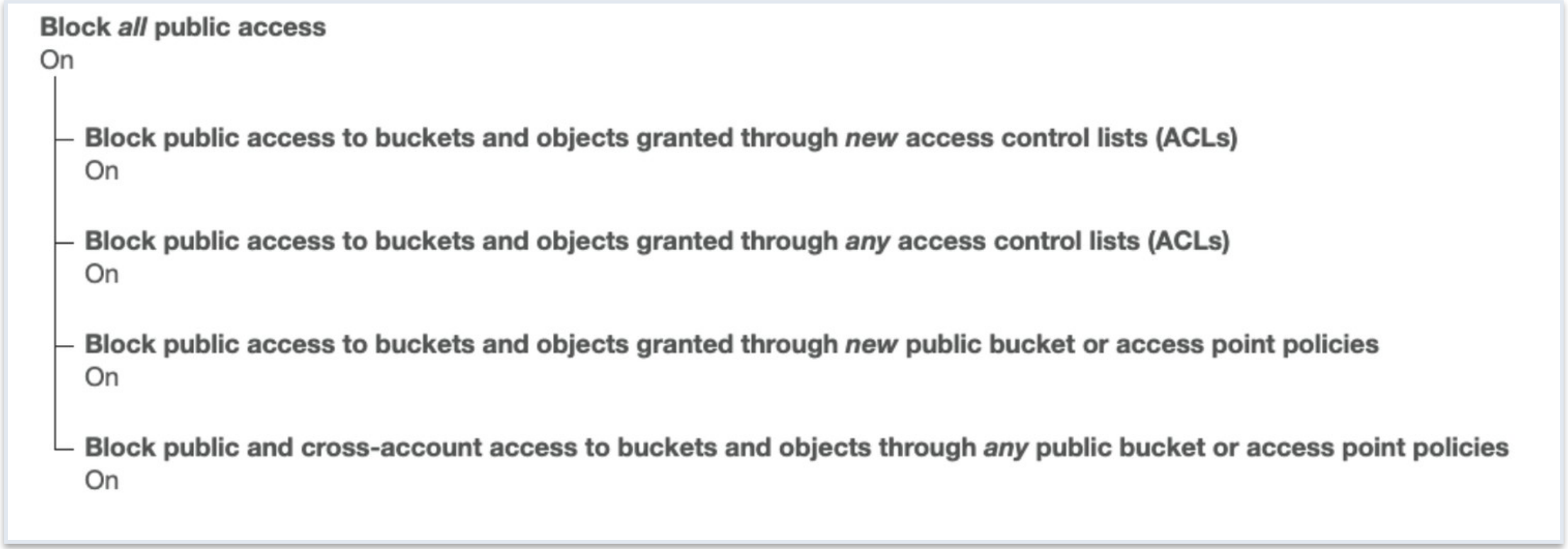
Bucket settings for Block Public Access



These settings were created to prevent company data leaks

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Amazon S3 – Static Website Hosting

- S3 can host static websites and have them accessible on the Internet
- The website URL will be (depending on the region) http://bucket-name.s3-website-aws-region.amazonaws.com OR http://bucket-name.s3-website.aws-region.amazonaws.com
- If you get a 403 Forbidden error, make sure the bucket policy allows public reads!

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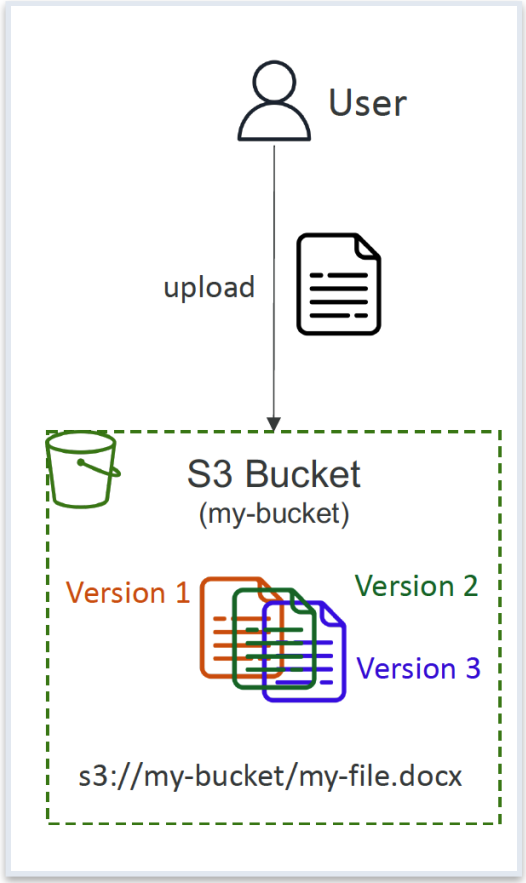
Amazon S3 - Versioning



You can version your files in Amazon S3

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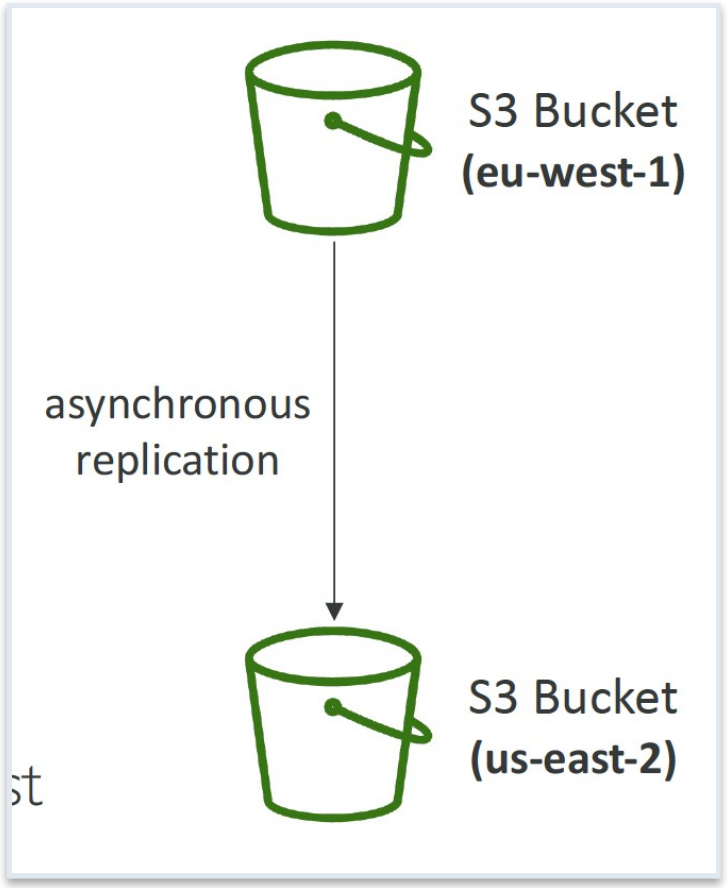
Amazon S3 – Replication (CRR & SRR)



Must enable Versioning in source and destination buckets

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S3 Storage Classes

- Amazon S3 Standard - General Purpose
- Amazon S3 Standard-Infrequent Access (IA)
- Amazon S3 One Zone-Infrequent Access
- Amazon S3 Glacier Instant Retrieval
- Amazon S3 Glacier Flexible Retrieval
- Amazon S3 Glacier Deep Archive
- Amazon S3 Intelligent Tiering
- Can move between classes manually or using S3 Lifecycle configurations

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S3 Durability and Availability

- Durability:
- High durability (99.999999999%, 11 9’s) of objects across multiple AZ
- If you store 10,000,000 objects with Amazon S3, you can on average expect to incur a loss of a single object once every 10,000 years
- Same for all storage classes
- Availability:
- Measures how readily available a service is
- Varies depending on storage class
- Example: S3 standard has 99.99% availability = not available 53 minutes a year

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S3 Standard – General Purpose

- 99.99% Availability
- Used for frequently accessed data
- Low latency and high throughput
- Sustain 2 concurrent facility failures
- Use Cases: Big Data analytics, mobile & gaming applications, content distribution…

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S3 Storage Classes – Infrequent Access

- For data that is less frequently accessed, but requires rapid access when needed
- Lower cost than S3 Standard
- Amazon S3 Standard-Infrequent Access (S3 Standard-IA)
- 99.9% Availability
- Use cases: Disaster Recovery, backups
- Amazon S3 One Zone-Infrequent Access (S3 One Zone-IA)
- High durability (99.999999999%) in a single AZ; data lost when AZ is destroyed
- 99.5% Availability
- Use Cases: Storing secondary backup copies of on-premise data, or data you can recreate

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Amazon S3 Glacier Storage Classes

- Low-cost object storage meant for archiving / backup
- Pricing: price for storage + object retrieval cost
- Amazon S3 Glacier Instant Retrieval
- Millisecond retrieval, great for data accessed once a quarter
- Minimum storage duration of 90 days
- Amazon S3 Glacier Flexible Retrieval (formerly Amazon S3 Glacier):
- Expedited (1 to 5 minutes), Standard (3 to 5 hours), Bulk (5 to 12 hours) – free
- Minimum storage duration of 90 days
- Amazon S3 Glacier Deep Archive – for long term storage:
- Standard (12 hours), Bulk (48 hours)
- Minimum storage duration of 180 days

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S3 Intelligent-Tiering

- Small monthly monitoring and auto-tiering fee
- Moves objects automatically between Access Tiers based on usage
- There are no retrieval charges in S3 Intelligent-Tiering
- Frequent Access tier (automatic): default tier
- Infrequent Access tier (automatic): objects not accessed for 30 days
- Archive Instant Access tier (automatic): objects not accessed for 90 days
- Archive Access tier (optional): configurable from 90 days to 700+ days
- Deep Archive Access tier (optional): config. from 180 days to 700+ days

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S3 Storage Classes Comparison

	Standard	Intelligent-Tiering	Standard-IA	One Zone-IA	Glacier Instant Retrieval	Glacier Flexible Retrieval	Glacier Deep Archive
Durability	99.999999999% == (11 9's)						
Availability	99.99%	99.9%	99.9%	99.5%	99.9%	99.99%	99.99%
Availability SLA	99.9%	99%	99%	99%	99%	99.9%	99.9%
Availability Zones	>= 3	>= 3	>= 3	1	>= 3	>= 3	>= 3
Min. Storage Duration Charge	None	None	30 Days	30 Days	90 Days	90 Days	180 Days
Min. Billable Object Size	None	None	128 KB	128 KB	128 KB	40 KB	40 KB
Retrieval Fee	None	None	Per GB retrieved	Per GB retrieved	Per GB retrieved	Per GB retrieved	Per GB retrieved

https://aws.amazon.com/s3/storage-classes/

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S3 Storage Classes – Price Comparison

	Standard	Intelligent-Tiering	Standard-IA	One Zone-IA	Glacier Instant Retrieval	Glacier Flexible Retrieval	Glacier Deep Archive
Storage Cost (per GB per month)	$0.023	$0.0025 - $0.023	$0.0125	$0.01	$0.004	$0.0036	$0.00099
Retrieval Cost (per 1000 request)	**GET:** $0.0004 **POST:** $0.005	**GET:** $0.0004 **POST:** $0.005	**GET:** $0.001 **POST:** $0.01	**GET:** $0.001 **POST:** $0.01	**GET:** $0.01 **POST:** $0.02	**GET:** $0.0004 **POST:** $0.03 **Expedited:** $10 **Standard:** $0.05 **Bulk:** free	**GET:** $0.0004 **POST:** $0.05 **Standard:** $0.10 **Bulk:** $0.025
Retrieval Time	Instantaneous					**Expedited** (1 – 5 mins) **Standard** (3 – 5 hours) Bulk (5 – 12 hours)	**Standard** (12 hours) **Bulk** (48 hours)
Monitoring Cost (pet 1000 objects)		$0.0025					

Example: us-east-1

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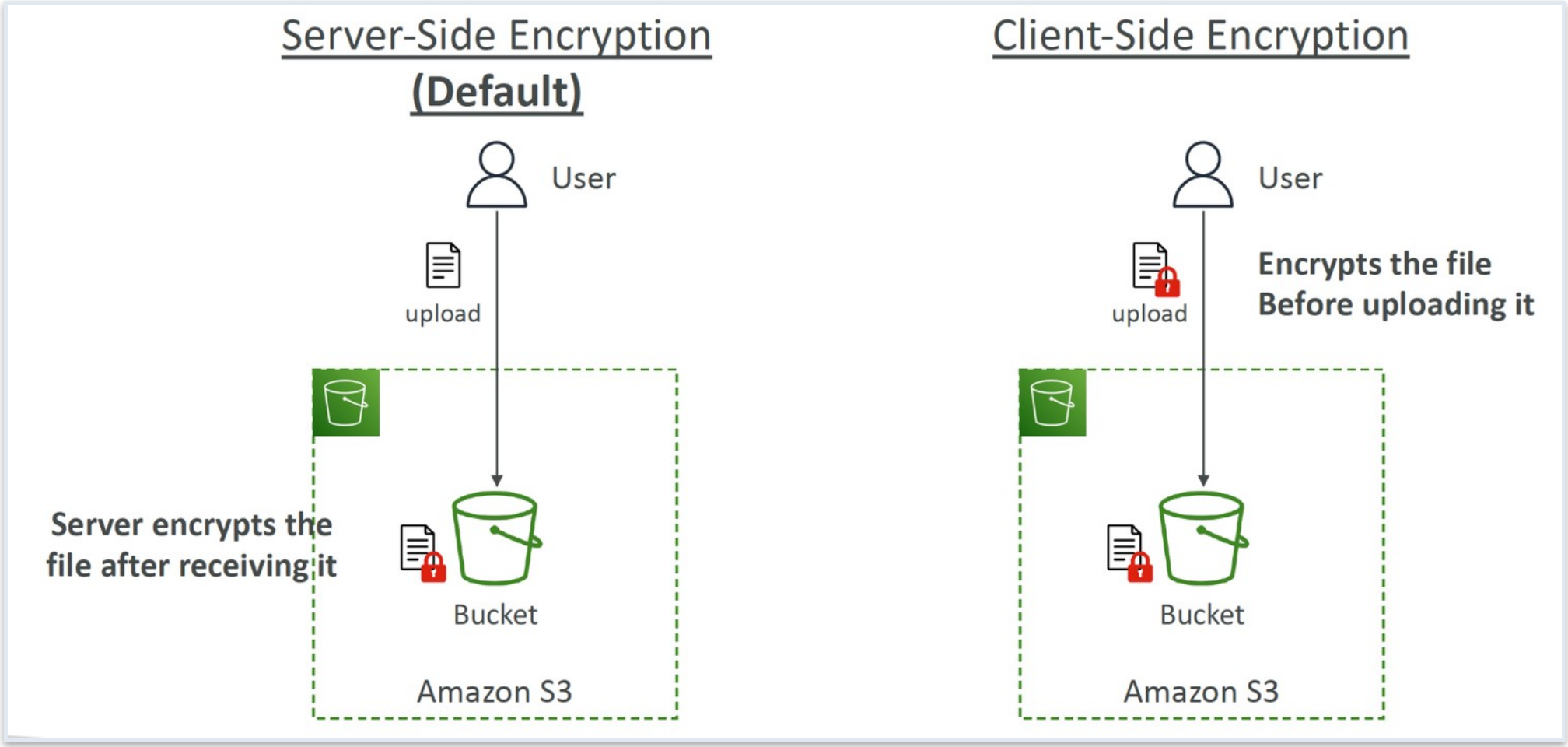
S3 Encryption



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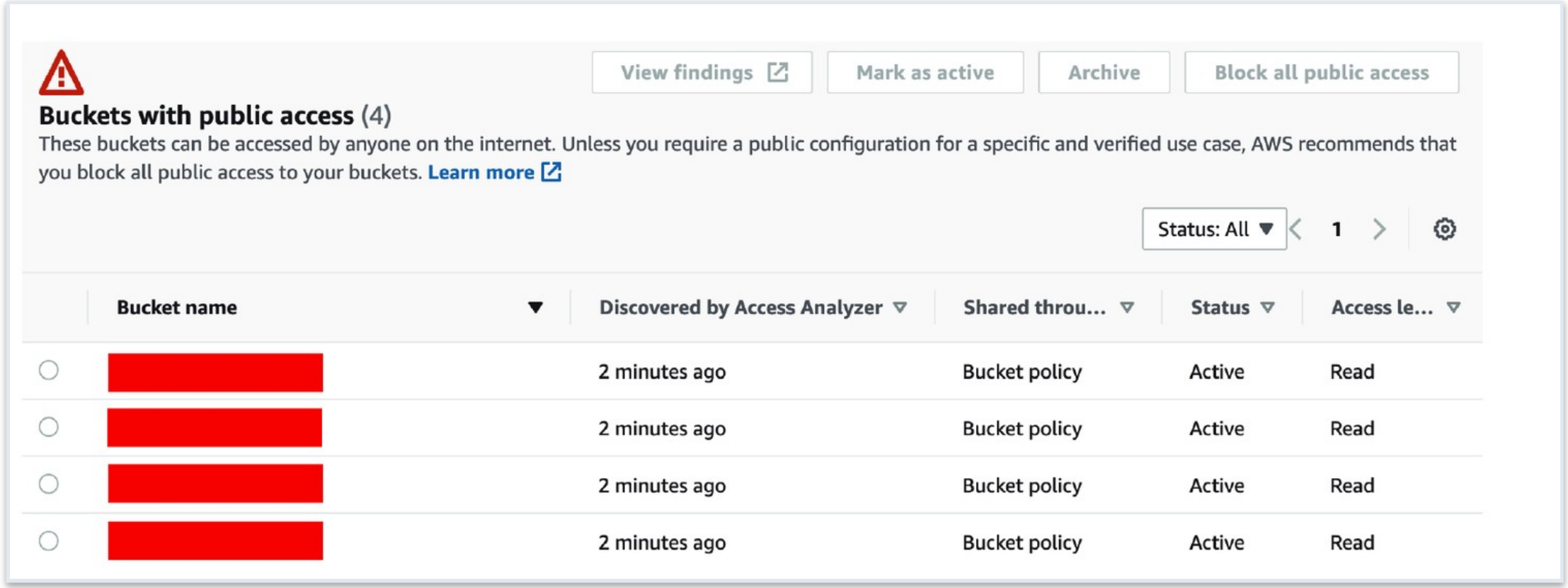
IAM Access Analyzer for S3



Ensures that only intended people have access to your S3 buckets

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Shared Responsibility Model for S3

- Infrastructure (global security, durability, availability, sustain concurrent loss of data in two facilities)
- Configuration and vulnerability analysis
- Compliance validation

- S3 Versioning
- S3 Bucket Policies
- S3 Replication Setup
- Logging and Monitoring
- S3 Storage Classes
- Data encryption at rest and in transit

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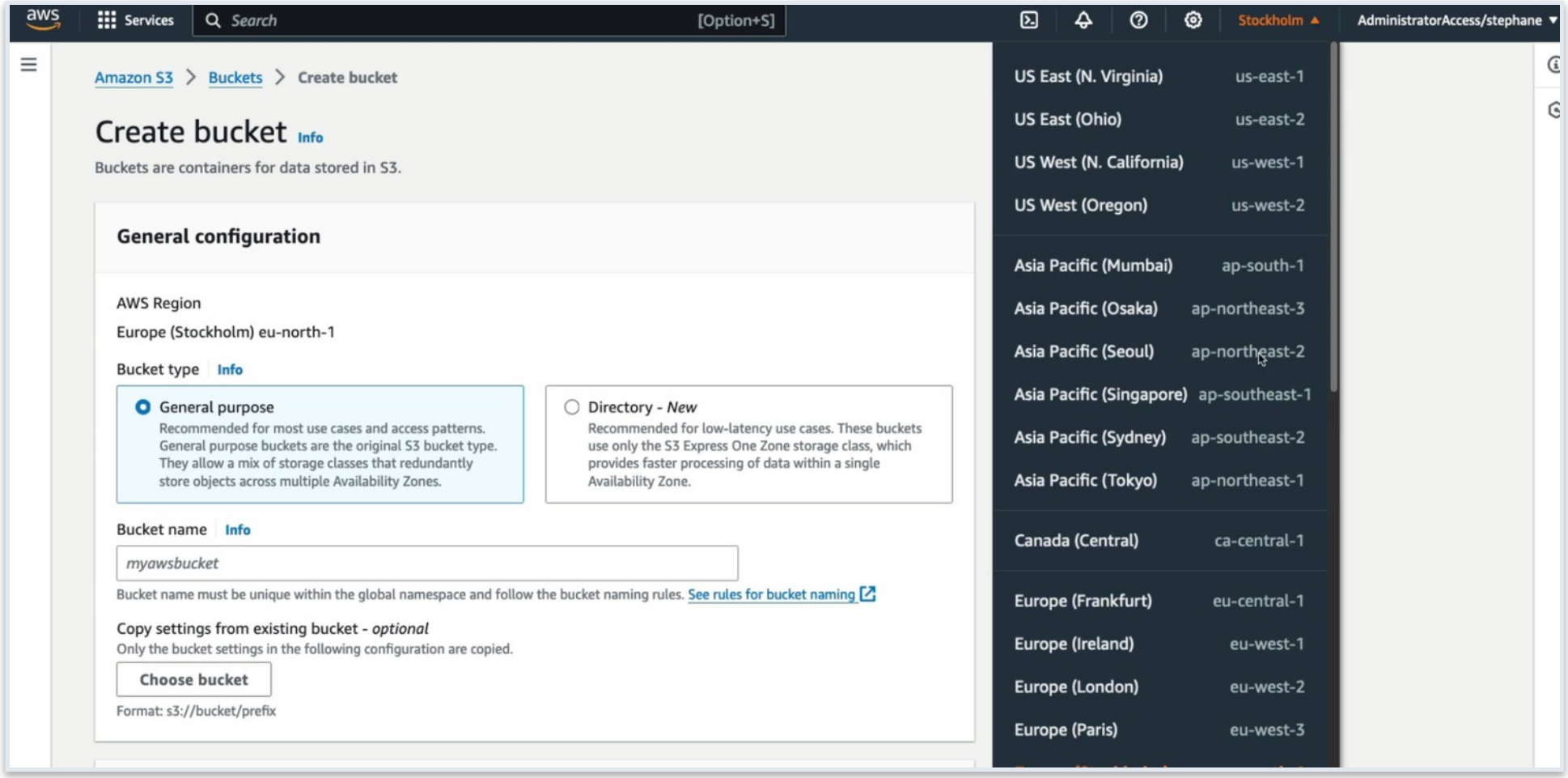
Activity: S3 Hand On



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AWS Snow Family

Highly-secure, portable devices to collect and process data at the edge, and migrate data into and out of AWS

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Data Migrations with AWS Snow Family

	Time to Transfer		
	100 Mbps	1Gbps	10Gbps
10 TB	12 days	30 hours	3 hours
100 TB	124 days	12 days	30 hours
1 PB	3 years	124 days	12 days

Challenges:

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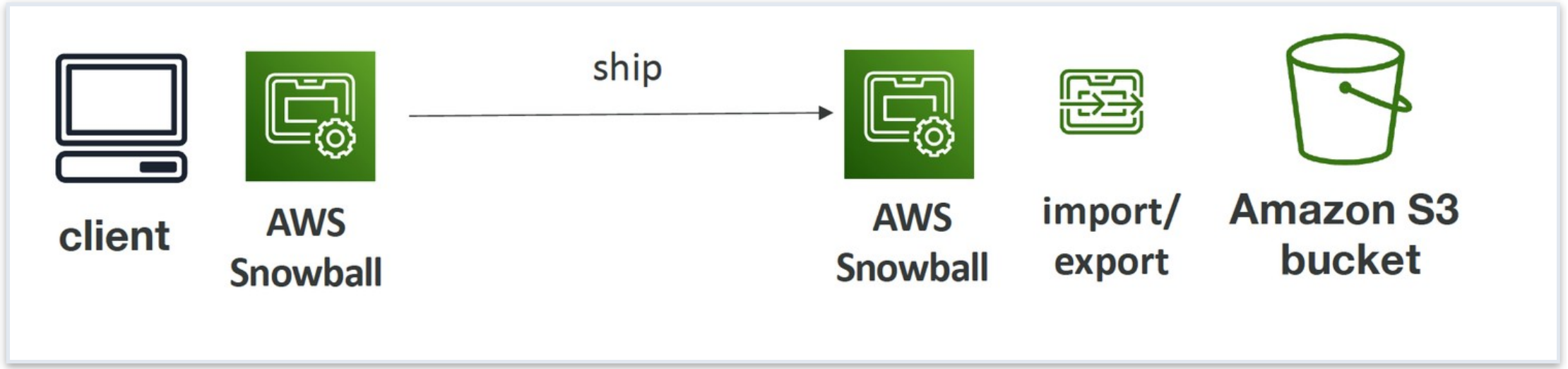
Diagrams



Direct upload to S3:

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Snowball Edge (for data transfers)

Physical data transport solution: move TBs or PBs of data in or out of AWS

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AWS Snowcone & Snowcone SSD

Small, portable computing, anywhere, rugged & secure, withstands harsh environments

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AWS Snowmobile

Transfer exabytes of data (1 EB = 1,000 PB = 1,000,000 TBs)

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AWS Snow Family for Data Migrations

Snowcone

Snowball Edge

Snowmobile

	Snowcone & Snowcone SSD	Snowball Edge Storage Optimized	Snowmobile
Storage Capacity	8 TB HDD 14 TB SSD	80 TB - 210 TB	< 100 PB
Migration Size	Up to 24 TB, online and offline	Up to petabytes, offline	Up to exabytes, offline
DataSync agent	Pre-installed		

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Snow Family – Usage Process

1. Request Snowball devices from the AWS console for delivery
2. Install the snowball client / AWS OpsHub on your servers
3. Connect the snowball to your servers and copy files using the client
4. Ship back the device when you're done (goes to the right AWS facility)
5. Data will be loaded into an S3 bucket
6. Snowball is completely wiped

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AWS DataSync

- Move large amount of data from on-premises to AWS
- Can synchronize to: Amazon S3 (any storage classes – including Glacier), Amazon EFS, Amazon FSx for Windows
- Replication tasks can be scheduled hourly, daily, weekly
- The replication tasks are incremental after the first full load

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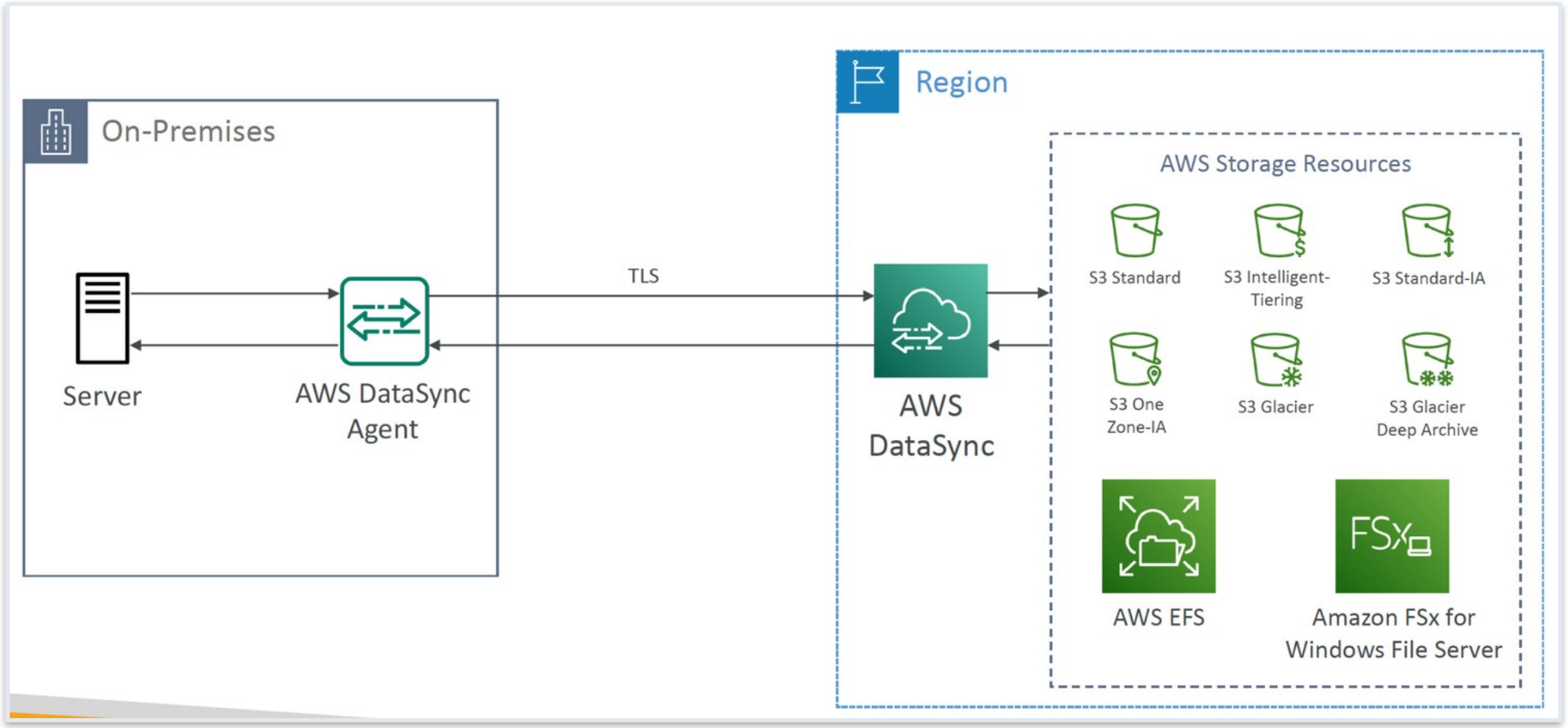
AWS DataSync



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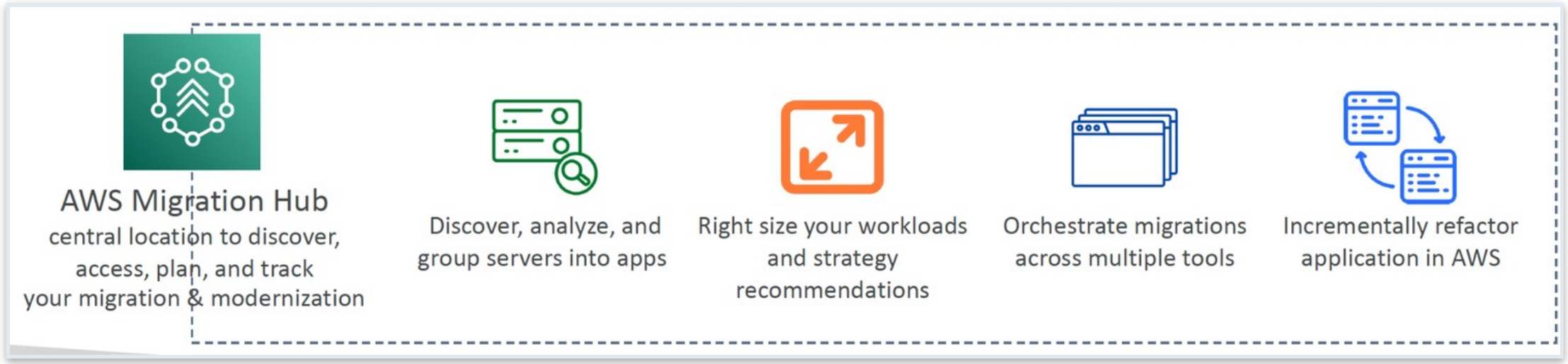
AWS Migration Hub



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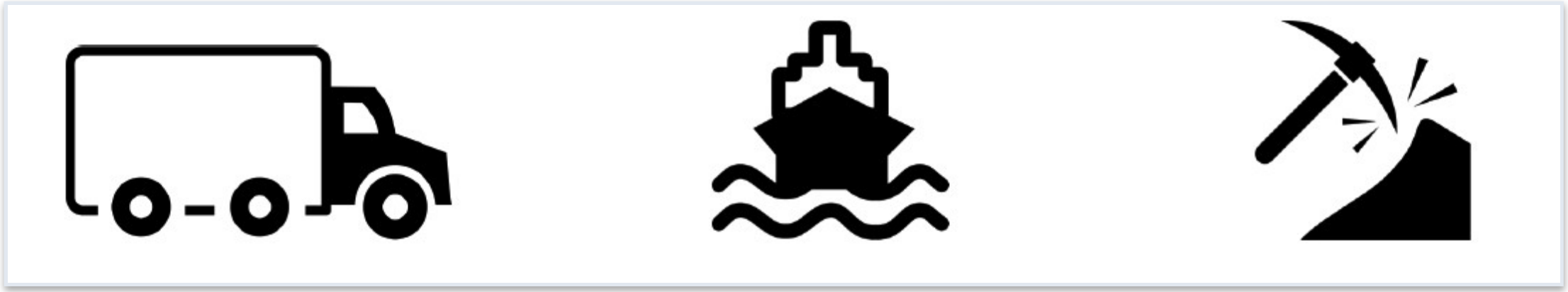
What is Edge Computing?



Process data while it’s being created on an edge location

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Snow Family – Edge Computing

- Snowcone & Snowcone SSD (smaller)
- 2 CPUs, 4 GB of memory, wired or wireless access
- USB-C power using a cord or the optional battery
- Snowball Edge – Compute Optimized
- 104 vCPUs, 416 GiB of RAM
- Optional GPU (useful for video processing or machine learning)
- 28 TB NVMe or 42TB HDD usable storage
- Storage Clustering available (up to 16 nodes)
- Snowball Edge – Storage Optimized
- Up to 40 vCPUs, 80 GiB of RAM, 80 TB storage
- Up to 104 vCPUs, 416 GiB of RAM, 210 TB NVMe storage
- All: Can run EC2 Instances & AWS Lambda functions (using AWS IoT Greengrass)
- Long-term deployment options: 1 and 3 years discounted pricing

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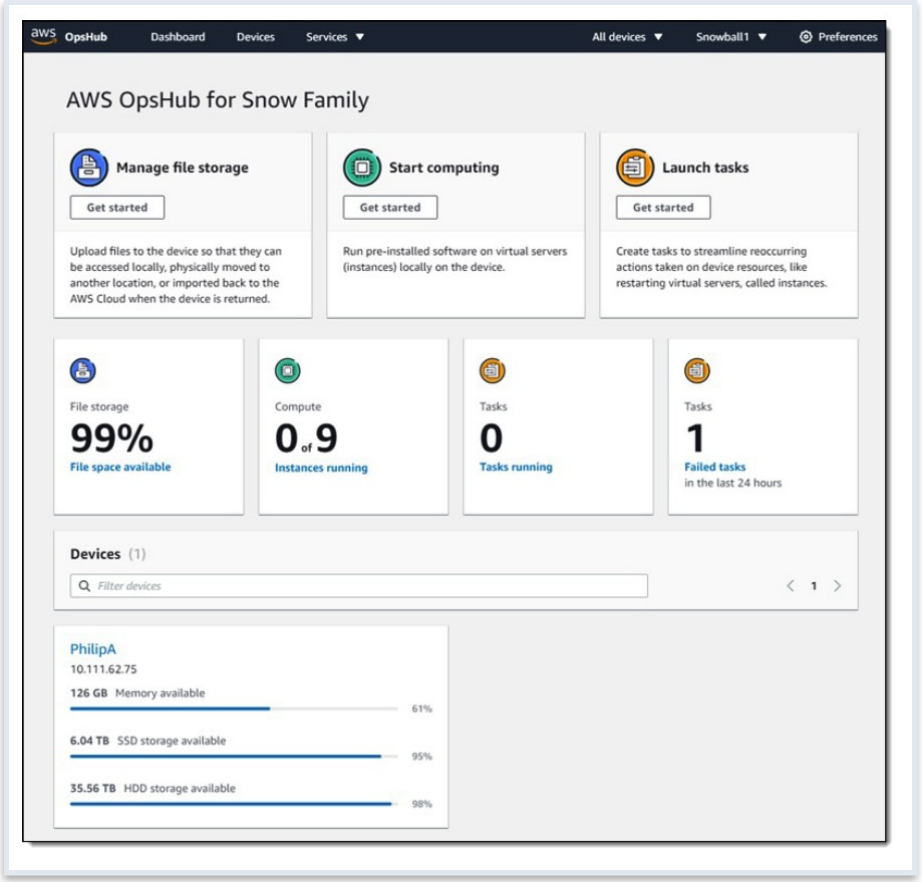
AWS OpsHub



Historically, to use Snow Family devices, you needed a CLI (Command Line Interface tool)

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Snowball Edge Pricing

- You pay for device usage and data transfer out of AWS
- Data transfer IN to Amazon S3 is $0.00 per GB
- On-Demand
- Includes a one-time service fee per job, which includes:
- 10 days of usage for Snowball Edge Storage Optimized 80TB
- 15 days of usage for Snowball Edge Storage Optimized 210TB
- Shipping days are NOT counted towards the included 10 or 15 days
- Pay per day for any additional days
- Committed Upfront
- Pay in advance for monthly, 1-year, and 3-years of usage (Edge Computing)
- Up to 62% discounted pricing

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Hybrid Cloud for Storage

- AWS is pushing for ”hybrid cloud”
- Part of your infrastructure is on-premises
- Part of your infrastructure is on the cloud
- This can be due to
- Long cloud migrations
- Security requirements
- Compliance requirements
- IT strategy
- S3 is a proprietary storage technology (unlike EFS / NFS), so how do you expose the S3 data on-premise?
- AWS Storage Gateway!

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AWS Storage Cloud Native Options

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AWS Storage Gateway

Bridge between on-premise data and cloud data in S3

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Amazon S3 – Summary

- Buckets vs Objects: global unique name, tied to a region
- S3 security: IAM policy, S3 Bucket Policy (public access), S3 Encryption
- S3 Websites: host a static website on Amazon S3
- S3 Versioning: multiple versions for files, prevent accidental deletes
- S3 Replication: same-region or cross-region, must enable versioning
- S3 Storage Classes: Standard, IA, 1Z-IA, Intelligent, Glacier (Instant, Flexible, Deep)
- Snow Family: import data onto S3 through a physical device, edge computing
- OpsHub: desktop application to manage Snow Family devices
- Storage Gateway: hybrid solution to extend on-premises storage to S3

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DATABASE

Database

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Databases Intro

- Storing data on disk (EFS, EBS, EC2 Instance Store, S3) can have its limits
- Sometimes, you want to store data in a database…
- You can structure the data
- You build indexes to efficiently query / search through the data
- You define relationships between your datasets
- Databases are optimized for a purpose and come with different features, shapes and constraints

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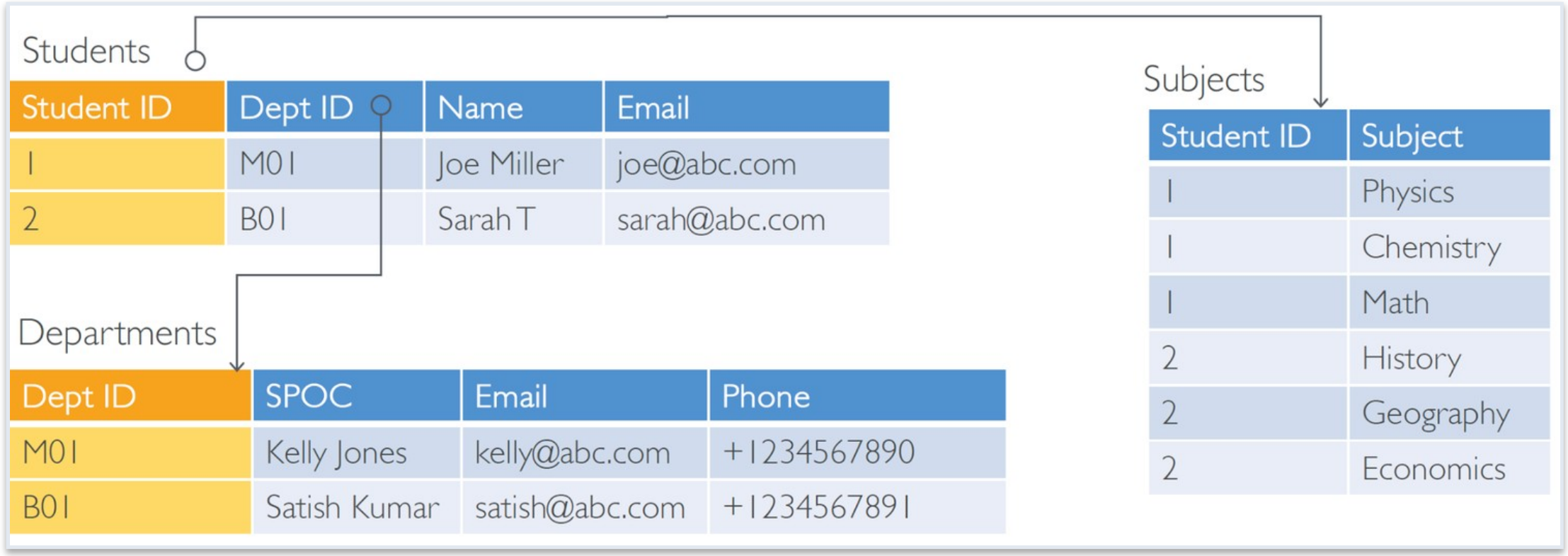
Relational Databases



Looks just like Excel spreadsheets, with links between them!

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NoSQL Databases

- NoSQL = non-SQL = non relational databases
- NoSQL databases are purpose built for specific data models and have flexible schemas for building modern applications.
- Benefits:
- Flexibility: easy to evolve data model
- Scalability: designed to scale-out by using distributed clusters
- High-performance: optimized for a specific data model
- Highly functional: types optimized for the data model
- Examples: Key-value, document, graph, in-memory, search databases

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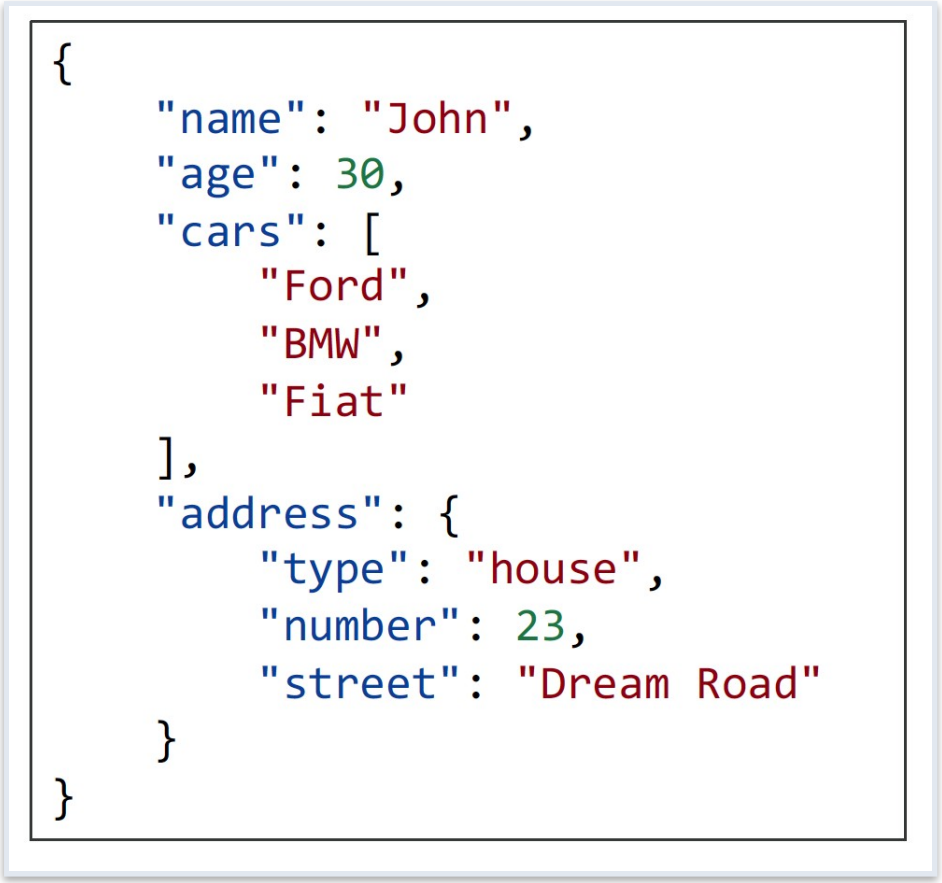
NoSQL data example: JSON



```
{
    "name": "John",
    "age": 30,
    "cars": [
        "Ford",
        "BMW",
        "Fiat"
    ],
    "address": {
        "type": "house",
        "number": 23,
        "street": "Dream Road"
    }
}
```

JSON = JavaScript Object Notation

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Databases & Shared Responsibility on AWS

- AWS offers use to manage different databases
- Benefits include:
- Quick Provisioning, High Availability, Vertical and Horizontal Scaling
- Automated Backup & Restore, Operations, Upgrades
- Operating System Patching is handled by AWS
- Monitoring, alerting
- Note: many databases technologies could be run on EC2, but you must handle yourself the resiliency, backup, patching, high availability, fault tolerance, scaling…

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Amazon RDS Overview

- RDS stands for Relational Database Service
- It’s a managed DB service for DB use SQL as a query language.
- It allows you to create databases in the cloud that are managed by AWS
- Postgres
- MySQL
- MariaDB
- Oracle
- Microsoft SQL Server
- IBM DB2
- Aurora (AWS Proprietary database)

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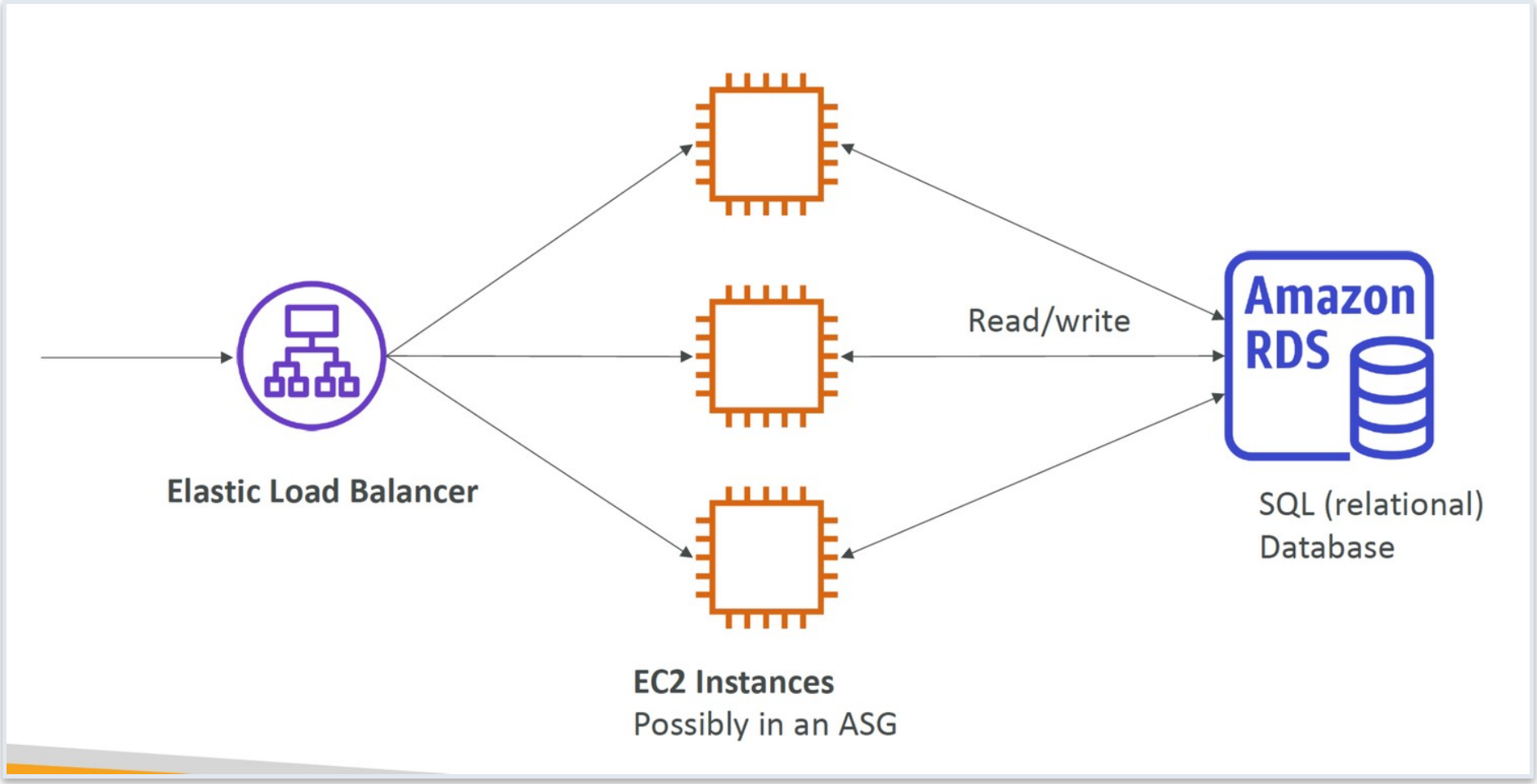
RDS Solution Architecture



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Advantage over using RDS versus deploying DB on EC2

- RDS is a managed service:
- Automated provisioning, OS patching
- Continuous backups and restore to specific timestamp (Point in Time Restore)!
- Monitoring dashboards
- Read replicas for improved read performance
- Multi AZ setup for DR (Disaster Recovery)
- Maintenance windows for upgrades
- Scaling capability (vertical and horizontal)
- Storage backed by EBS
- BUT you can’t SSH into your instances

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Amazon Aurora

Aurora is a proprietary technology from AWS (not open sourced)

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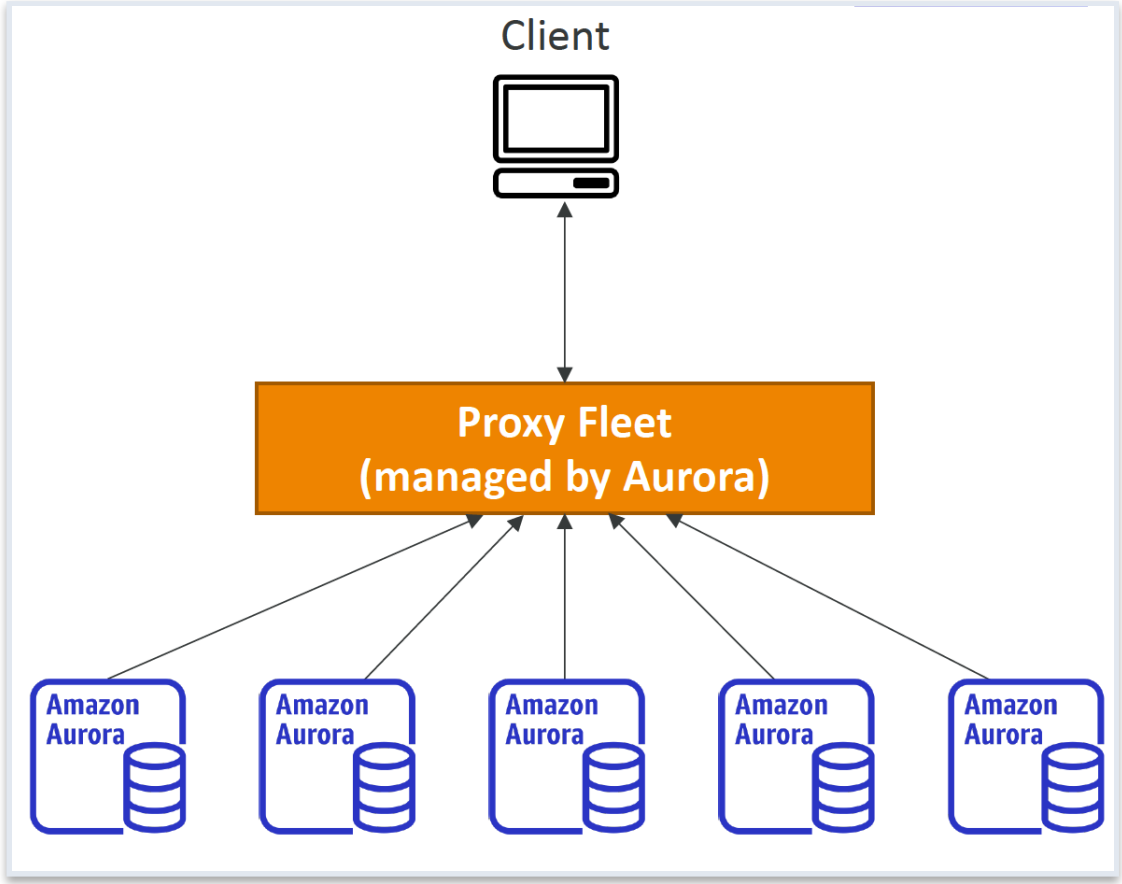
Amazon Aurora Serverless



Automated database instantiation and Client auto-scaling based on actual usage

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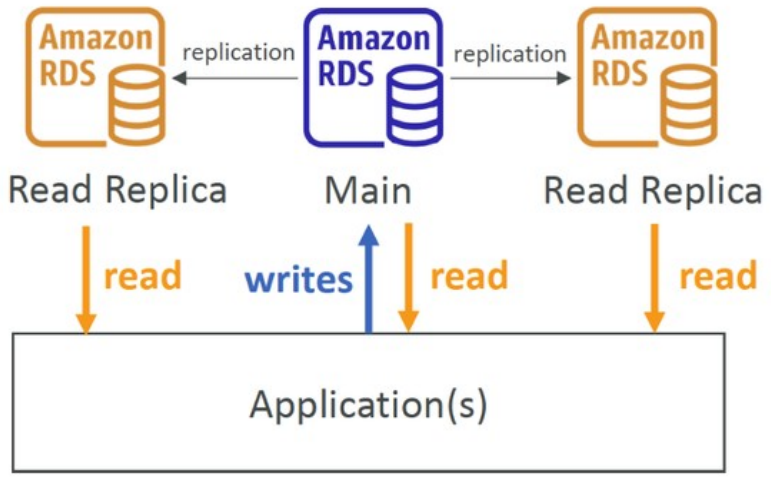
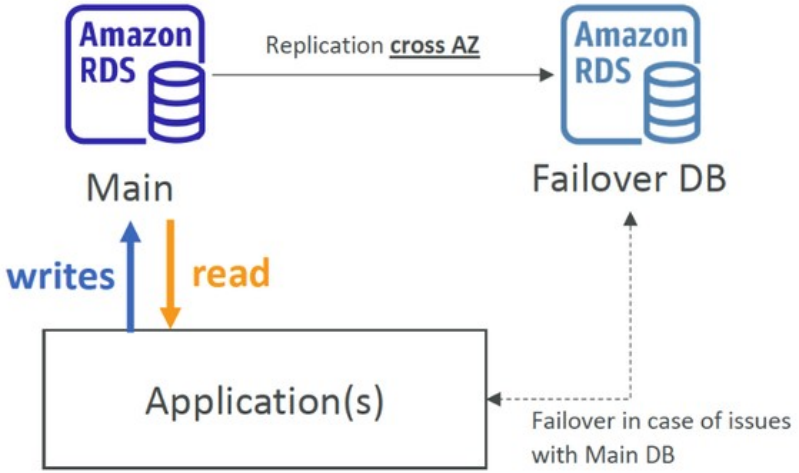
RDS Deployments: Read Replicas, Multi-AZ

- Read Replicas:
  - **Scale** the read workload of your DB
  - Can create up to 15 Read Replicas
  - Data is only written to the main DB



- Multi-AZ:
  - **Failover** in case of AZ outage (high availability)
  - Data is only read/written to the main database
  - Can only have 1 other AZ as failover



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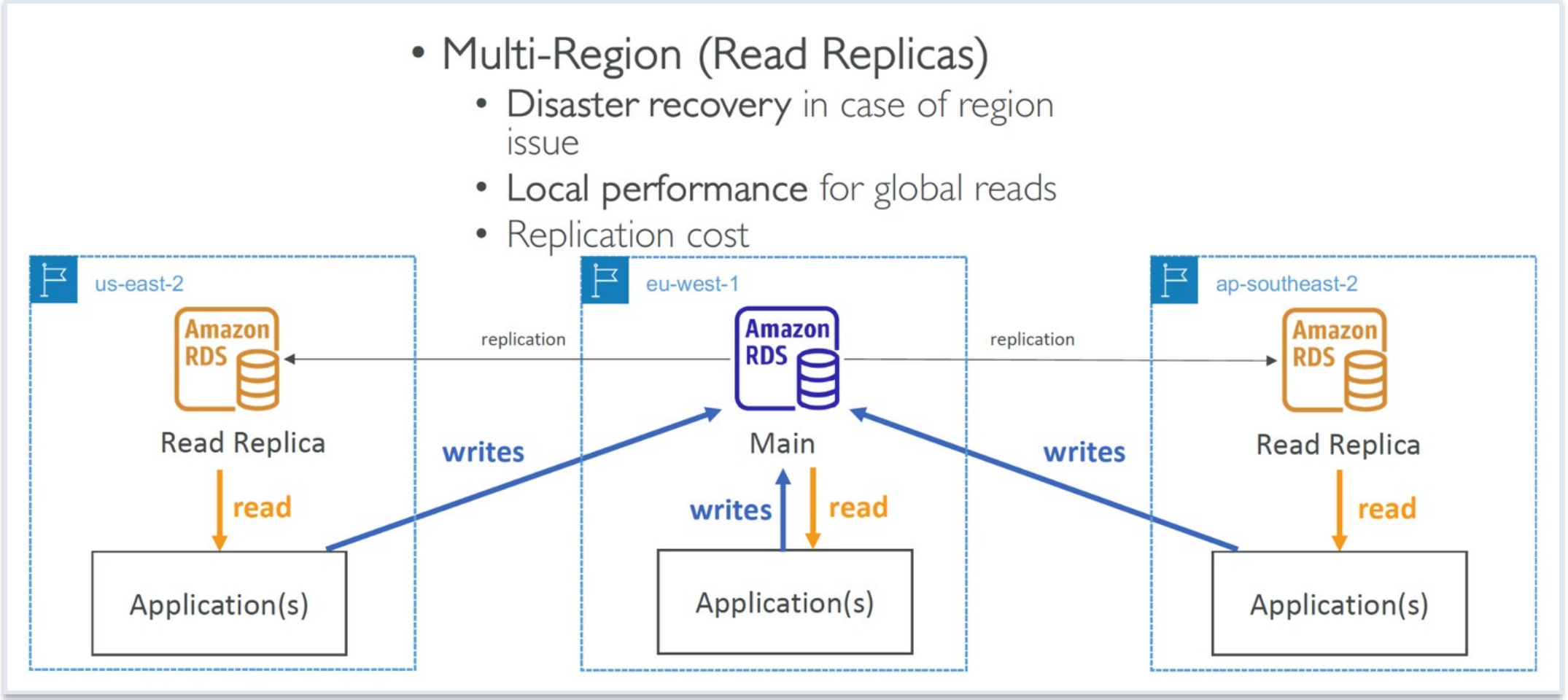
RDS Deployments: Multi-Region



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Amazon ElastiCache Overview

- The same way RDS is to get managed Relational Databases…
- ElastiCache is to get managed Redis or Memcached
- Caches are in-memory databases with high performance, low latency
- Helps reduce load off databases for read intensive workloads
- AWS takes care of OS maintenance / patching, optimizations, setup, configuration, monitoring, failure recovery and backups

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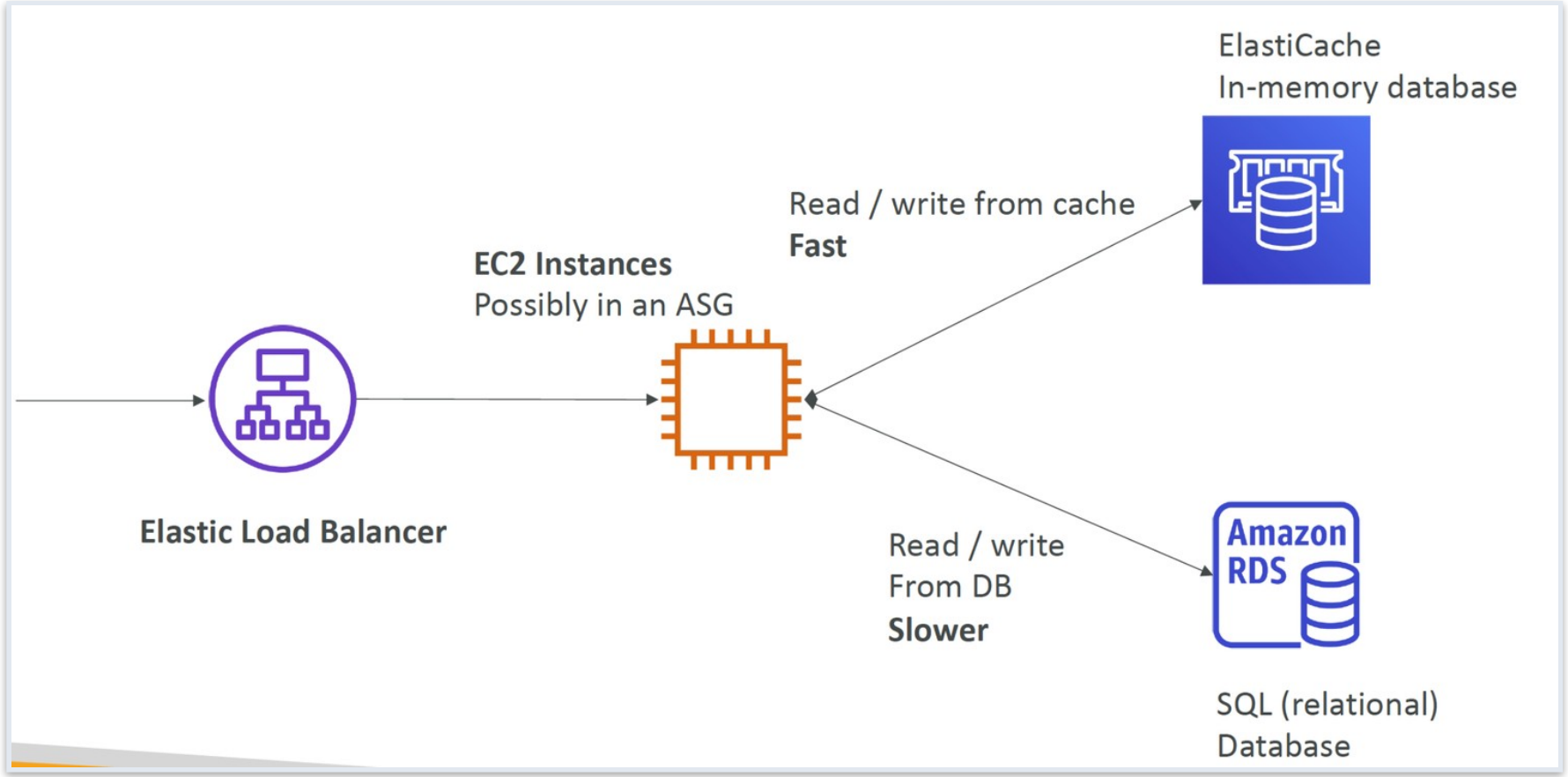
ElastiCache



Solution Architecture - Cache

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DynamoDB

- Fully Managed Highly available with replication across 3 AZ
- NoSQL database - not a relational database
- Scales to massive workloads, distributed “serverless” database
- Millions of requests per seconds, trillions of row, 100s of TB of storage
- Fast and consistent in performance
- Single-digit millisecond latency – low latency retrieval
- Integrated with IAM for security, authorization and administration
- Low cost and auto scaling capabilities
- Standard & Infrequent Access (IA) Table Class

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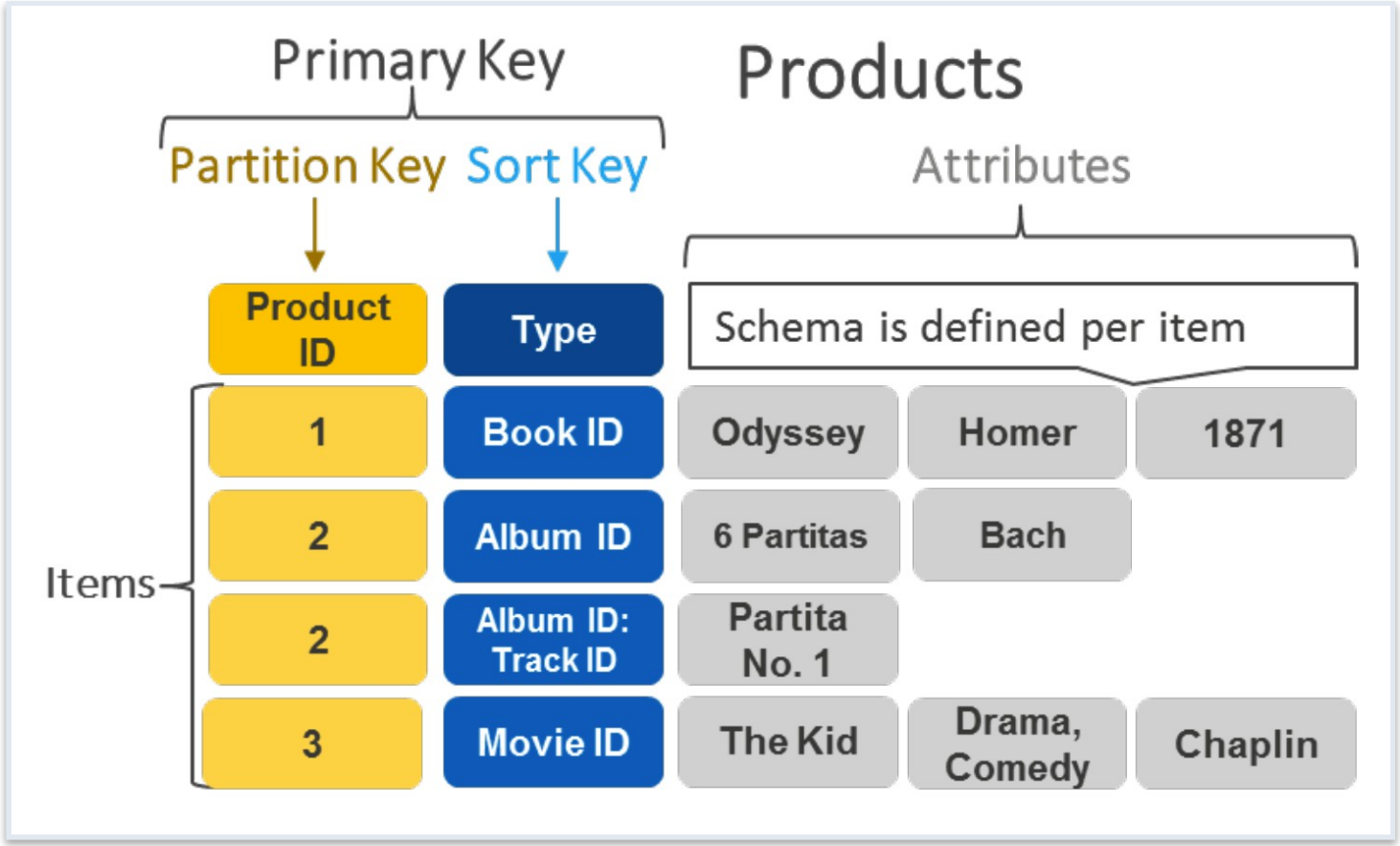
DynamoDB – type of data



https://aws.amazon.com/nosql/key-value/

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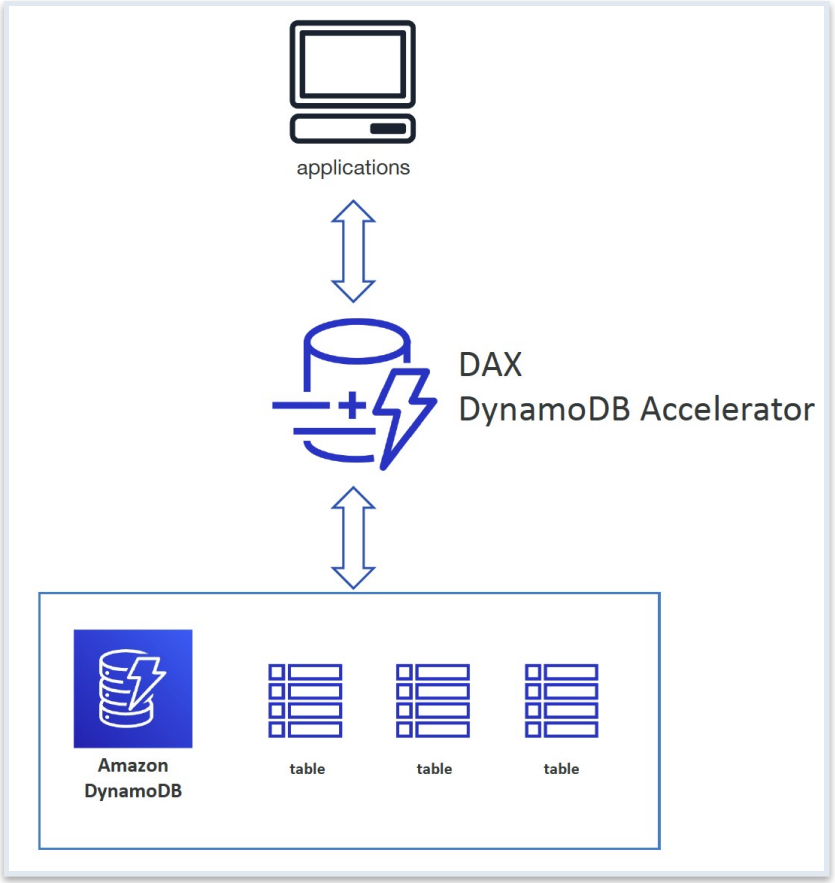
DynamoDB Accelerator - DAX



Fully Managed in-memory cache for DynamoDB

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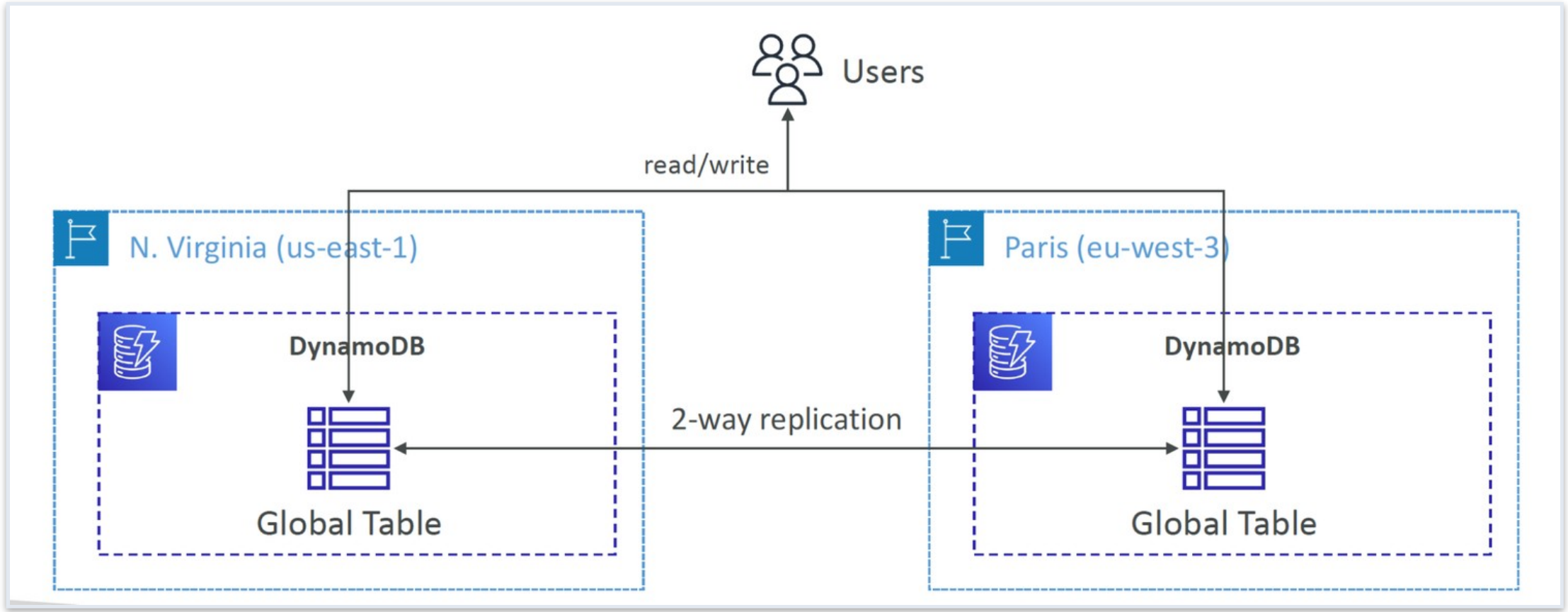
DynamoDB – Global Tables



Make a DynamoDB table accessible with low latency in multiple-regions

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Redshift Overview

- Redshift is based on PostgreSQL, but it’s not used for OLTP
- It’s OLAP – online analytical processing (analytics and data warehousing)
- Load data once every hour, not every second
- 10x better performance than other data warehouses, scale to PBs of data
- Columnar storage of data (instead of row based)
- Massively Parallel Query Execution (MPP), highly available
- Pay as you go based on the instances provisioned
- Has a SQL interface for performing the queries
- BI tools such as AWS Quicksight or Tableau integrate with it

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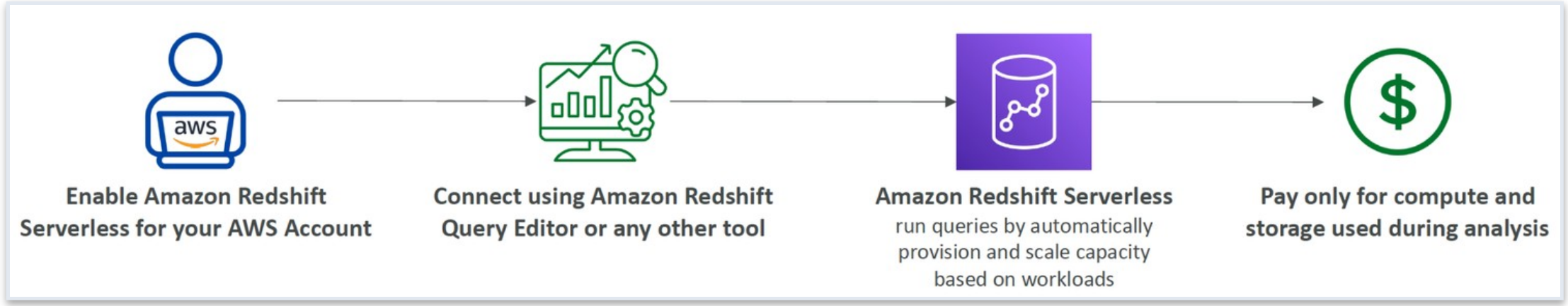
Redshift Serverless



Automatically provisions and scales data warehouse underlying capacity

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Amazon EMR

- EMR stands for “Elastic MapReduce”
- EMR helps creating Hadoop clusters (Big Data) to analyze and process vast amount of data
- The clusters can be made of hundreds of EC2 instances
- Also supports Apache Spark, HBase, Presto, Flink…
- EMR takes care of all the provisioning and configuration
- Auto-scaling and integrated with Spot instances
- Use cases: data processing, machine learning, web indexing, big data…

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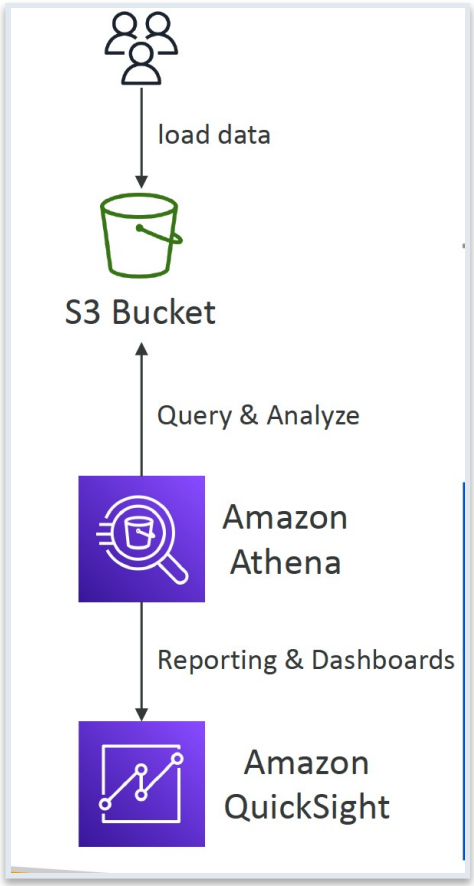
Amazon Athena



Serverless query service to analyze data stored in Amazon S3

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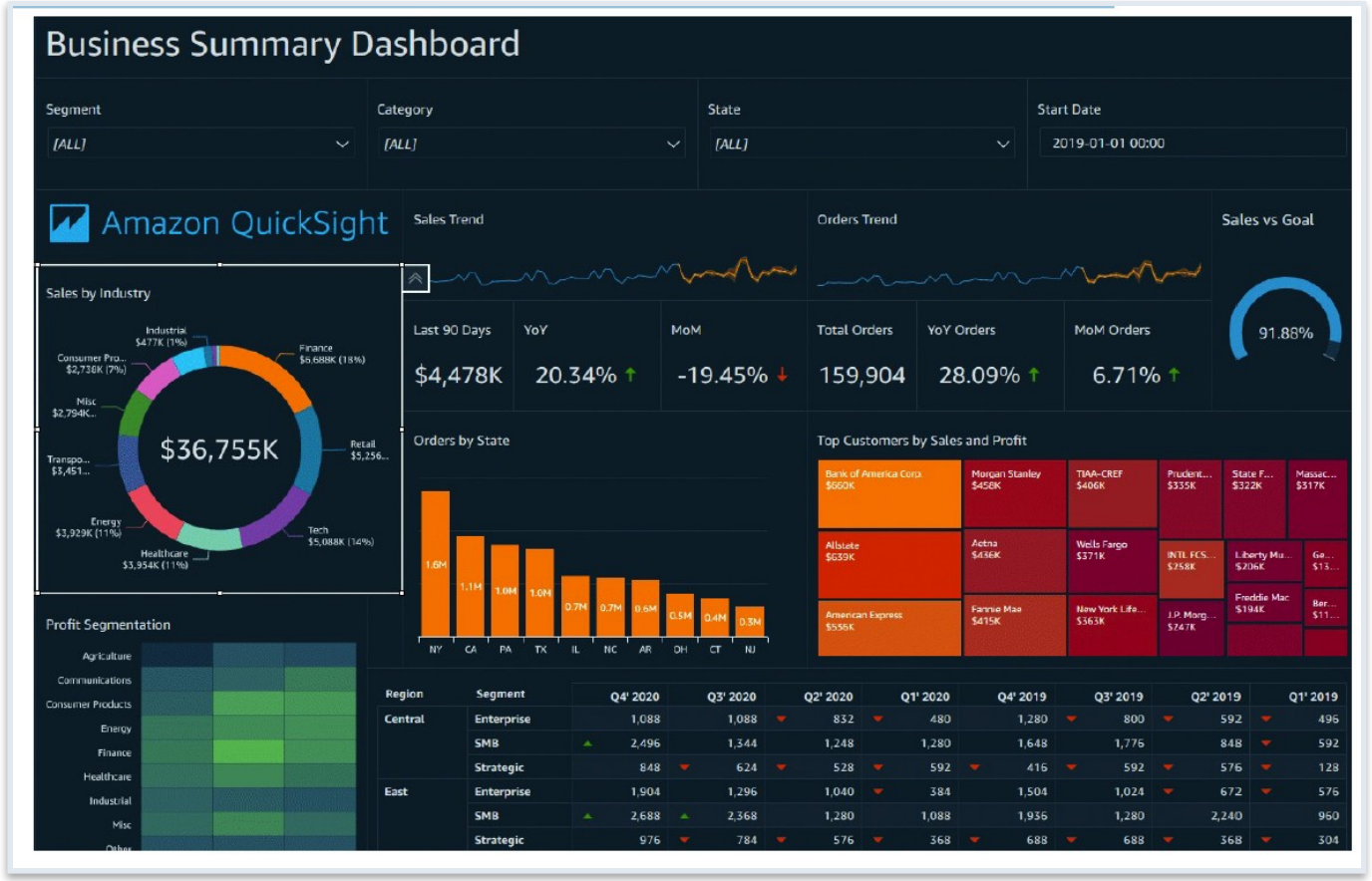
Amazon QuickSight



Serverless machine learning-powered business intelligence service to create interactive dashboards

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DocumentDB

- Aurora is an “AWS-implementation” of PostgreSQL / MySQL …
- DocumentDB is the same for MongoDB (which is a NoSQL database)
- MongoDB is used to store, query, and index JSON data
- Similar “deployment concepts” as Aurora
- Fully Managed, highly available with replication across 3 AZ
- DocumentDB storage automatically grows in increments of 10GB
- Automatically scales to workloads with millions of requests per seconds

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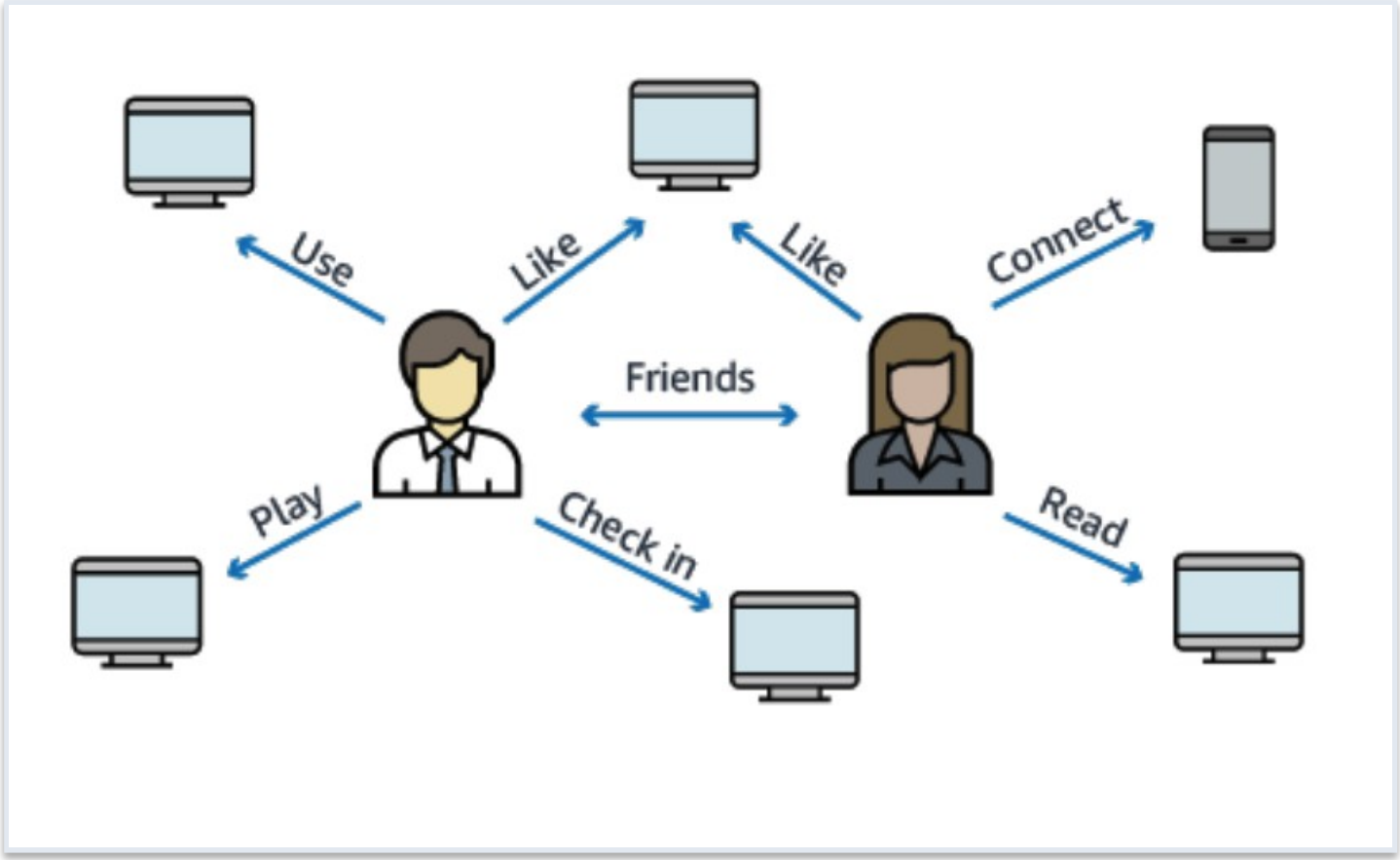
Amazon Neptune



Fully managed graph database

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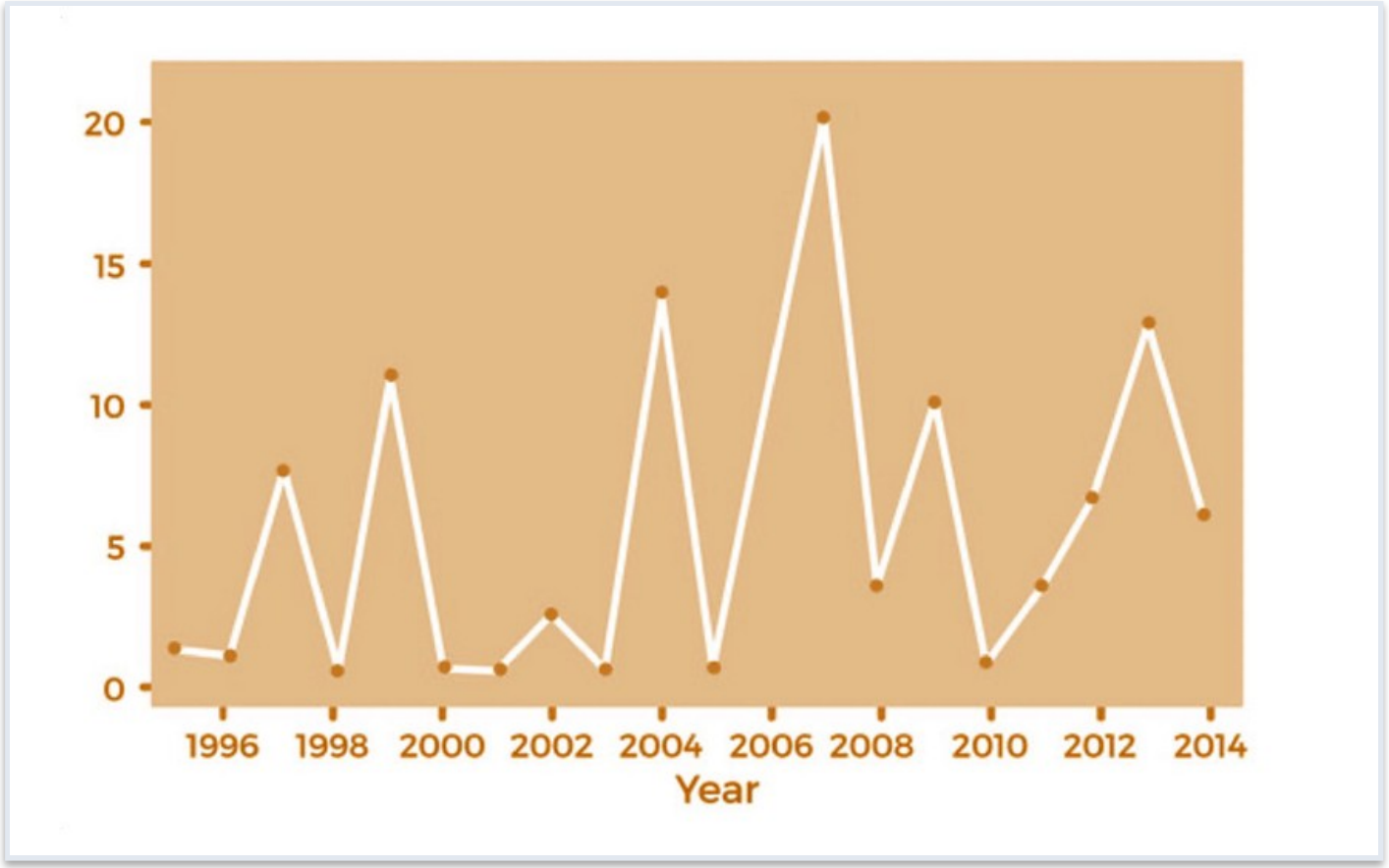
Amazon Timestream



Fully managed, fast, scalable, serverless time series database

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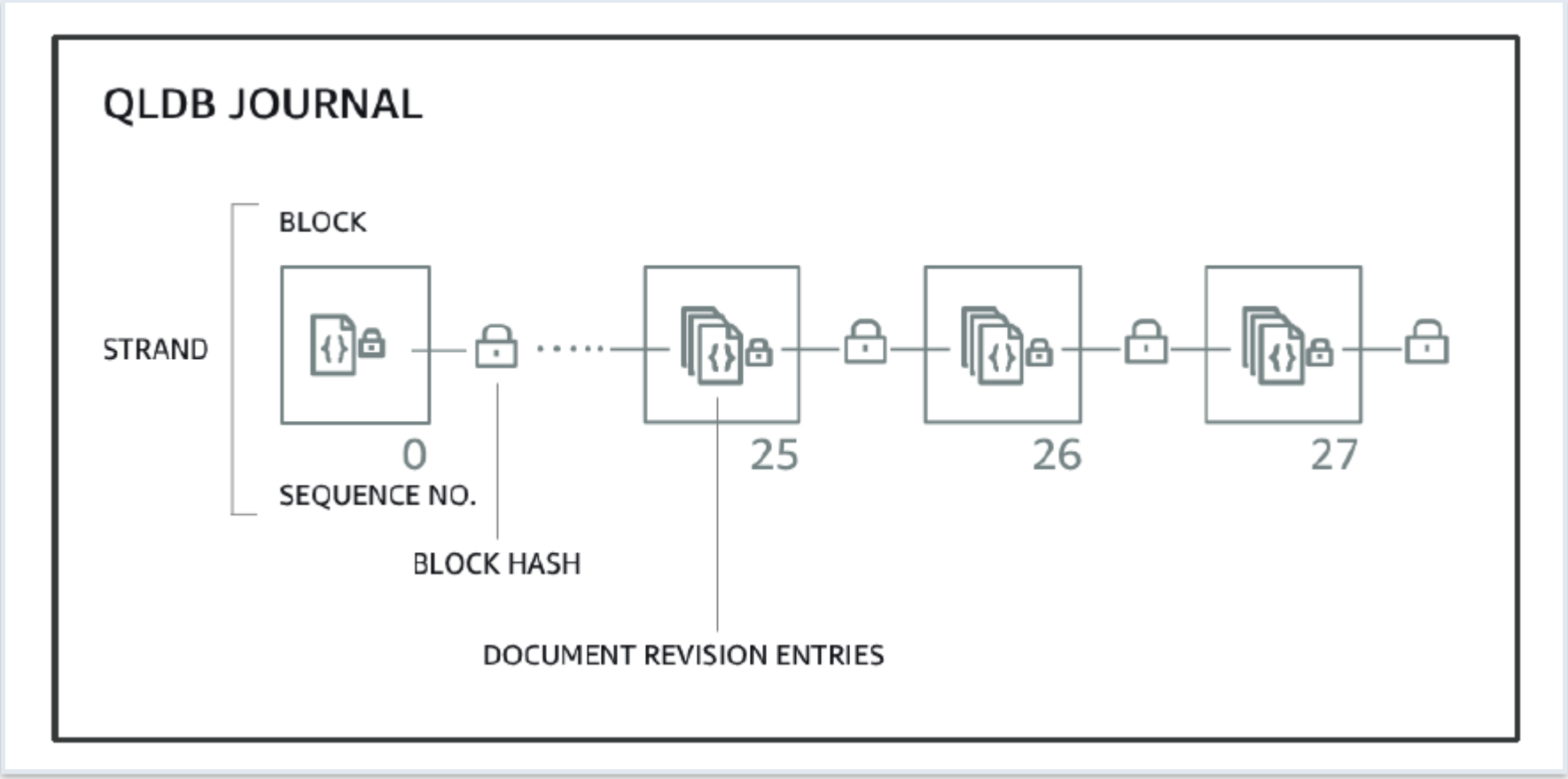
Amazon QLDB



QLDB stands for ”Quantum Ledger Database”

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Amazon Managed Blockchain

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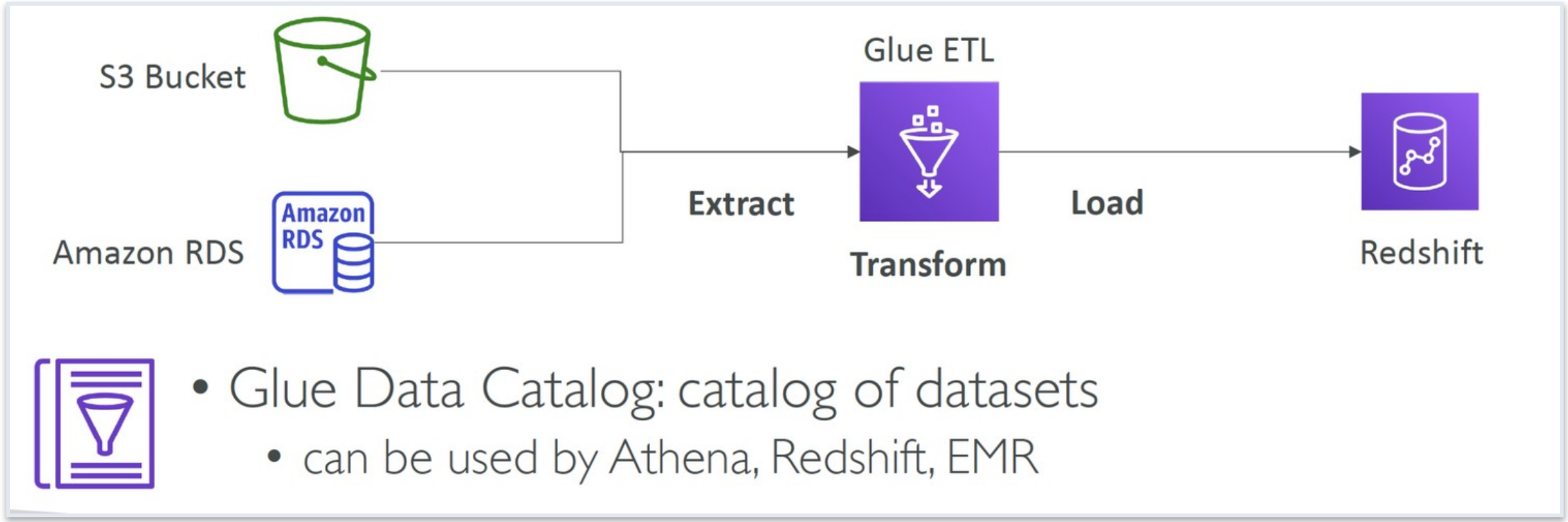
AWS Glue



Managed extract, transform, and load (ETL) service

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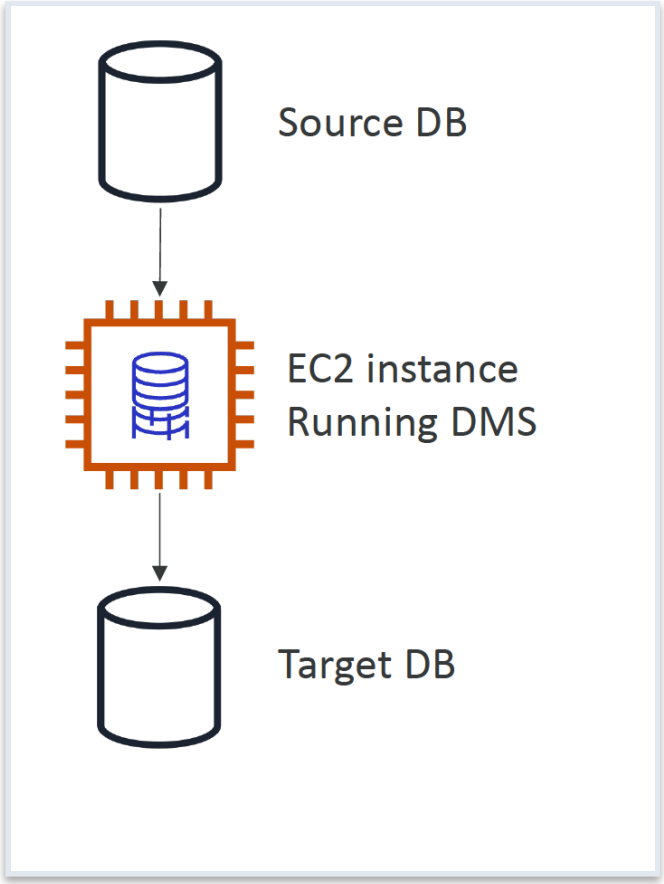
DMS – Database Migration Service



Quickly and securely migrate databases to AWS, resilient, self healing

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Databases & Analytics Summary

- Relational Databases - OLTP: RDS & Aurora (SQL)
- Differences between Multi-AZ, Read Replicas, Multi-Region
- In-memory Database: ElastiCache
- Key/Value Database: DynamoDB (serverless) & DAX (cache for DynamoDB)
- Warehouse - OLAP: Redshift (SQL)
- Hadoop Cluster: EMR
- Athena: query data on Amazon S3 (serverless & SQL)
- QuickSight: dashboards on your data (serverless)
- DocumentDB: “Aurora for MongoDB” (JSON – NoSQL database)
- Amazon QLDB: Financial Transactions Ledger (immutable journal, cryptographically verifiable)
- Amazon Managed Blockchain: managed Hyperledger Fabric & Ethereum blockchains
- Glue: Managed ETL (Extract Transform Load) and Data Catalog service
- Database Migration: DMS

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AWS Networking and Other Services

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TOPIC 5

VPC

- VPC

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Amazon Virtual Private Cloud (Amazon VPC)

- VPC is something you should know in depth for the AWS Certified Solutions Architect Associate & AWS Certified SysOps Administrator
- At the AWS Certified Cloud Practitioner Level, you should know about:
- VPC, Subnets, Internet Gateways & NAT Gateways
- Security Groups, Network ACL (NACL), VPC Flow Logs
- VPC Peering, VPC Endpoints
- Site to Site VPN & Direct Connect
- Transit Gateway

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IP Addresses in AWS

- IPv4 – Internet Protocol version 4 (4.3 Billion Addresses)
- Public IPv4 – can be used on the Internet
- EC2 instance gets a new a public IP address every time you stop then start it (default)
- Private IPv4 – can be used on private networks (LAN) such as internal AWS networking (e.g., 192.168.1.1)
- Private IPv4 is fixed for EC2 Instances even if you start/stop them
- Elastic IP – allows you to attach a fixed public IPv4 address to EC2 instance
- Note: all public IPv4 on AWS will be charged $0.005 per hour
- Free Tier: 750 hours usage per month
- IPv6 – Internet Protocol version 6 (3.4 × 10!" Addresses)
- Every IP address is public in AWS (no private range)
- Example: 2001:db8:3333:4444:cccc:dddd:eeee:ffff
- Free

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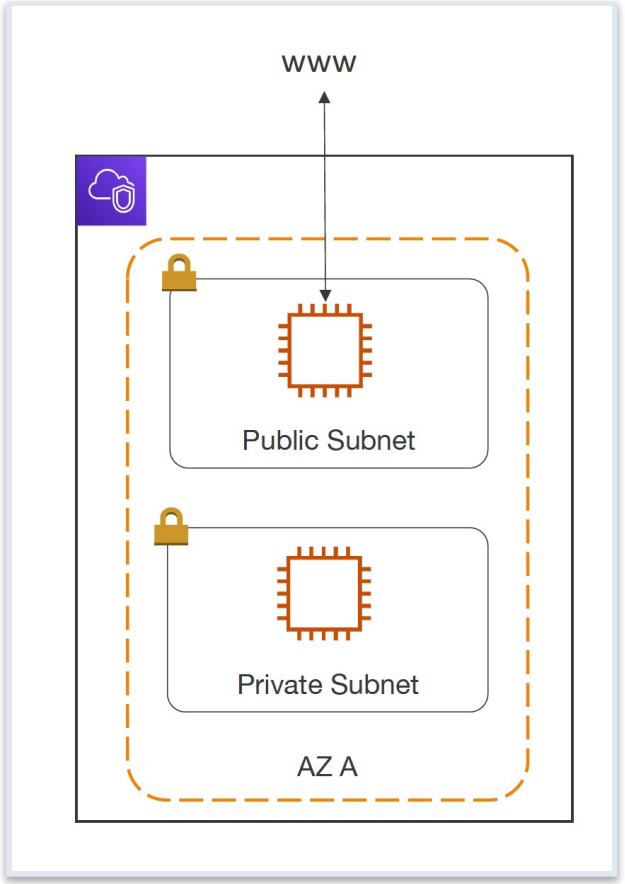
VPC & Subnets Primer



VPC - Virtual Private Cloud: private network to deploy your resources (regional resource)

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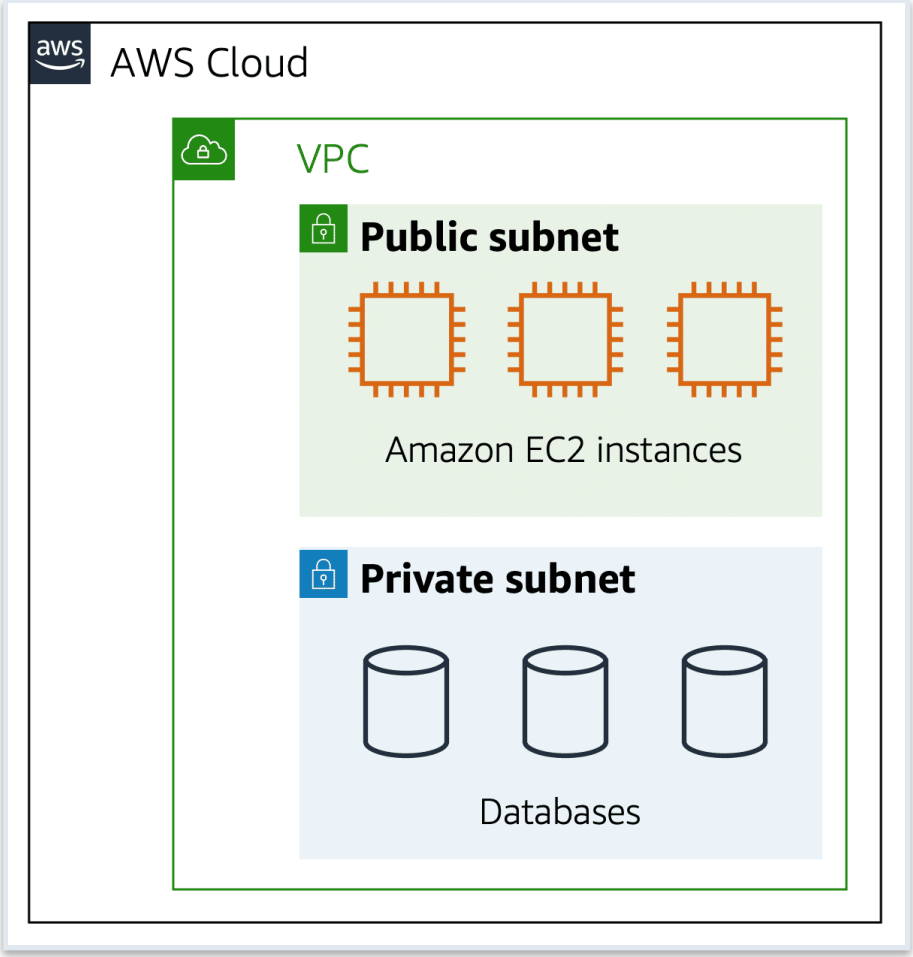
Subnets



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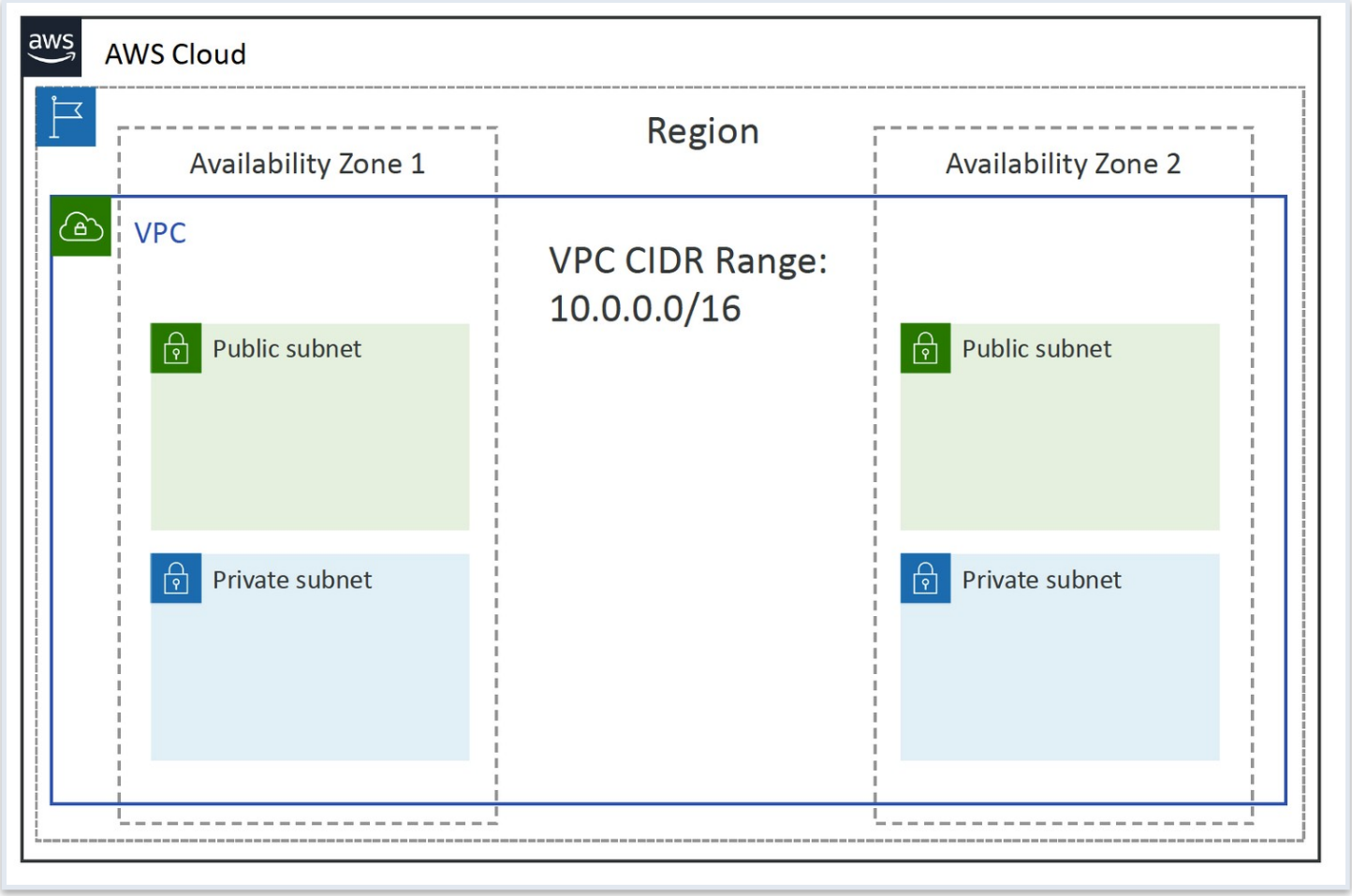
VPC Diagram



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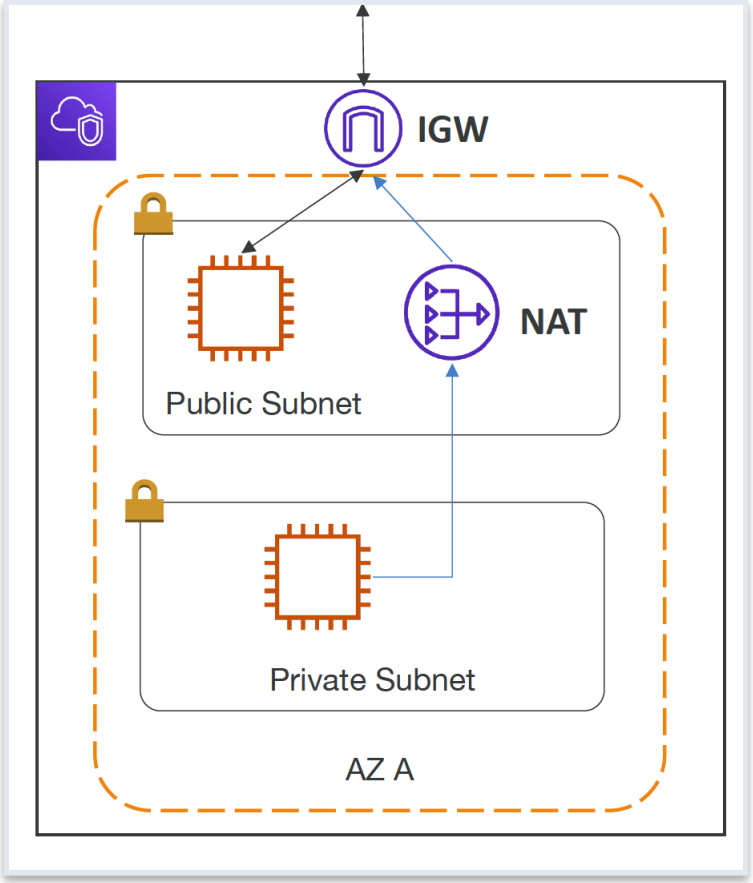
Internet Gateway & NAT Gateways



Internet Gateways helps our VPC instances connect with the internet

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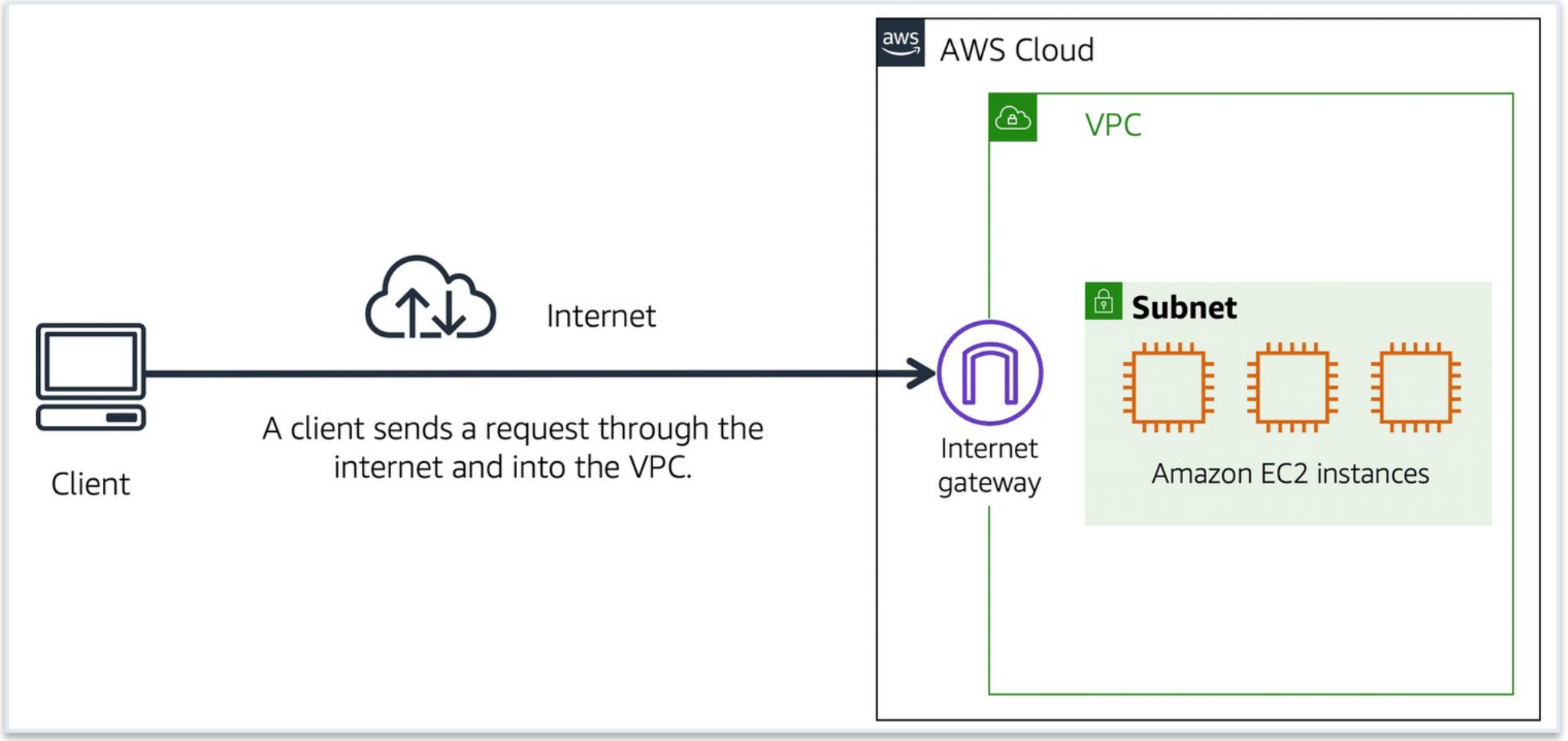
Internet Gateway



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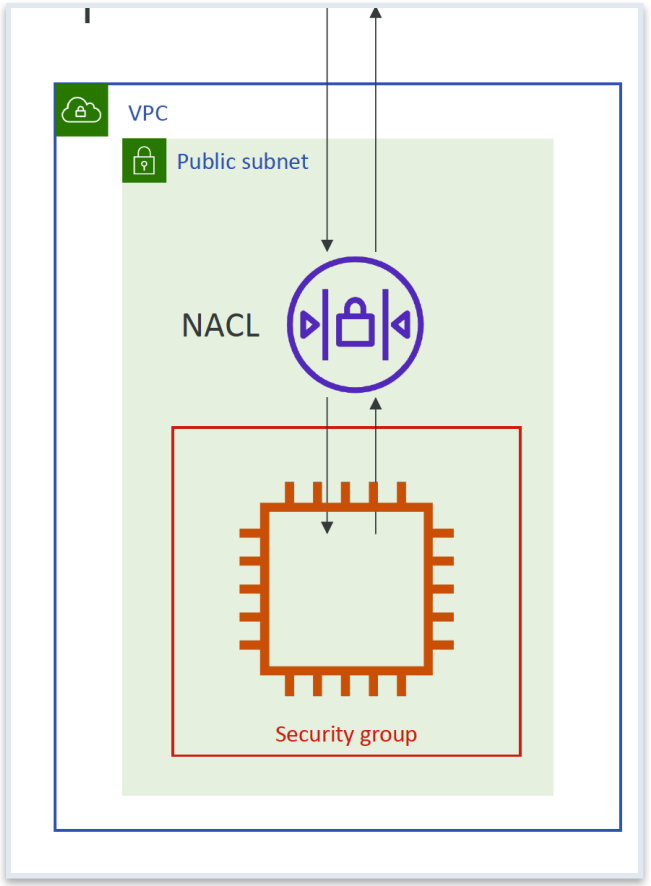
Network ACL & Security Groups



NACL (Network ACL)

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Network ACLs vs Security Groups

Security Group	Network ACL
Operates at the instance level	Operates at the subnet level
Supports allow rules only	Supports allow rules and deny rules
Is stateful: Return traffic is automatically allowed, regardless of any rules	Is stateless: Return traffic must be explicitly allowed by rules
We evaluate all rules before deciding whether to allow traffic	We process rules in number order when deciding whether to allow traffic
Applies to an instance only if someone specifies the security group when launching the instance, or associates the security group with the instance later on	Automatically applies to all instances in the subnets it's associated with (therefore, you don't have to rely on users to specify the security group)

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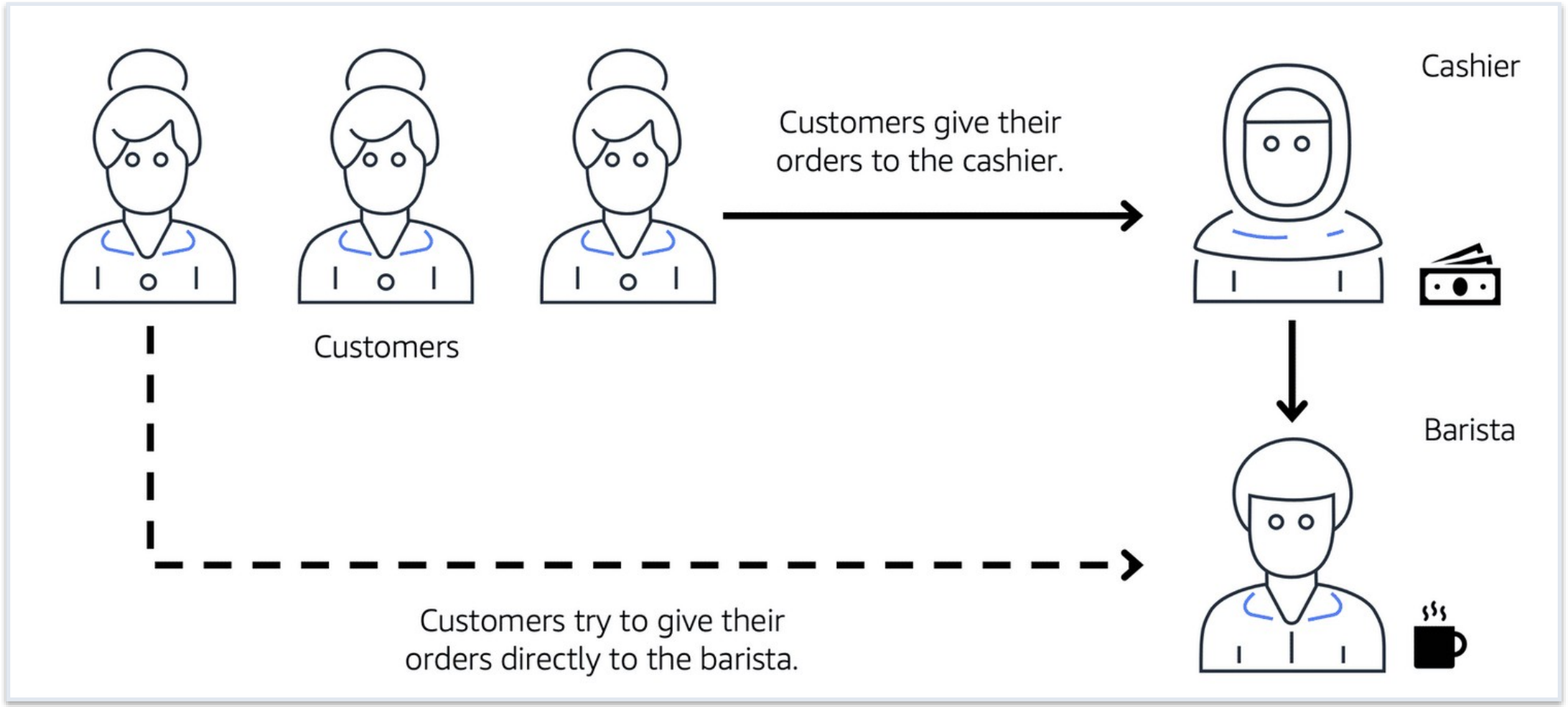
Subnets and Network Access Control Lists



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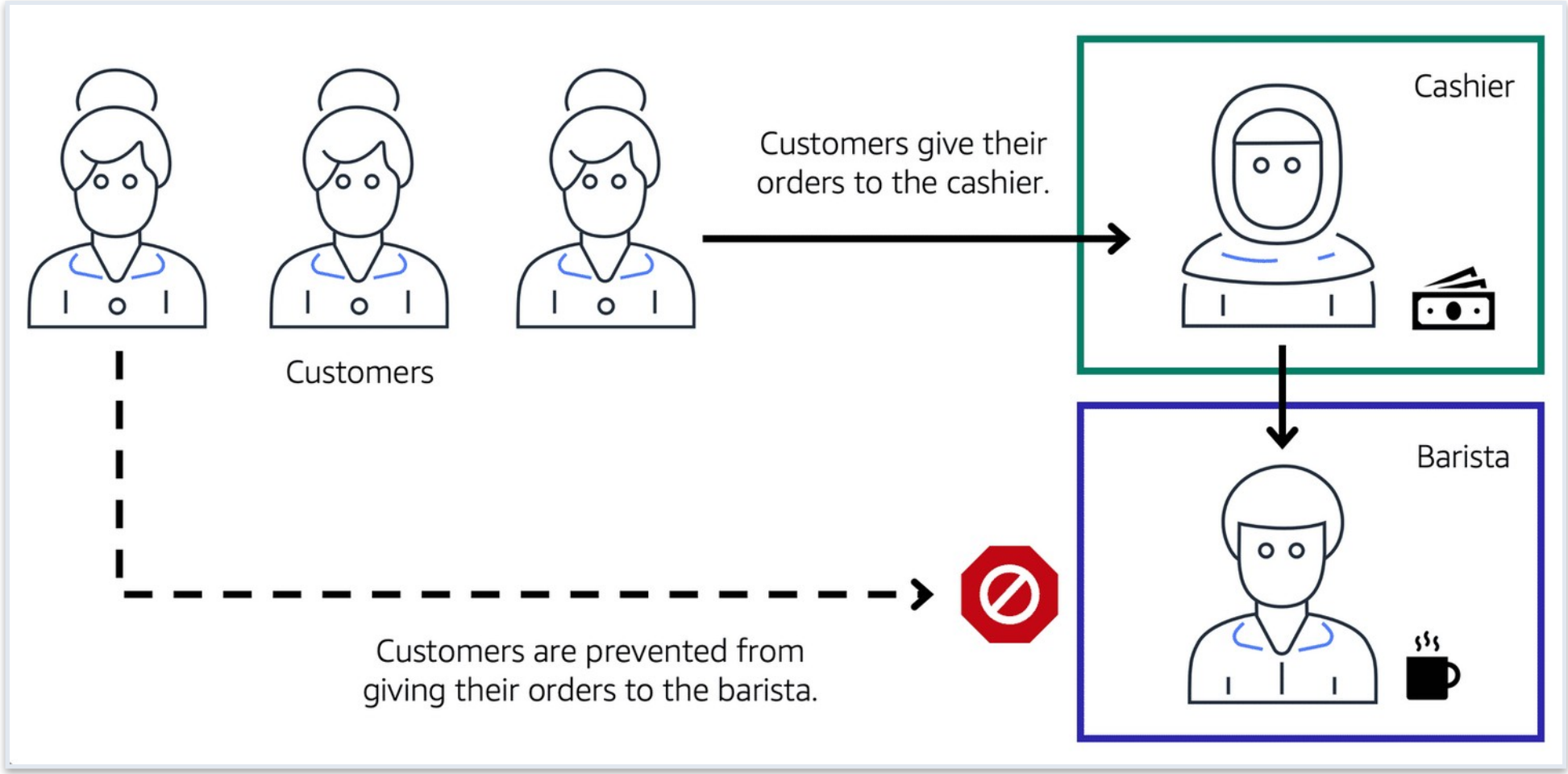
Subnets and Network Access Control Lists



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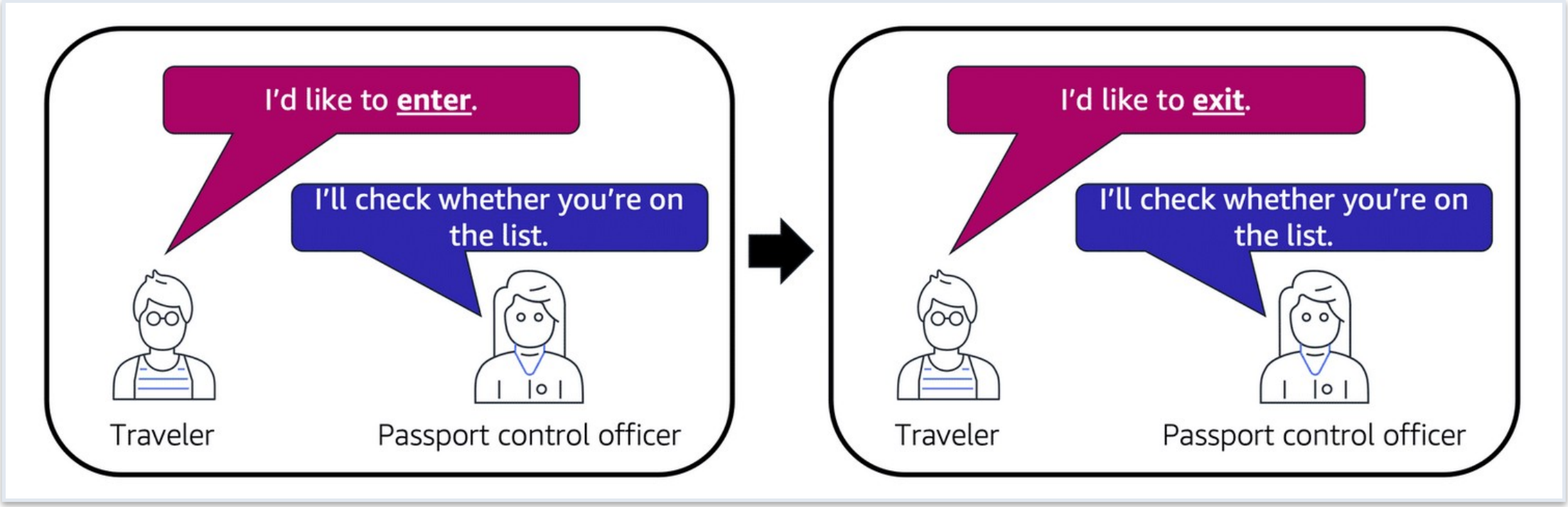
Stateless packet filtering



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Security groups

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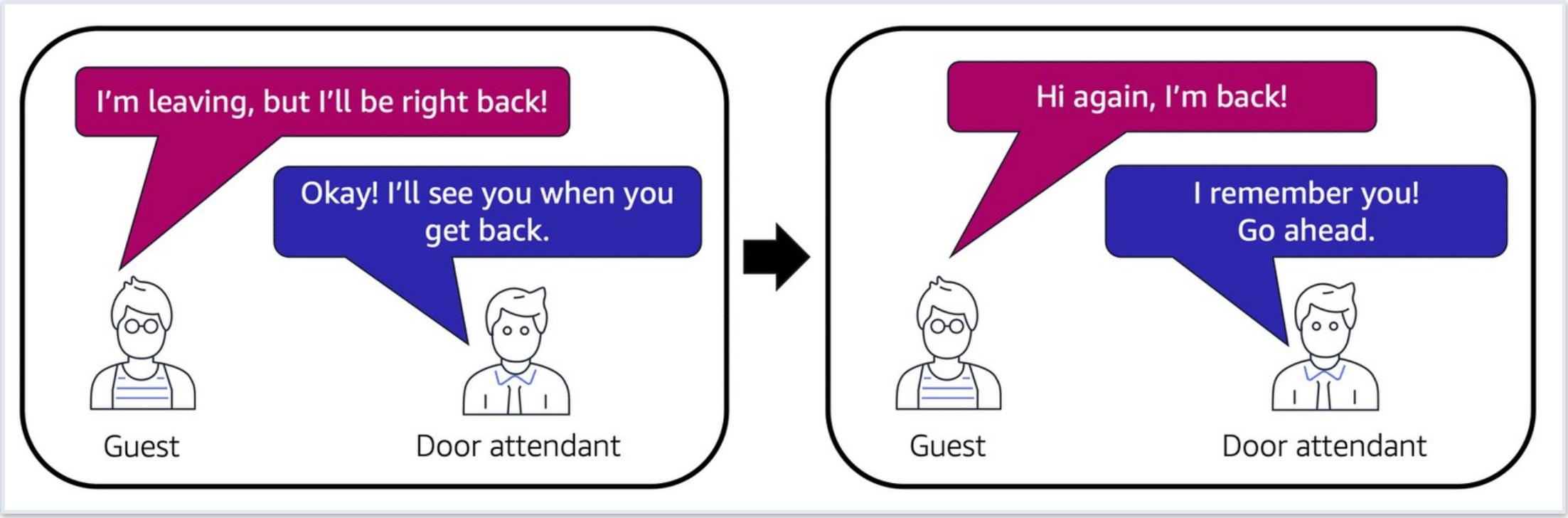
Stateful packet filtering



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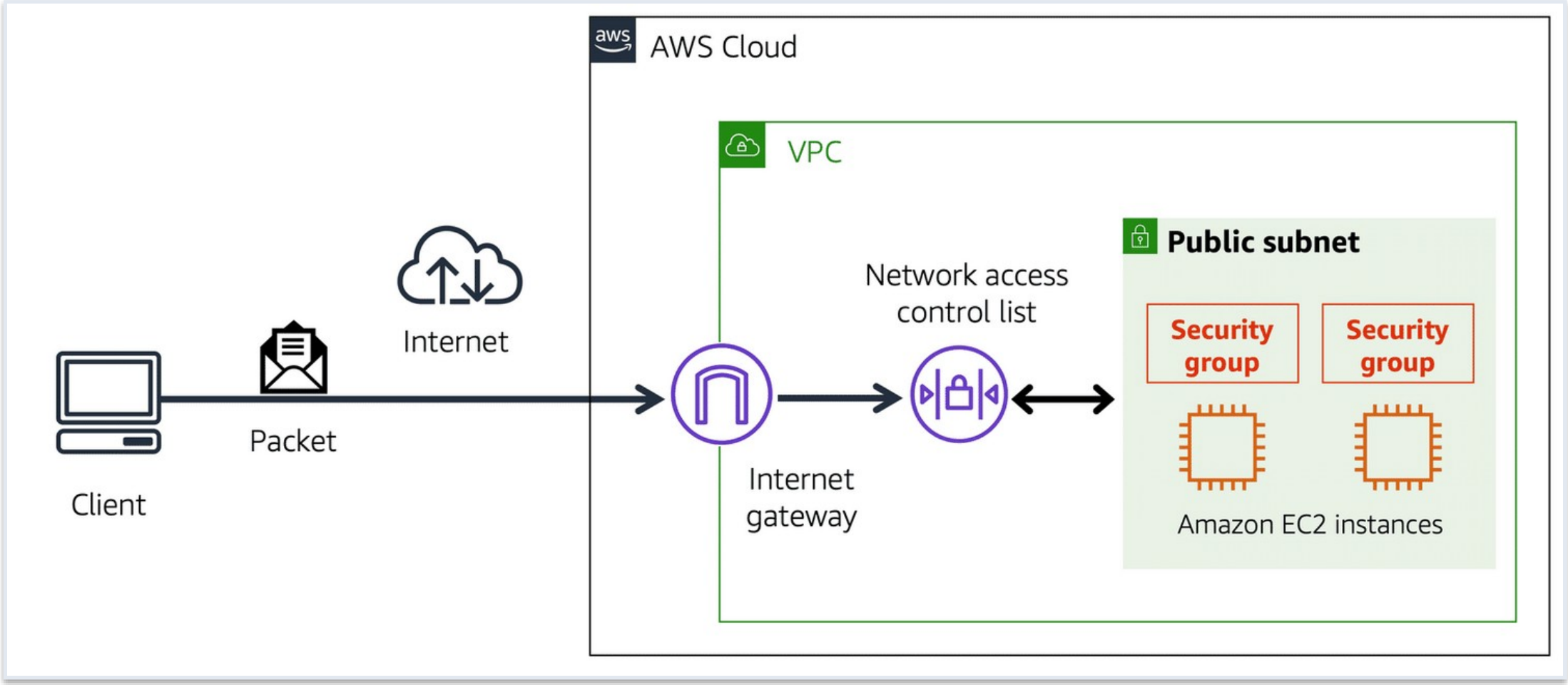
Stateful packet filtering



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VPC Flow Logs

- Capture information about IP traffic going into your interfaces:
- VPC Flow Logs
- Subnet Flow Logs
- Elastic Network Interface Flow Logs
- Helps to monitor & troubleshoot connectivity issues. Example:
- Subnets to internet
- Subnets to subnets
- Internet to subnets
- Captures network information from AWS managed interfaces too: Elastic Load Balancers, ElastiCache, RDS, Aurora, etc…
- VPC Flow logs data can go to S3, CloudWatch Logs, and Kinesis Data Firehose

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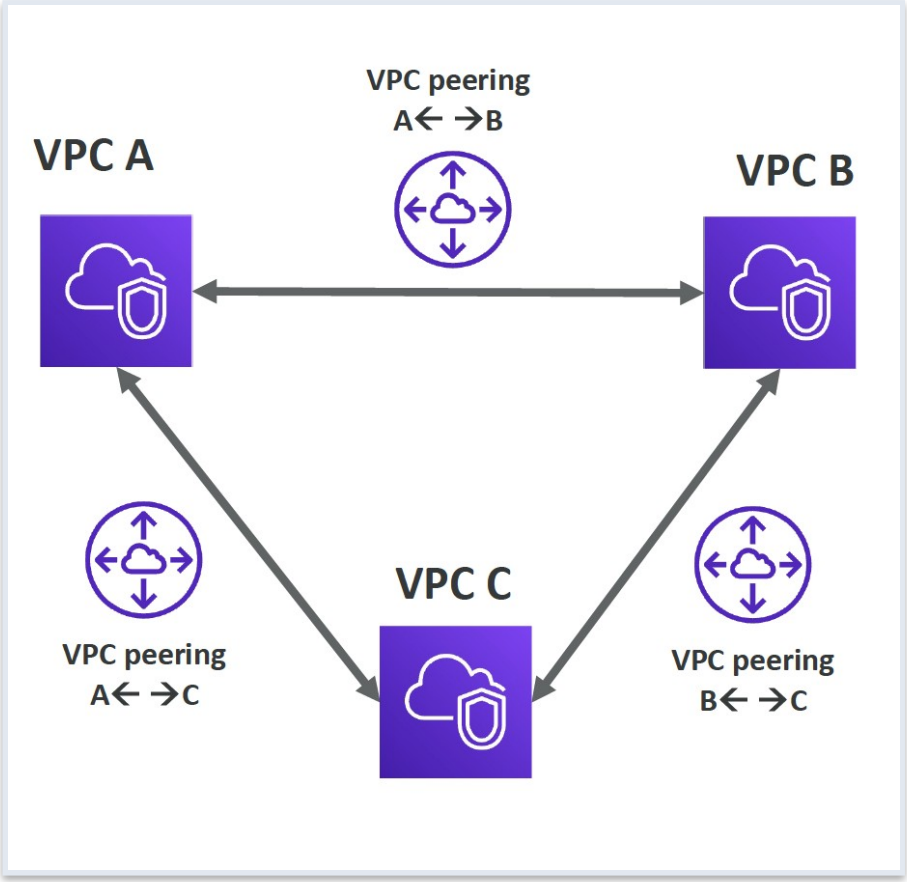
VPC Peering



Connect two VPC, privately using AWS' network

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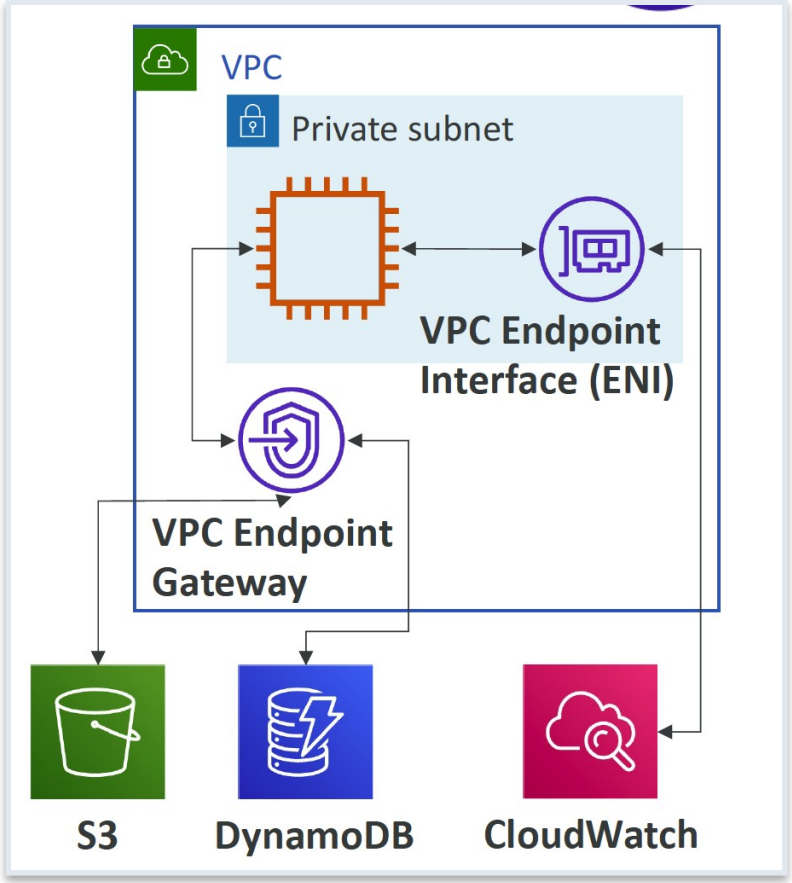
VPC Endpoints



Endpoints allow you to connect to AWS Services using a private network instead of the public www network

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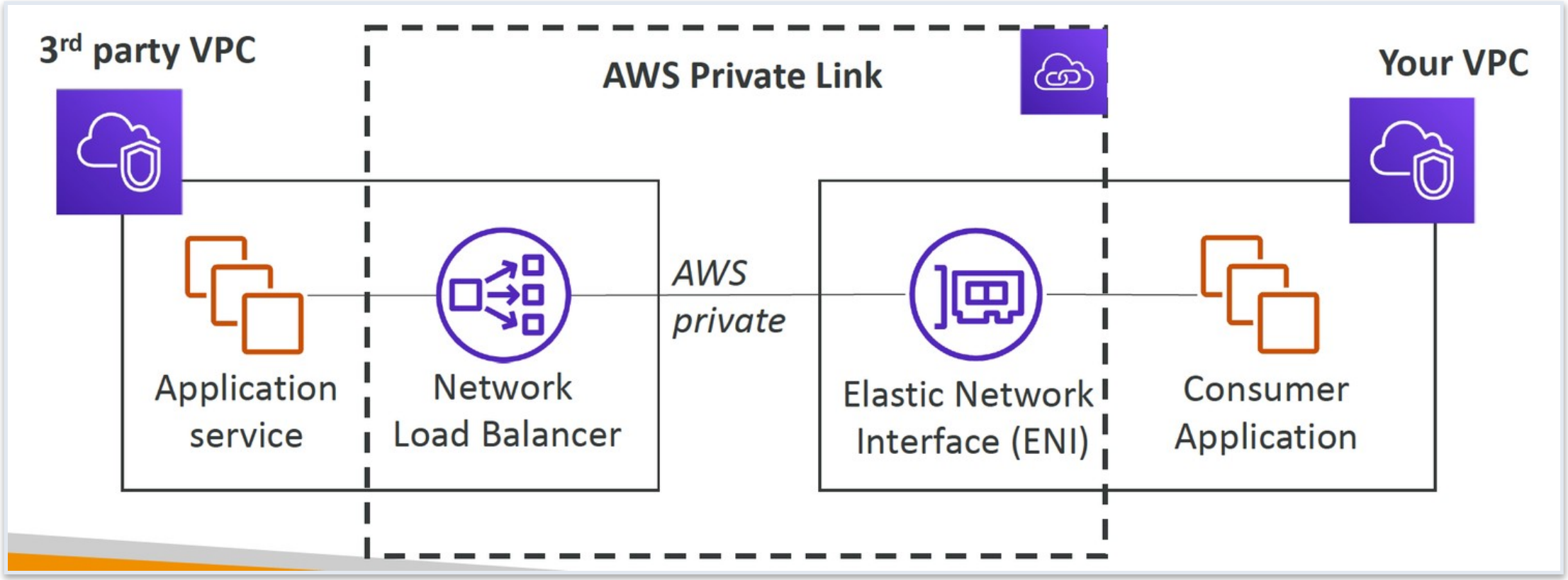
AWS PrivateLink (VPC Endpoint Services)



Most secure & scalable way to expose a service to 1000s of VPCs

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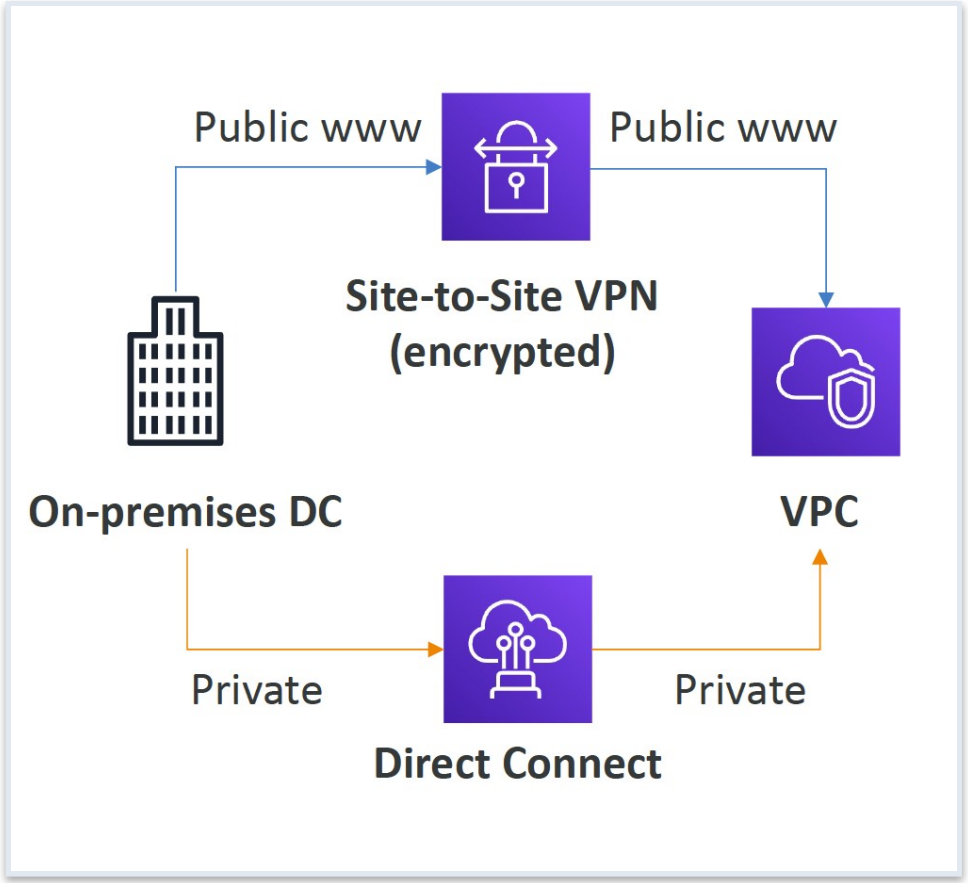
Site to Site VPN & Direct Connect



Site to Site VPN

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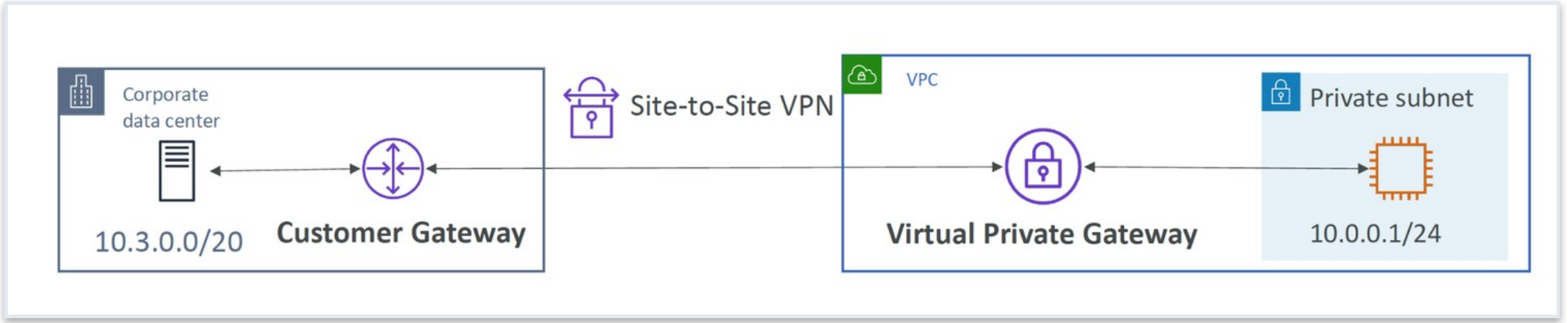
Site to Site VPN



On-premises: must use a Customer Gateway (CGW)

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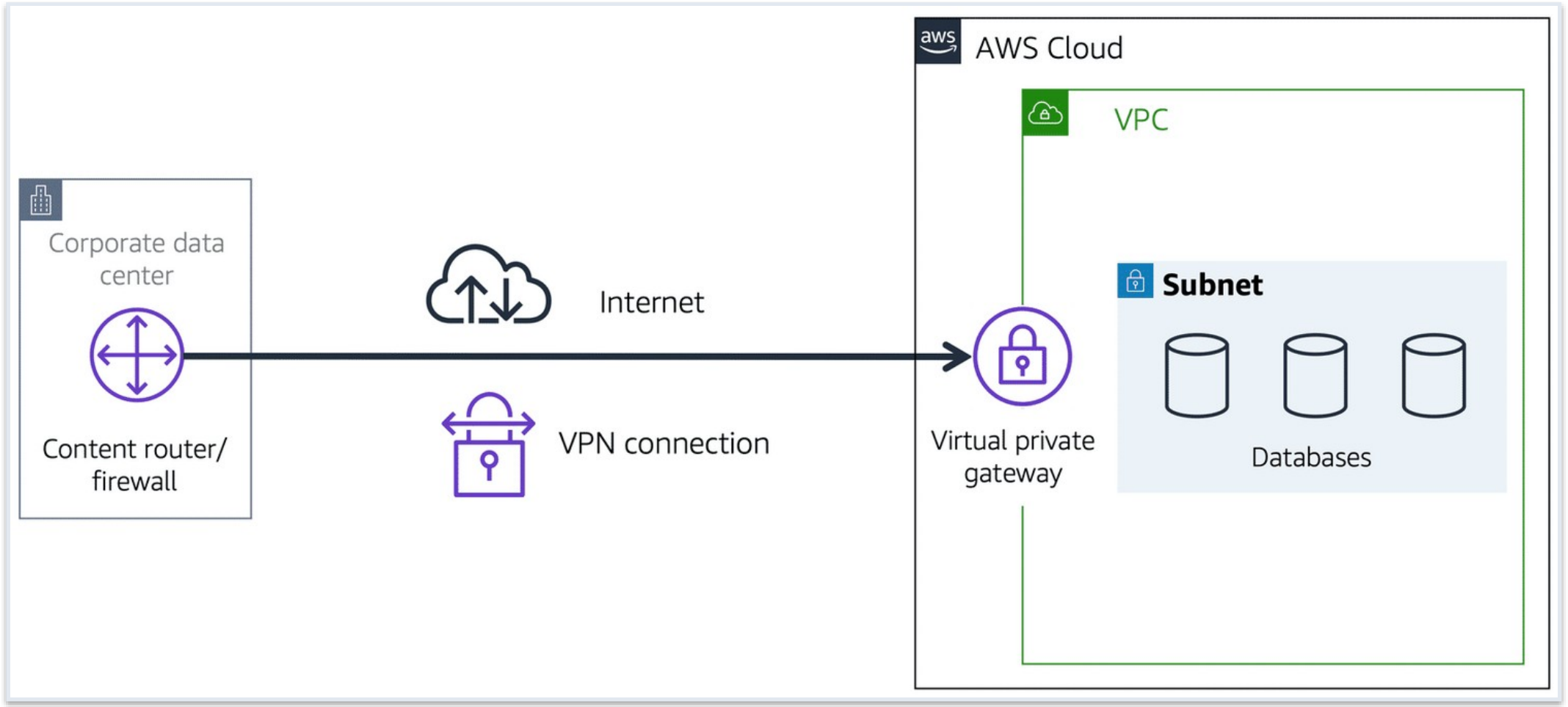
Virtual Private Gateway



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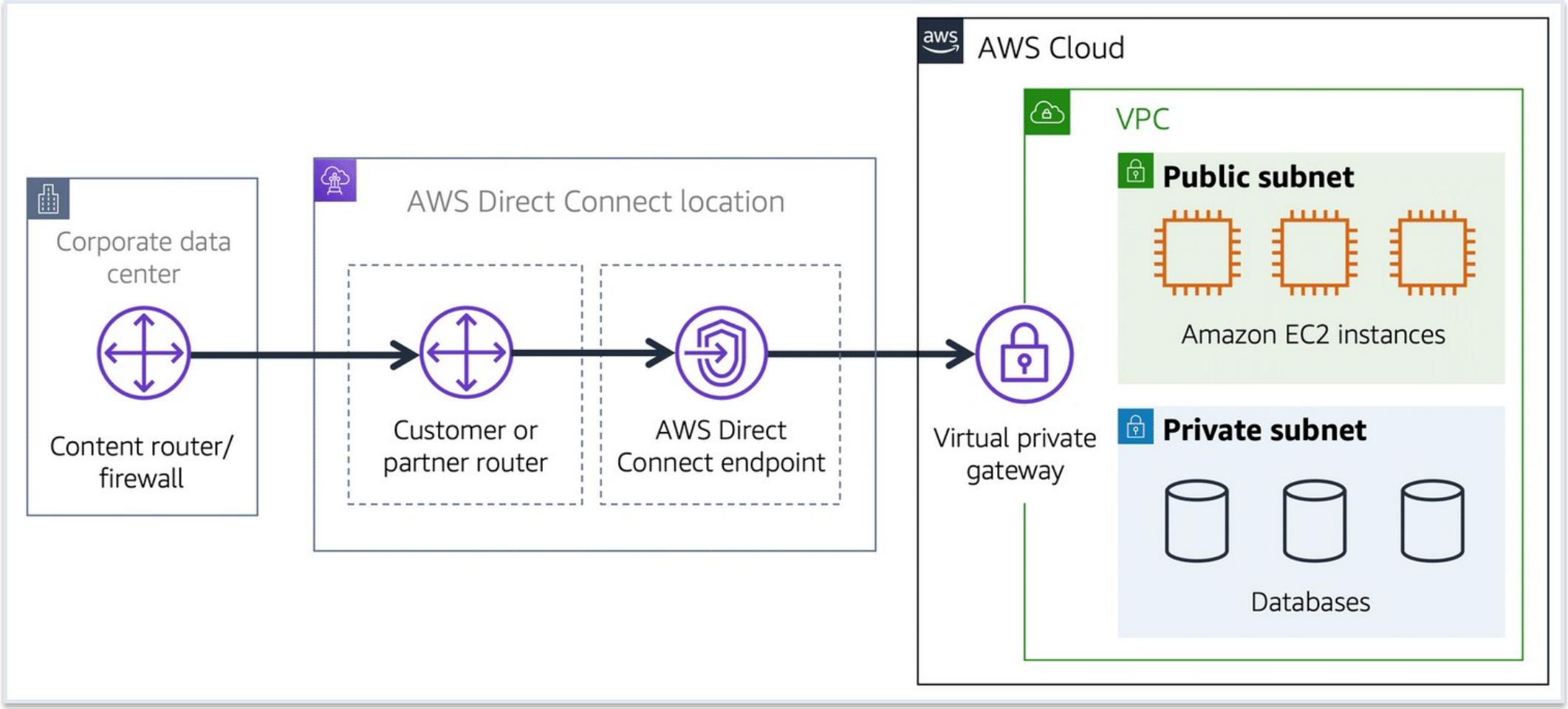
AWS Direct Connect



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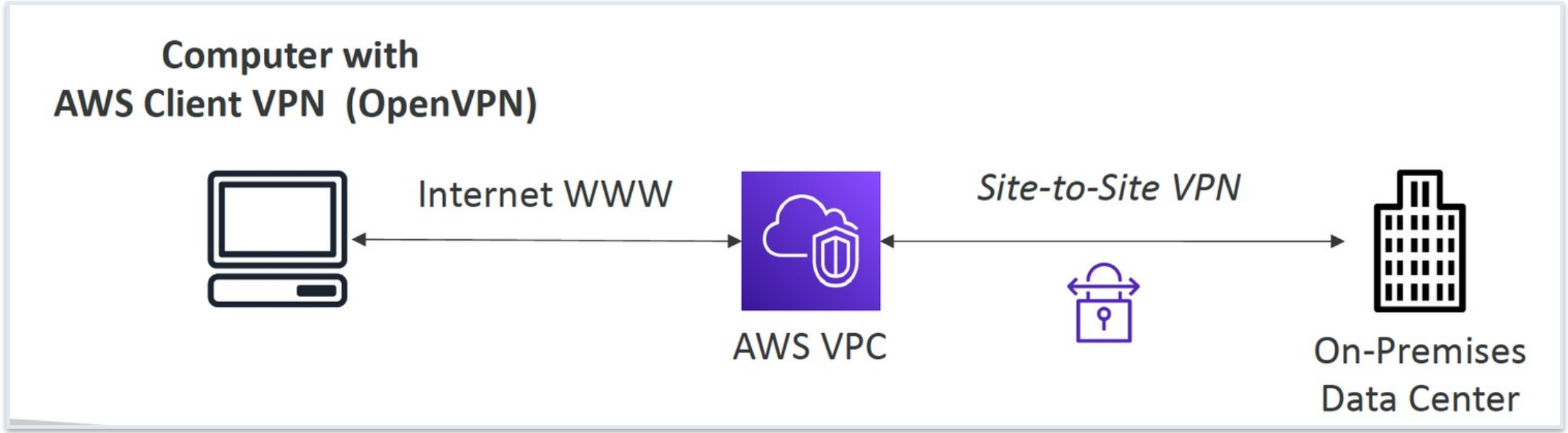
AWS Client VPN



Connect from your computer using OpenVPN to your private network in AWS and on-premises

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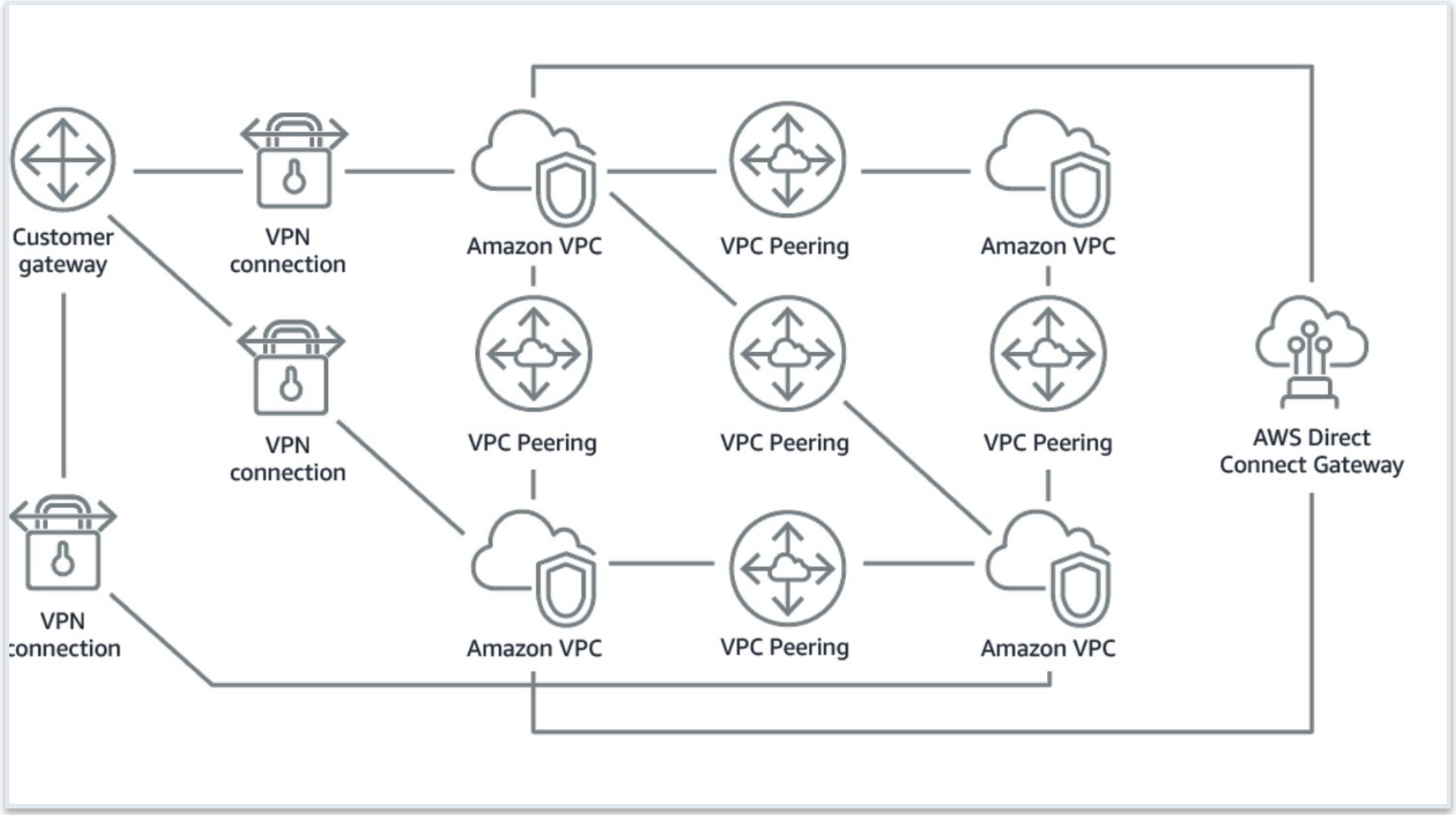
Network topologies can become complicated



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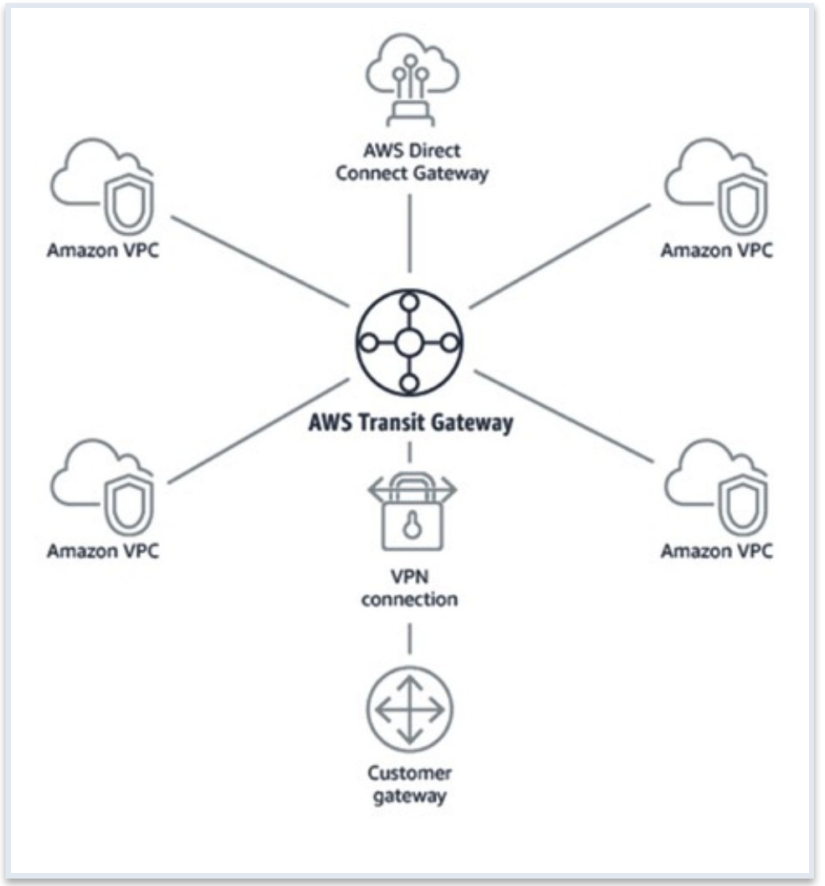
Transit Gateway



For having transitive peering between thousands of VPC and on-premises, hub-and-spoke (star) connection

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VPC Summary

- VPC – Virtual Private Cloud
- Subnets – Tied to an AZ, network partition of the VPC
- Internet Gateway – at the VPC level, provide Internet Access
- NAT Gateway / Instances – give internet access to private subnets
- NACL – Stateless, subnet rules for inbound and outbound
- Security Groups – Stateful, operate at the EC2 instance level or ENI
- VPC Peering – Connect two VPC with non overlapping IP ranges, nontransitive
- Elastic IP –fixed public IPv4, ongoing cost if not in-use

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VPC Summary

- VPC Endpoints – Provide private access to AWS Services within VPC
- PrivateLink – Privately connect to a service in a 3rd party VPC
- VPC Flow Logs – network traffic logs
- Site to Site VPN – VPN over public internet between on-premises DC and AWS
- Client VPN – OpenVPN connection from your computer into your VPC
- Direct Connect – direct private connection to AWS
- Transit Gateway – Connect thousands of VPC and on-premises networks together

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Elastic Load Balancing & Auto

- Scaling Groups

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Scalability & High Availability

- Scalability means that an application / system can handle greater loads by adapting.
- There are two kinds of scalability:
- Vertical Scalability
- Horizontal Scalability (= elasticity)
- Scalability is linked but different to High Availability
- Let’s deep dive into the distinction, using a call center as an example

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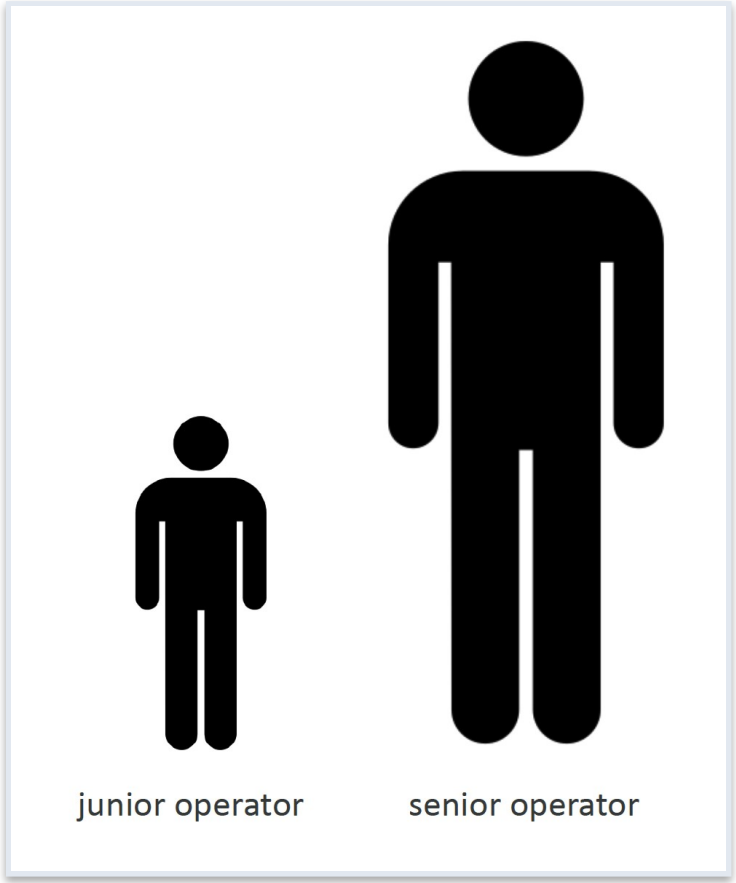
Vertical Scalability



Vertical Scalability means increasing the size of the instance

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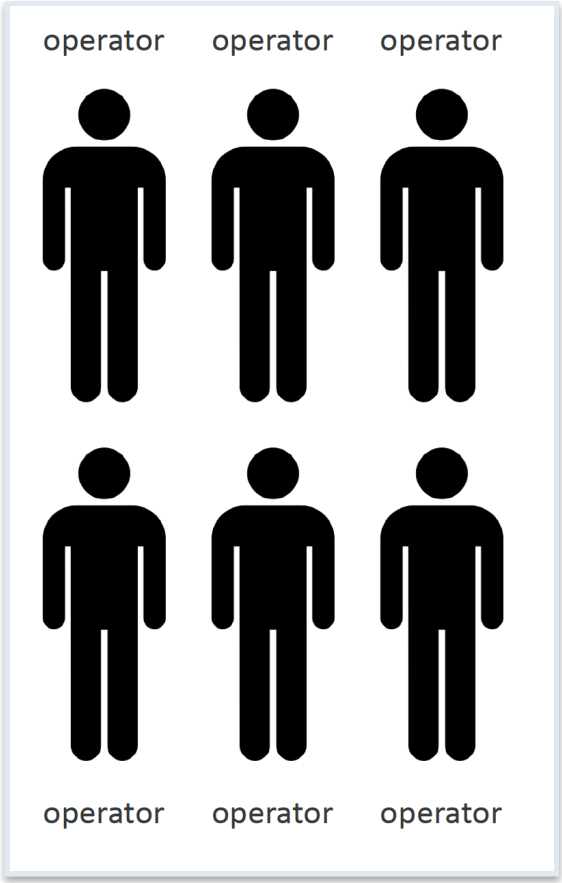
Horizontal Scalability



Horizontal Scalability means increasing the

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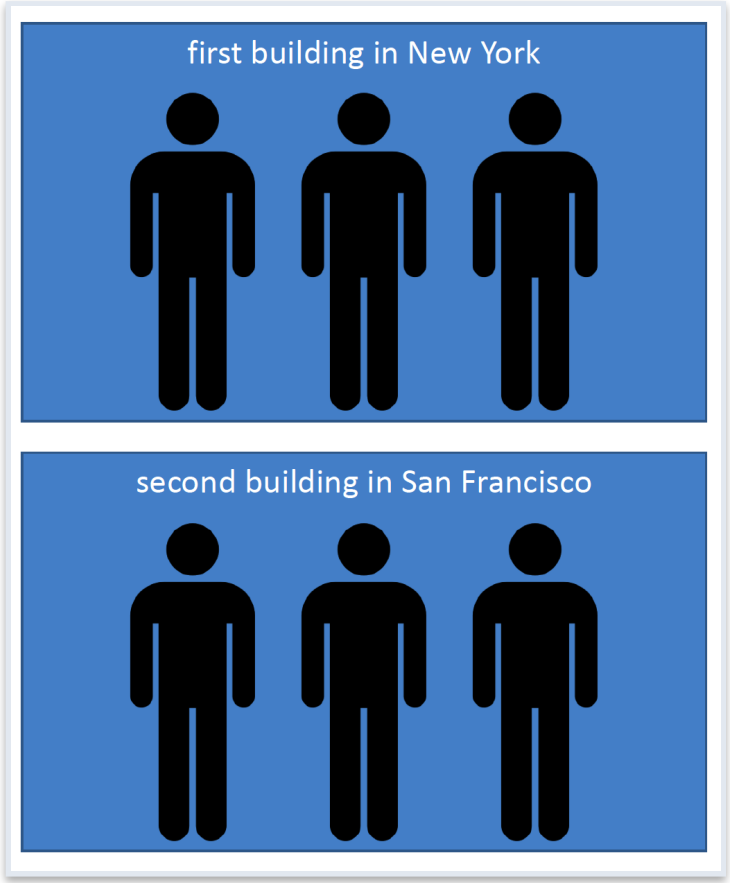
High Availability



High Availability usually goes hand in hand with horizontal scaling

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High Availability & Scalability For EC2

- Vertical Scaling: Increase instance size (= scale up / down)
- From: t2.nano - 0.5G of RAM, 1 vCPU
- To: u-12tb1.metal – 12.3 TB of RAM, 448 vCPUs
- Horizontal Scaling: Increase number of instances (= scale out / in)
- Auto Scaling Group
- Load Balancer
- High Availability: Run instances for the same application across multi AZ
- Auto Scaling Group multi AZ
- Load Balancer multi AZ

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TOPIC 5

Scalability vs Elasticity (vs Agility)

- Scalability: ability to accommodate a larger load by making the hardware stronger (scale up), or by adding nodes (scale out)
- Elasticity: once a system is scalable, elasticity means that there will be some “auto-scaling” so that the system can scale based on the load. This is “cloud-friendly”: pay-per-use, match demand, optimize costs
- Agility: (not related to scalability - distractor) new IT resources are only a click away, which means that you reduce the time to make those resources available to your developers from weeks to just minutes.

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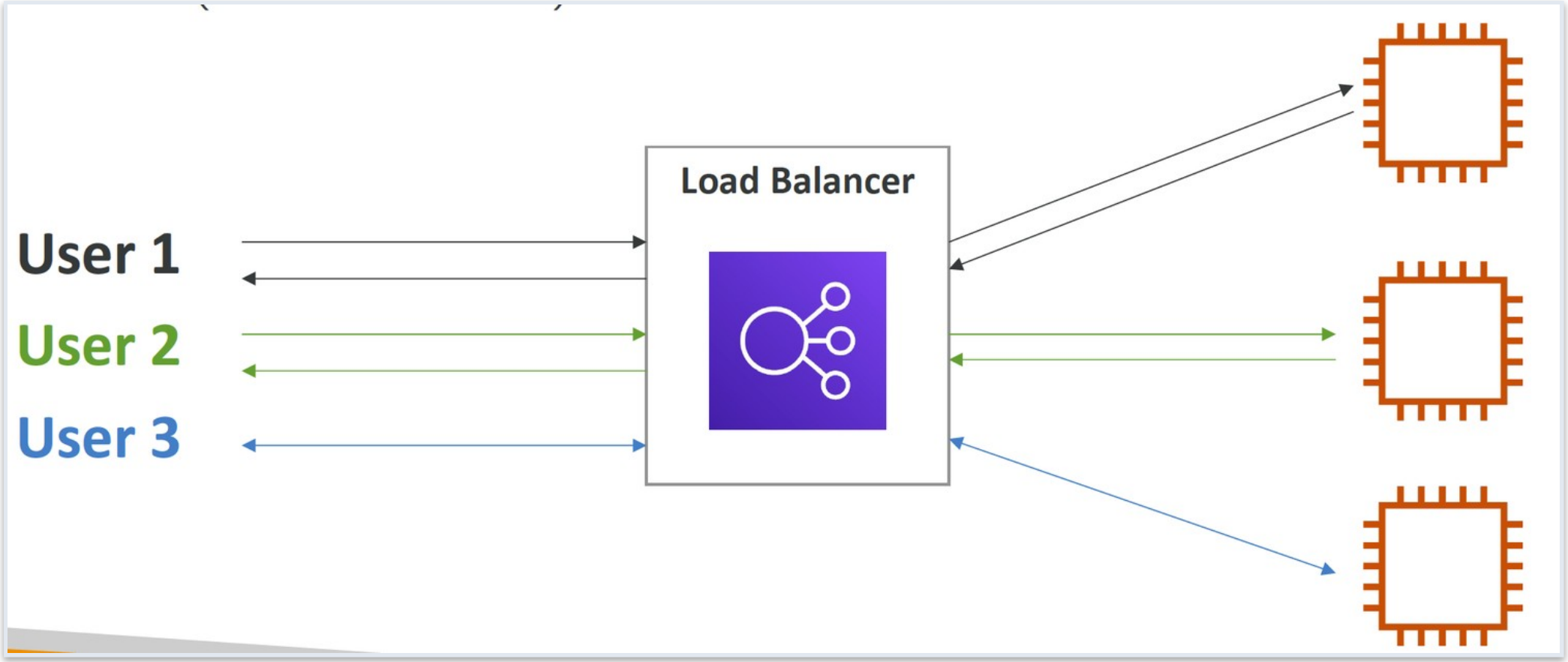
TOPIC 5

What is load balancing?



Load balancers are servers that forward internet traffic to multiple servers (EC2 Instances) downstream.

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Why use an Elastic Load Balancer?

- An ELB (Elastic Load Balancer) is a managed load balancer
- AWS guarantees that it will be working
- AWS takes care of upgrades, maintenance, high availability
- AWS provides only a few configuration knobs
- It costs less to setup your own load balancer but it will be a lot more effort on your end (maintenance, integrations)
- 4 kinds of load balancers offered by AWS:
- Application Load Balancer (HTTP / HTTPS only) – Layer 7
- Network Load Balancer (ultra-high performance, allows for TCP) – Layer 4
- Gateway Load Balancer – Layer 3
- Classic Load Balancer (retired in 2023) – Layer 4 & 7

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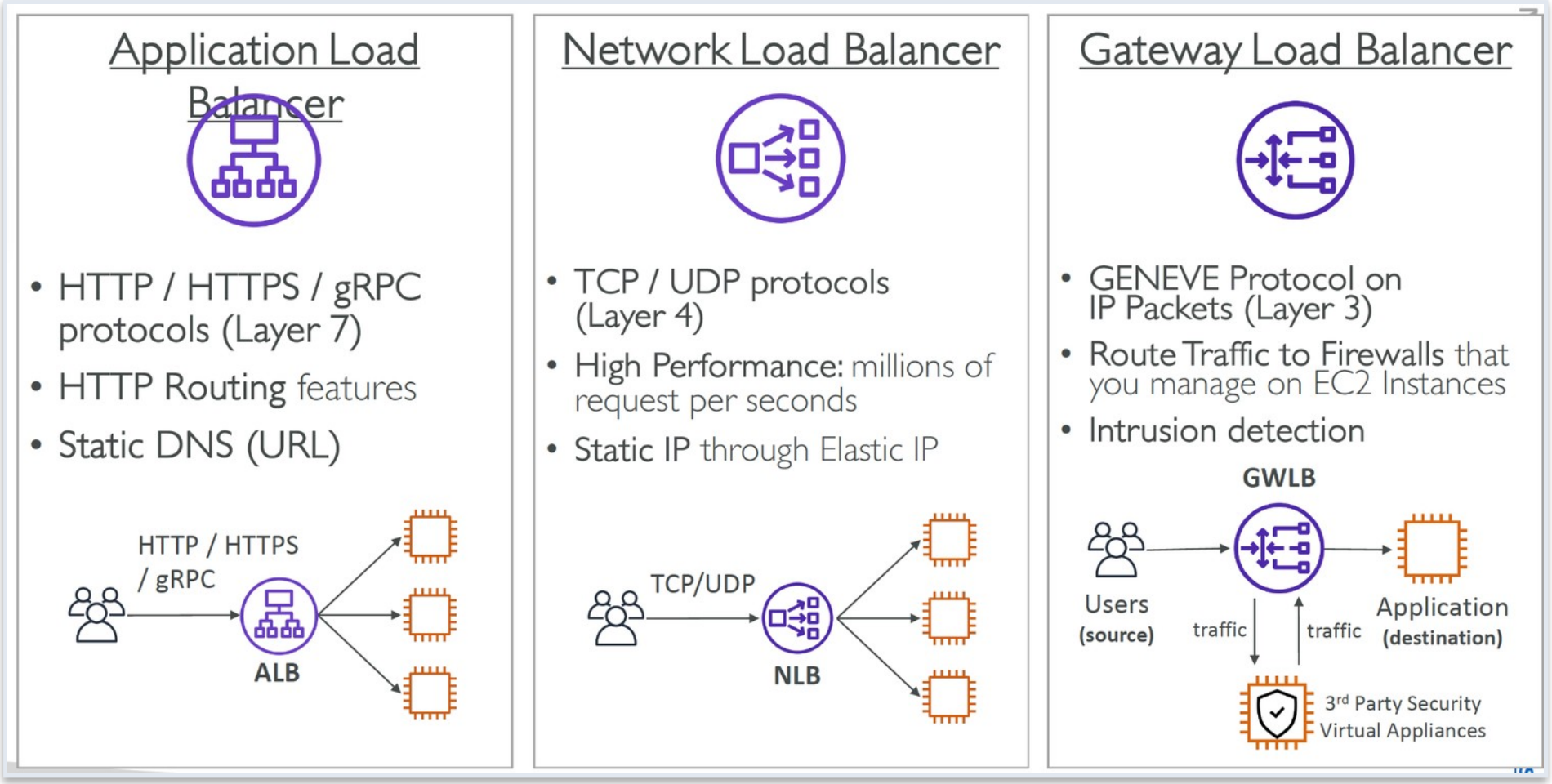
Reference



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TOPIC 5

What’s an Auto Scaling Group?

- In real-life, the load on your websites and application can change
- In the cloud, you can create and get rid of servers very quickly
- The goal of an Auto Scaling Group (ASG) is to:
- Scale out (add EC2 instances) to match an increased load
- Scale in (remove EC2 instances) to match a decreased load
- Ensure we have a minimum and a maximum number of machines running
- Automatically register new instances to a load balancer
- Replace unhealthy instances
- Cost Savings: only run at an optimal capacity (principle of the cloud)

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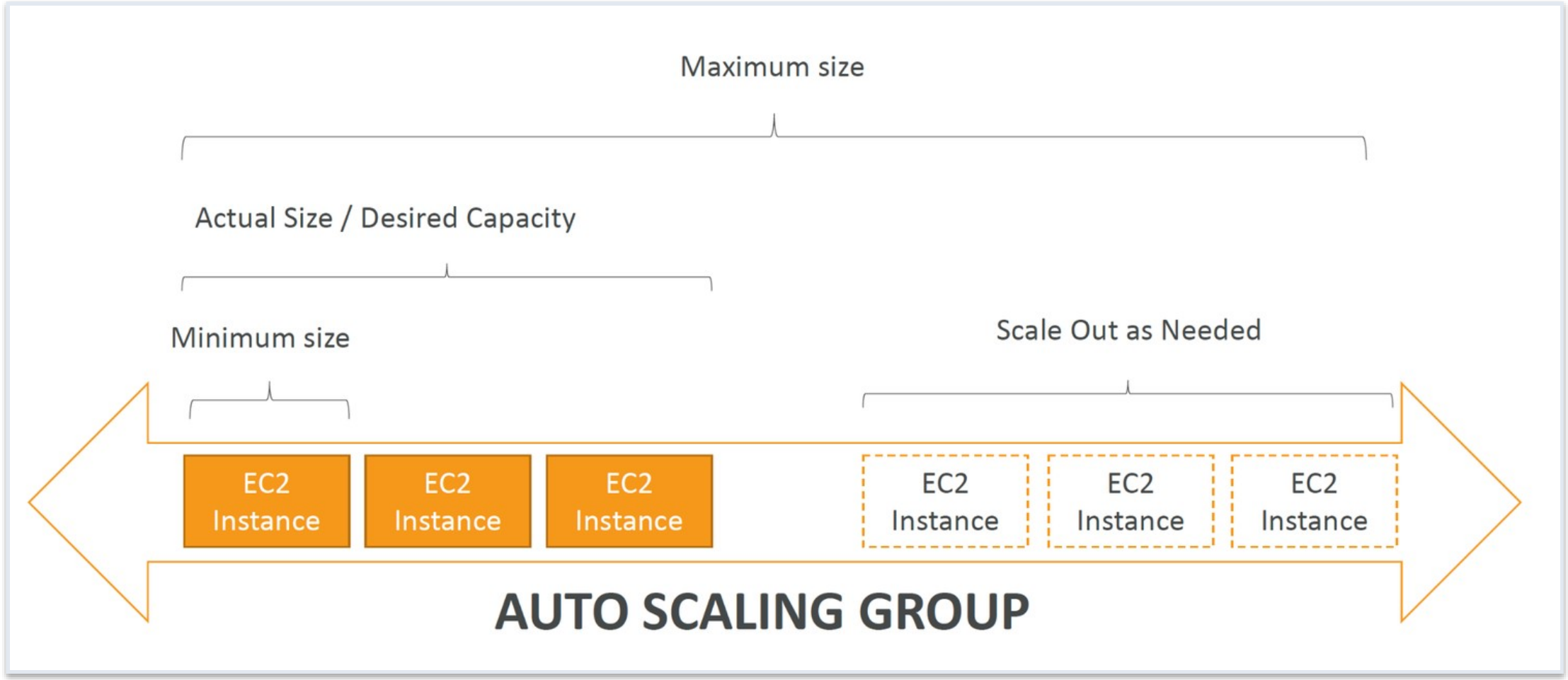
Auto Scaling Group in AWS



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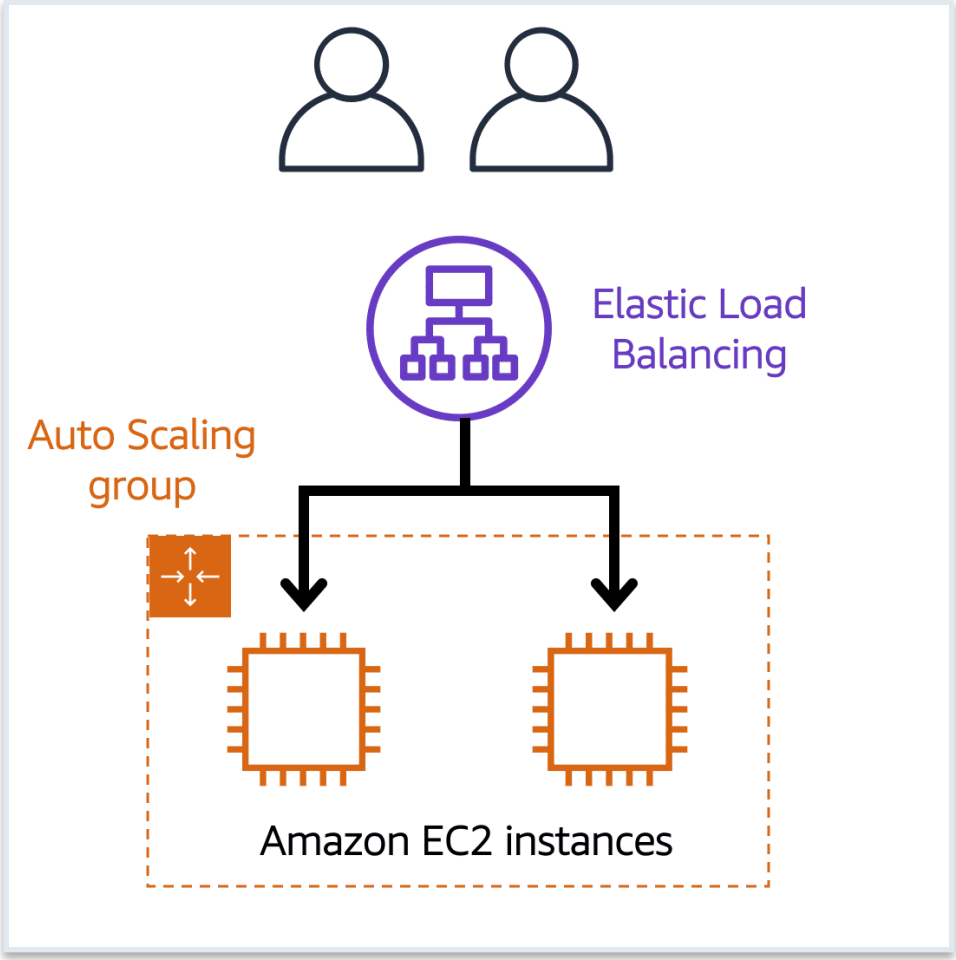
Example: Elastic Load Balancing



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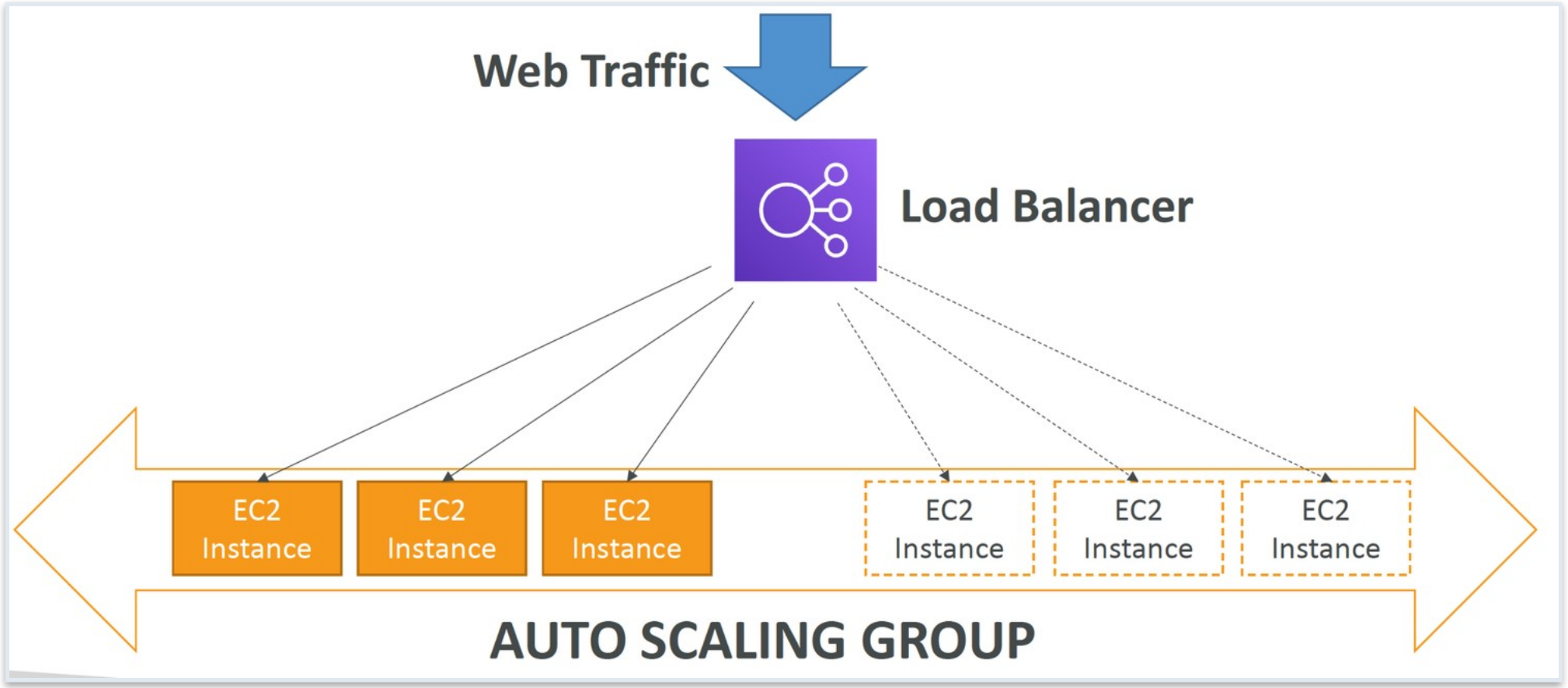
Auto Scaling Group in AWS



With Load Balancer

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Auto Scaling Groups – Scaling Strategies

- Manual Scaling: Update the size of an ASG manually
- Dynamic Scaling: Respond to changing demand
- Simple / Step Scaling
- When a CloudWatch alarm is triggered (example CPU > 70%), then add 2 units
- When a CloudWatch alarm is triggered (example CPU < 30%), then remove 1
- Target Tracking Scaling
- Example: I want the average ASG CPU to stay at around 40%
- Scheduled Scaling
- Anticipate a scaling based on known usage patterns
- Example: increase the min. capacity to 10 at 5 pm on Fridays

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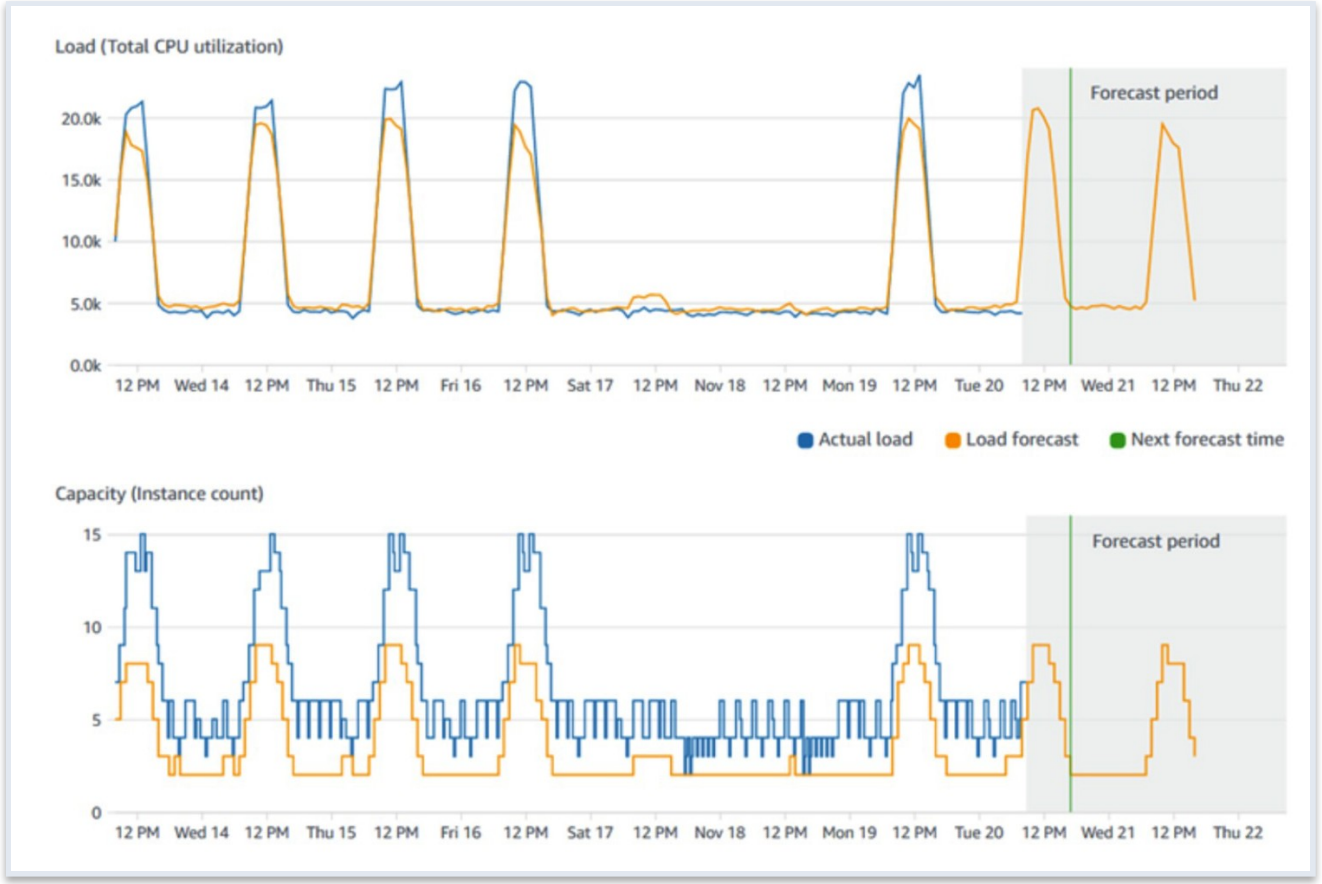
Auto Scaling Groups – Scaling Strategies



Predictive Scaling

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TOPIC 5

ELB & ASG – Summary

- High Availability vs Scalability (vertical and horizontal) vs Elasticity vs Agility in the Cloud
- Elastic Load Balancers (ELB)
- Distribute traffic across backend EC2 instances, can be Multi-AZ
- Supports health checks
- 4 types: Classic (old), Application (HTTP – L7), Network (TCP – L4), Gateway (L3)
- Auto Scaling Groups (ASG)
- Implement Elasticity for your application, across multiple AZ
- Scale EC2 instances based on the demand on your system, replace unhealthy
- Integrated with the ELB

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GLOBAL INFRASTRUCTURE

Global Infrastructure

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Why make a global application?

- A global application is an application deployed in multiple geographies
- On AWS: this could be Regions and / or Edge Locations
- Decreased Latency
- Latency is the time it takes for a network packet to reach a server
- It takes time for a packet from Asia to reach the US
- Deploy your applications closer to your users to decrease latency, better experience
- Disaster Recovery (DR)
- If an AWS region goes down (earthquake, storms, power shutdown, politics)…
- You can fail-over to another region and have your application still working
- A DR plan is important to increase the availability of your application
- Attack protection: distributed global infrastructure is harder to attack

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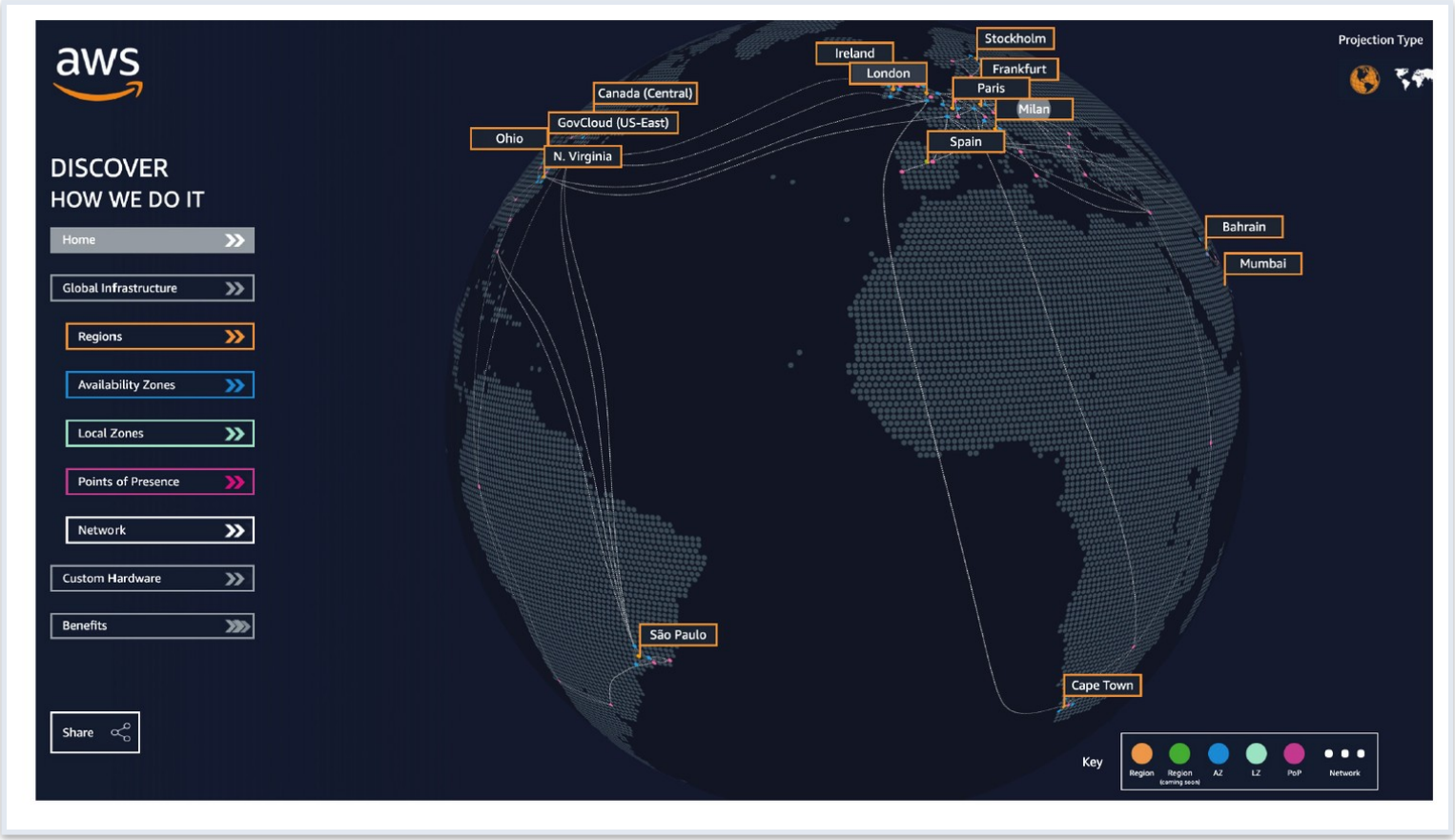
Global AWS Infrastructure



Regions: For deploying applications and infrastructure

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Global Applications in AWS

- Global DNS: Route 53
- Great to route users to the closest deployment with least latency
- Great for disaster recovery strategies
- Global Content Delivery Network (CDN): CloudFront
- Replicate part of your application to AWS Edge Locations – decrease latency
- Cache common requests – improved user experience and decreased latency
- S3 Transfer Acceleration
- Accelerate global uploads & downloads into Amazon S3
- AWS Global Accelerator:
- Improve global application availability and performance using the AWS global network

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Amazon Route 53 Overview

- Route53 is a Managed DNS (Domain Name System)
- DNS is a collection of rules and records which helps clients understand how to reach a server through URLs.
- In AWS, the most common records are:
- www.google.com => 12.34.56.78 == A record (IPv4)
- www.google.com => 2001:0db8:85a3:0000:0000:8a2e:0370:7334 == AAAA IPv6
- search.google.com => www.google.com == CNAME: hostname to hostname
- example.com => AWS resource == Alias (ex: ELB, CloudFront, S3, RDS, etc…)

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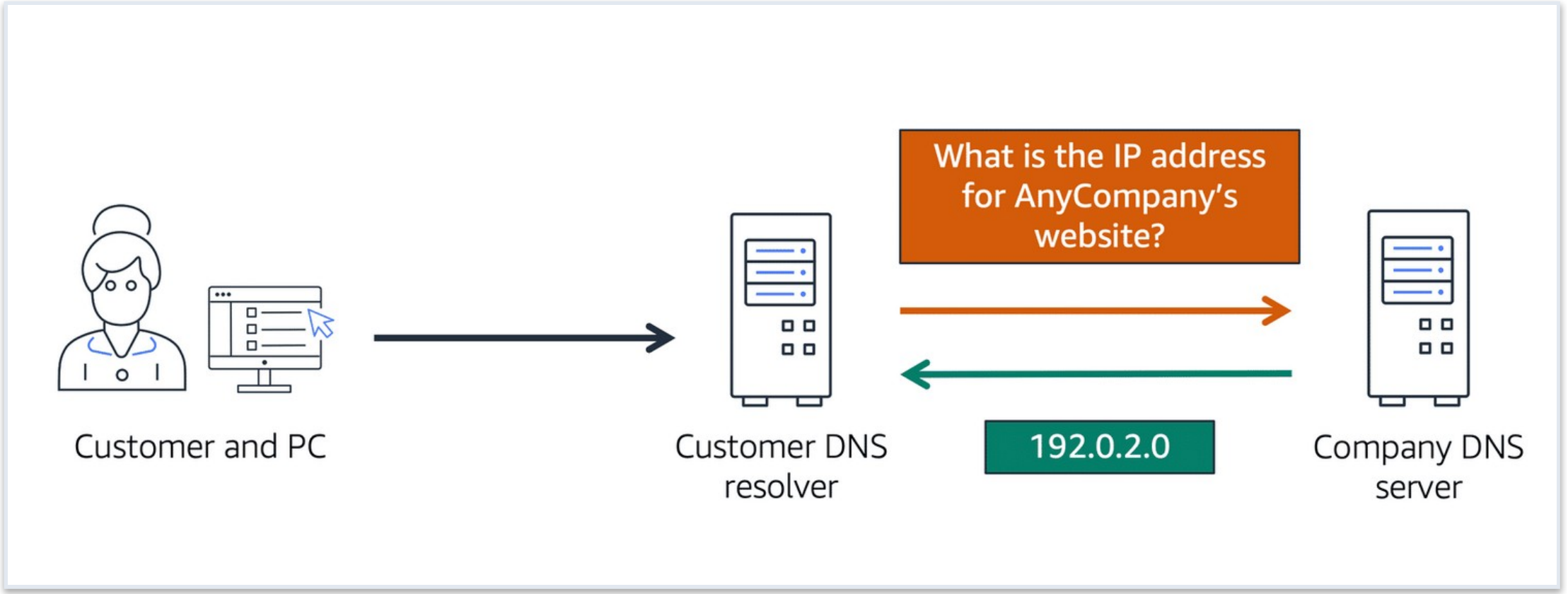
Domain Name System (DNS)



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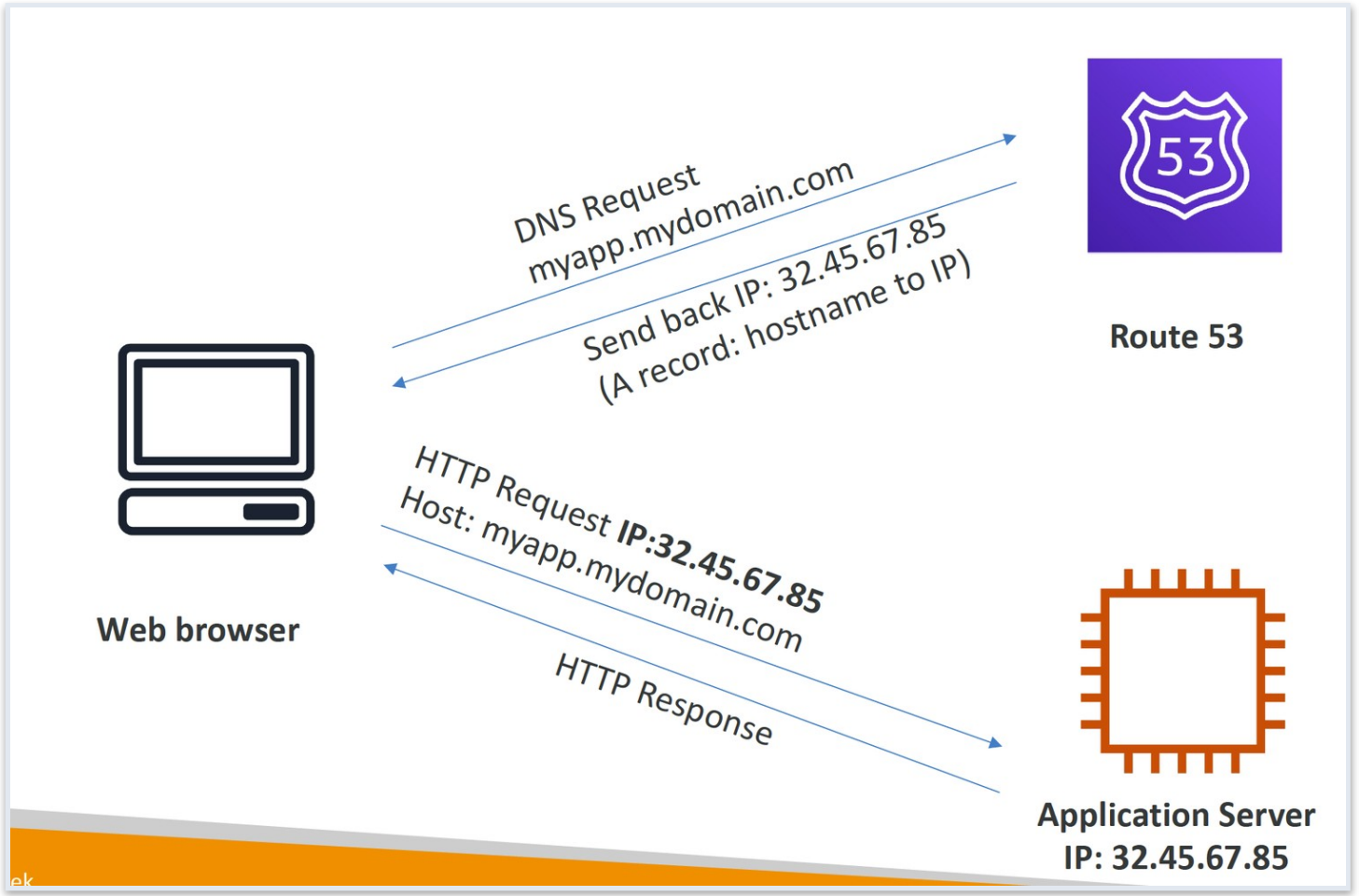
Route 53 – Diagram for A Record



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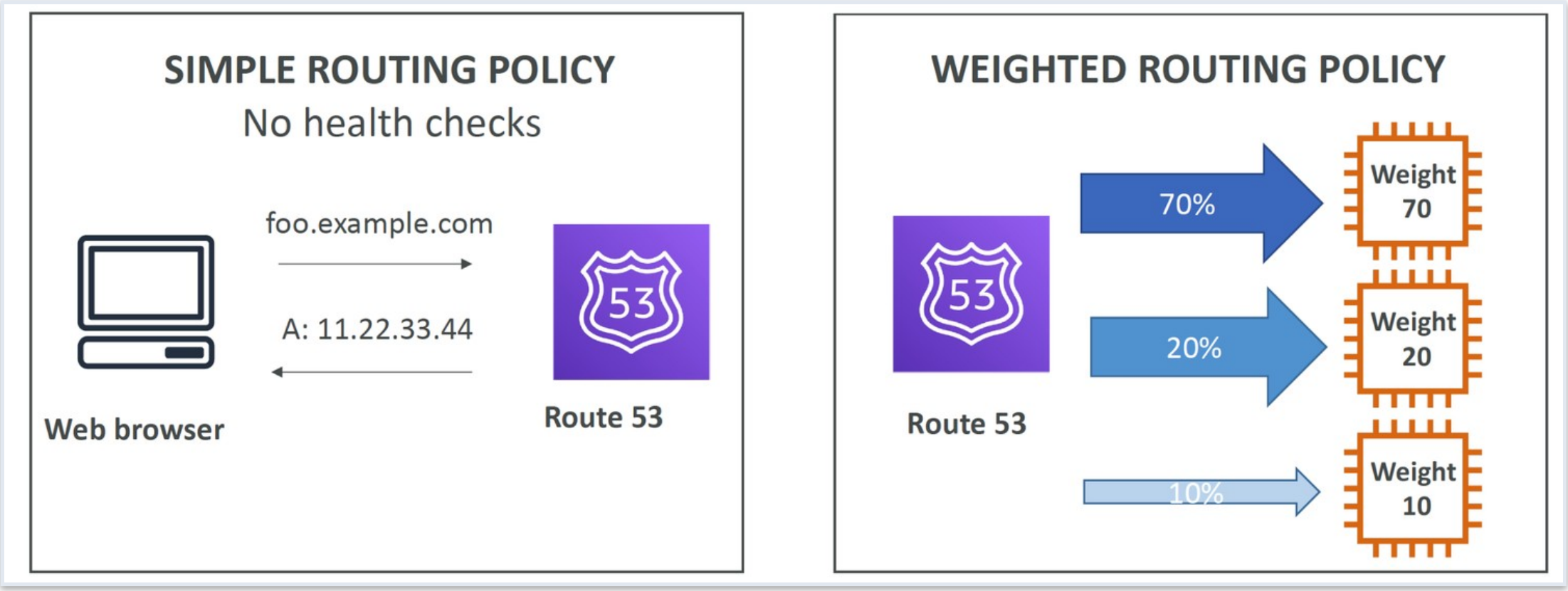
Route 53 Routing Policies



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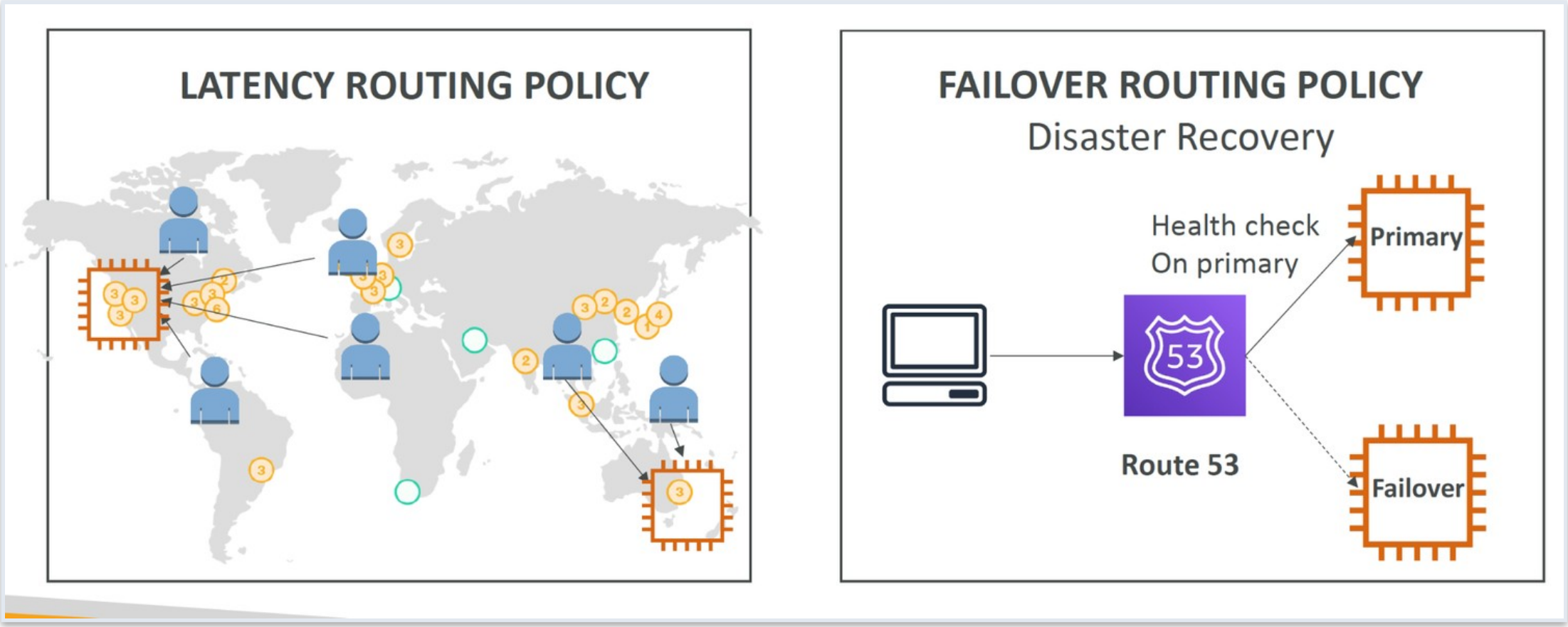
Route 53 Routing Policies



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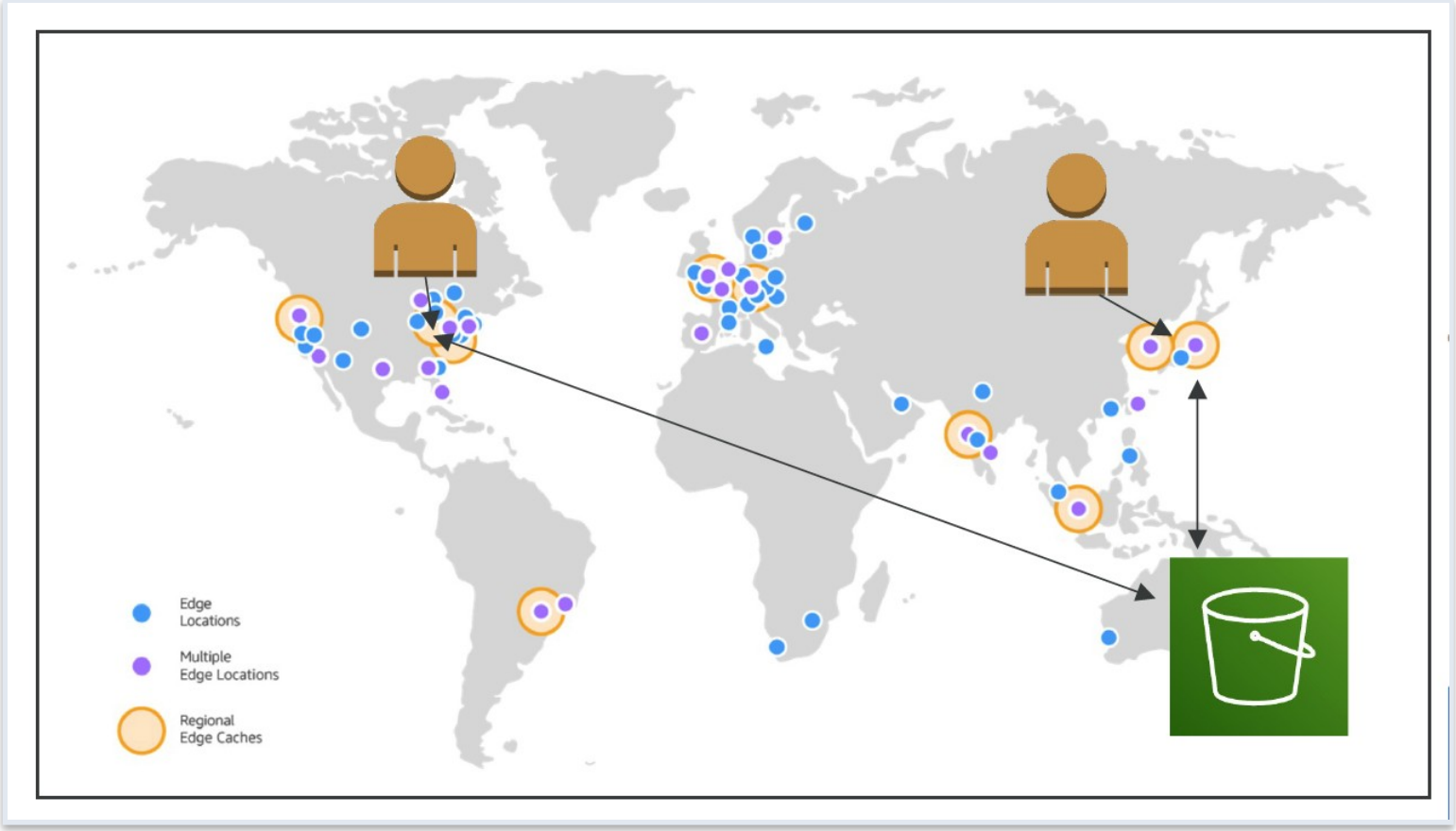
Amazon CloudFront



Content Delivery Network (CDN)

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CloudFront – Origins

- S3 bucket
- For distributing files and caching them at the edge
- Enhanced security with CloudFront Origin Access Control (OAC)
- OAC is replacing Origin Access Identity (OAI)
- CloudFront can be used as an ingress (to upload files to S3)
- Custom Origin (HTTP)
- Application Load Balancer
- EC2 instance
- S3 website (must first enable the bucket as a static S3 website)
- Any HTTP backend you want

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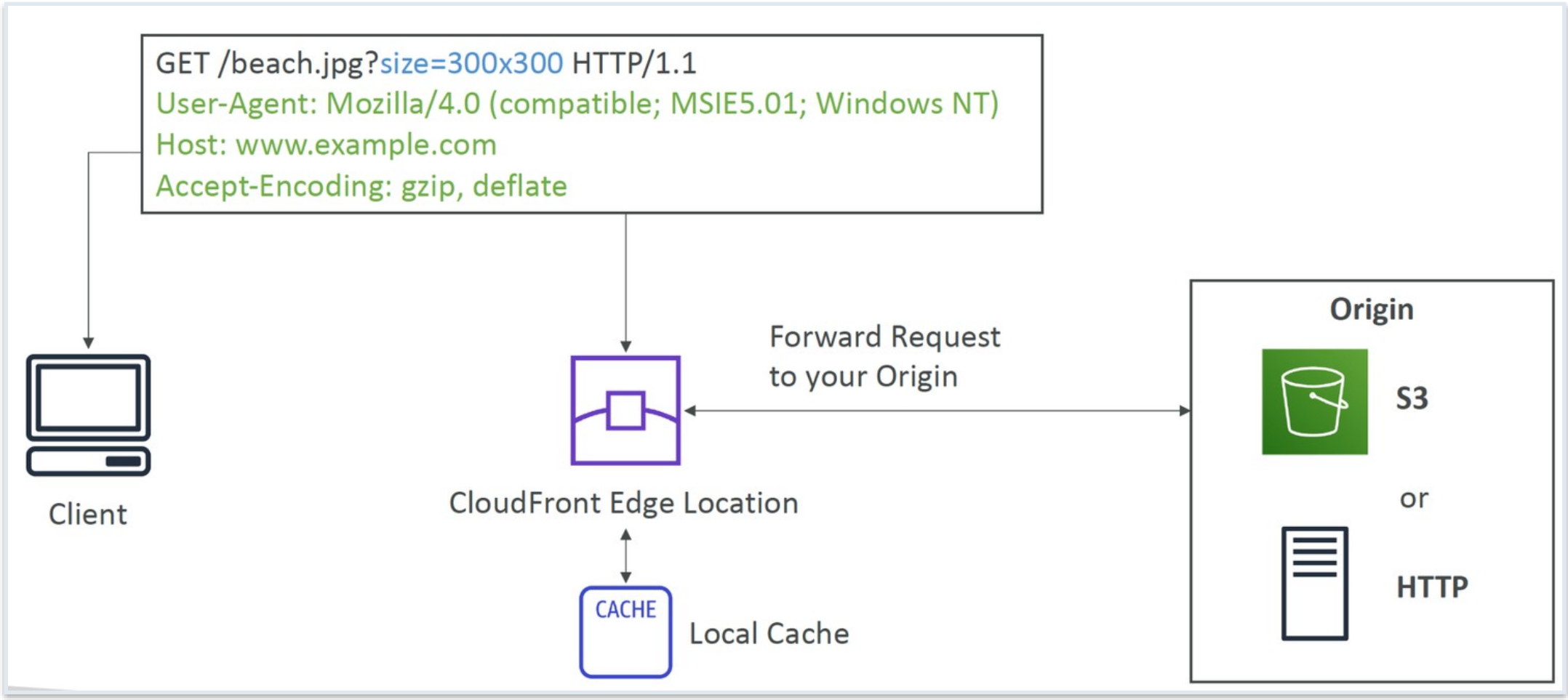
CloudFront at a high level



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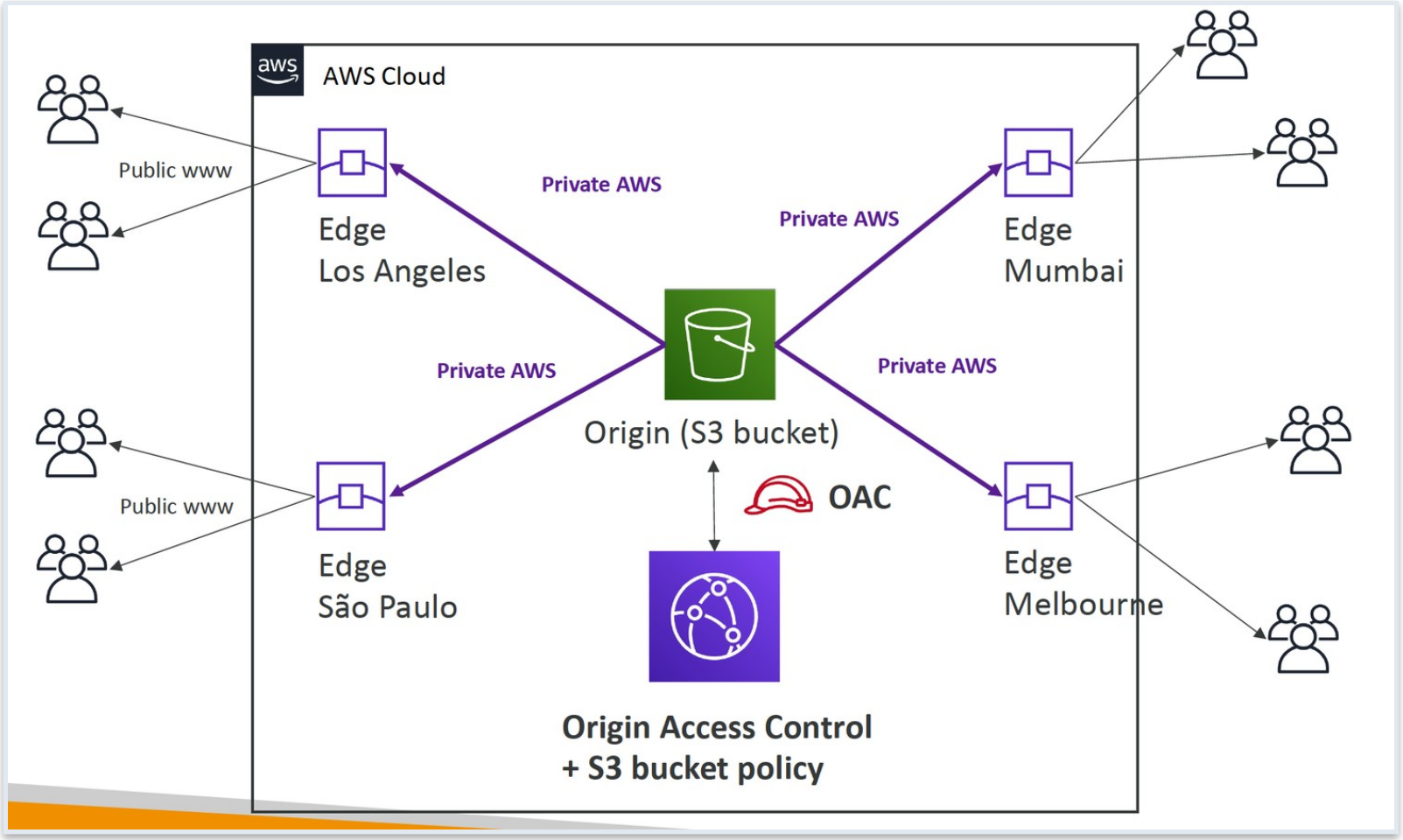
CloudFront – S3 as an Origin



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CloudFront vs S3 Cross Region Replication

- CloudFront:
- Global Edge network
- Files are cached for a TTL (maybe a day)
- Great for static content that must be available everywhere
- S3 Cross Region Replication:
- Must be setup for each region you want replication to happen
- Files are updated in near real-time
- Read only
- Great for dynamic content that needs to be available at low-latency in few regions

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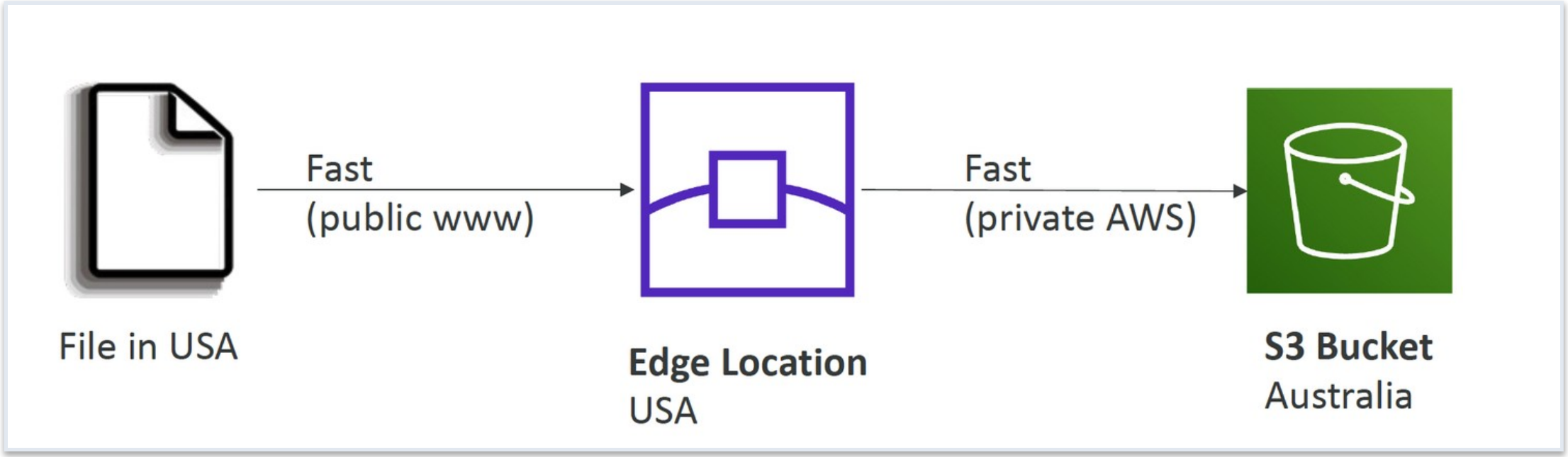
S3 Transfer Acceleration



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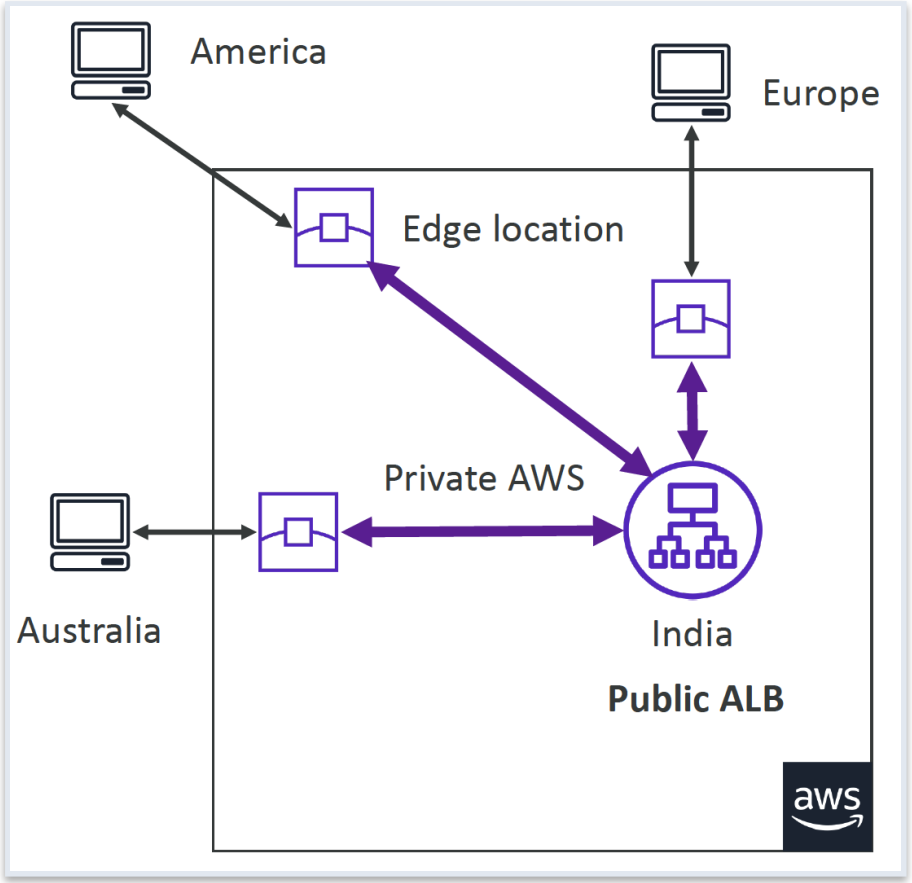
AWS Global Accelerator



Improve global application availability and performance using the AWS global network

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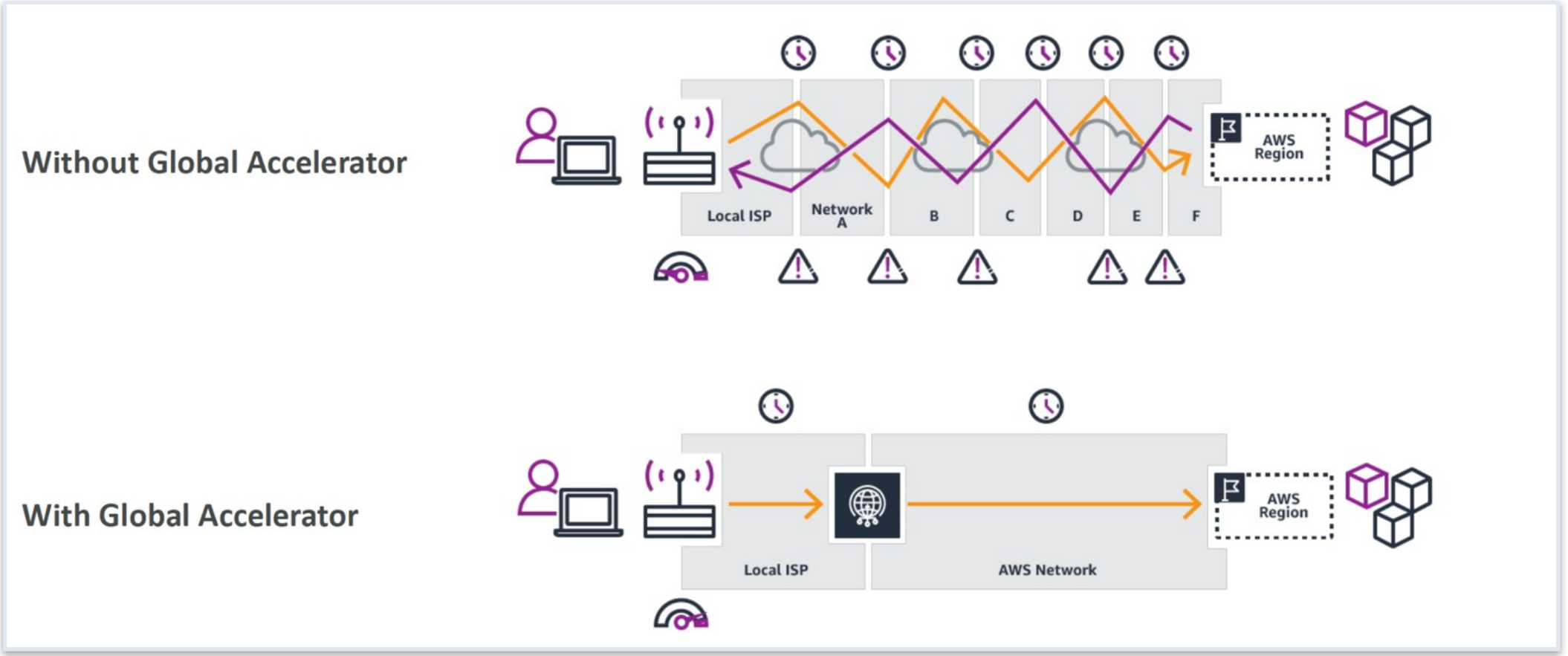
AWS Global Accelerator



https://aws.amazon.com/global-accelerator

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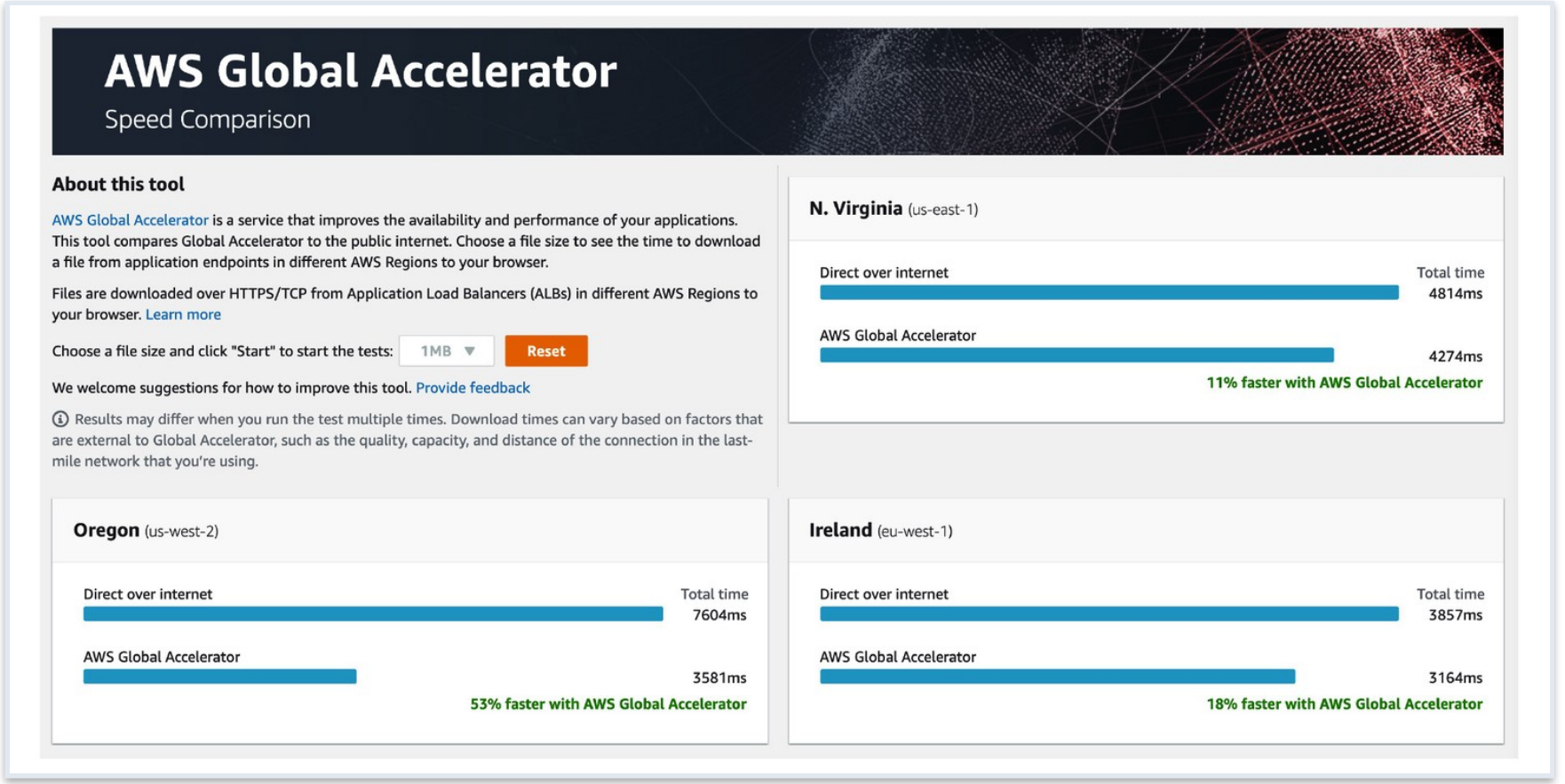
AWS Global Accelerator vs CloudFront



https://speedtest.globalaccelerator.aws/#/

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AWS Global Accelerator vs CloudFront

- They both use the AWS global network and its edge locations around the world
- Both services integrate with AWS Shield for DDoS protection.
- CloudFront – Content Delivery Network
- Improves performance for your cacheable content (such as images and videos)
- Content is served at the edge
- Global Accelerator
- No caching, proxying packets at the edge to applications running in one or more AWS Regions.
- Improves performance for a wide range of applications over TCP or UDP
- Good for HTTP use cases that require static IP addresses
- Good for HTTP use cases that required deterministic, fast regional failover

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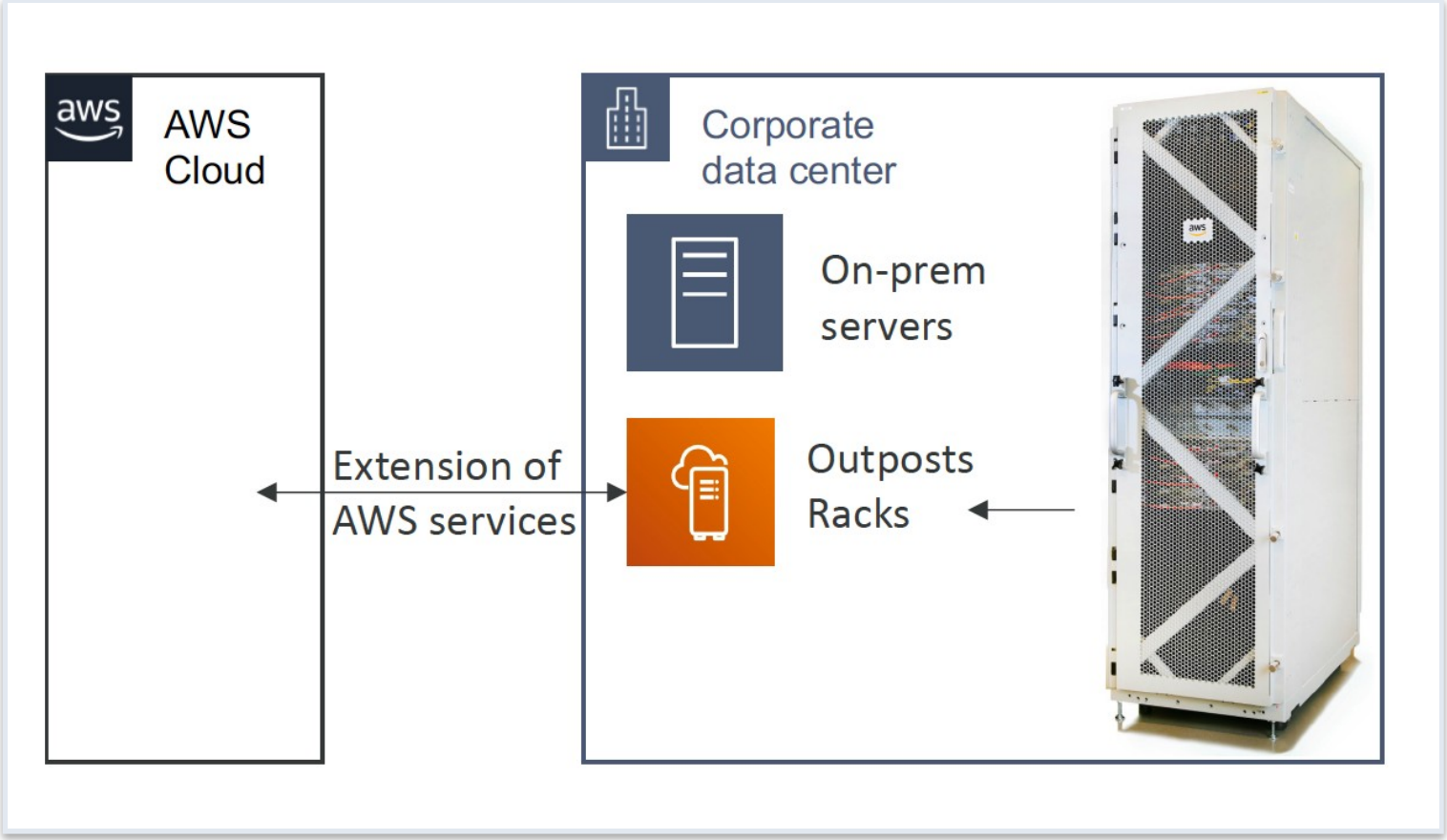
AWS Outposts



Hybrid Cloud: businesses that keep an on-premises infrastructure alongside a cloud infrastructure

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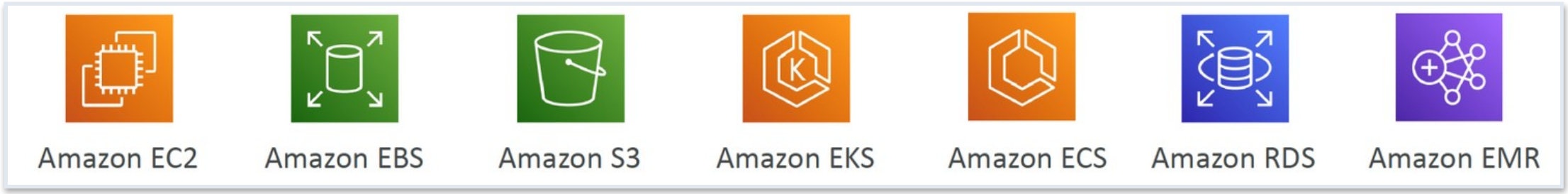
AWS Outposts



Benefits:

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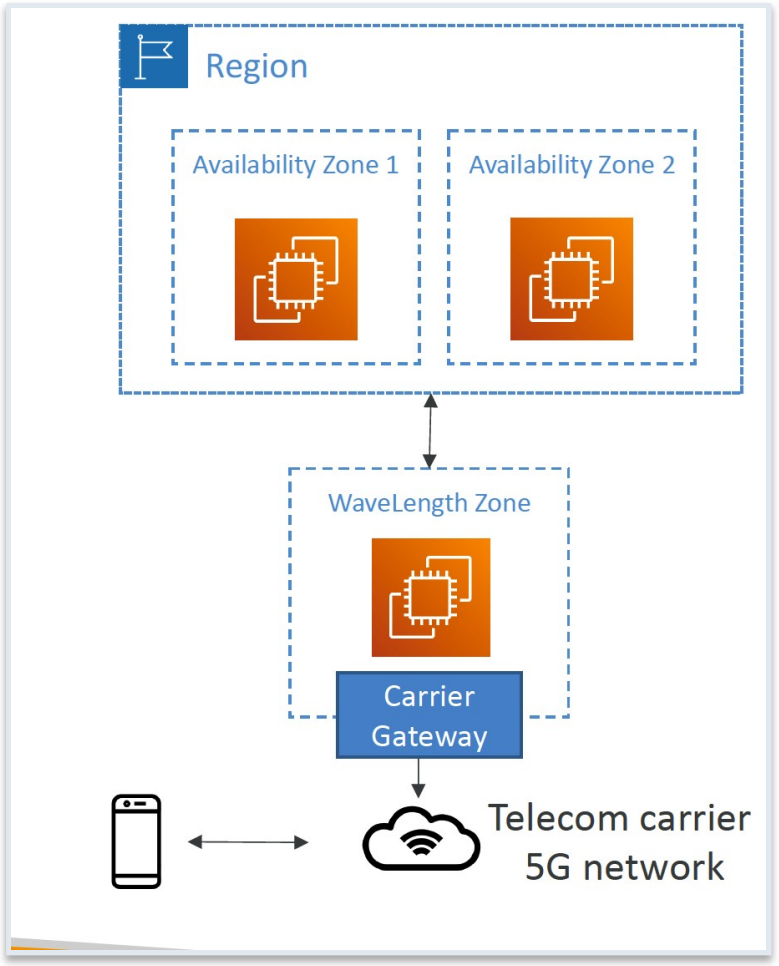
AWS WaveLength



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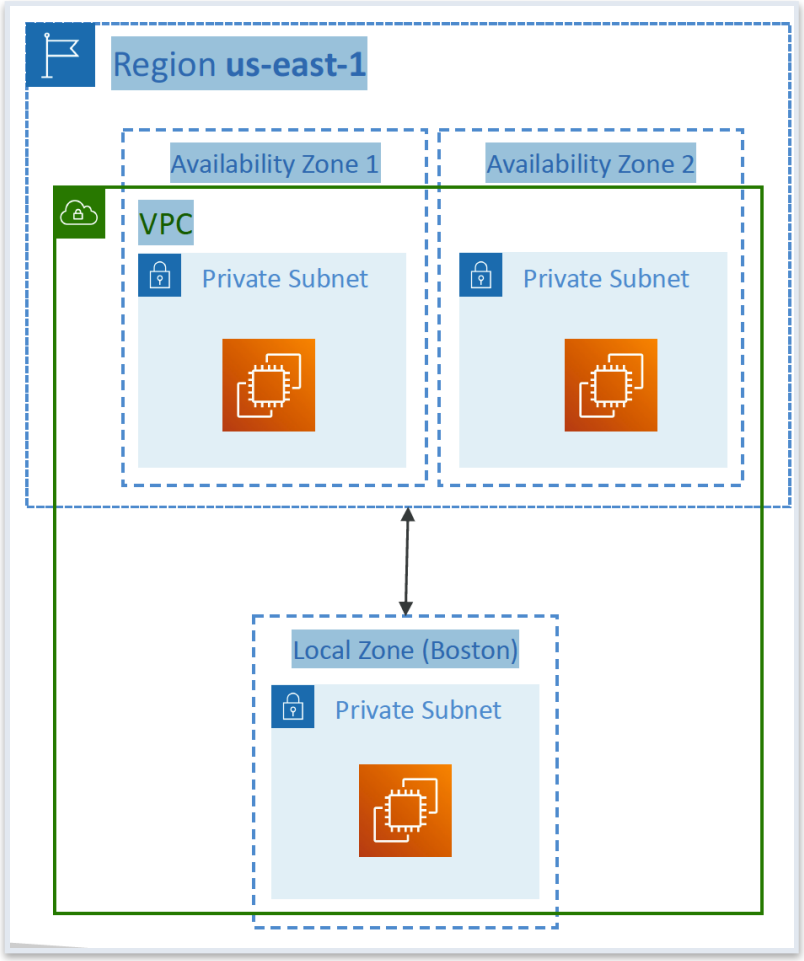
AWS Local Zones



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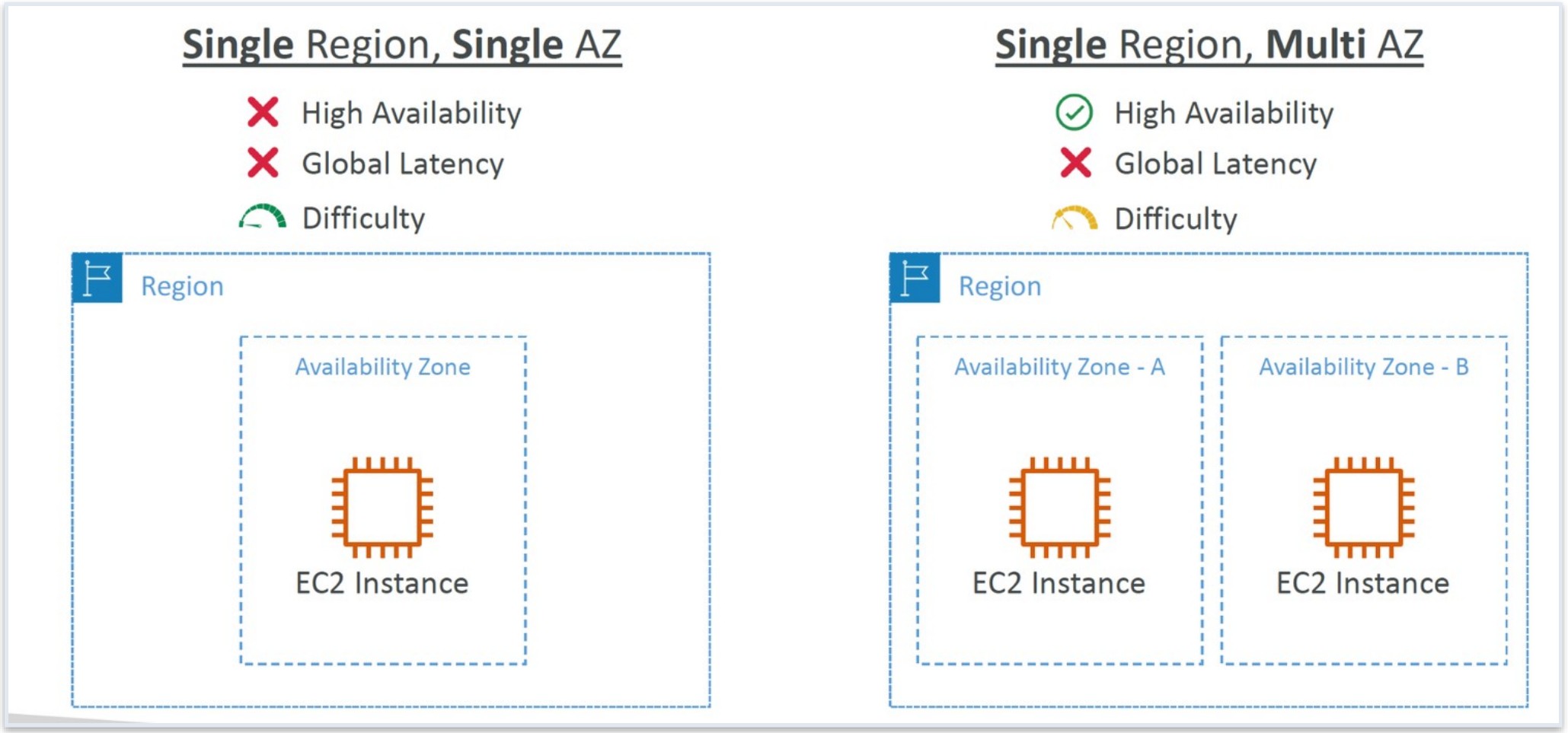
Global Applications Architecture



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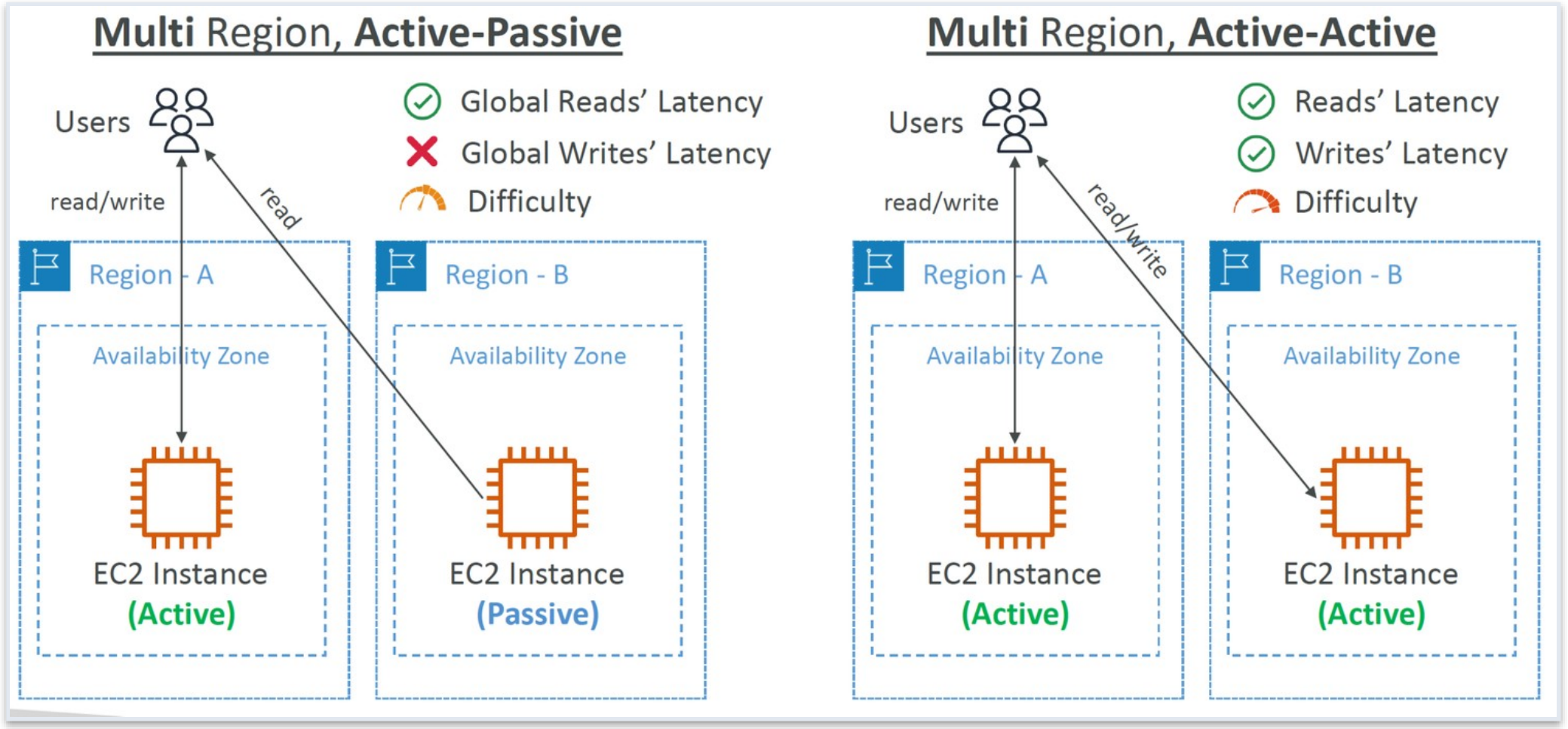
Global Applications Architecture



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Global Applications in AWS - Summary

- AWS Outposts
- Deploy Outposts Racks in your own Data Centers to extend AWS services
- AWS WaveLength
- Brings AWS services to the edge of the 5G networks
- Ultra-low latency applications
- AWS Local Zones
- Bring AWS resources (compute, database, storage, …) closer to your users
- Good for latency-sensitive applications

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Global Applications in AWS - Summary

- Global DNS: Route 53
- Great to route users to the closest deployment with least latency
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Cloud Integration

- Cloud Integration

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Sync vs Async Communications

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Sync vs Async Communications

- Synchronous between applications can be problematic if there are sudden spikes of traffic
- What if you need to suddenly encode 1000 videos but usually it’s 10?
- In that case, it’s better to decouple your applications:
- using SQS: queue model
- using SNS: pub/sub model
- using Kinesis: real-time data streaming model
- These services can scale independently from our application!

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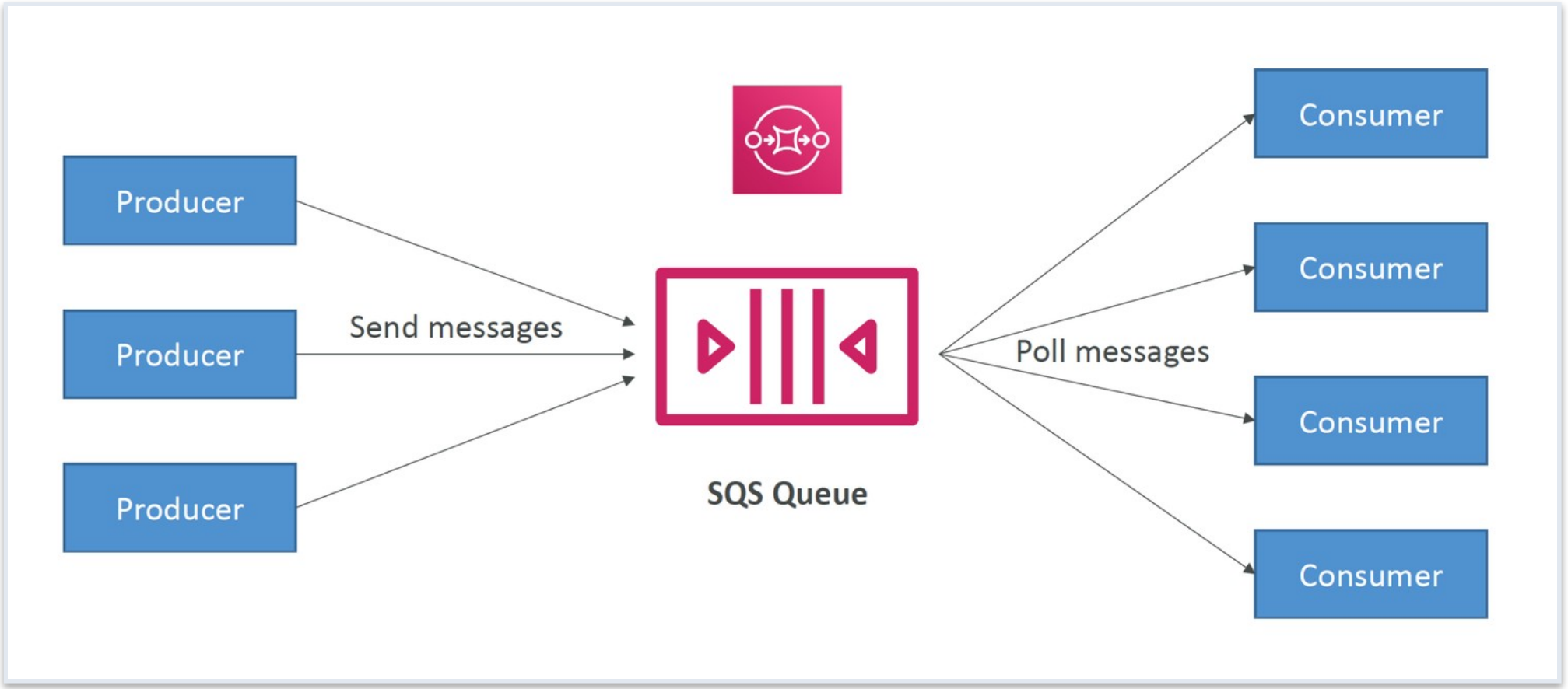
Amazon SQS – Simple Queue Service



What's a queue?

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Amazon SQS – Standard Queue

- Oldest AWS offering (over 10 years old)
- Fully managed service (~serverless), use to decouple applications
- Scales from 1 message per second to 10,000s per second
- Default retention of messages: 4 days, maximum of 14 days
- No limit to how many messages can be in the queue
- Messages are deleted after they’re read by consumers
- Low latency (<10 ms on publish and receive)
- Consumers share the work to read messages & scale horizontally

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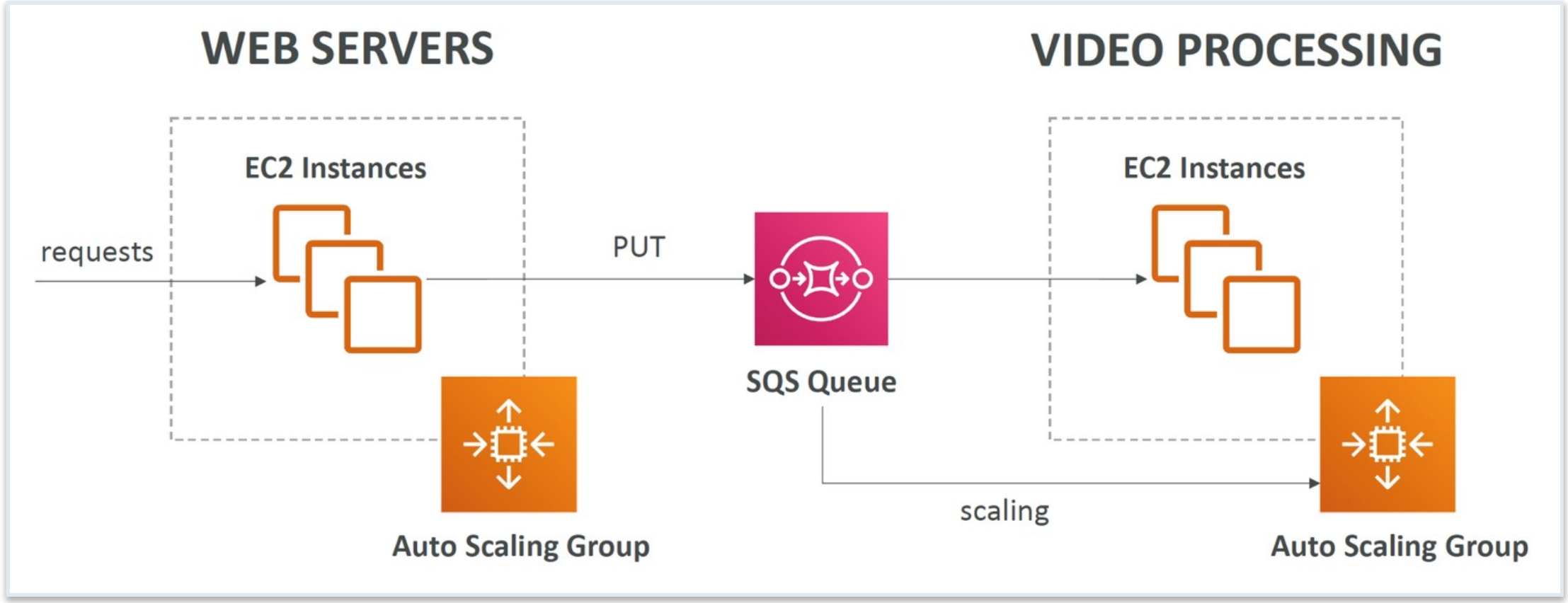
SQS to decouple between application tiers



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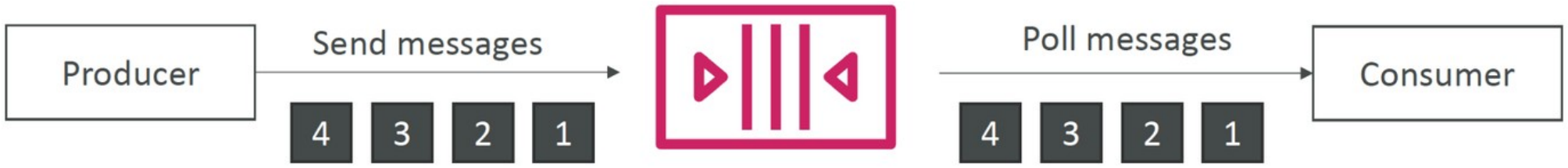
Amazon SQS – FIFO Queues

- FIFO = First In First Out (ordering of messages in the queue)



- Messages are processed in order by the consumer

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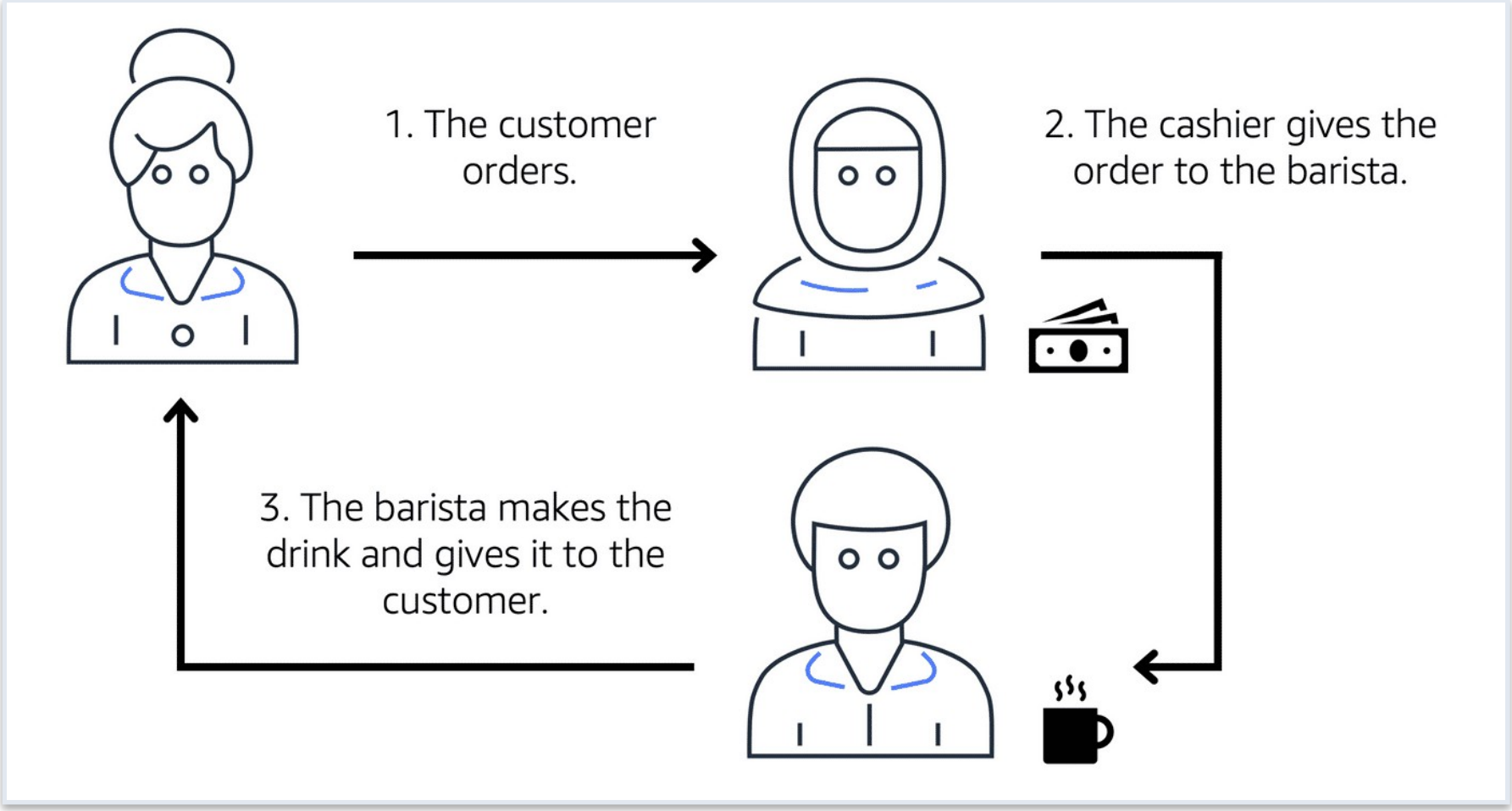
Example 1: Fulfilling an order



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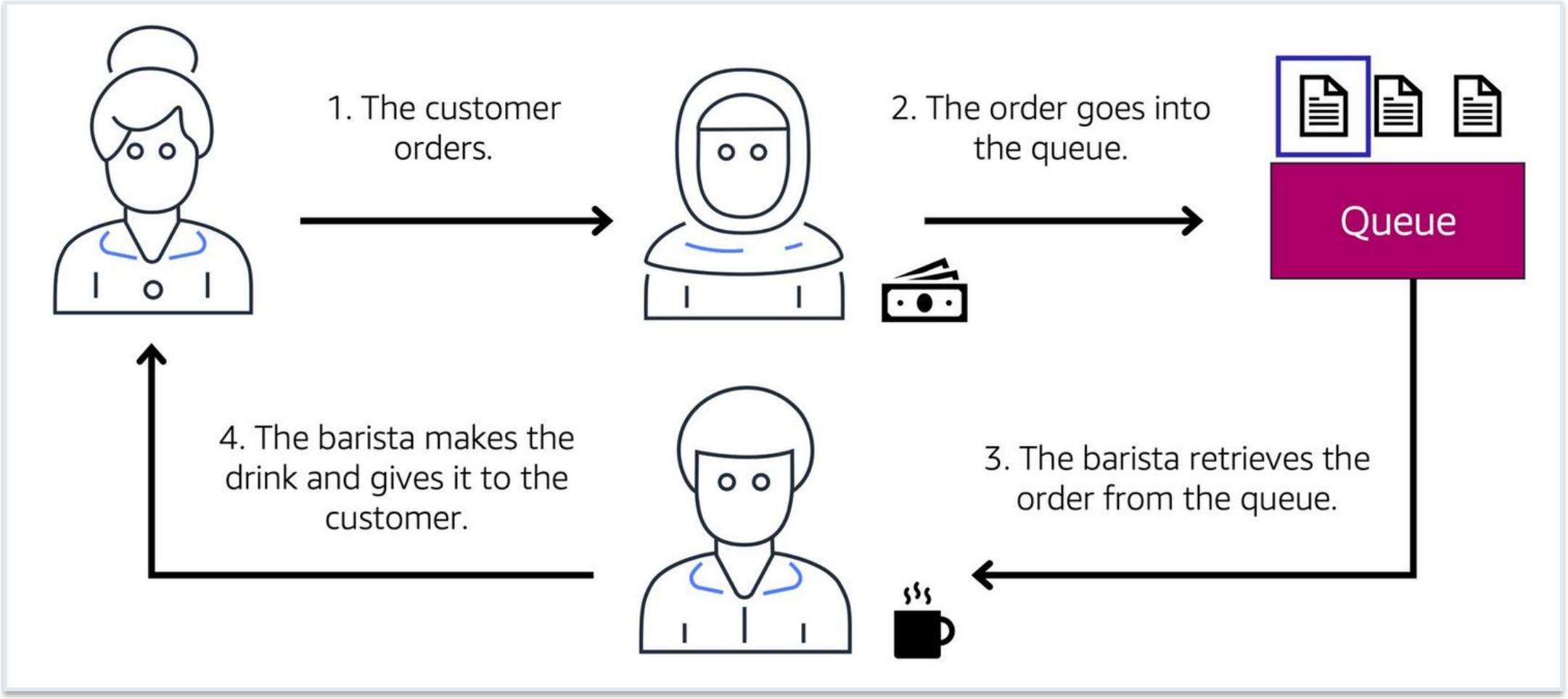
Example 2: Orders in a queue



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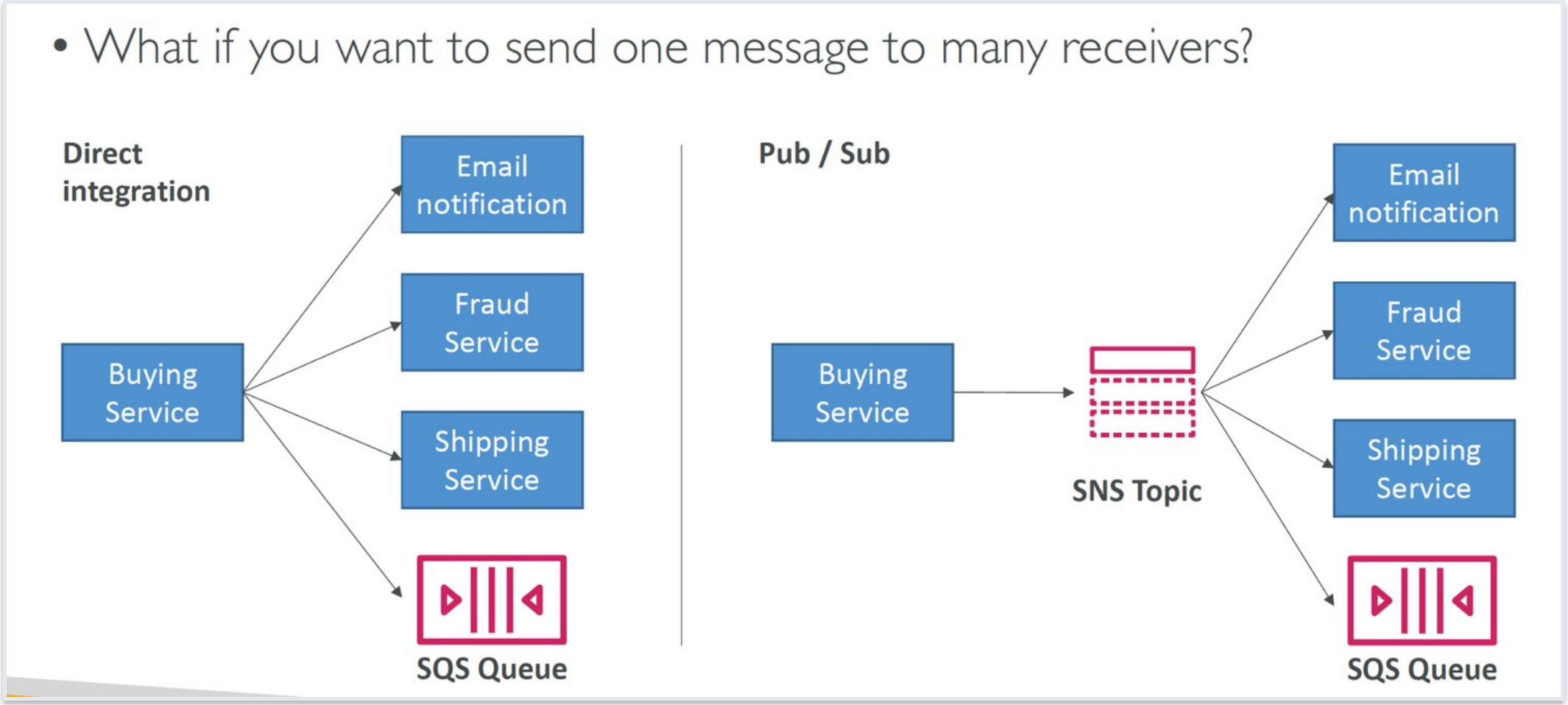
Amazon SNS



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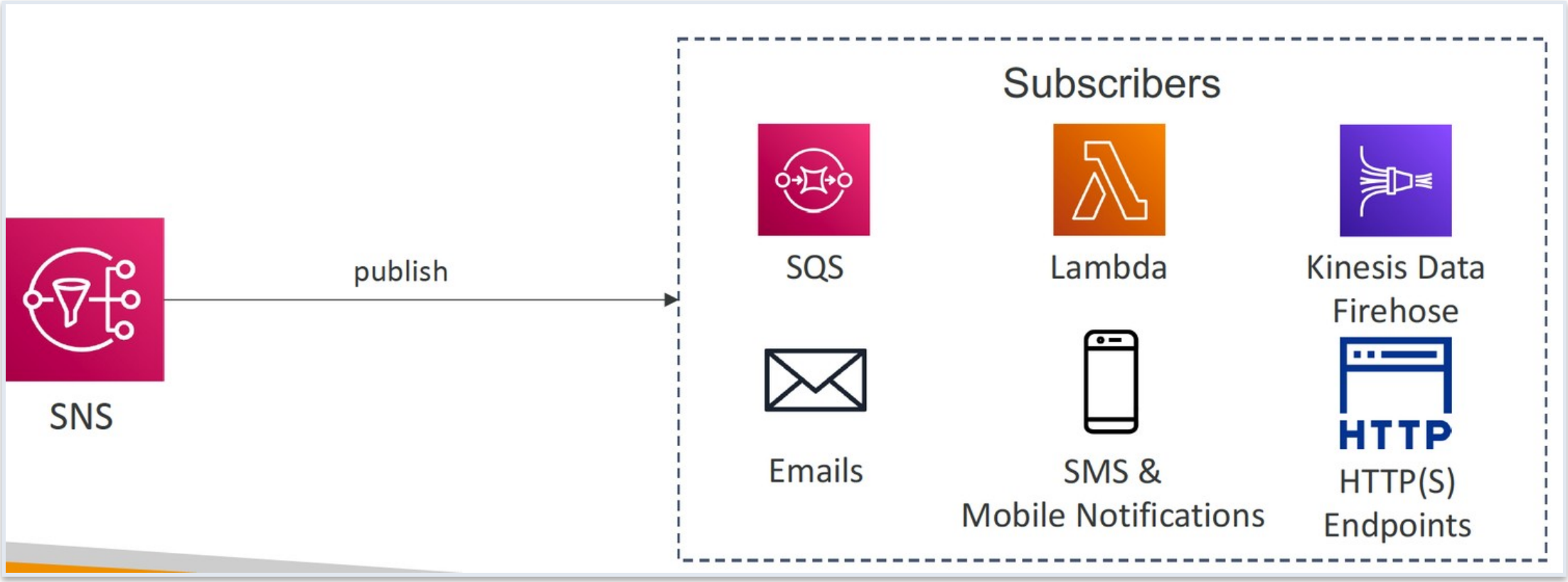
Amazon SNS



The “event publishers” only sends message to one SNS topic

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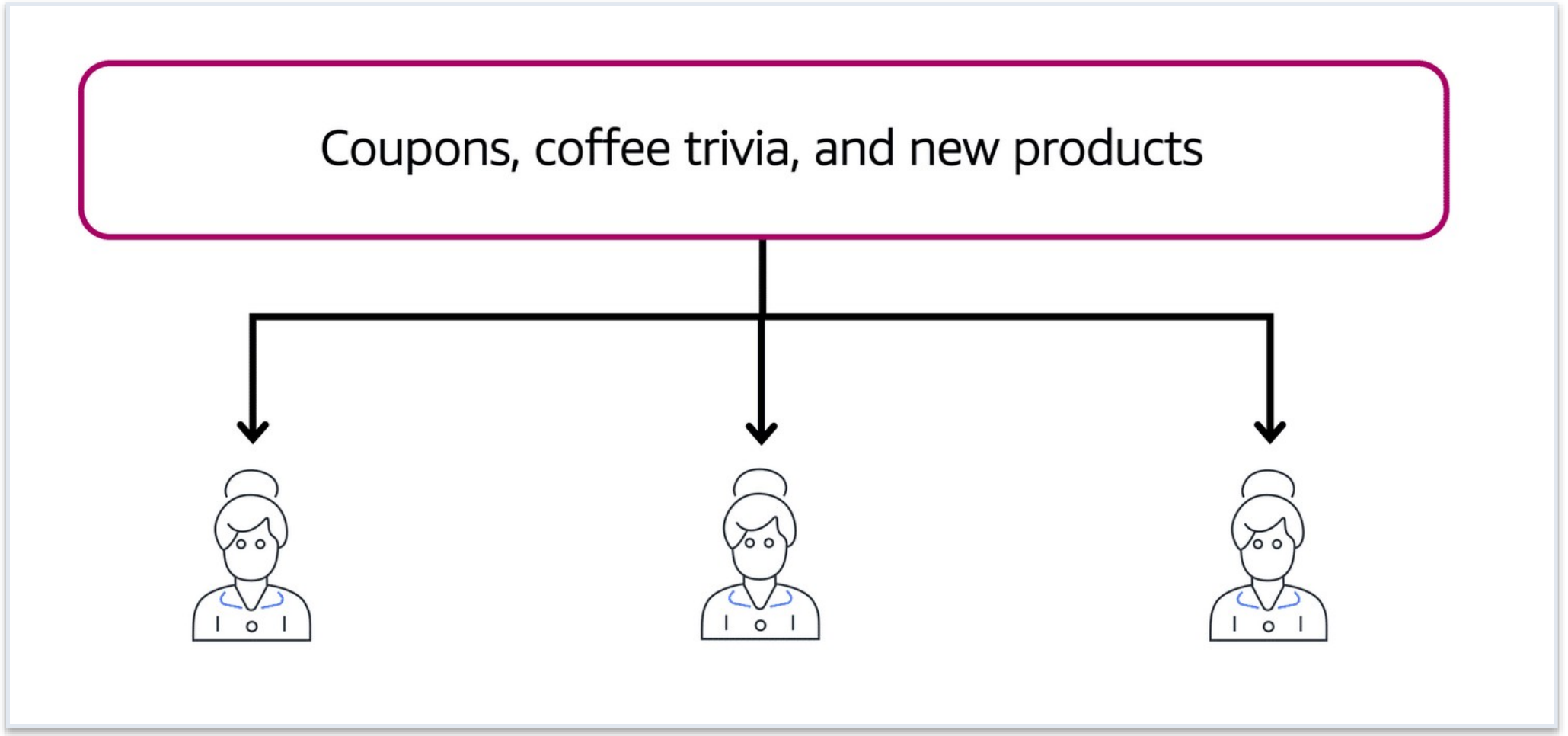
Publishing updates from a single topic



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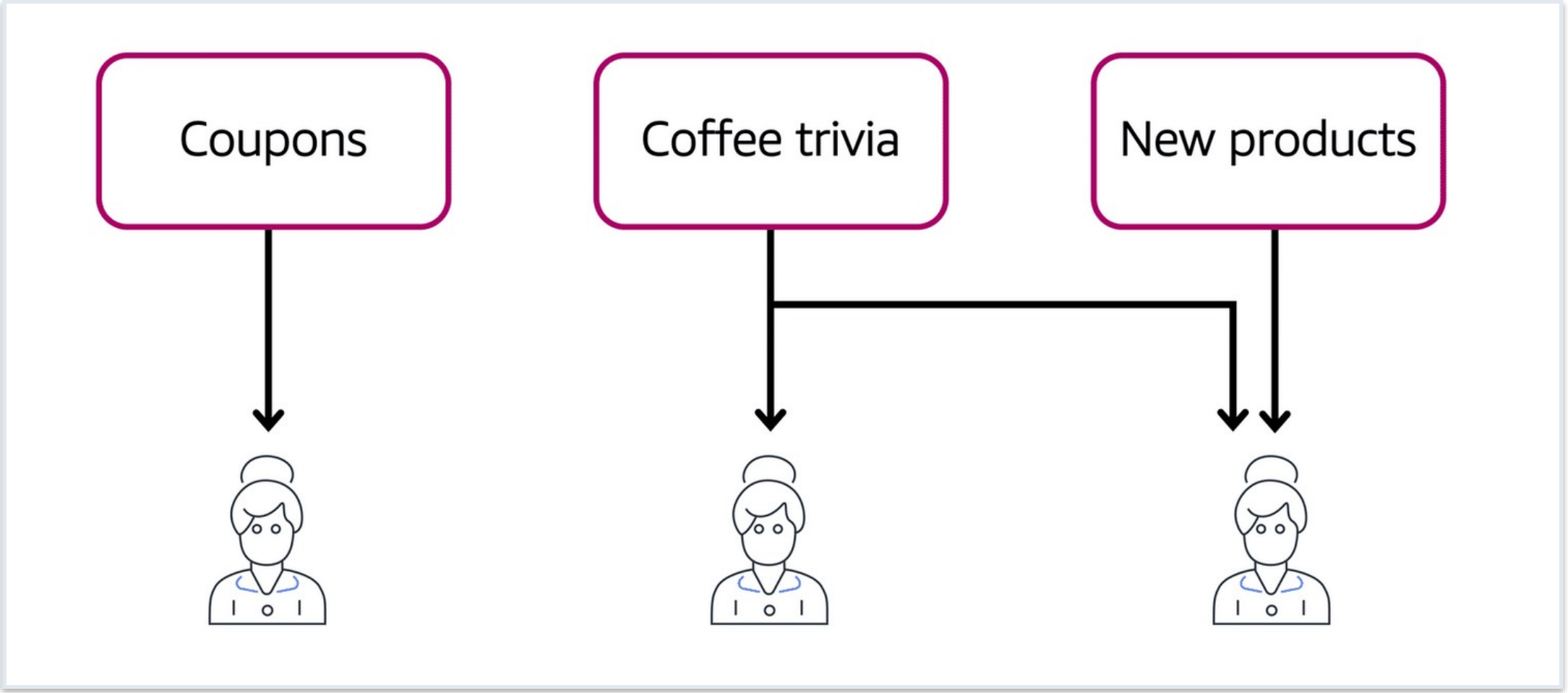
Publishing updates from multiple topics



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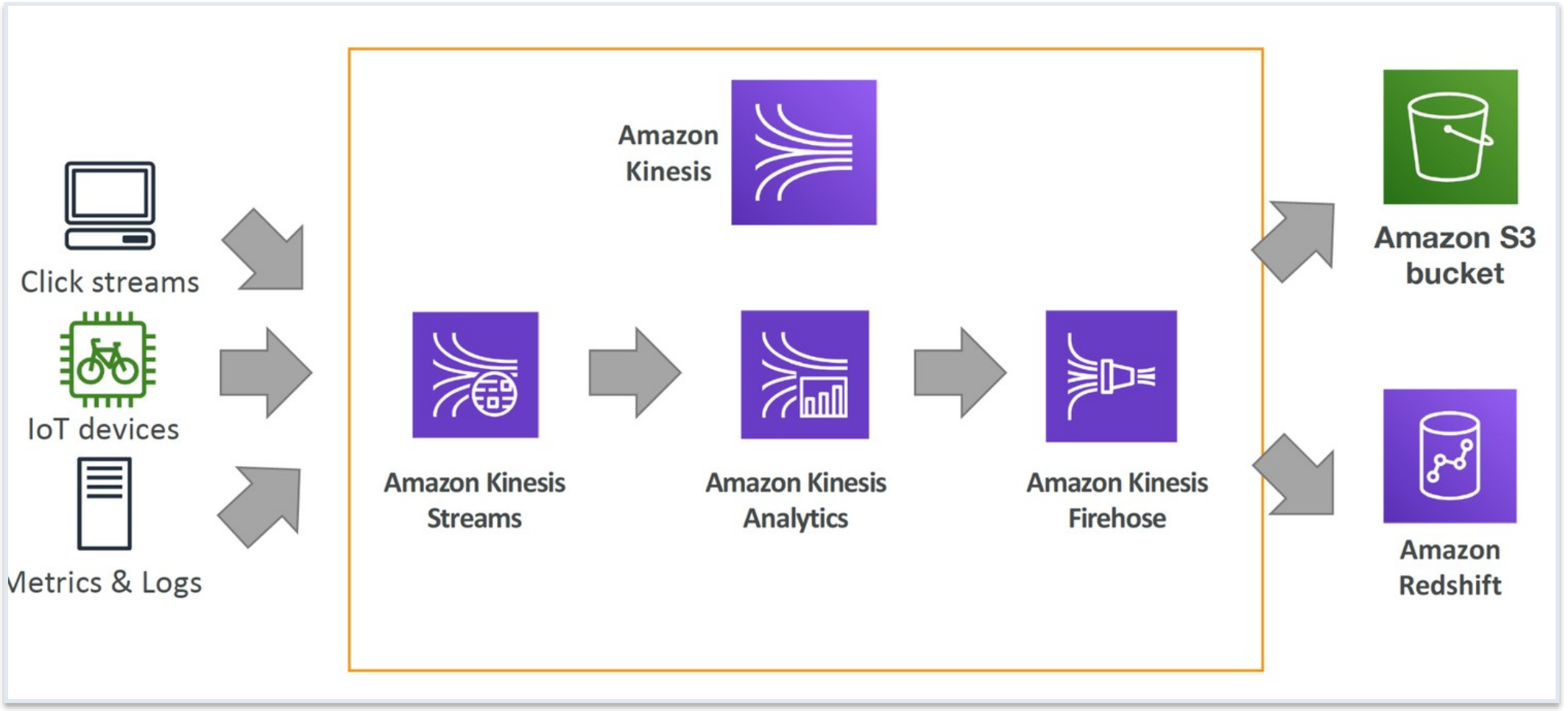
Kinesis (high level overview)



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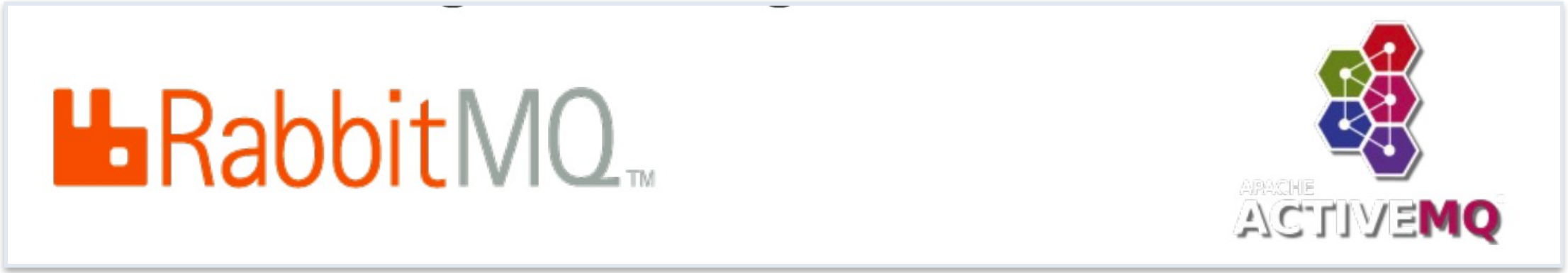
Amazon MQ



SQS, SNS are “cloud-native” services: proprietary protocols from AWS

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Integration Summary

- SQS:
- Queue service in AWS
- Multiple Producers, messages are kept up to 14 days
- Multiple Consumers share the read and delete messages when done
- Used to decouple applications in AWS
- SNS:
- Notification service in AWS
- Subscribers: Email, Lambda, SQS, HTTP, Mobile…
- Multiple Subscribers, send all messages to all of them
- No message retention
- Kinesis: real-time data streaming, persistence and analysis
- Amazon MQ: managed message broker for ActiveMQ and RabbitMQ in the cloud (MQTT, AMQP.. protocols)

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TOPIC 6

AWS Analytics Services

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CLOUD MONITORING

Cloud Monitoring

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Amazon CloudWatch Metrics

1. CloudWatch provides metrics for every services in AWS
2. Metric is a variable to monitor (CPUUtilization, NetworkIn…)
3. Metrics have timestamps
4. Can create CloudWatch dashboards of metrics

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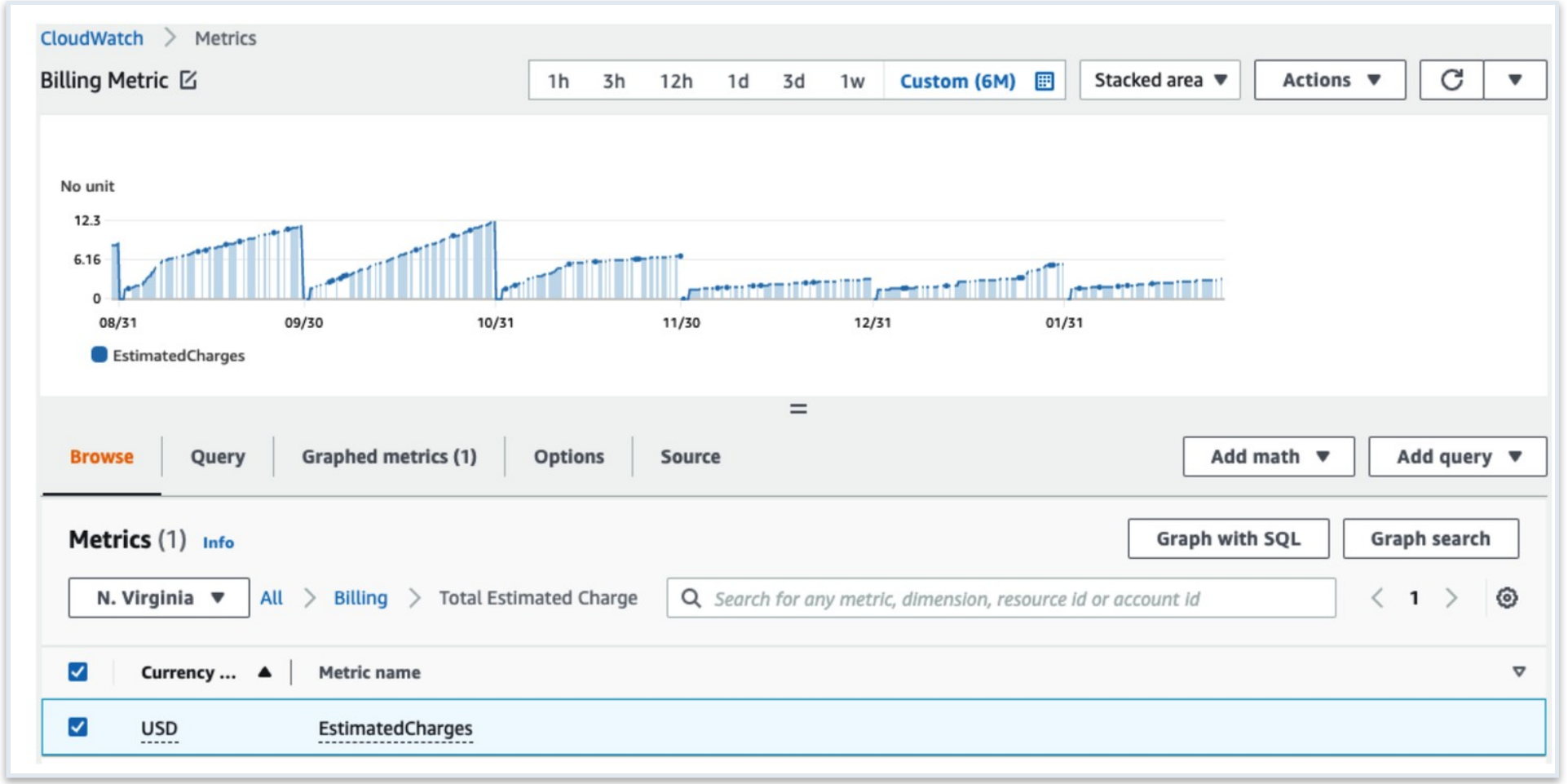
Example: CloudWatch Billing metric



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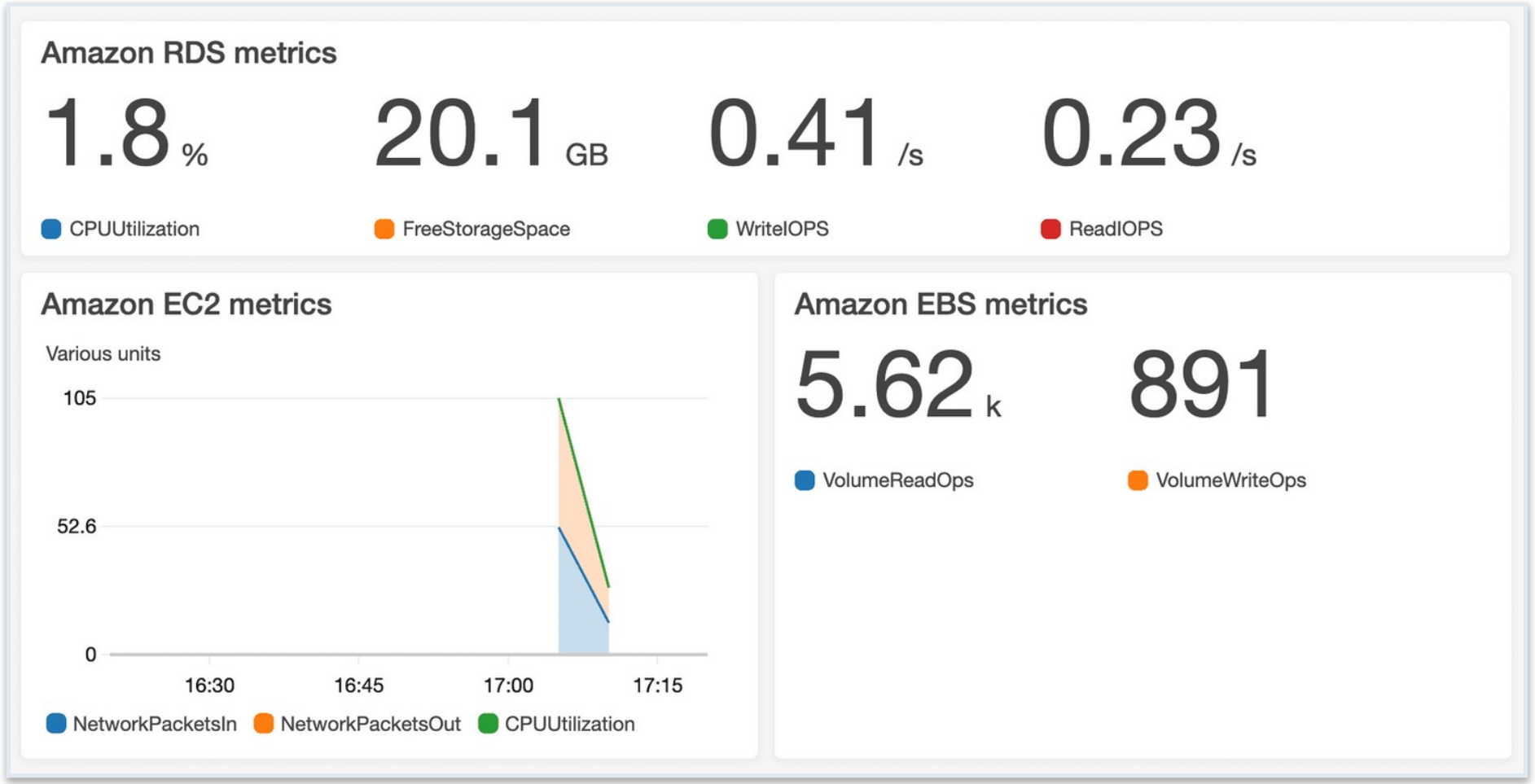
CloudWatch dashboard



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Important Metrics

- EC2 instances: CPU Utilization, Status Checks, Network (not RAM)
- Default metrics every 5 minutes
- Option for Detailed Monitoring ($$$): metrics every 1 minute
- EBS volumes: Disk Read/Writes
- S3 buckets: BucketSizeBytes, NumberOfObjects, AllRequests
- Billing: Total Estimated Charge (only in us-east-1)
- Service Limits: how much you've been using a service API
- Custom metrics: push your own metrics

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Amazon CloudWatch Alarms

- Alarms are used to trigger notifications for any metric
- Alarms actions…
- Auto Scaling: increase or decrease EC2 instances “desired” count
- EC2 Actions: stop, terminate, reboot or recover an EC2 instance
- SNS notifications: send a notification into an SNS topic
- Various options (sampling, %, max, min, etc…)
- Can choose the period on which to evaluate an alarm
- Example: create a billing alarm on the CloudWatch Billing metric
- Alarm States: OK. INSUFFICIENT_DATA, ALARM

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Amazon CloudWatch Logs

- CloudWatch Logs can collect log from:
- Elastic Beanstalk: collection of logs from application
- ECS: collection from containers
- AWS Lambda: collection from function logs
- CloudTrail based on filter
- CloudWatch log agents: on EC2 machines or on-premises servers
- Route53: Log DNS queries
- Enables real-time monitoring of logs
- Adjustable CloudWatch Logs retention

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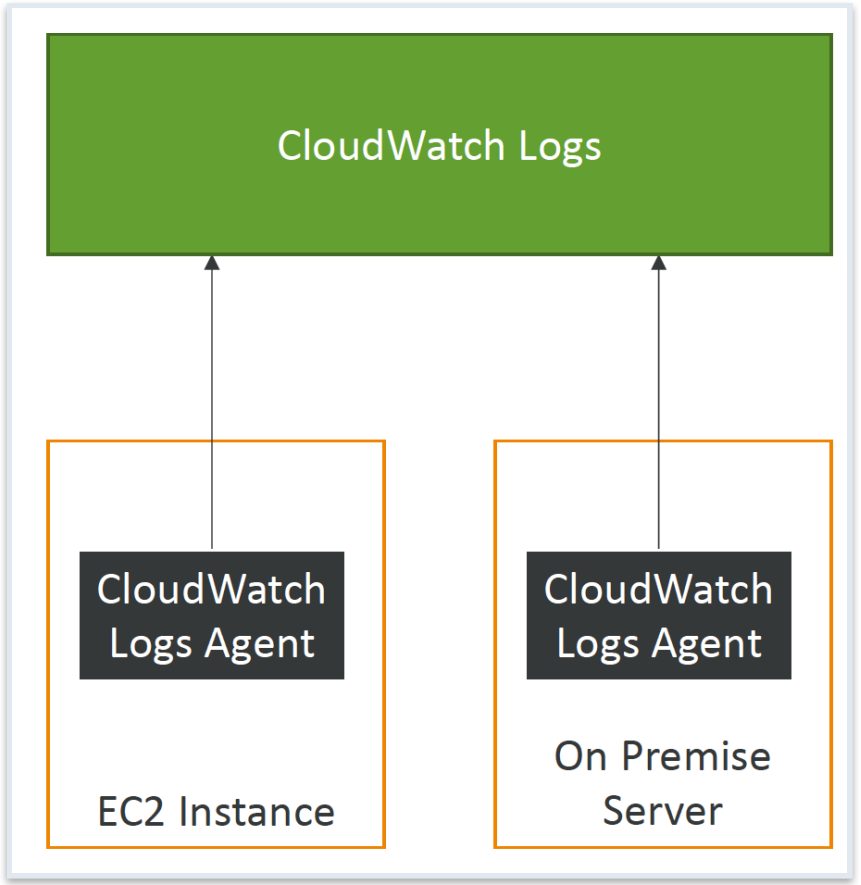
CloudWatch Logs for EC2



By default, no logs from your EC2

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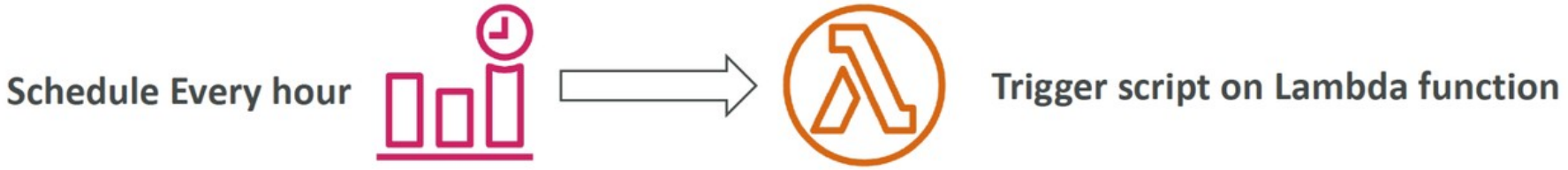
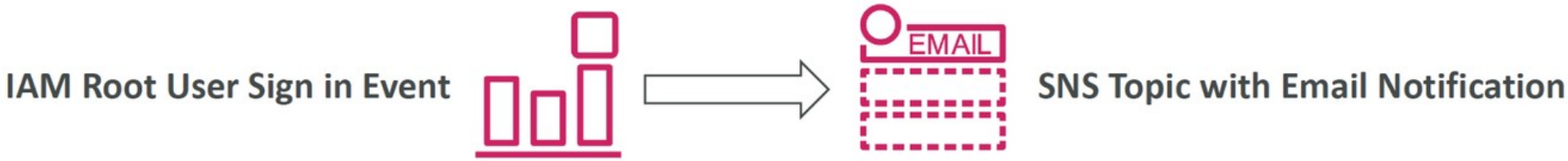
Amazon EventBridge

- Schedule: Cron jobs (scheduled scripts)



- Event Pattern: Event rules to react to a service doing something



- Trigger Lambda functions, send SQS/SNS messages…

(formerly CloudWatch Events)

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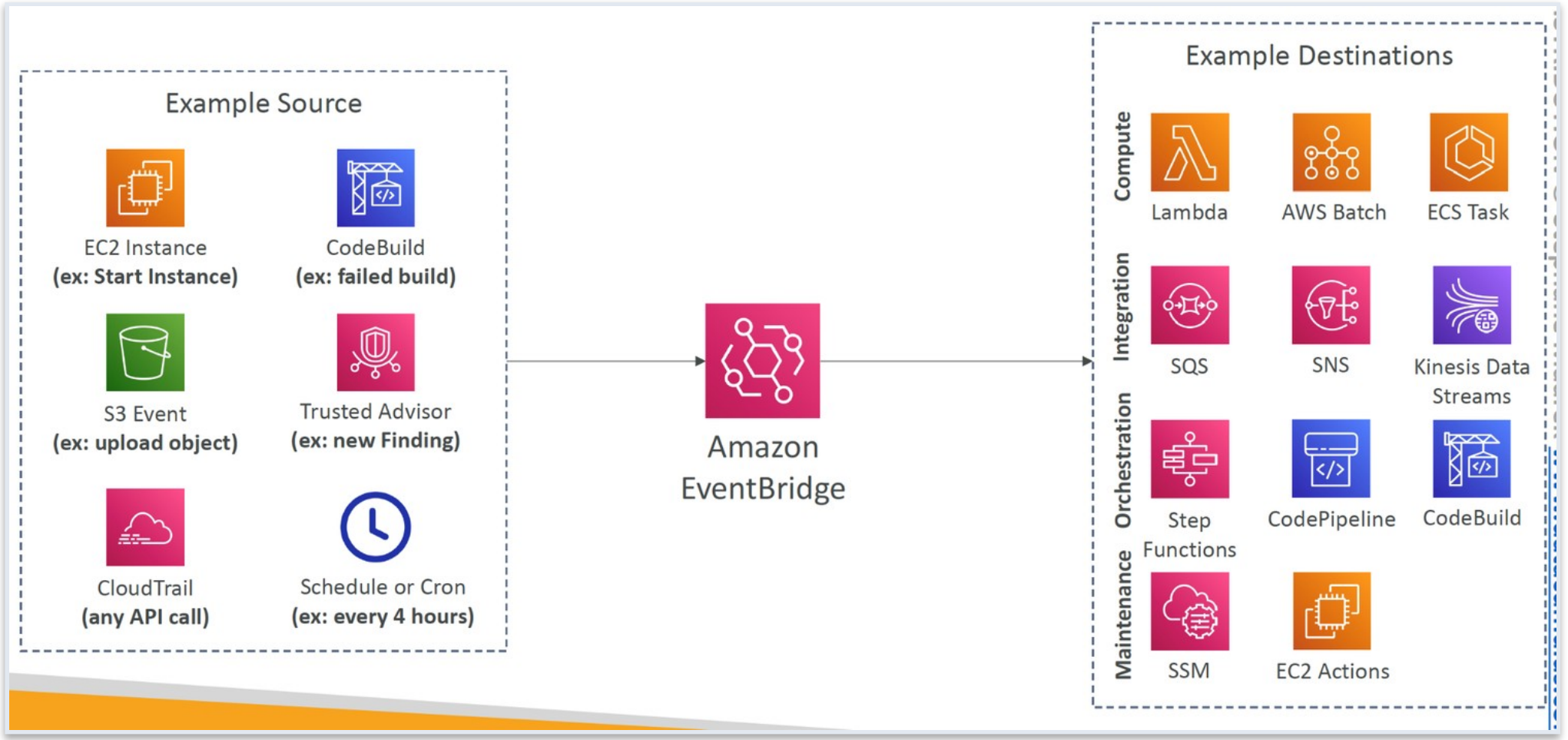
Amazon EventBridge Rules



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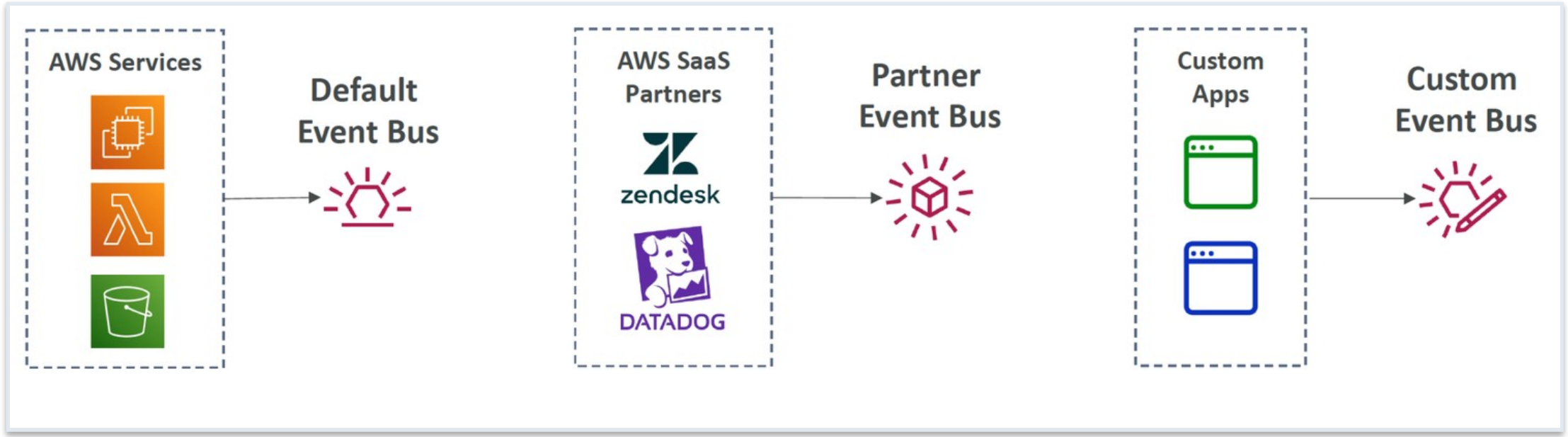
Amazon EventBridge



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AWS CloudTrail

- Provides governance, compliance and audit for your AWS Account
- CloudTrail is enabled by default!
- Get an history of events / API calls made within your AWS Account by:
- Console
- SDK
- CLI
- AWS Services
- Can put logs from CloudTrail into CloudWatch Logs or S3
- A trail can be applied to All Regions (default) or a single Region.
- If a resource is deleted in AWS, investigate CloudTrail first!

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Example: AWS CloudTrail event

What happened?	A new IAM user (Mary) was created.
Who made the request?	IAM user John
When did this occur?	January 1, 2020 at 9:00 AM
How was the request made?	Through the AWS Management Console

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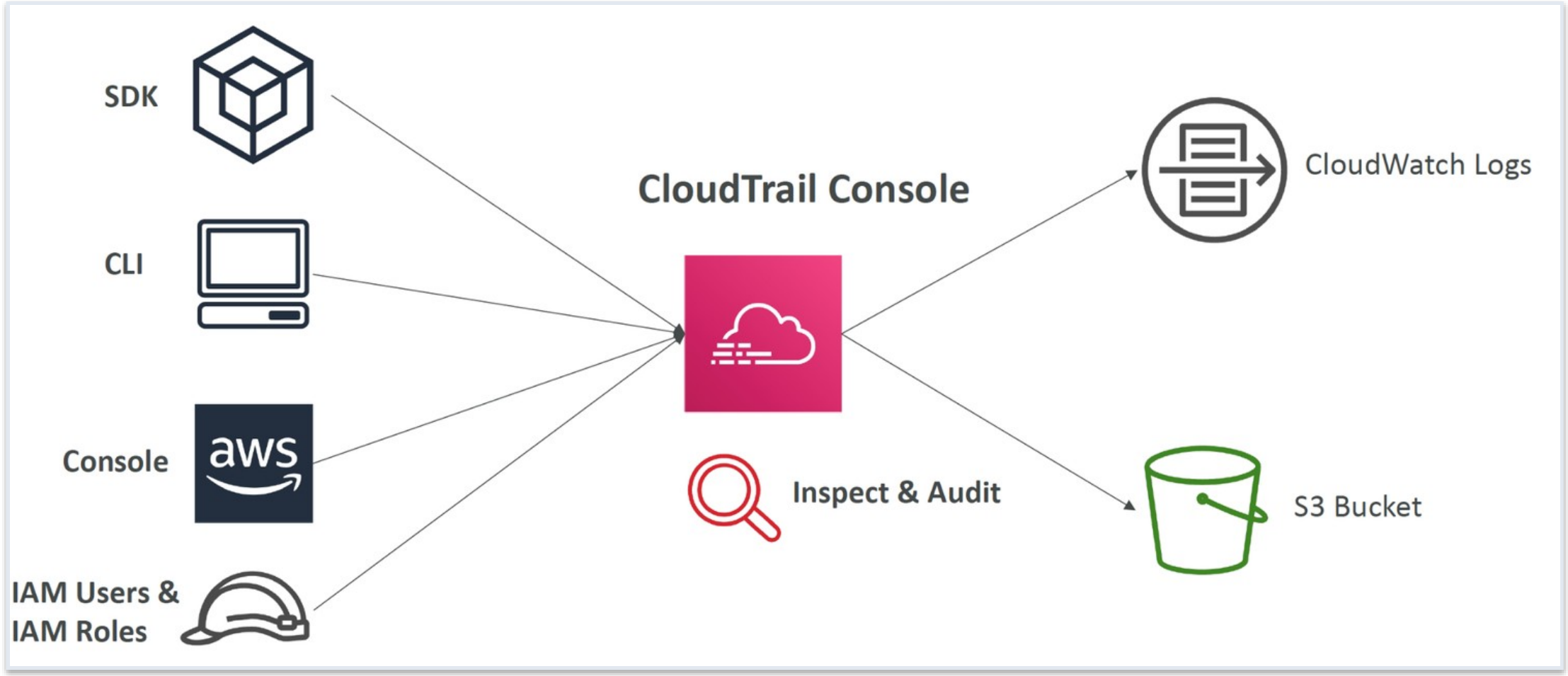
CloudTrail Diagram



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AWS X-Ray

- Debugging in Production, the good old way:
- Test locally
- Add log statements everywhere
- Re-deploy in production
- Log formats differ across applications and log analysis is hard.
- Debugging: one big monolith “easy”, distributed services “hard”
- No common views of your entire architecture
- Enter… AWS X-Ray!

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AWS X-Ray

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AWS X-Ray advantages

- Troubleshooting performance (bottlenecks)
- Understand dependencies in a microservice architecture
- Pinpoint service issues
- Review request behavior
- Find errors and exceptions
- Are we meeting time SLA?
- Where I am throttled?
- Identify users that are impacted

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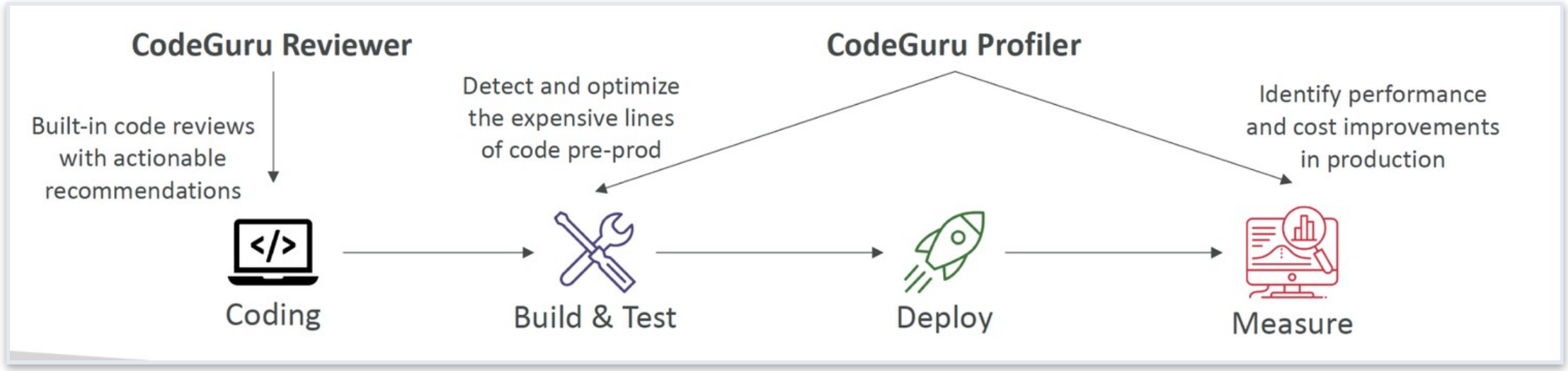
Amazon CodeGuru



An ML-powered service for automated code reviews and application performance recommendations

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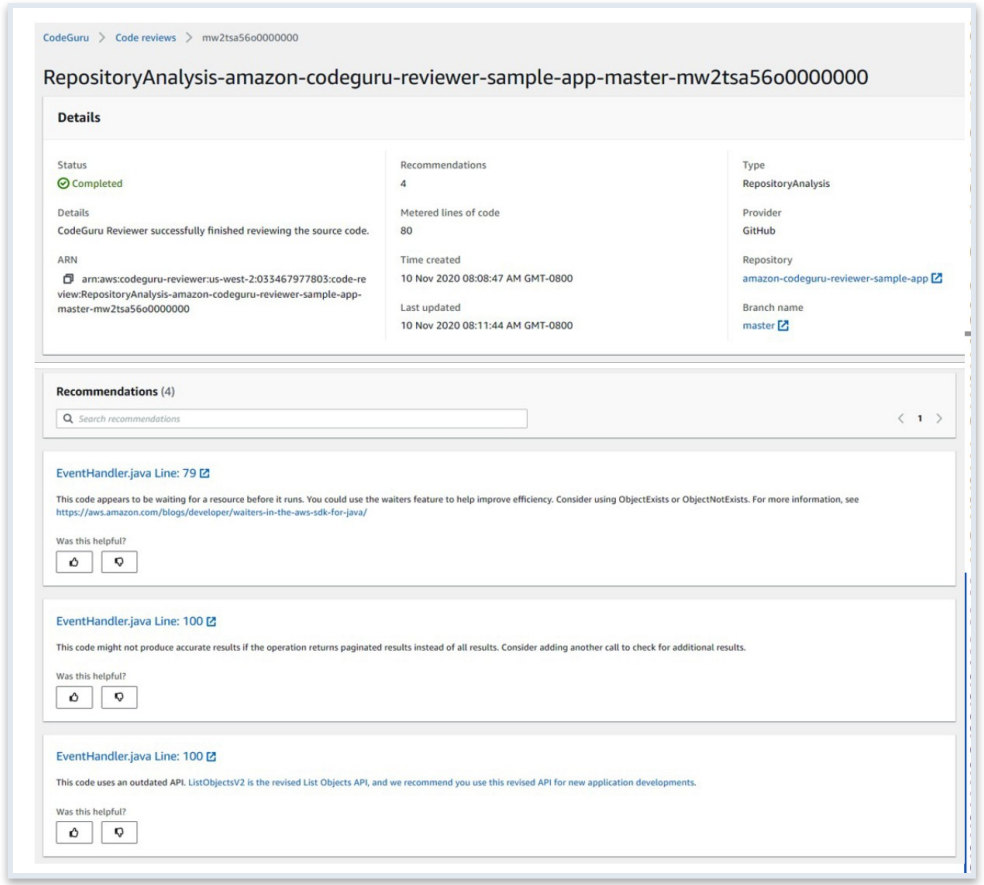
Amazon CodeGuru Reviewer



Identify critical issues, security vulnerabilities, and hard-to-find bugs

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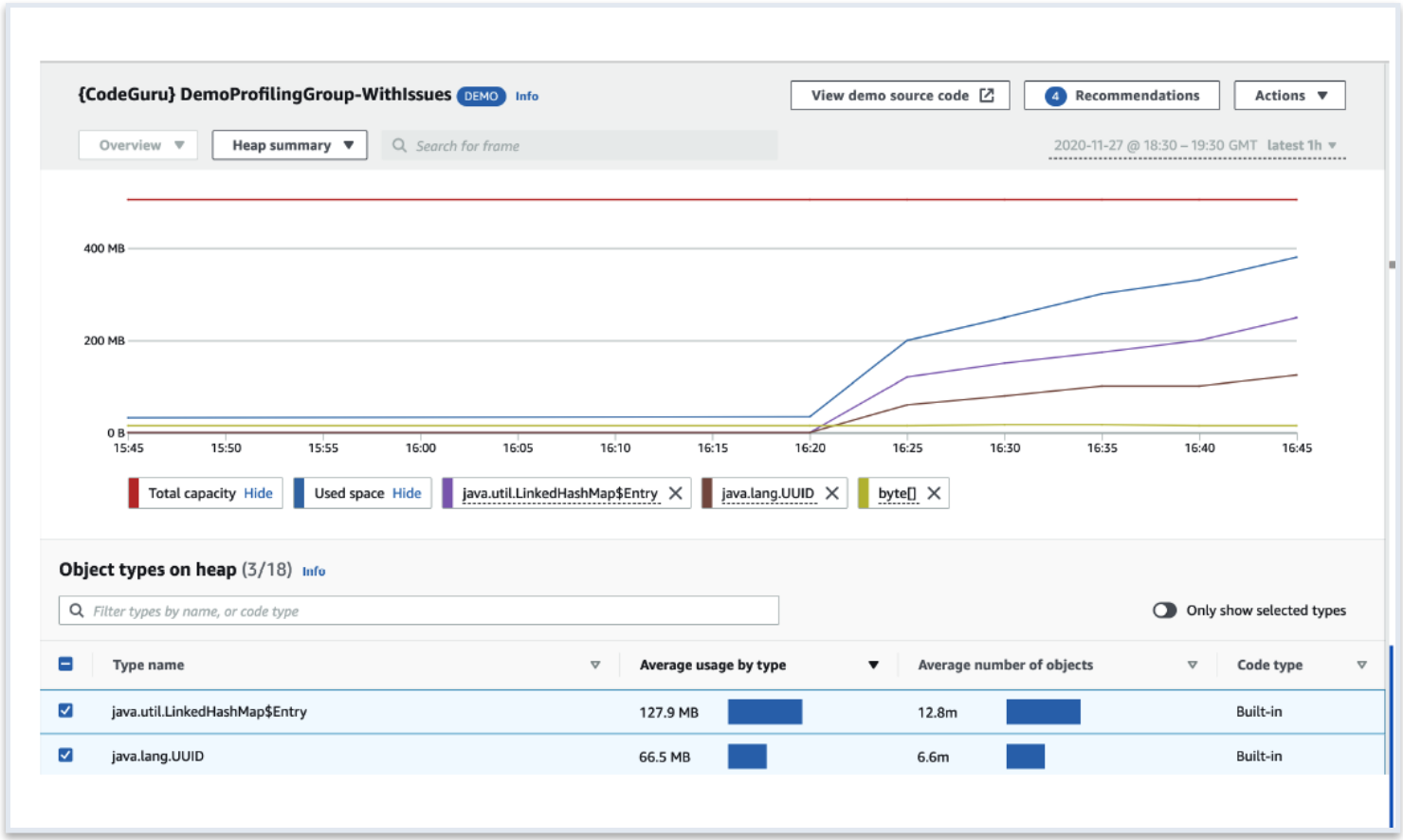
Amazon CodeGuru Profiler



Helps understand the runtime behavior of your application

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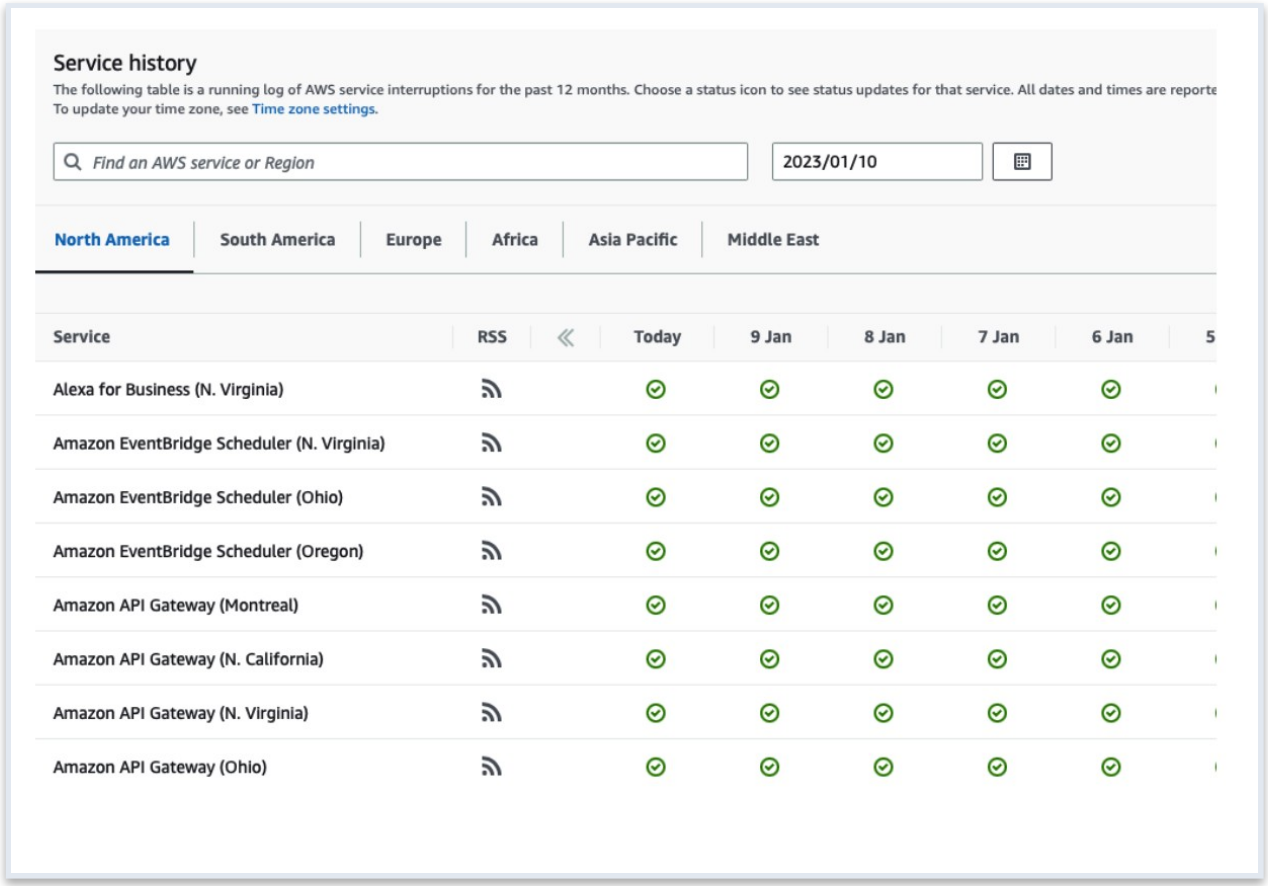
AWS Health Dashboard - Service History



Shows all regions, all services health

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AWS Health Dashboard – Your Account

- Previously called AWS Personal Health Dashboard (PHD)
- AWS Account Health Dashboard provides alerts and remediation guidance when AWS is experiencing events that may impact you.
- While the Service Health Dashboard displays the general status of AWS services, Account Health Dashboard gives you a personalized view into the performance and availability of the AWS services underlying your AWS resources.
- The dashboard displays relevant and timely information to help you manage events in progress and provides proactive notification to help you plan for scheduled activities.
- Can aggregate data from an entire AWS Organization

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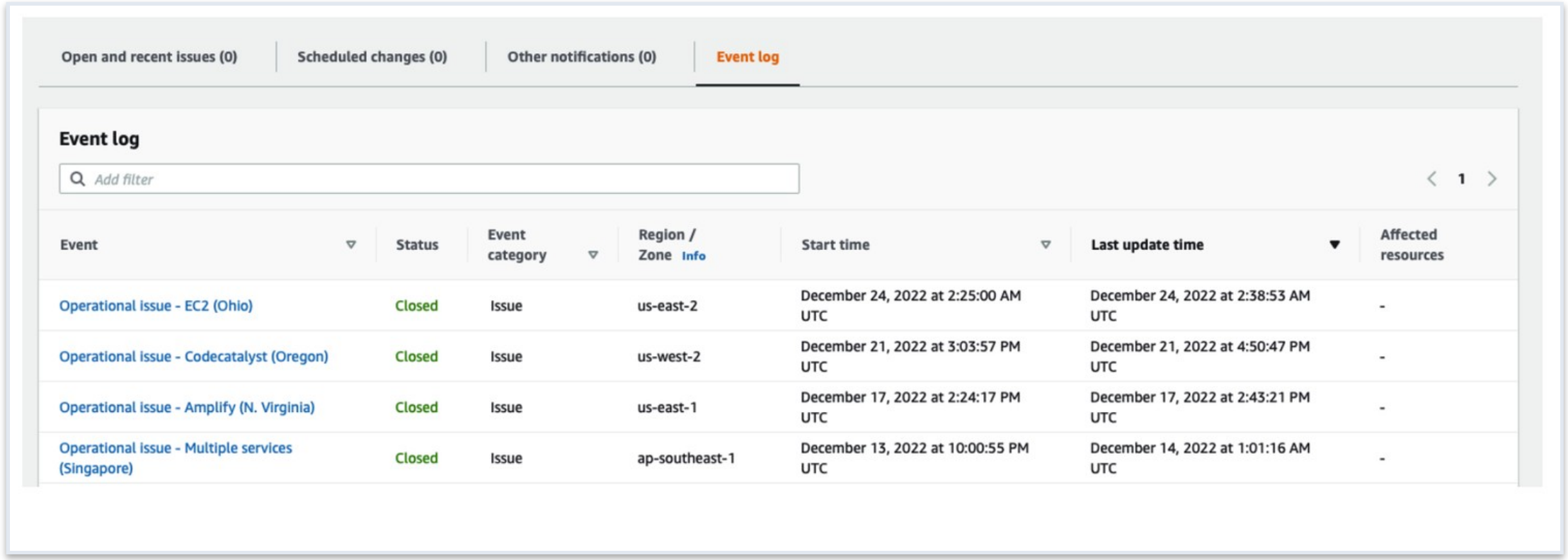
AWS Health Dashboard – Your Account



Global service

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Monitoring Summary

- CloudWatch:
- Metrics: monitor the performance of AWS services and billing metrics
- Alarms: automate notification, perform EC2 action, notify to SNS based on metric
- Logs: collect log files from EC2 instances, servers, Lambda functions…
- Events (or EventBridge): react to events in AWS, or trigger a rule on a schedule
- CloudTrail: audit API calls made within your AWS account
- CloudTrail Insights: automated analysis of your CloudTrail Events
- X-Ray: trace requests made through your distributed applications
- AWS Health Dashboard: status of all AWS services across all regions
- AWS Account Health Dashboard: AWS events that impact your infrastructure
- Amazon CodeGuru: automated code reviews and application performance recommendations

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MACHINE LEARNING

Machine Learning

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Amazon Rekognition

- Find objects, people, text, scenes in images and videos using ML
- Facial analysis and facial search to do user verification, people counting
- Create a database of “familiar faces” or compare against celebrities
- Use cases:
- Labeling
- Content Moderation
- Text Detection
- Face Detection and Analysis (gender, age range, emotions…)
- Face Search and Verification
- Celebrity Recognition
- Pathing (ex: for sports game analysis)
- https://aws.amazon.com/rekognition/

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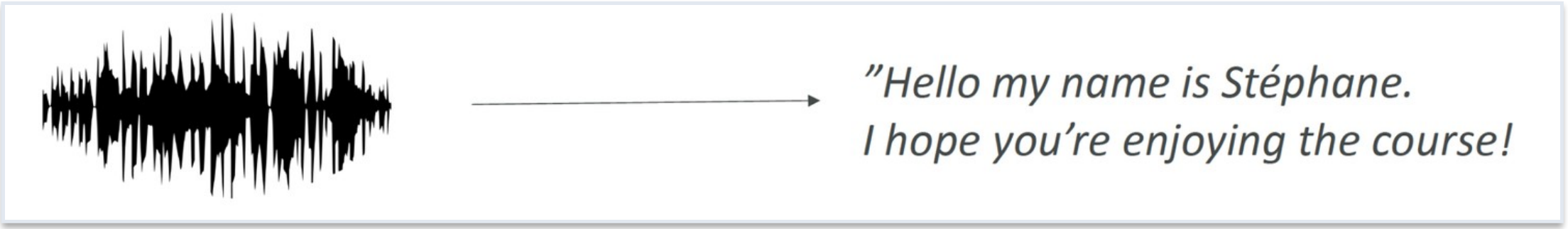
Amazon Transcribe



Automatically convert speech to text

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Amazon Polly

Turn text into lifelike speech using deep learning

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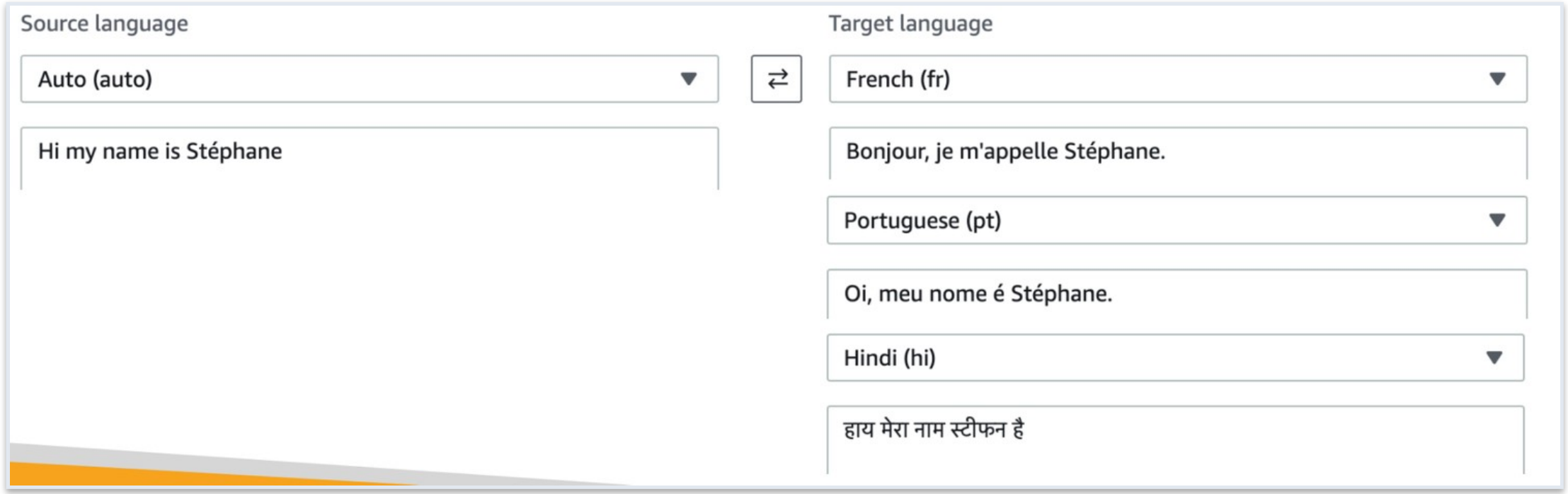
Amazon Translate



Natural and accurate language translation

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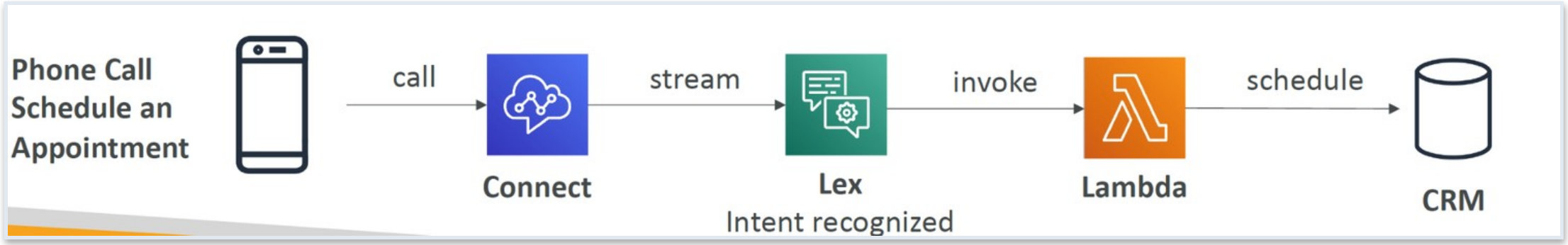
Amazon Lex & Connect



Amazon Lex: (same technology that powers Alexa)

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Amazon Comprehend

- For Natural Language Processing – NLP
- Fully managed and serverless service
- Uses machine learning to find insights and relationships in text
- Language of the text
- Extracts key phrases, places, people, brands, or events
- Understands how positive or negative the text is
- Analyzes text using tokenization and parts of speech
- Automatically organizes a collection of text files by topic
- Sample use cases:
- analyze customer interactions (emails) to find what leads to a positive or negative experience
- Create and groups articles by topics that Comprehend will uncover

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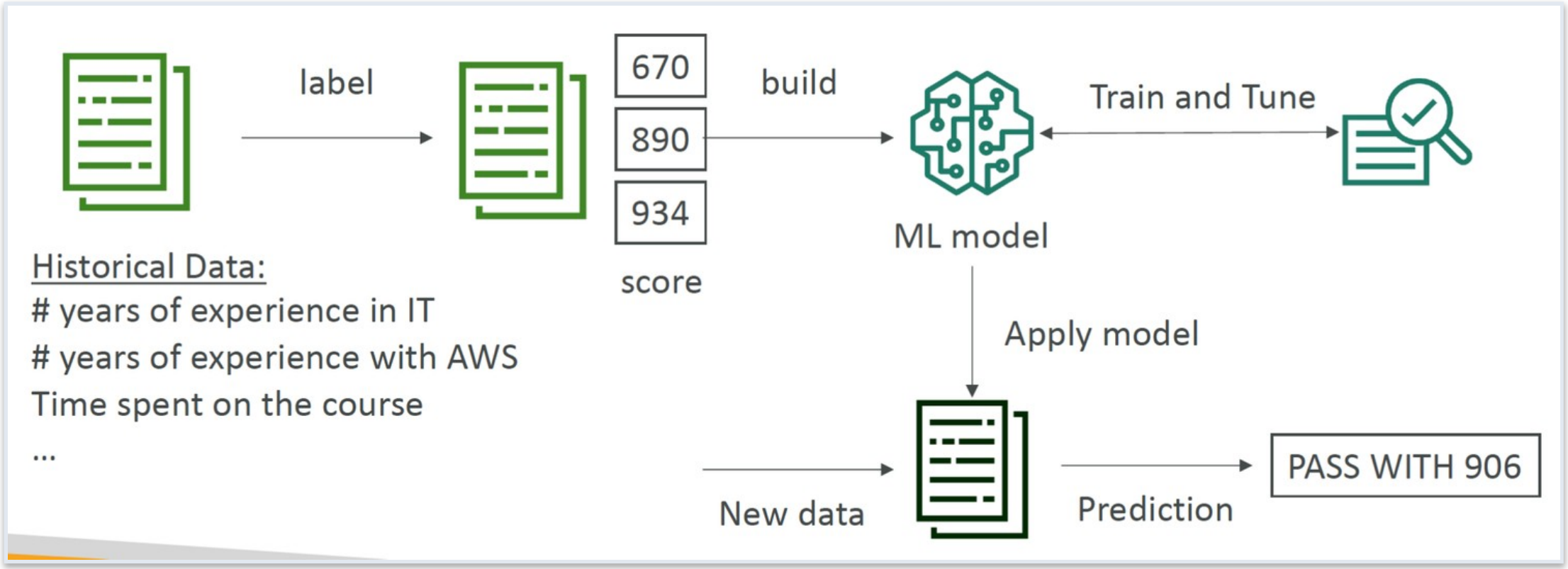
Amazon SageMaker



Fully managed service for developers / data scientists to build ML models

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Amazon Forecast

Fully managed service that uses ML to deliver highly accurate forecasts

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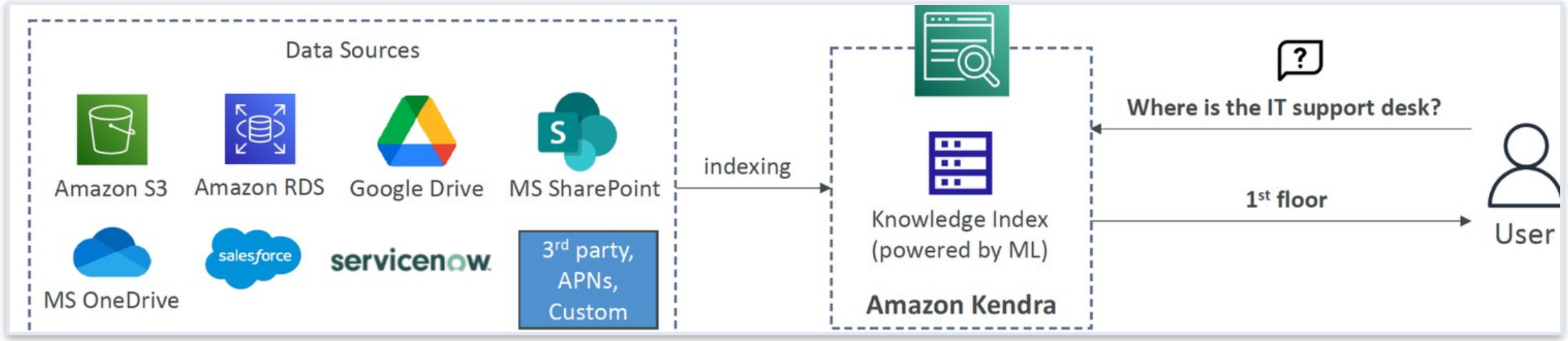
Amazon Kendra



Fully managed document search service powered by Machine Learning

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Amazon Personalize

Fully managed ML-service to build apps with real-time personalized recommendations

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Amazon Textract

Automatically extracts text, handwriting, and data from any scanned documents using AI and ML

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Machine Learning Summary

- Rekognition: face detection, labeling, celebrity recognition
- Transcribe: audio to text (ex: subtitles)
- Polly: text to audio
- Translate: translations
- Lex: build conversational bots – chatbots
- Connect: cloud contact center
- Comprehend: natural language processing
- SageMaker: machine learning for every developer and data scientist
- Forecast: build highly accurate forecasts
- Kendra: ML-powered search engine
- Personalize: real-time personalized recommendations
- Textract: detect text and data in documents

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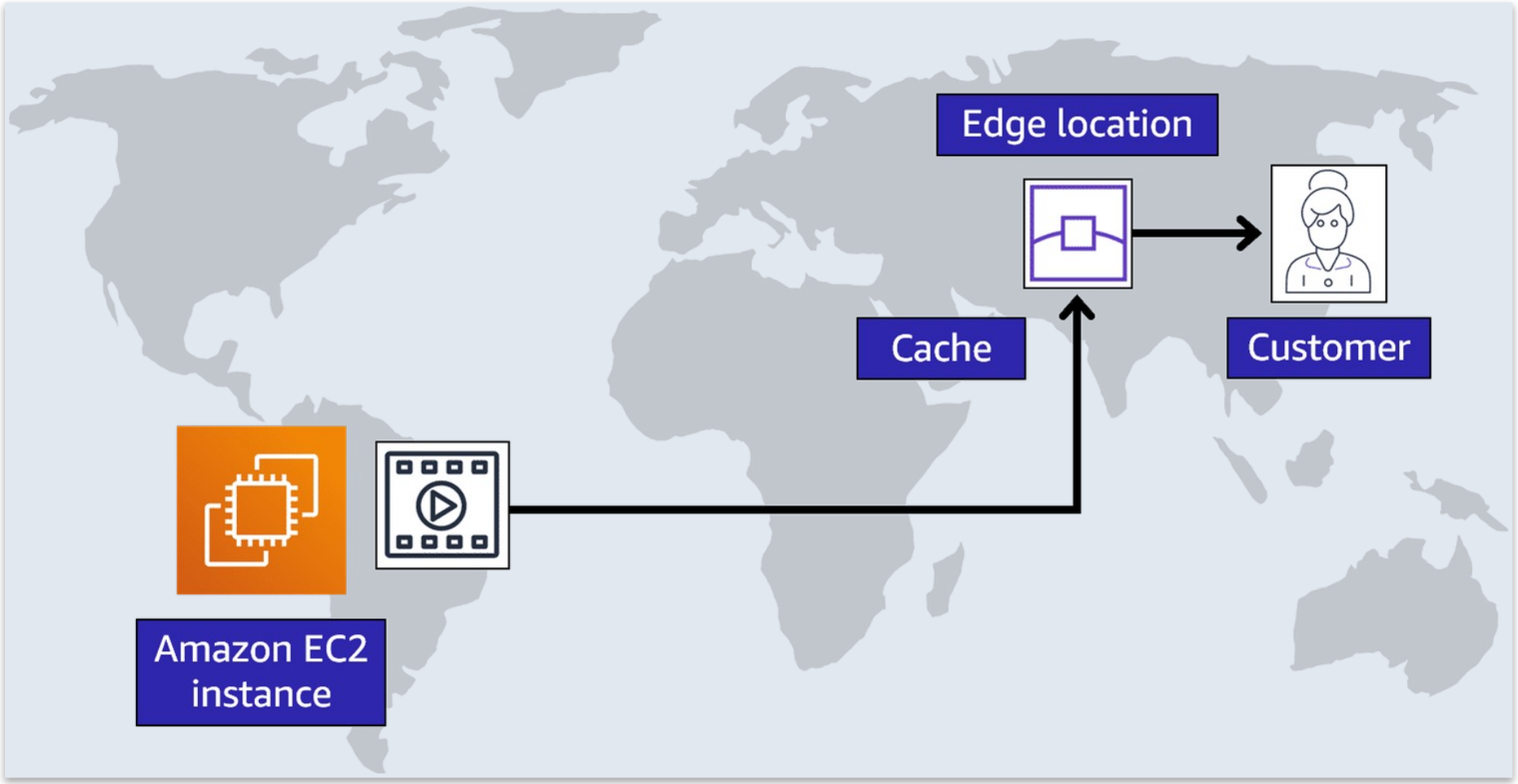
Edge locations



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Support

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TOPIC 7

Security & Compliance

- Security & Compliance

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TOPIC 7

AWS Shared Responsibility Model

- AWS responsibility - Security of the Cloud
- Protecting infrastructure (hardware, software, facilities, and networking) that runs
- all the AWS services
- Managed services like S3, DynamoDB, RDS, etc.
- Customer responsibility - Security in the Cloud
- For EC2 instance, customer is responsible for management of the guest OS
- (including security patches and updates), firewall & network configuration, IAM
- Encrypting application data
- Shared controls:
- Patch Management, Configuration Management, Awareness & Training

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Example, for RDS

- AWS responsibility:
- Manage the underlying EC2 instance, disable SSH access
- Automated DB patching
- Automated OS patching
- Audit the underlying instance and disks & guarantee it functions
- Your responsibility:
- Check the ports / IP / security group inbound rules in DB’s SG
- In-database user creation and permissions
- Creating a database with or without public access
- Ensure parameter groups or DB is configured to only allow SSL connections
- Database encryption setting

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Example, for S3

- AWS responsibility:
- Guarantee you get unlimited storage
- Guarantee you get encryption
- Ensure separation of the data between different customers
- Ensure AWS employees can’t access your data
- Your responsibility:
- Bucket configuration
- Bucket policy / public setting
- IAM user and roles
- Enabling encryption

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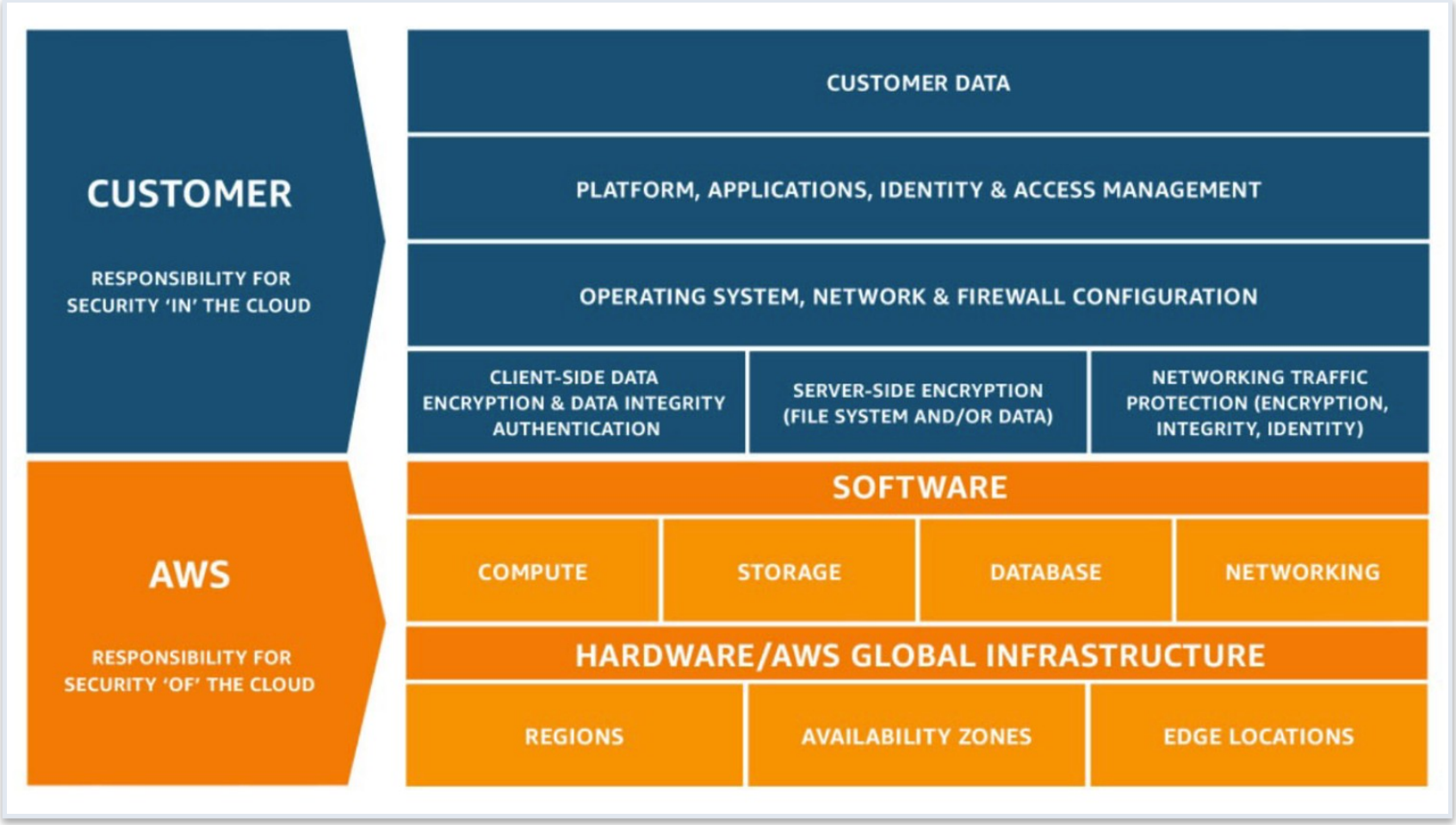
Shared Responsibility Model diagram



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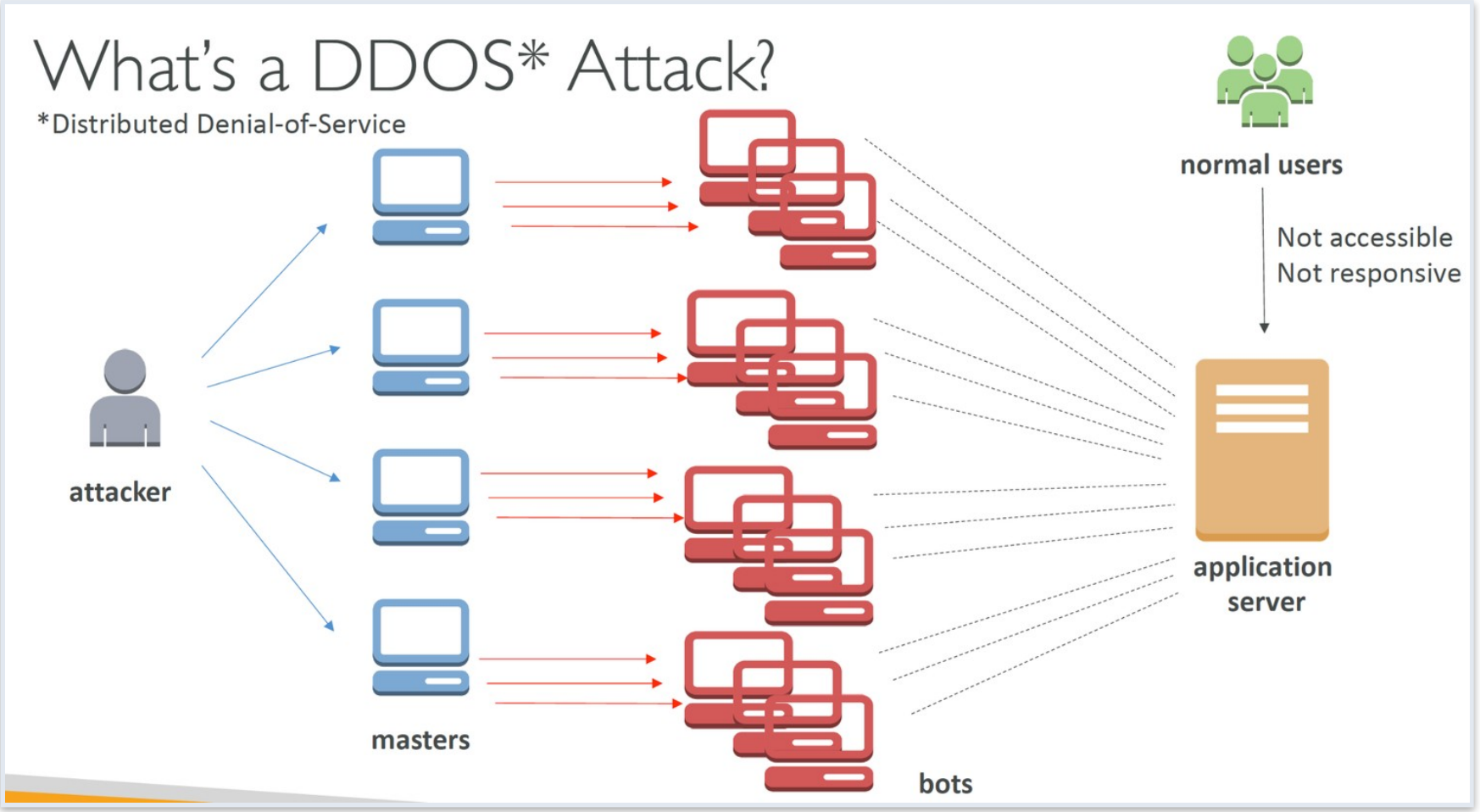
Reference



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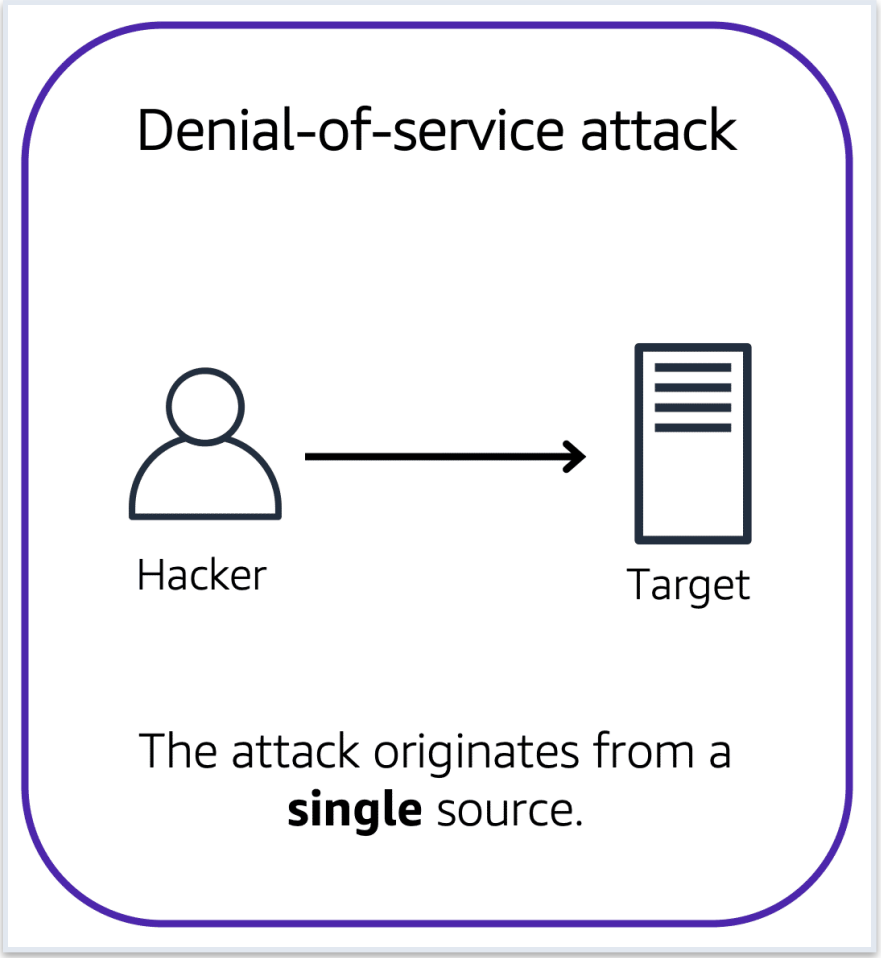
DDoS Protection on AWS



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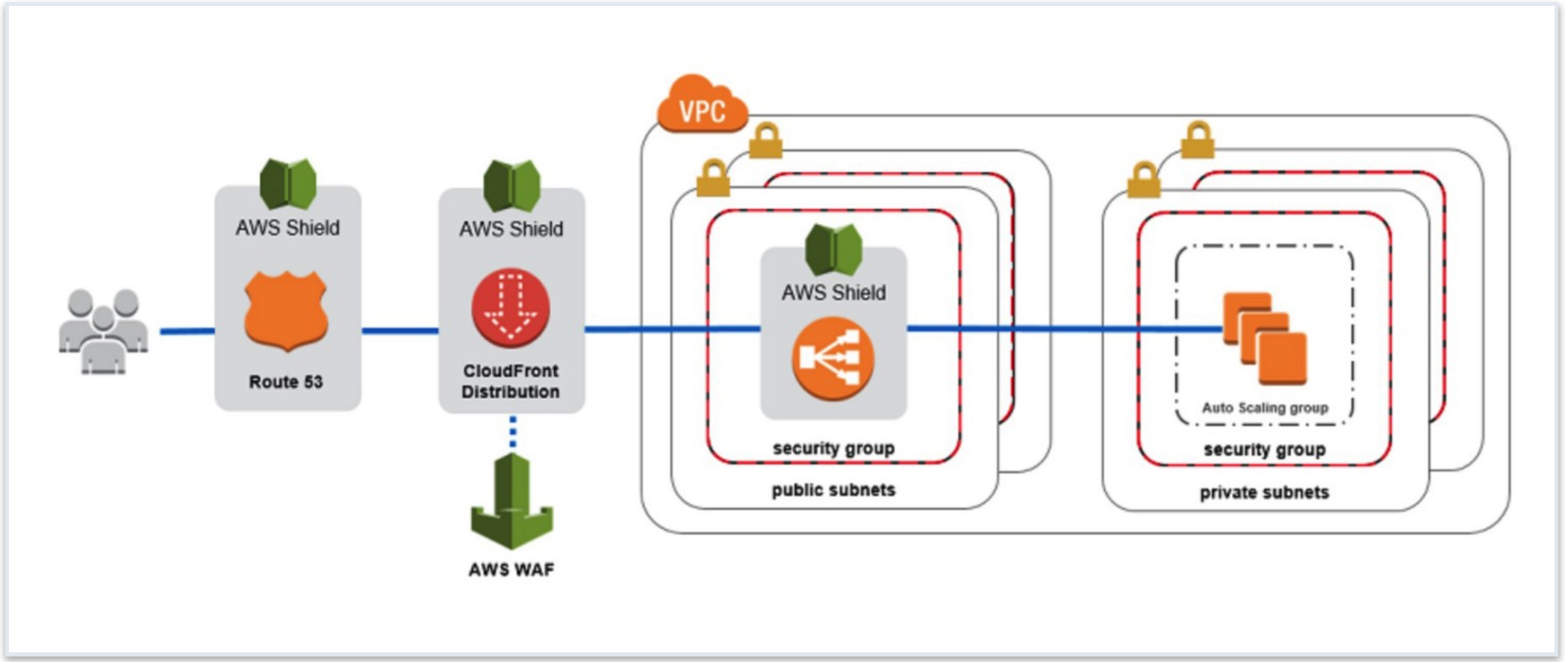
Sample Reference Architecture



for DDoS Protection

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AWS Shield

- AWS Shield Standard:
- Free service that is activated for every AWS customer
- Provides protection from attacks such as SYN/UDP Floods, Reflection attacks and other layer 3/layer 4 attacks
- AWS Shield Advanced:
- Optional DDoS mitigation service ($3,000 per month per organization)
- Protect against more sophisticated attack on Amazon EC2, Elastic Load
- Balancing (ELB), Amazon CloudFront, AWS Global Accelerator, and Route 53
- 24/7 access to AWS DDoS response team (DRP)
- Protect against higher fees during usage spikes due to DDoS

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AWS WAF – Web Application Firewall

- Protects your web applications from common web exploits (Layer 7)
- Layer 7 is HTTP (vs Layer 4 is TCP)
- Deploy on Application Load Balancer, API Gateway, CloudFront
- Define Web ACL (Web Access Control List):
- Rules can include IP addresses, HTTP headers, HTTP body, or URI strings
- Protects from common attack - SQL injection and Cross-Site Scripting (XSS)
- Size constraints, geo-match (block countries)
- Rate-based rules (to count occurrences of events) – for DDoS protection

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AWS WAF

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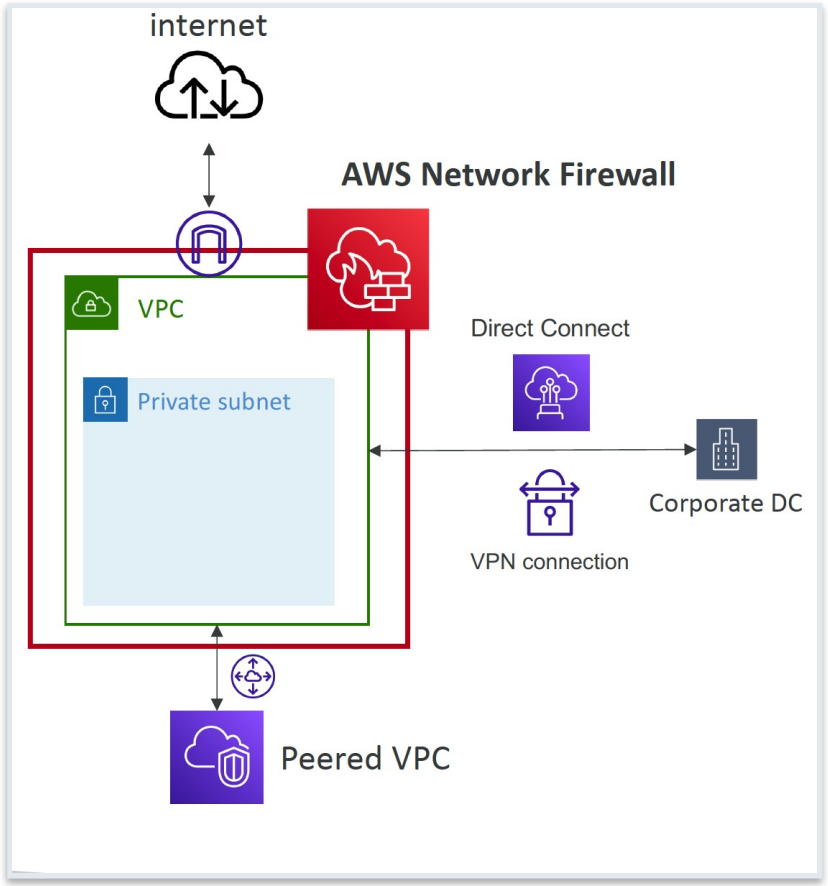
AWS Network Firewall



Protect your entire Amazon VPC

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AWS Firewall Manager

- Manage security rules in all accounts of an AWS Organization
- Security policy: common set of security rules
- VPC Security Groups for EC2, Application Load Balancer, etc…
- WAF rules
- AWS Shield Advanced
- AWS Network Firewall
- Rules are applied to new resources as they are created (good for compliance) across all and future accounts in your Organization

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Penetration Testing on AWS Cloud

- AWS customers are welcome to carry out security assessments o penetration tests against their AWS infrastructure without prior approval for 8 services:
- Amazon EC2 instances, NAT Gateways, and Elastic Load Balancers
- Amazon RDS
- Amazon CloudFront
- Amazon Aurora
- Amazon API Gateways
- AWS Lambda and Lambda Edge functions
- Amazon Lightsail resources
- Amazon Elastic Beanstalk environments

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Penetration Testing on AWS Cloud

- Prohibited Activities
- DNS zone walking via Amazon Route 53 Hosted Zones
- Denial of Service (DoS), Distributed Denial of Service (DDoS), Simulated DoS, Simulated DDoS
- Port flooding
- Protocol flooding
- Request flooding (login request flooding, API request flooding)
- For any other simulated events, contact aws-security-simulatedevent@amazon.com
- Read more: https://aws.amazon.com/security/penetration-testing/

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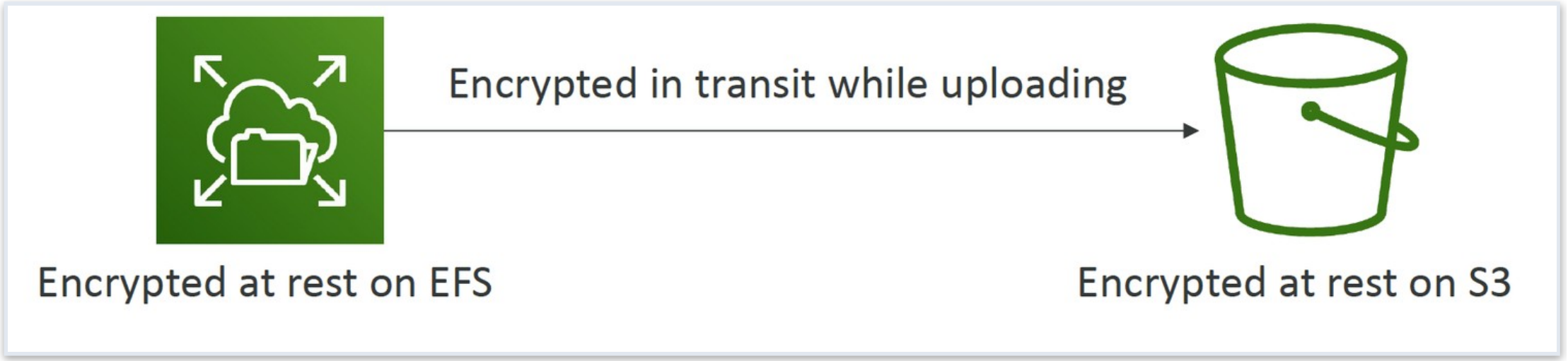
Data at rest vs. Data in transit



At rest: data stored or archived on a device

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AWS KMS (Key Management Service)

- Anytime you hear “encryption” for an AWS service, it’s most likely KMS
- KMS = AWS manages the encryption keys for us
- Encryption Opt-in:
- EBS volumes: encrypt volumes
- S3 buckets: Server-side encryption of objects
- Redshift database: encryption of data
- RDS database: encryption of data
- EFS drives: encryption of data
- Encryption Automatically enabled:
- CloudTrail Logs
- S3 Glacier
- Storage Gateway

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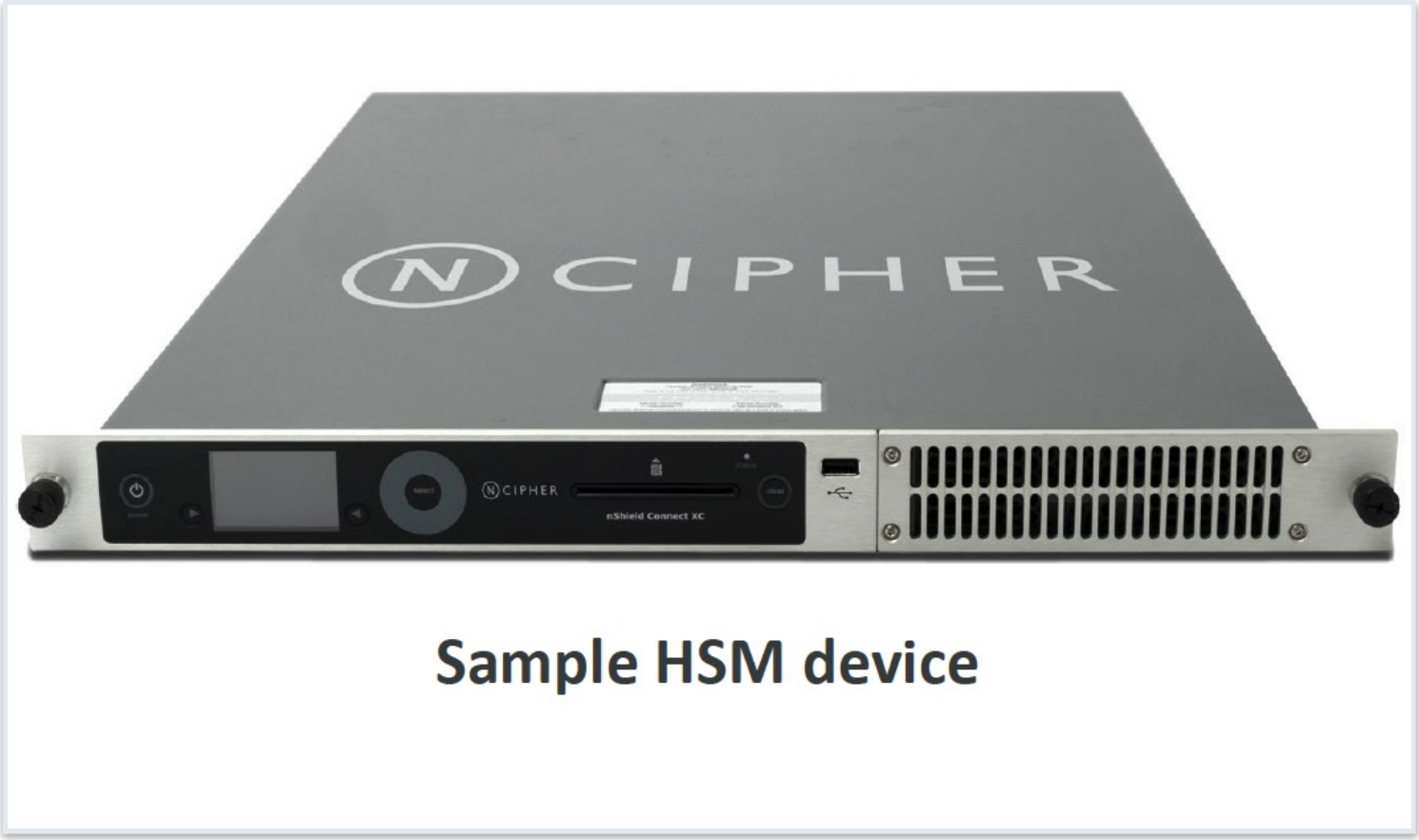
CloudHSM



Sample HSM device

KMS => AWS manages the software for encryption

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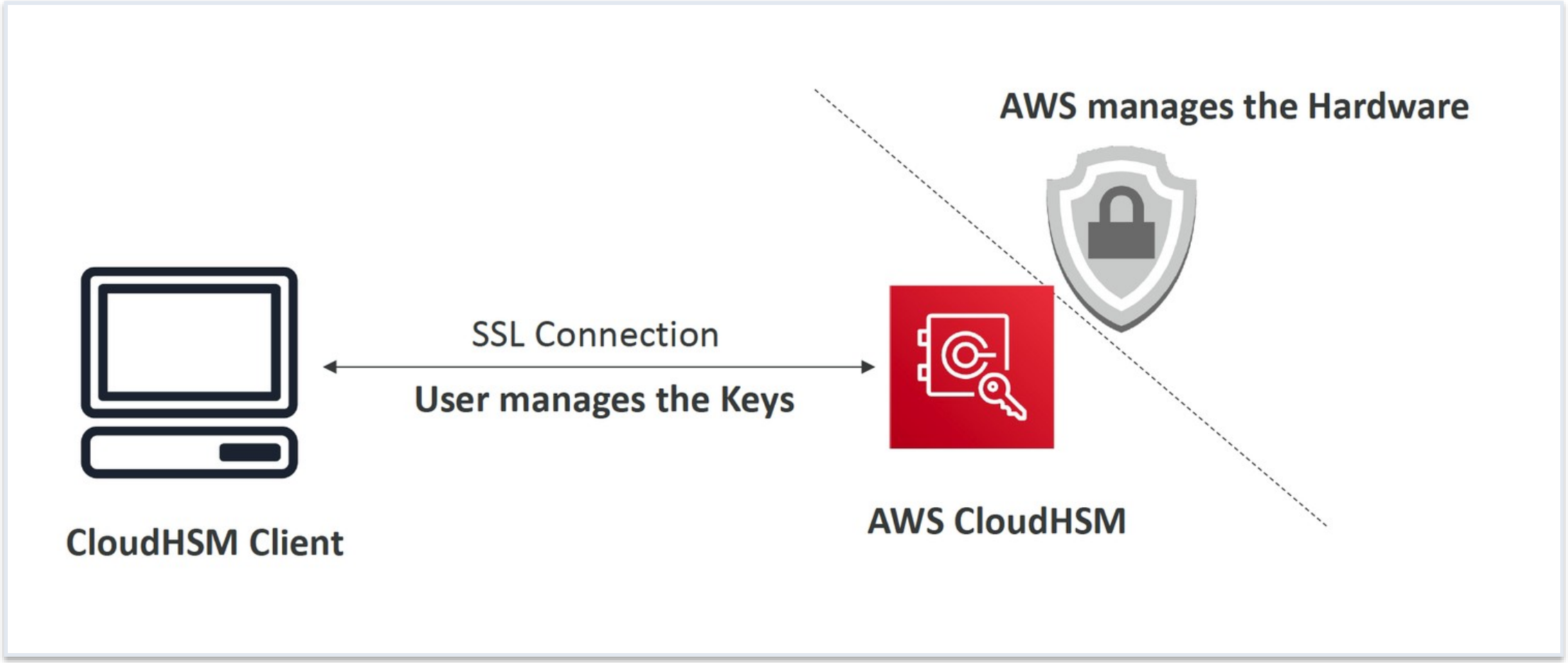
CloudHSM Diagram



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Types of KMS Keys

- Customer Managed Key:
- Create, manage and used by the customer, can enable or disable
- Possibility of rotation policy (new key generated every year, old key preserved)
- Possibility to bring-your-own-key
- AWS Managed Key:
- Created, managed and used on the customer’s behalf by AWS
- Used by AWS services (aws/s3, aws/ebs, aws/redshift)
- AWS Owned Key:
- Collection of CMKs that an AWS service owns and manages to use in multiple accounts
- AWS can use those to protect resources in your account (but you can’t view the keys)
- CloudHSM Keys (custom keystore):
- Keys generated from your own CloudHSM hardware device
- Cryptographic operations are performed within the CloudHSM cluster

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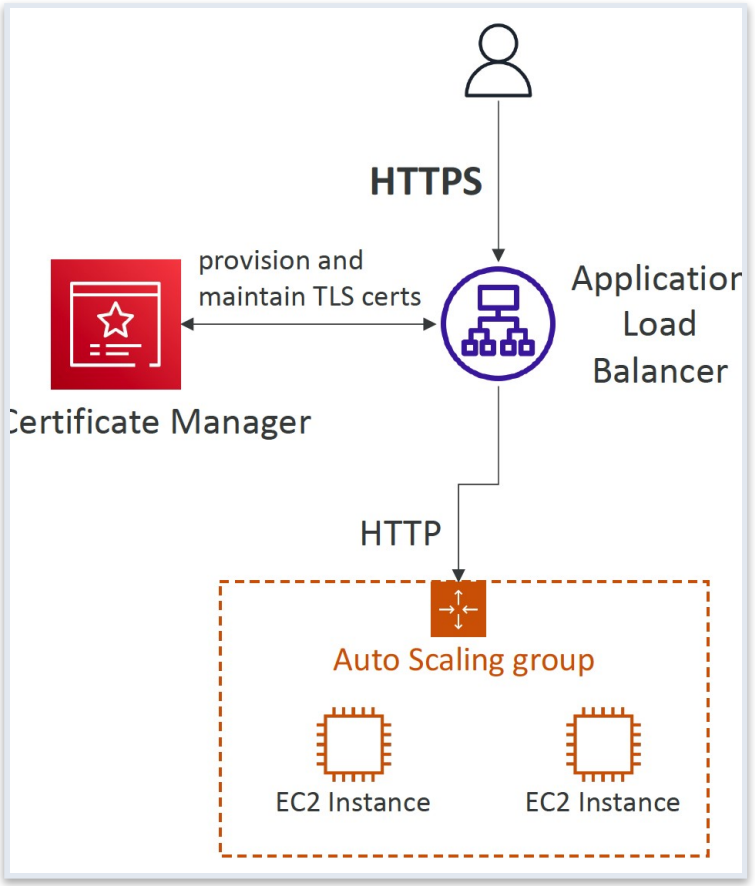
AWS Certificate Manager (ACM)



Let's you easily provision, manage, and deploy SSL/TLS Certificates

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AWS Secrets Manager

1. Newer service, meant for storing secrets
2. Capability to force rotation of secrets every X days
3. Automate generation of secrets on rotation (uses Lambda)
4. Integration with Amazon RDS (MySQL, PostgreSQL, Aurora)
5. Secrets are encrypted using KMS
6. Mostly meant for RDS integration

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AWS Artifact

- Portal that provides customers with on-demand access to AWS compliance documentation and AWS agreements
- Artifact Reports - Allows you to download AWS security and compliance documents from third-party auditors, like AWS ISO certifications, Payment
- Card Industry (PCI), and System and Organization Control (SOC) reports
- Artifact Agreements - Allows you to review, accept, and track the status of AWS agreements such as the Business Associate Addendum (BAA) or the
- Health Insurance Portability and Accountability Act (HIPAA) for an individual account or in your organization
- Can be used to support internal audit or compliance

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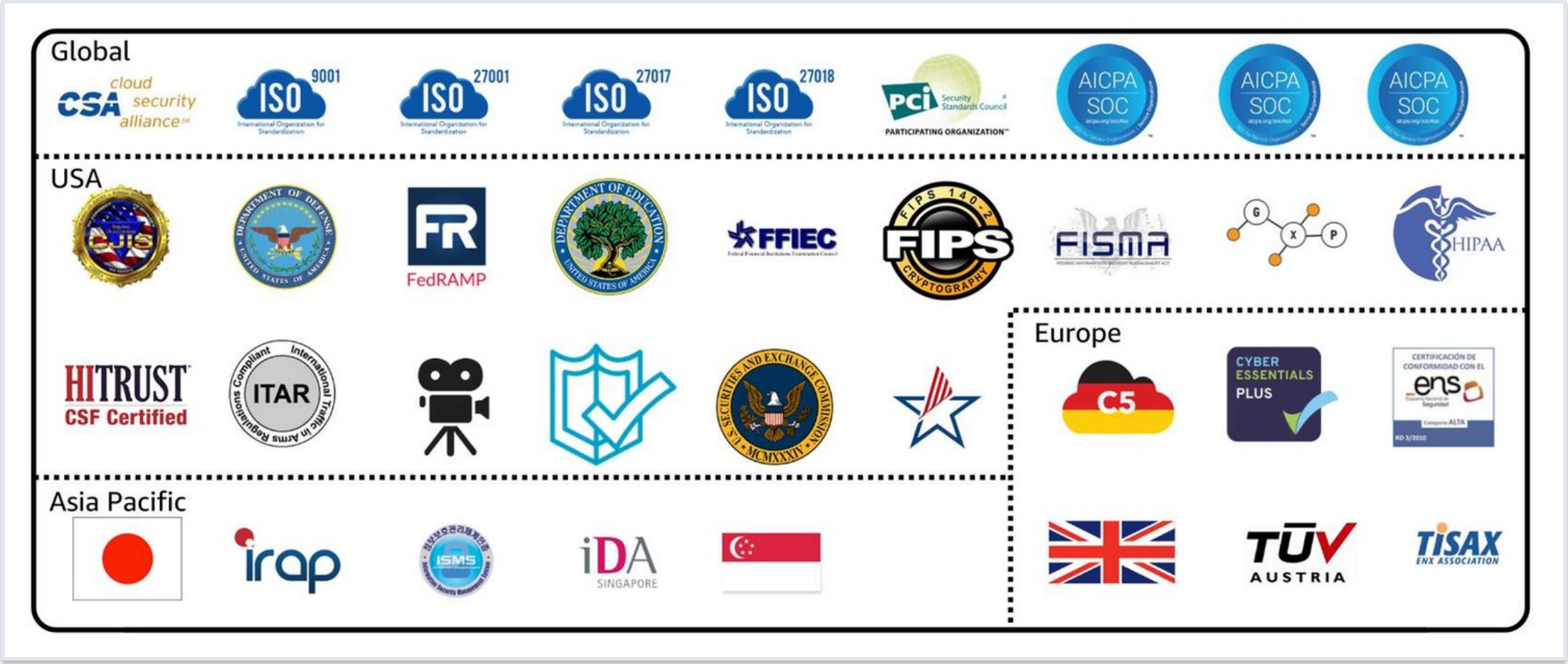
AWS Artifact



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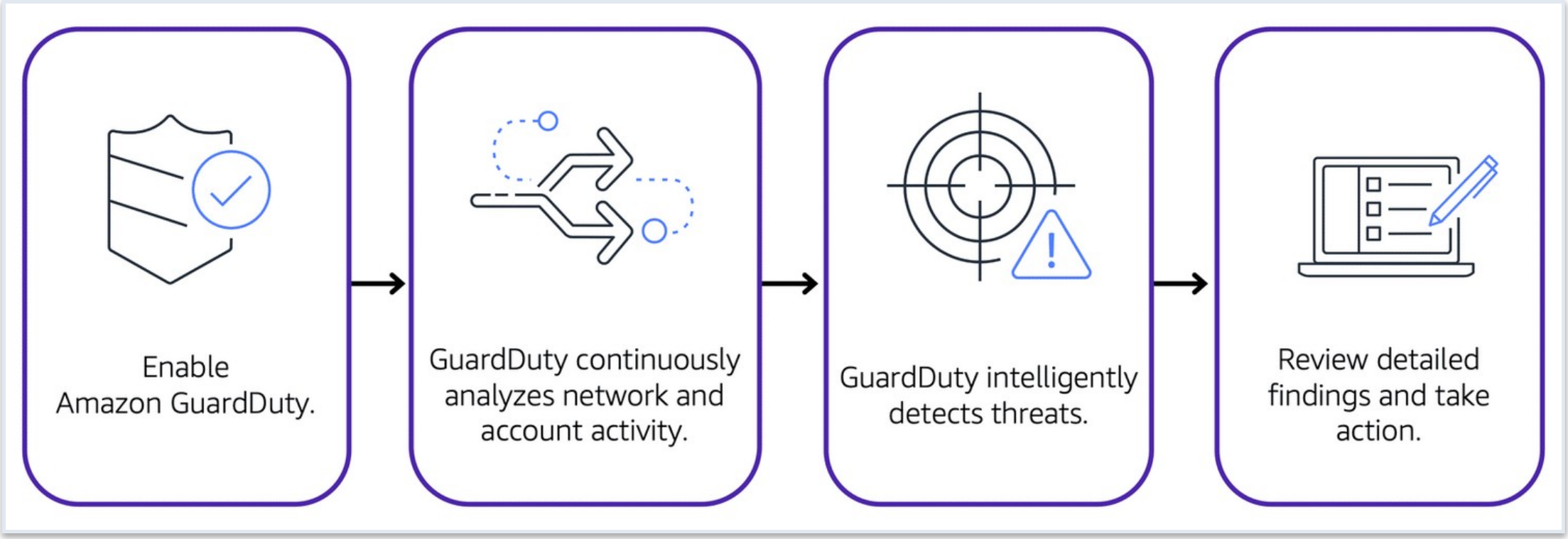
Amazon GuardDuty



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Amazon GuardDuty

- Intelligent Threat discovery to protect your AWS Account
- Uses Machine Learning algorithms, anomaly detection, 3rd party data
- One click to enable (30 days trial), no need to install software
- Input data includes:
- CloudTrail Events Logs – unusual API calls, unauthorized deployments
- CloudTrail Management Events – create VPC subnet, create trail, …
- CloudTrail S3 Data Events – get object, list objects, delete object, …
- VPC Flow Logs – unusual internal traffic, unusual IP address
- DNS Logs – compromised EC2 instances sending encoded data within DNS queries
- Optional Features – EKS Audit Logs, RDS & Aurora, EBS, Lambda, S3 Data Events…
- Can setup EventBridge rules to be notified in case of findings
- EventBridge rules can target AWS Lambda or SNS
- Can protect against CryptoCurrency attacks (has a dedicated “finding” for it)

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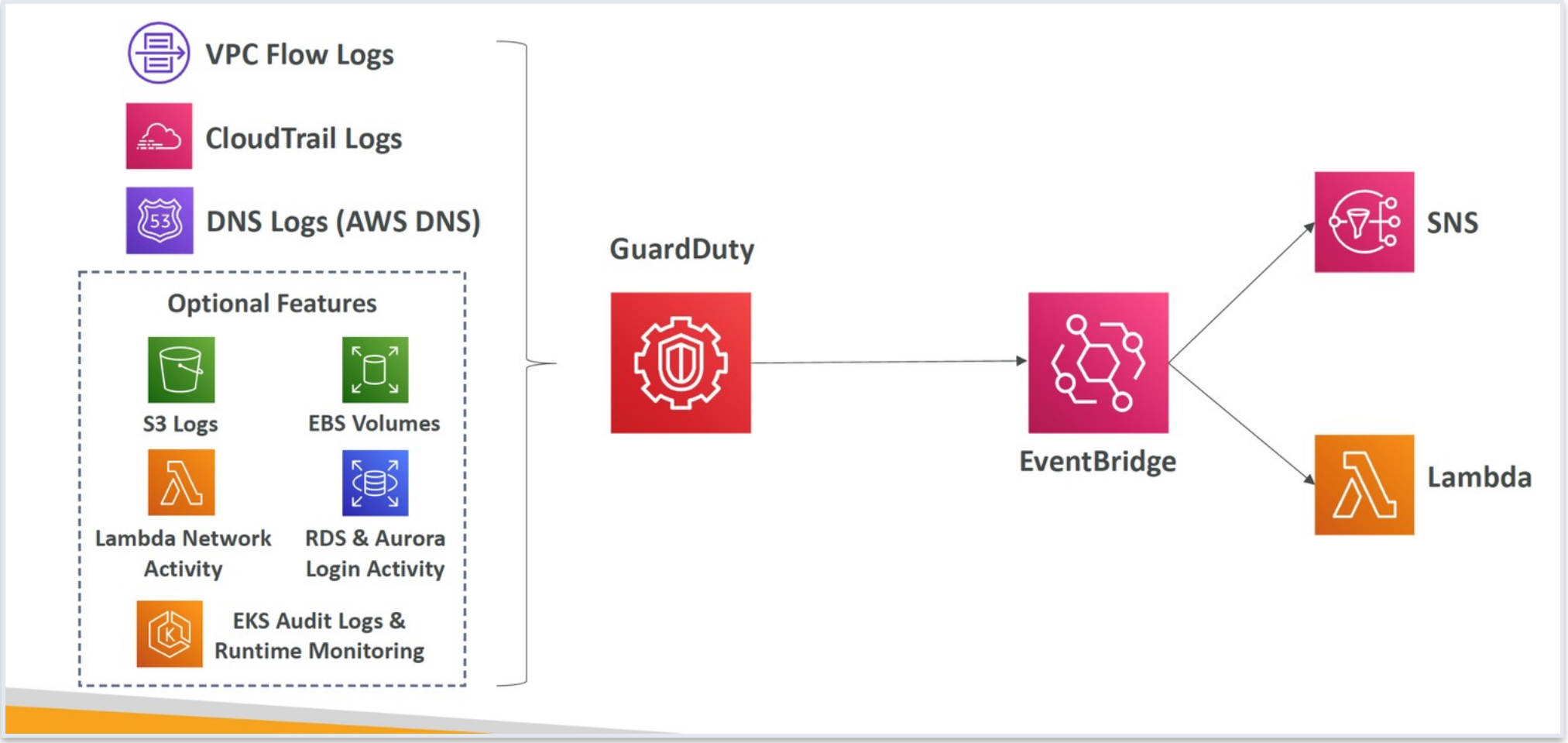
Amazon GuardDuty



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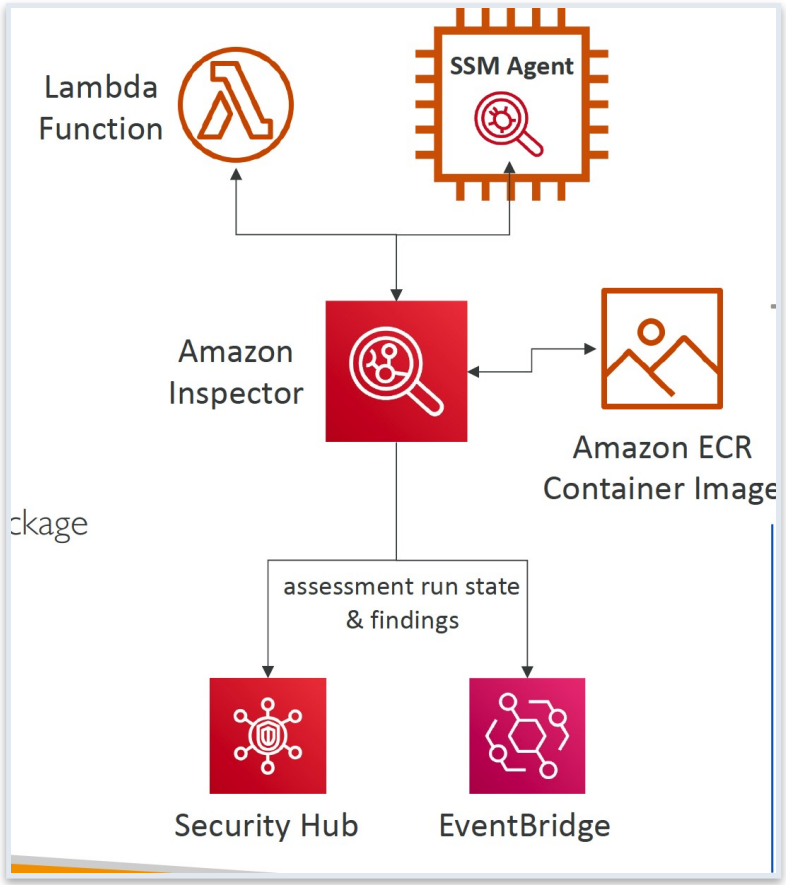
Amazon Inspector



Automated Security Assessments

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What does Amazon Inspector evaluate?

- Remember: only for EC2 instances, Container Images & Lambda functions
- Continuous scanning of the infrastructure, only when needed
- Package vulnerabilities (EC2, ECR & Lambda) – database of CVE
- Network reachability (EC2)
- A risk score is associated with all vulnerabilities for prioritization

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AWS Config

- Helps with auditing and recording compliance of your AWS resources
- Helps record configurations and changes over time
- Possibility of storing the configuration data into S3 (analyzed by Athena)
- Questions that can be solved by AWS Config:
- Is there unrestricted SSH access to my security groups?
- Do my buckets have any public access?
- How has my ALB configuration changed over time?
- You can receive alerts (SNS notifications) for any changes
- AWS Config is a per-region service
- Can be aggregated across regions and accounts

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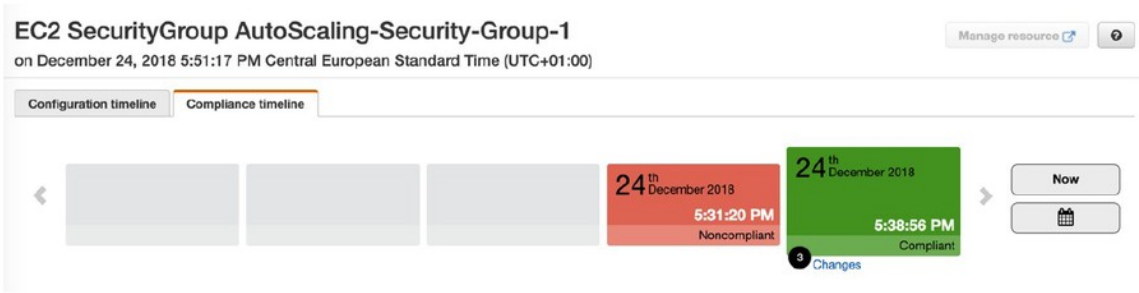
AWS Config Resource

- View compliance of a resource over time



- View configuration of a resource over time

- View CloudTrail API calls if enabled

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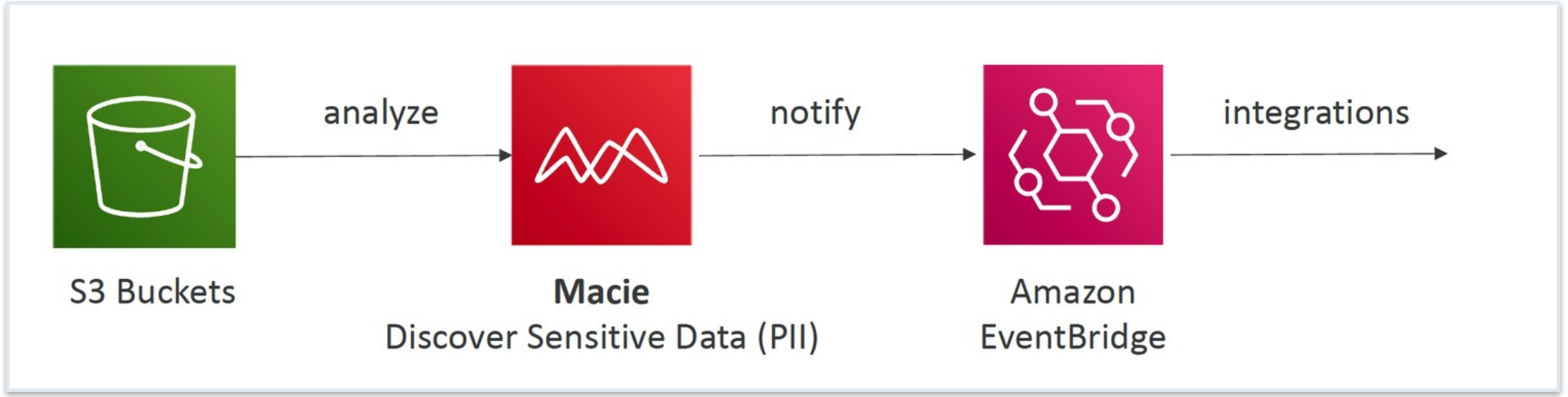
AWS Macie



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AWS Security Hub

- Central security tool to manage security across several AWS accounts and automate security checks
- Integrated dashboards showing current security and compliance status to quickly take actions
- Automatically aggregates alerts in predefined or personal findings formats from various AWS services & AWS partner tools:
- Config
- GuardDuty
- Inspector
- Macie
- IAM Access Analyzer
- AWS Systems Manager
- AWS Firewall Manager
- AWS Health
- AWS Partner Network Solutions

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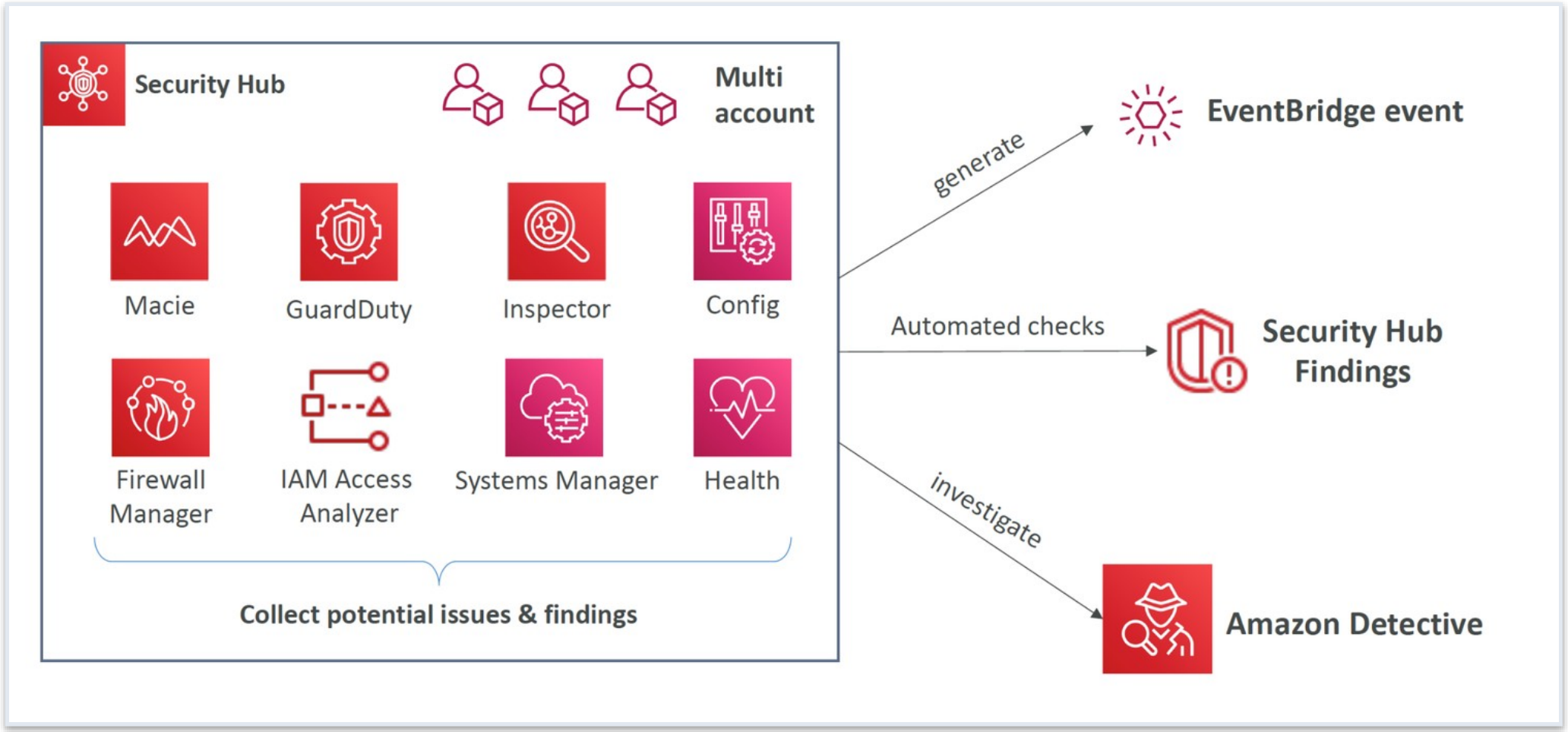
AWS Security Hub



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Amazon Detective

- GuardDuty, Macie, and Security Hub are used to identify potential security issues, or findings
- Sometimes security findings require deeper analysis to isolate the root cause and take action – it's a complex process
- Amazon Detective analyzes, investigates, and quickly identifies the root cause of security issues or suspicious activities (using ML and graphs)
- Automatically collects and processes events from VPC Flow Logs, CloudTrail, GuardDuty and create a unified view
- Produces visualizations with details and context to get to the root cause

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AWS Abuse

- Report suspected AWS resources used for abusive or illegal purposes
- Abusive & prohibited behaviors are:
- Spam – receiving undesired emails from AWS-owned IP address, websites & forum spammed by AWS resources
- Port scanning – sending packets to your ports to discover the unsecured ones
- DoS or DDoS attacks – AWS-owned IP addresses attempting to overwhlem or crash your servers/softwares
- Intrusion attempts – logging in on your resources
- Hosting objectionable or copyrighted content – distributing illegal or copyrighted content without consent
- Distributing malware – AWS resources distributing softwares to harm computers or Machines
- Contact the AWS Abuse team: AWS abuse form, or abuse@amazonaws.com

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Root user privileges

- Root user = Account Owner (created when the account is created)
- Has complete access to all AWS services and resources
- Lock away your AWS account root user access keys!
- Do not use the root account for everyday tasks, even administrative tasks
- Actions that can be performed only by the root user:
- Change account settings (account name, email address, root user password, root user access keys)
- View certain tax invoices
- Close your AWS account
- Restore IAM user permissions
- Change or cancel your AWS Support plan
- Register as a seller in the Reserved Instance Marketplace
- Configure an Amazon S3 bucket to enable MFA
- Edit or delete an Amazon S3 bucket policy that includes an invalid VPC ID or VPC endpoint ID

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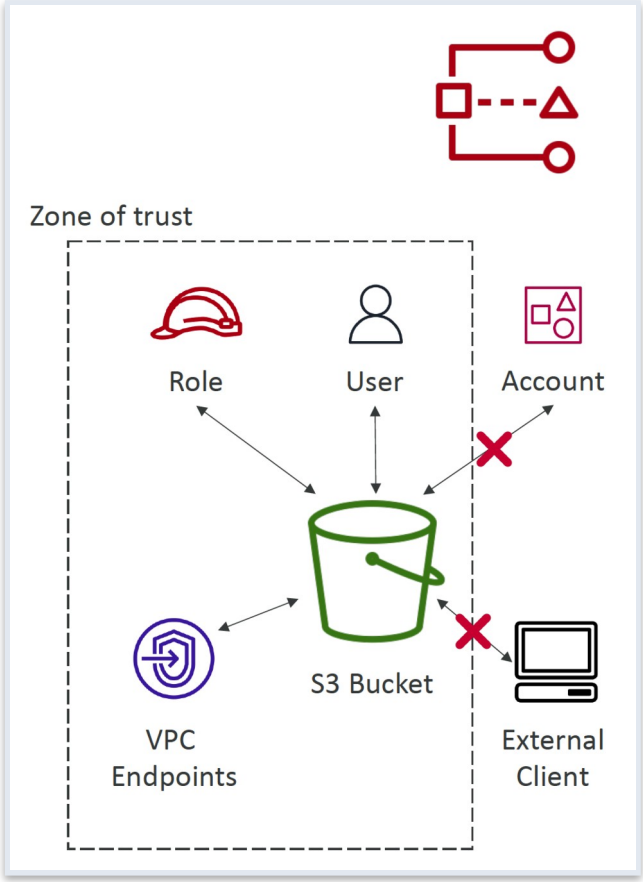
IAM Access Analyzer



Find out which resources are shared externally

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Security & Compliance Summary

- Shared Responsibility on AWS
- Shield: Automatic DDoS Protection + 24/7 support for advanced
- WAF: Firewall to filter incoming requests based on rules
- KMS: Encryption keys managed by AWS
- CloudHSM: Hardware encryption, we manage encryption keys
- AWS Certificate Manager: provision, manage, and deploy SSL/TLS Certificates
- Artifact: Get access to compliance reports such as PCI, ISO, etc…
- GuardDuty: Find malicious behavior with VPC, DNS & CloudTrail Logs
- Inspector: find software vulnerabilities in EC2, ECR Images, and Lambda functions
- Network Firewall: Protect VPC against network attacks

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Security & Compliance Summary

- Config: Track config changes and compliance against rules
- Macie: Find sensitive data (ex: PII data) in Amazon S3 buckets
- CloudTrail: Track API calls made by users within account
- AWS Security Hub: gather security findings from multiple AWS accounts
- Amazon Detective: find the root cause of security issues or suspicious activities
- AWS Abuse: Report AWS resources used for abusive or illegal purposes
- Root user privileges:
- Change account settings
- Close your AWS account
- Change or cancel your AWS Support plan
- Register as a seller in the Reserved Instance Marketplace
- IAM Access Analyzer: identify which resources are shared externally
- Firewall Manager: manage security rules across an Organization (WAF, Shield…)

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AWS Trusted Advisor

- AWS Trusted Advisor is a web service that inspects your AWS environment and provides real-time recommendations in accordance with AWS best practices.

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AWS Trusted Advisor dashboard

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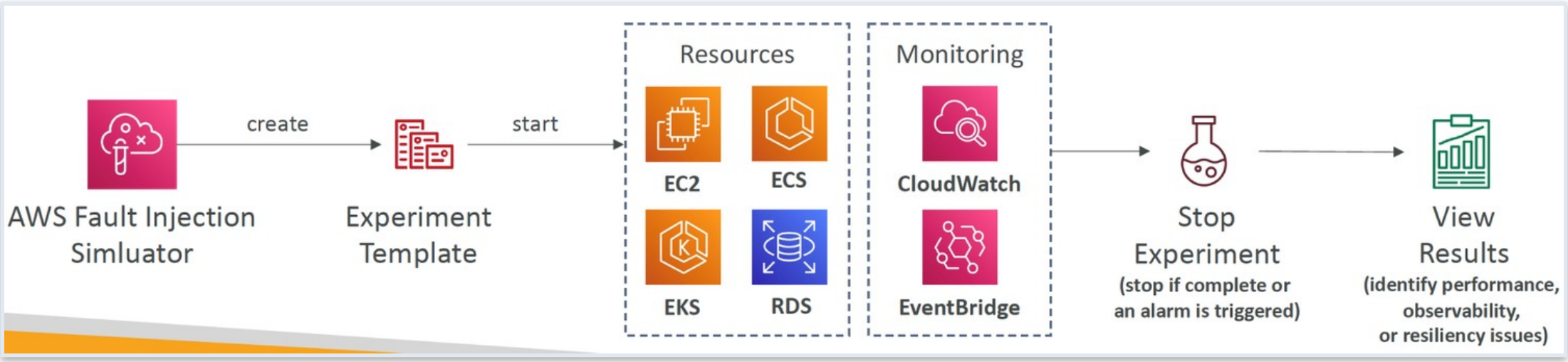
AWS Fault Injection Simulator (FIS)



A fully managed service for running fault injection experiments on AWS workloads

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Account Management, Billing &

- Support

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AWS Organizations

- Global service
- Allows to manage multiple AWS accounts
- The main account is the master account
- Cost Benefits:
- Consolidated Billing across all accounts - single payment method
- Pricing benefits from aggregated usage (volume discount for EC2, S3…)
- Pooling of Reserved EC2 instances for optimal savings
- API is available to automate AWS account creation
- Restrict account privileges using Service Control Policies (SCP)

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Multi Account Strategies

- Create accounts per department, per cost center, per dev / test / prod, based on regulatory restrictions (using SCP), for better resource isolation (ex: VPC), to have separate per-account service limits, isolated account for logging
- Multi Account vs One Account Multi VPC
- Use tagging standards for billing purposes
- Enable CloudTrail on all accounts, send logs to central S3 account
- Send CloudWatch Logs to central logging account

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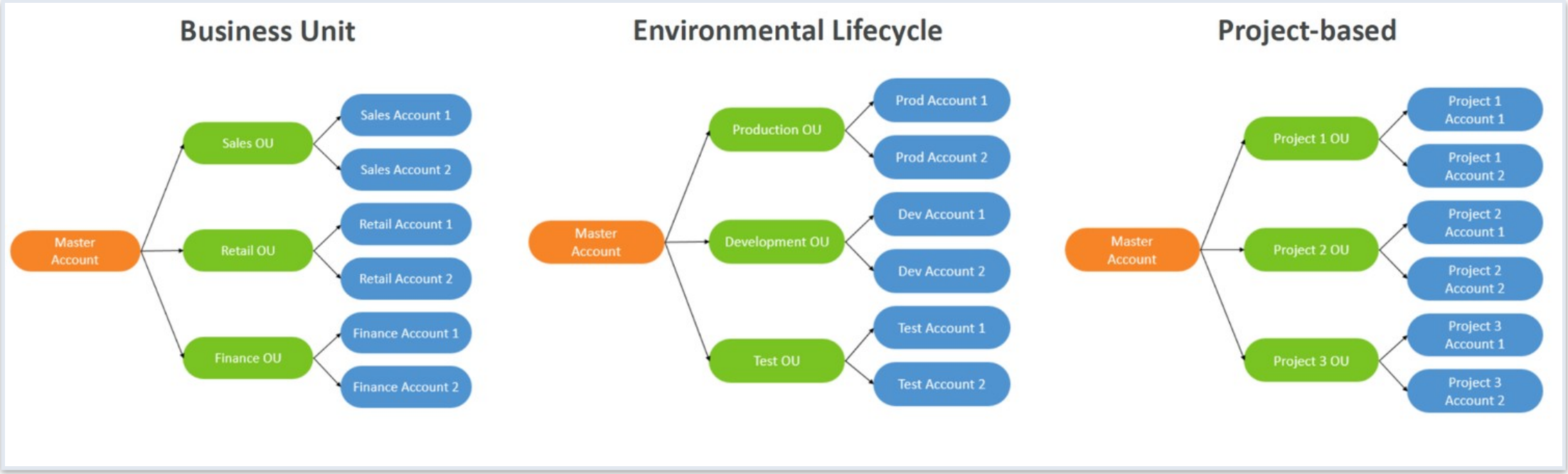
Organizational Units (OU) - Examples



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AWS Organizations

- In AWS Organizations, you can centrally control permissions for the accounts in your organization by using service control policies.
- In AWS Organizations, you can group accounts into organizational units (OUs) to make it easier to manage accounts with similar business or security requirements.

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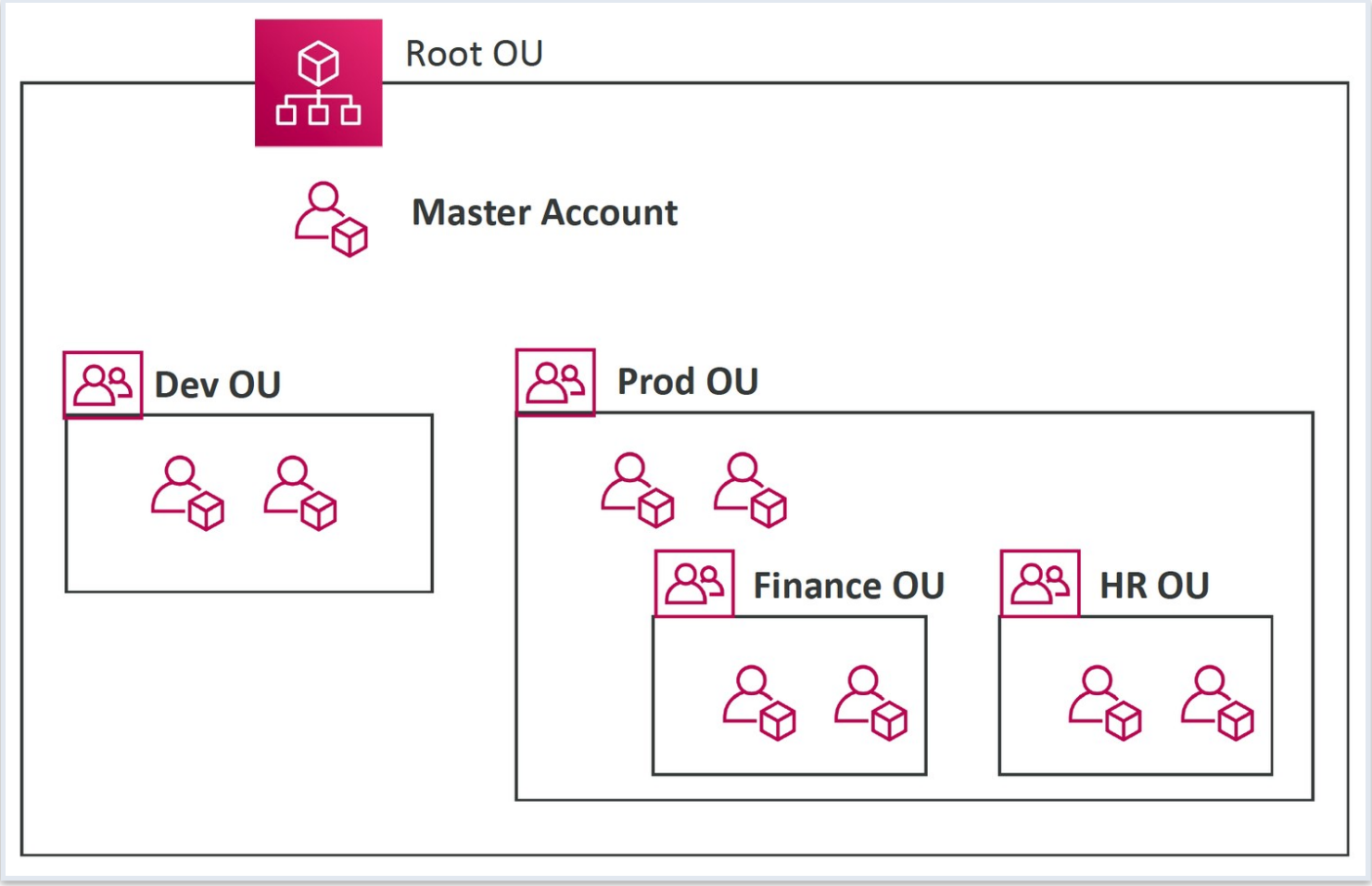
AWS Organization



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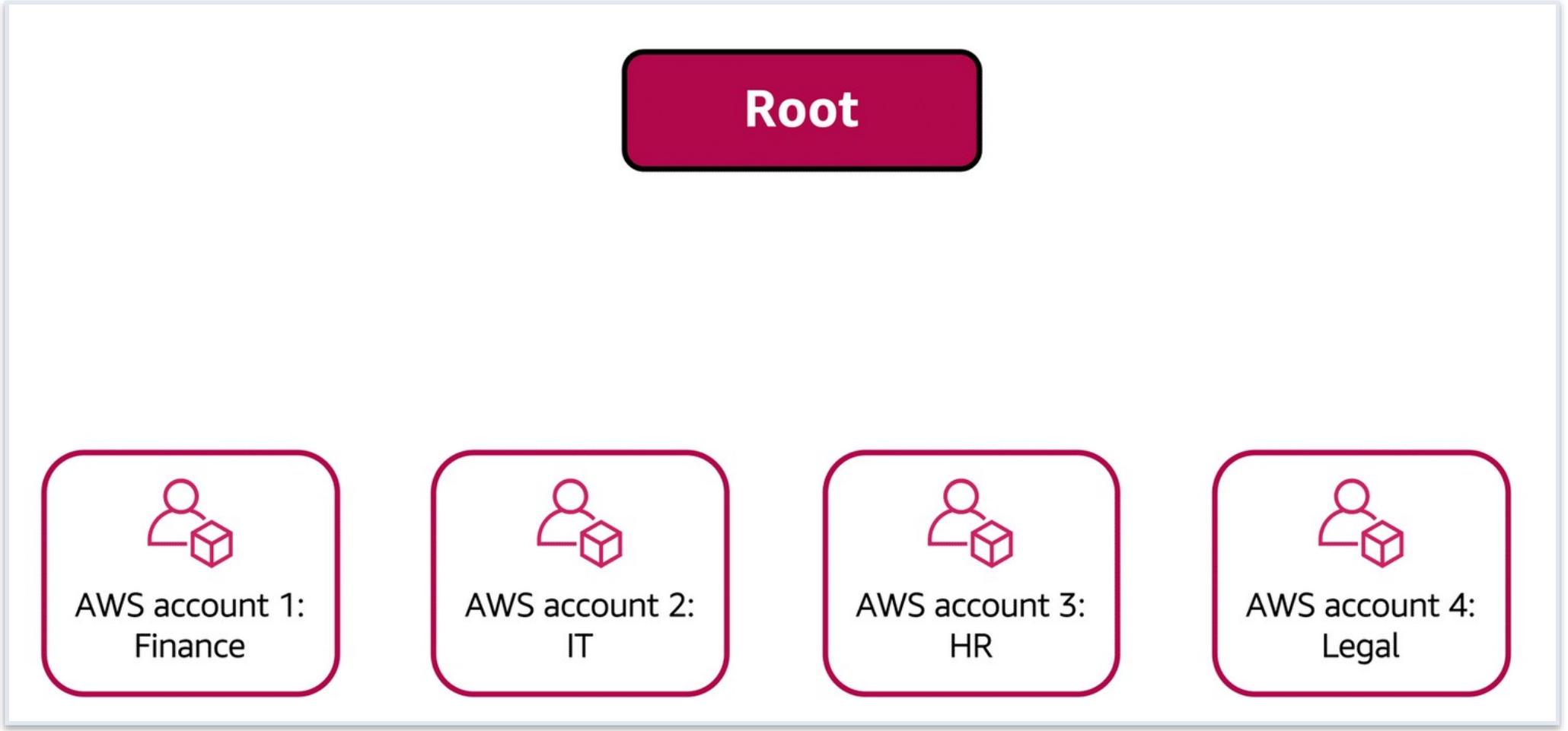
Organizational units



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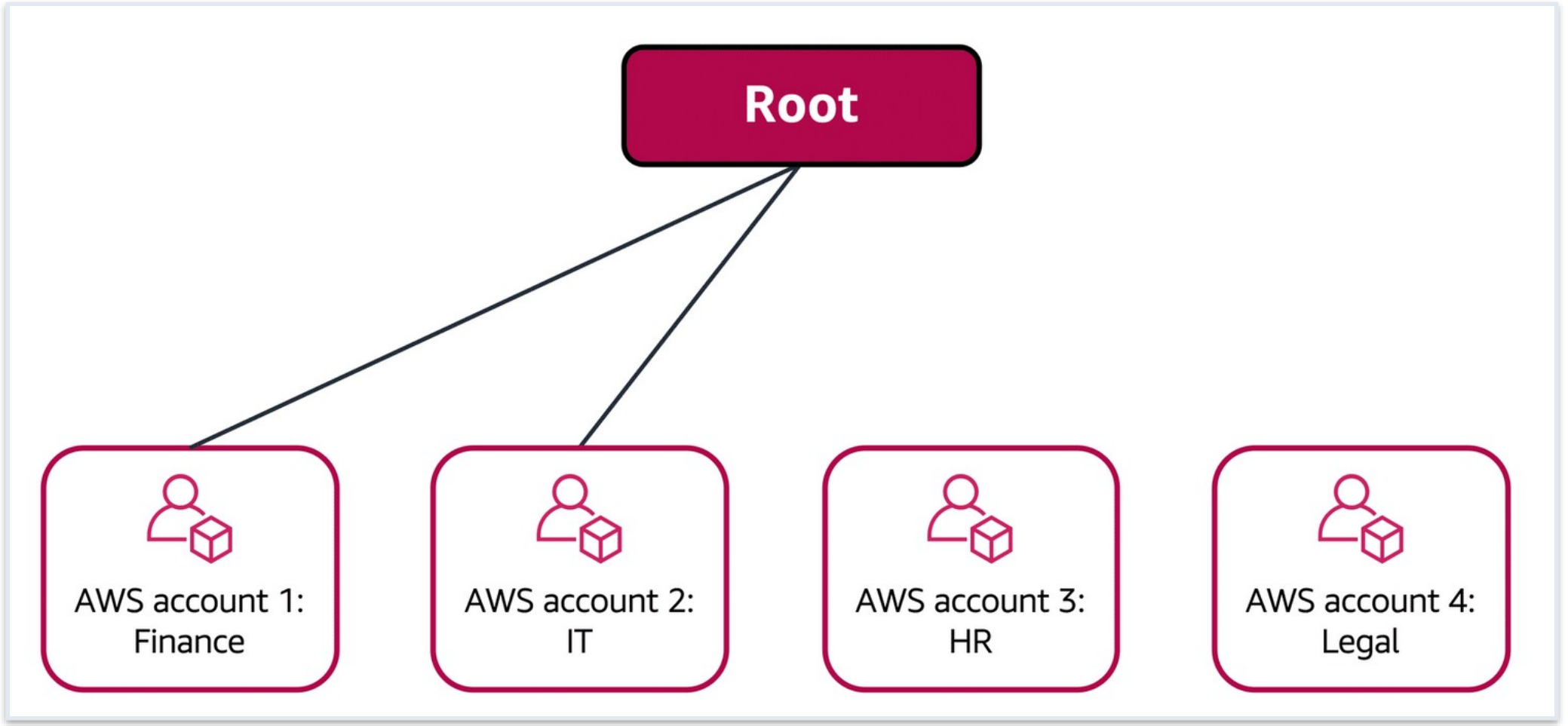
Organizational units



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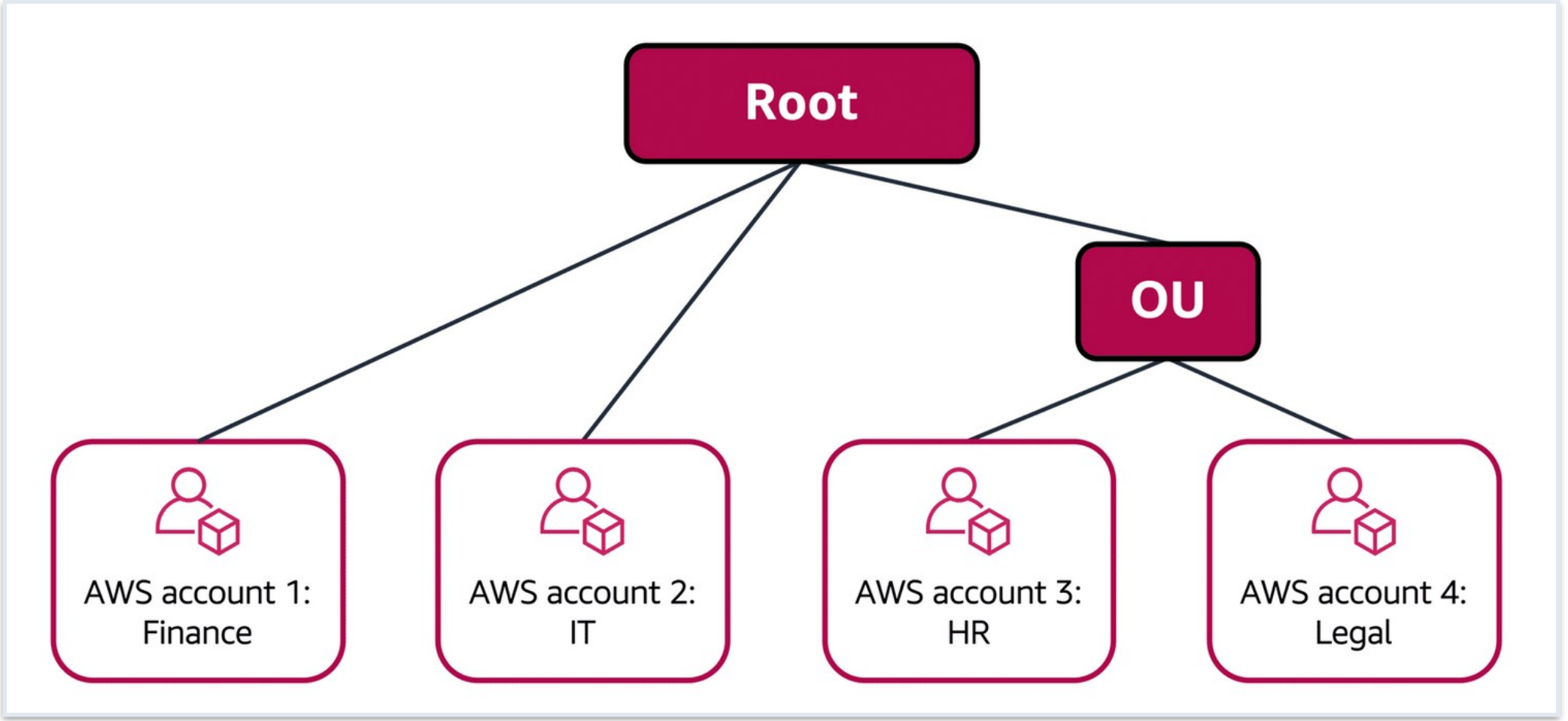
Organizational units



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Service Control Policies (SCP)

- Whitelist or blacklist IAM actions
- Applied at the OU or Account level
- Does not apply to the Master Account
- SCP is applied to all the Users and Roles of the Account, including Root user
- The SCP does not affect service-linked roles
- Service-linked roles enable other AWS services to integrate with AWS Organizations and can't be restricted by SCPs.
- SCP must have an explicit Allow (does not allow anything by default)
- Use cases:
- Restrict access to certain services (for example: can’t use EMR)
- Enforce PCI compliance by explicitly disabling services

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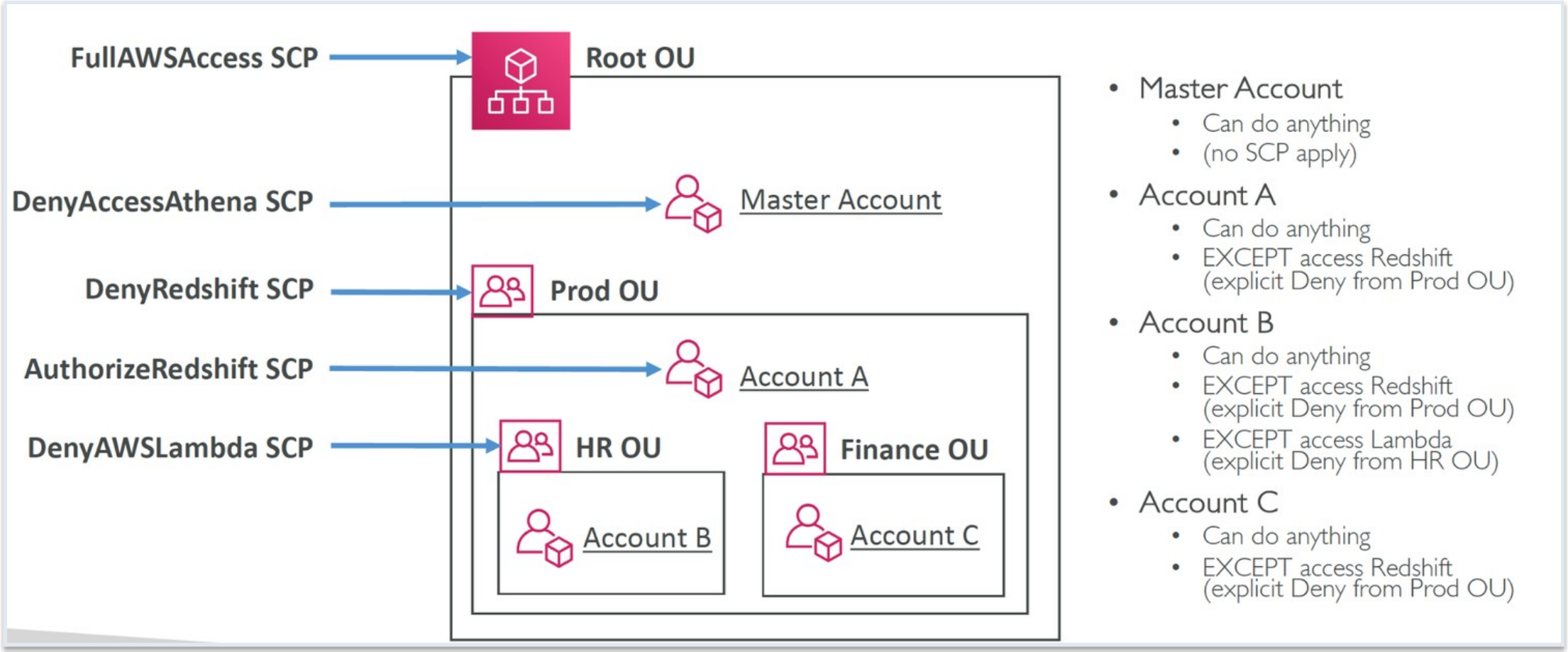
SCP Hierarchy



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SCP Examples

```
{
    "Version": "2012-10-17",
    "Statement": [
        {
            "Sid": "AllowsAllActions",
            "Effect": "Allow",
            "Action": "*",
            "Resource": "*"
        },
        {
            "Sid": "DenyDynamoDB",
            "Effect": "Deny",
            "Action": "dynamodb:*",
            "Resource": "*"
        }
    ]
}
```

```
{
    "Version": "2012-10-17",
    "Statement": [
        {
            "Effect": "Allow",
            "Action": [
                "ec2:*",
                "cloudwatch:*"
            ],
            "Resource": "*"
        }
    ]
}
```


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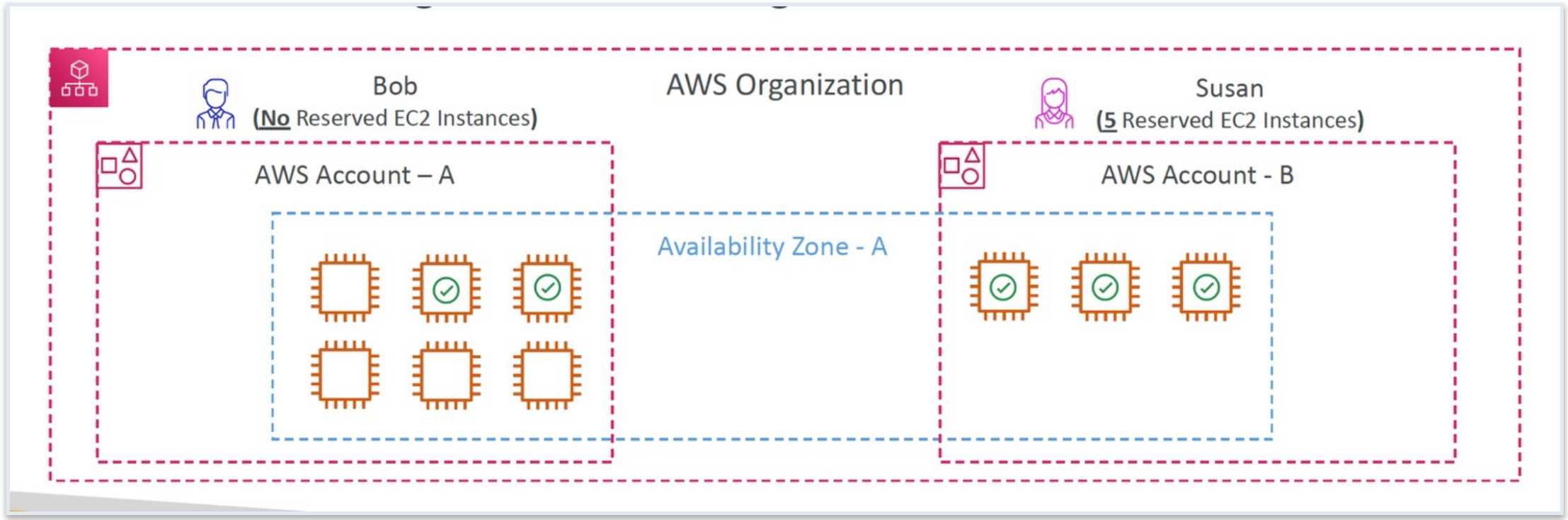
AWS Organization – Consolidated Billing



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AWS Control Tower

- Easy way to set up and govern a secure and compliant multi-account AWS environment based on best practices
- Benefits:
- Automate the set up of your environment in a few clicks
- Automate ongoing policy management using guardrails
- Detect policy violations and remediate them
- Monitor compliance through an interactive dashboard
- AWS Control Tower runs on top of AWS Organizations:
- It automatically sets up AWS Organizations to organize accounts and implement SCPs (Service Control Policies)

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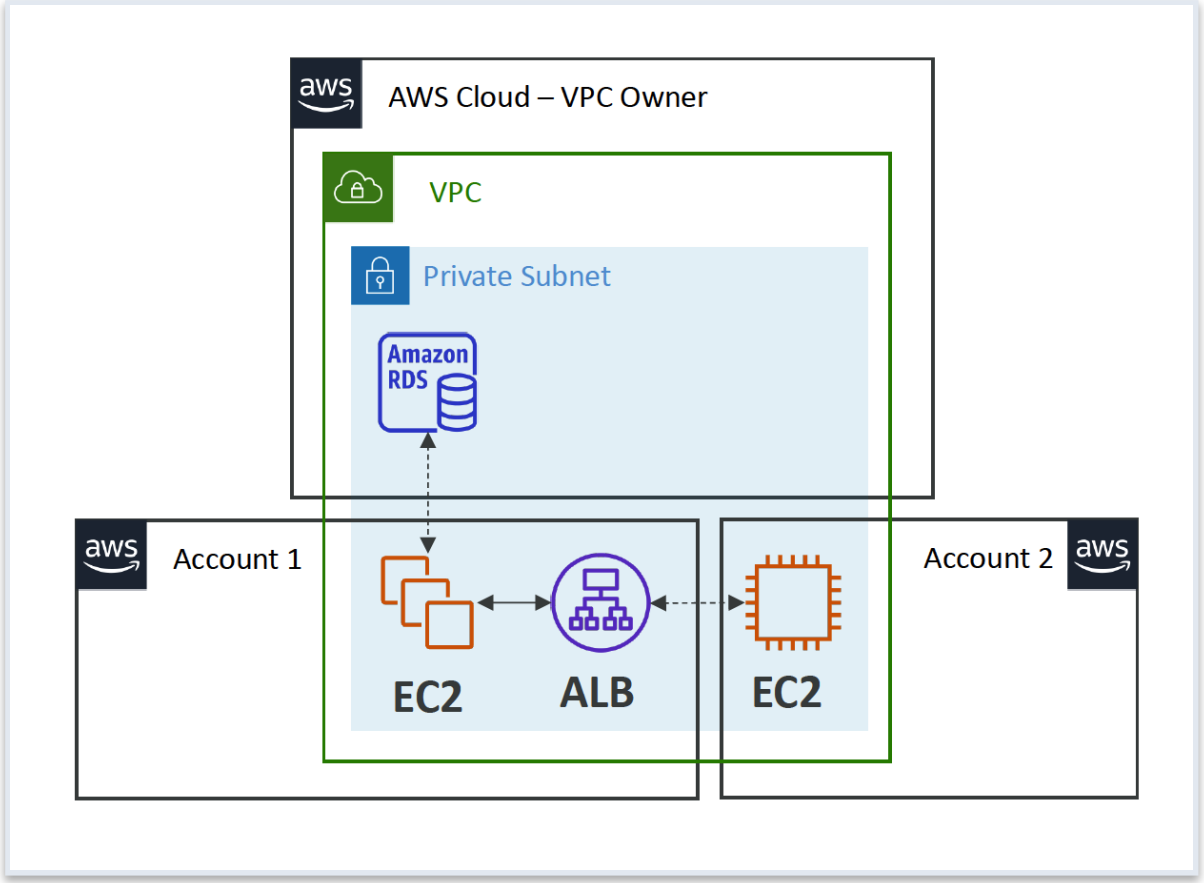
AWS Resource Access Manager (AWS RAM)



Share AWS resources that you own with other AWS accounts

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AWS Service Catalog

- Users that are new to AWS have too many options, and may create stacks that are not compliant / in line with the rest of the organization
- Some users just want a quick self-service portal to launch a set of authorized products pre-defined by admins
- Includes: virtual machines, databases, storage options, etc…
- Enter AWS Service Catalog!

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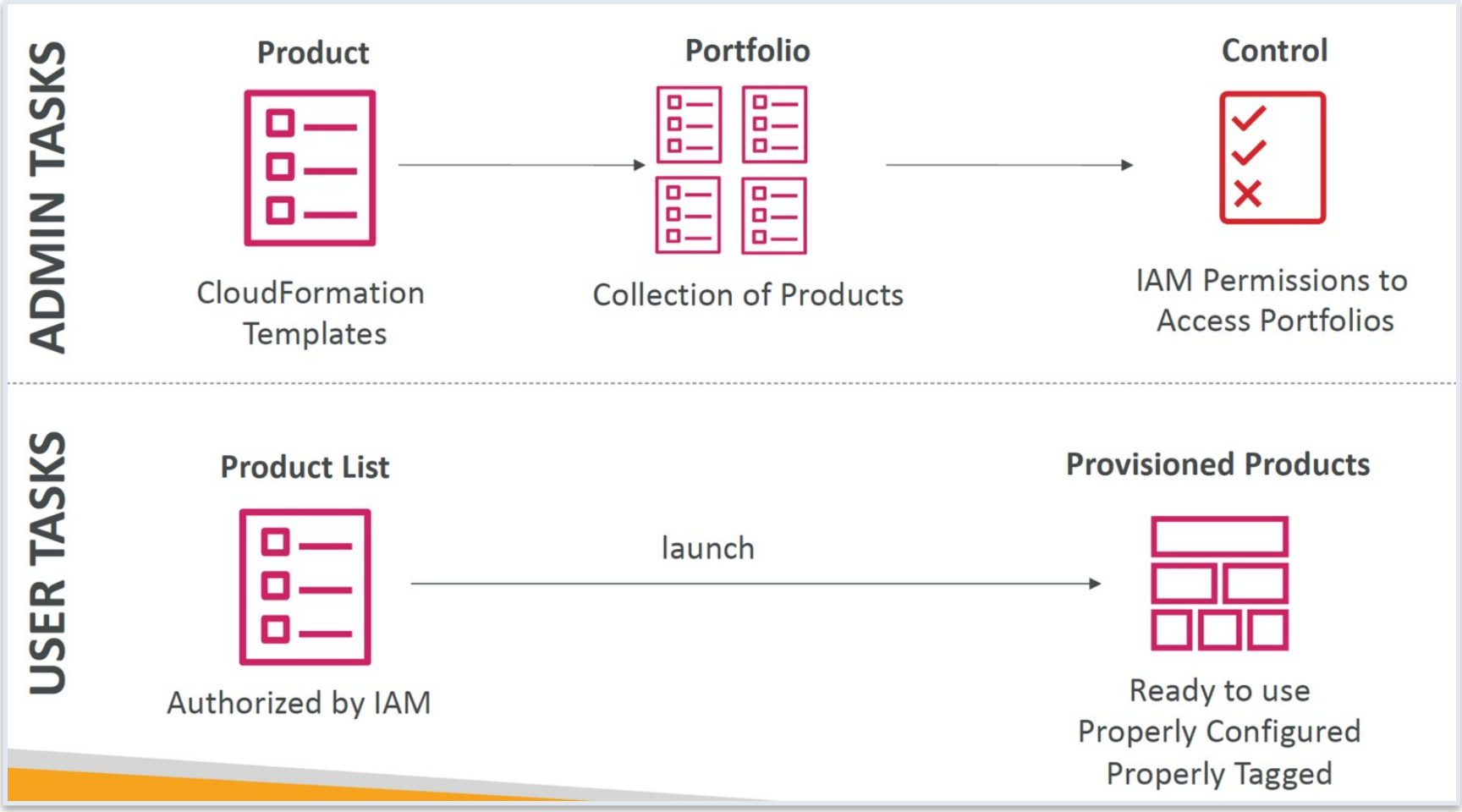
Service Catalog diagram



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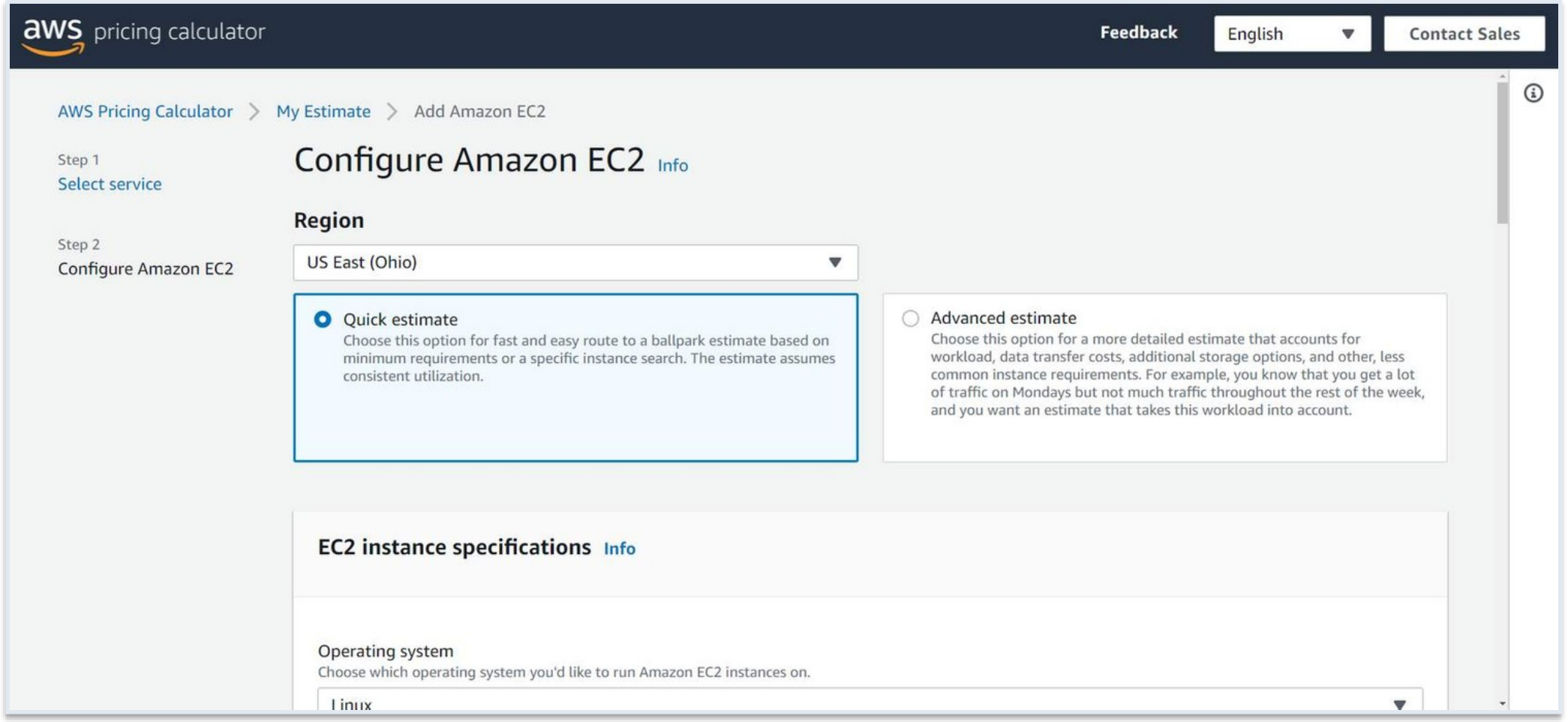
AWS Pricing Calculator



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Pricing Example

▾ **Lambda**		**$0.00**
▾ **US East (N. Virginia)**		**$0.00**
AWS Lambda Lambda-GB-Second		$0.00
AWS Lambda - Compute Free Tier - 400,000 GB-Seconds - US East (Northern Virginia)	254.575 seconds	$0.00
AWS Lambda Request		$0.00
AWS Lambda - Requests Free Tier - 1,000,000 Requests - US East (Northern Virginia)	680.000 Requests	$0.00

If you have used AWS Lambda in multiple AWS Regions, you can view the itemized charges by Region on your bill.

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Pricing Example

▾ **Elastic Compute Cloud**		**$0.00**
▾ **US East (N. Virginia)**		**$0.00**
Amazon Elastic Compute Cloud running Linux/UNIX		$0.00
$0.00 per Linux t2.micro instance-hour (or partial hour) under monthly free tier	106.512 Hrs	$0.00
EBS		$0.00
$0.00 per GB-month of General Purpose (SSD) provisioned storage under monthly free tier	11.294 GB-Mo	$0.00
Elastic Load Balancing - Application		$0.00
$0.00 per Application LoadBalancer-hour (or partial hour) under monthly free tier	268.000 Hrs	$0.00

Take an Amazon EC2 instance that has been deployed in the Northern Virginia Region for example.

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Amazon EC3 Pricing

1. Storage
2. Requests and data retrievals
3. Data transfer
4. Management and replication

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Pricing Example

Simple Storage Service		**$0.00**
US East (N. Virginia)		**$0.00**
Amazon Simple Storage Service Requests-Tier1		$0.00
$0.00 per request - PUT, COPY, POST, or LIST requests under the monthly global free tier	185.000 Requests	$0.00
Amazon Simple Storage Service Requests-Tier2		$0.00
$0.00 per request - GET and all other requests under the monthly global free tier	923.000 Requests	$0.00
Amazon Simple Storage Service TimedStorage-ByteHrs		$0.00
$0.000 per GB - storage under the monthly global free tier	0.159 GB-Mo	$0.00
US East (Ohio)		**$0.00**
Amazon Simple Storage Service USE2-Requests-Tier2		$0.00
$0.00 per request - GET and all other requests under the monthly global free tier	4.000 Requests	$0.00
Amazon Simple Storage Service USE2-TimedStorage-ByteHrs		$0.00
$0.000 per GB - storage under the monthly global free tier	0.000001 GB-Mo	$0.00

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Pricing Models in AWS

- AWS has 4 pricing models:
- Pay as you go: pay for what you use, remain agile, responsive, meet scale demands
- Save when you reserve: minimize risks, predictably manage budgets, comply with long-terms requirements
- Reservations are available for EC2 Reserved Instances, DynamoDB Reserved Capacity, ElastiCache Reserved Nodes, RDS Reserved Instance, Redshift Reserved Nodes
- Pay less by using more: volume-based discounts
- Pay less as AWS grows

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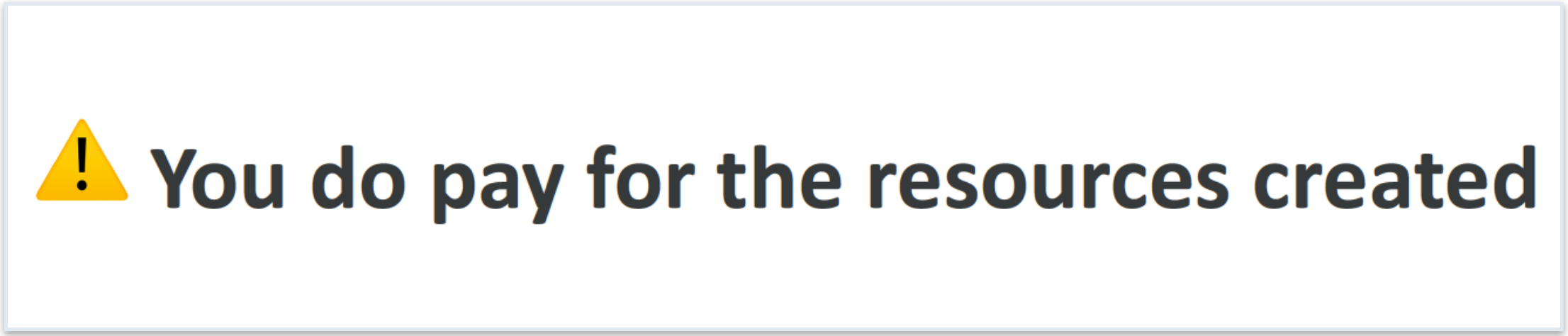
Free services & free tier in AWS



IAM

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Compute Pricing – EC2s

- Only charged for what you use
- Number of instances
- Instance configuration:
- Physical capacity
- Region
- OS and software
- Instance type
- Instance size
- ELB running time and amount of data processed
- Detailed monitoring

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Compute Pricing – EC2

- On-demand instances:
- Minimum of 60s
- Pay per second (Linux/Windows) or per hour (other)
- Reserved instances:
- Up to 75% discount compared to On-demand on hourly rate
- 1- or 3-years commitment
- All upfront, partial upfront, no upfront
- Spot instances:
- Up to 90% discount compared to On-demand on hourly rate
- Bid for unused capacity
- Dedicated Host:
- On-demand
- Reservation for 1 year or 3 years commitment

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Compute Pricing – Lambda & ECS

- Lambda:
- Pay per call
- Pay per duration
- ECS:
- EC2 Launch Type Model: No additional fees, you pay for
- AWS resources stored and created in your application
- Fargate:
- Fargate Launch Type Model: Pay for vCPU and memory
- Rresources allocated to your applications in your containers

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Storage Pricing – S3

- Storage class: S3 Standard, S3 Infrequent Access, S3 One-Zone IA, S3 Intelligent Tiering, S3 Glacier and S3 Glacier Deep Archive
- Number and size of objects: Price can be tiered (based on volume)
- Number and type of requests
- Data transfer OUT of the S3 region
- S3 Transfer Acceleration
- Lifecycle transitions
- Similar service: EFS (pay per use, has infrequent access & lifecycle rules)

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Content Delivery – CloudFront

Per Month	United States & Canada	Europe & Israel	South Africa, Kenya, & Middle East	South America	Japan	Australia	Singapore, South Korea, Taiwan, Hong Kong, & Philippines	India
First 10TB	$0.085	$0.085	$0.110	$0.110	$0.114	$0.114	$0.140	$0.170
Next 40TB	$0.080	$0.080	$0.105	$0.105	$0.089	$0.098	$0.135	$0.130
Next 100TB	$0.060	$0.060	$0.090	$0.090	$0.086	$0.094	$0.120	$0.110

Pricing is different across different geographic regions

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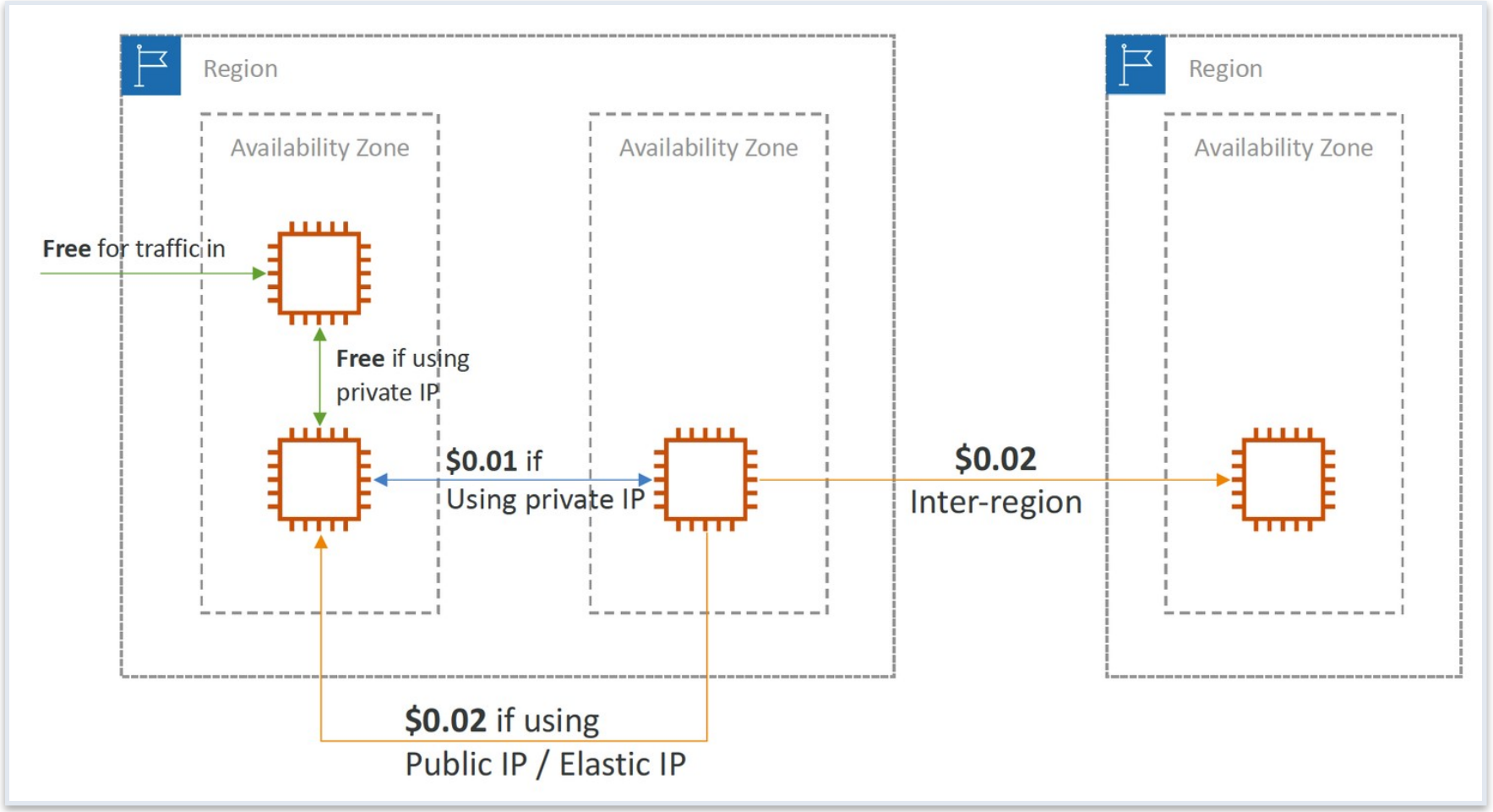
Networking Costs in AWS per GB - Simplified



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Savings Plan

- Commit a certain $ amount per hour for 1 or 3 years
- Easiest way to setup long-term commitments on AWS
- EC2 Savings Plan
- Up to 72% discount compared to On-Demand
- Commit to usage of individual instance families in a region (e.g. C5 or M5)
- Regardless of AZ, size (m5.xl to m5.4xl), OS (Linux/Windows) or tenancy
- All upfront, partial upfront, no upfront
- Compute Savings Plan
- Up to 66% discount compared to On-Demand
- Regardless of Family, Region, size, OS, tenancy, compute options
- Compute Options: EC2, Fargate, Lambda
- Machine Learning Savings Plan: SageMaker…
- Setup from the AWS Cost Explorer console

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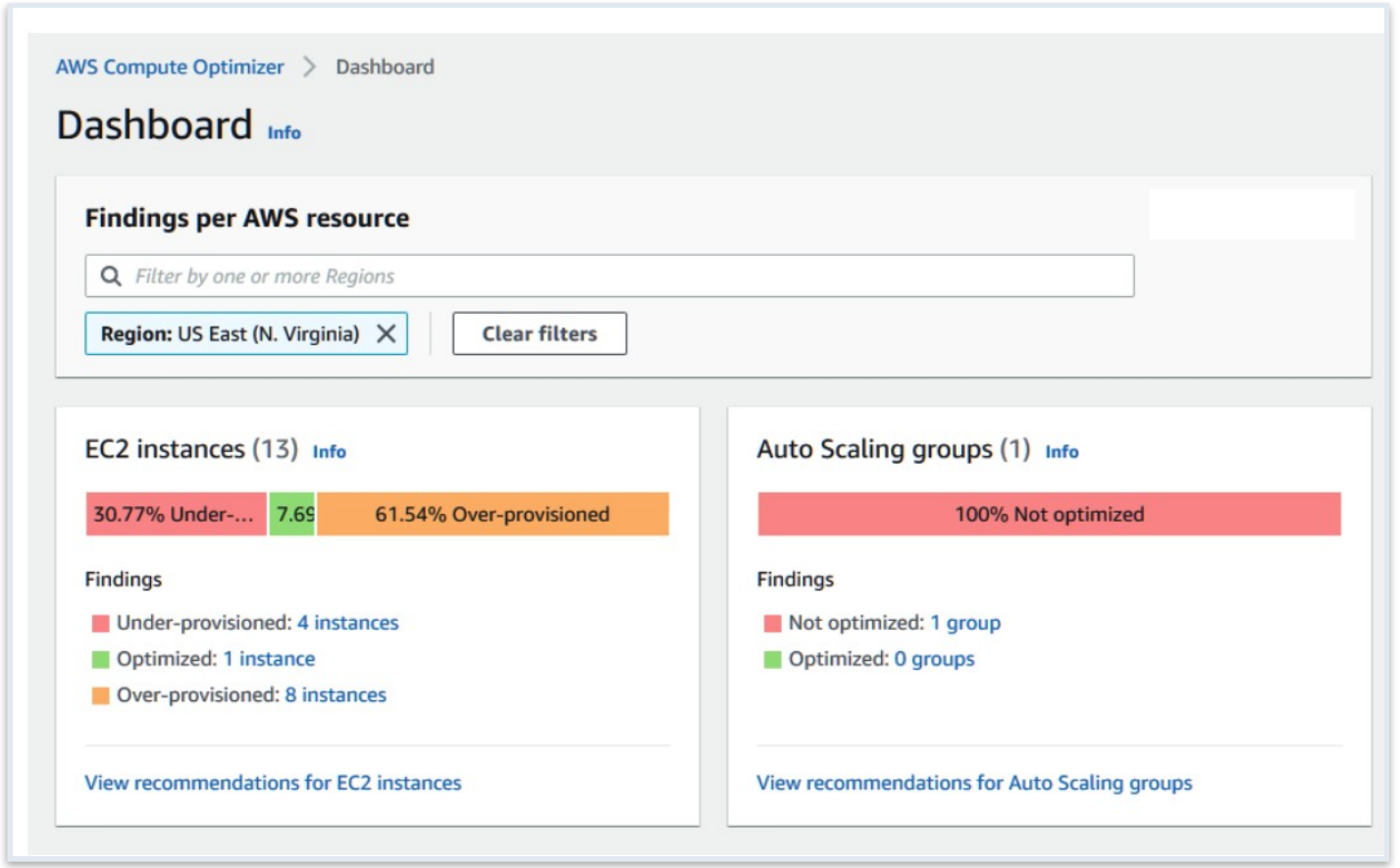
AWS Compute Optimizer



Reduce costs and improve performance by recommending optimal AWS resources for your workloads

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Billing and Costing Tools

- Estimating costs in the cloud:
- Pricing Calculator
- Tracking costs in the cloud:
- Billing Dashboard
- Cost Allocation Tags
- Cost and Usage Reports
- Cost Explorer
- Monitoring against costs plans:
- Billing Alarms
- Budgets

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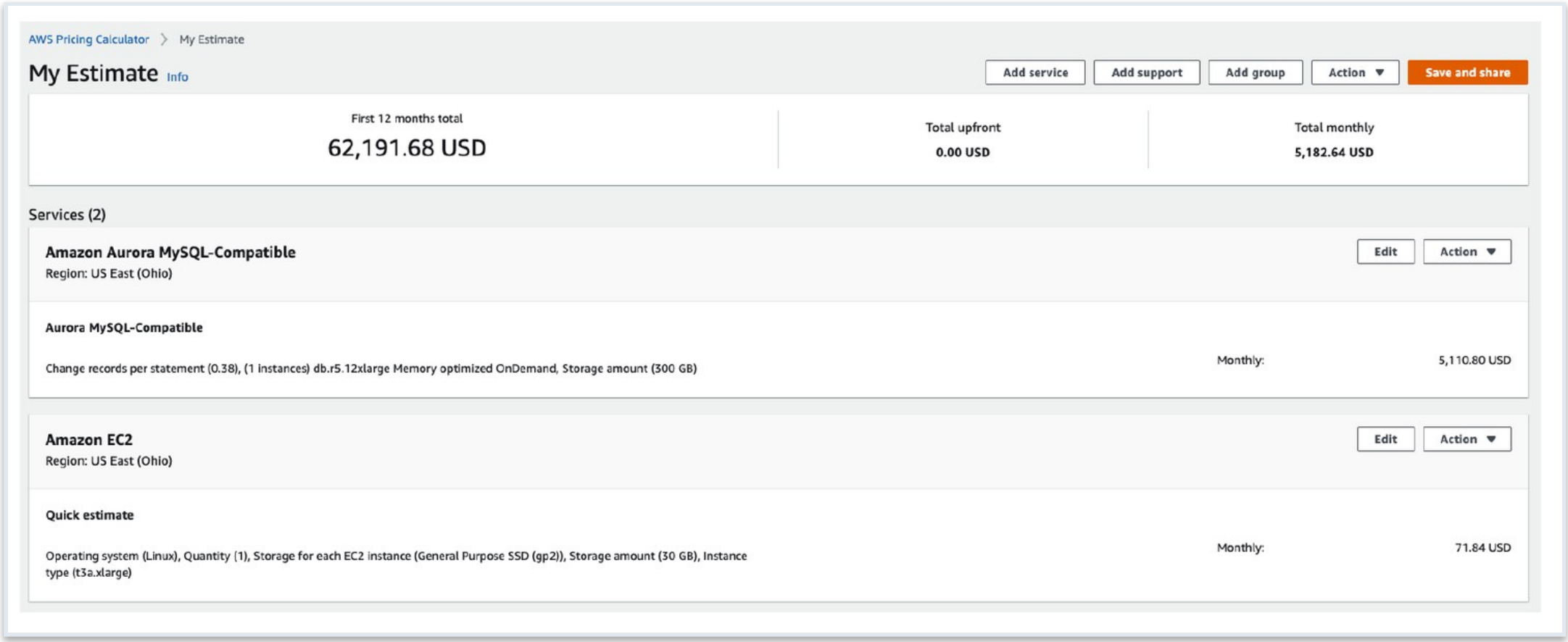
AWS Pricing Calculator



Available at https://calculator.aws/

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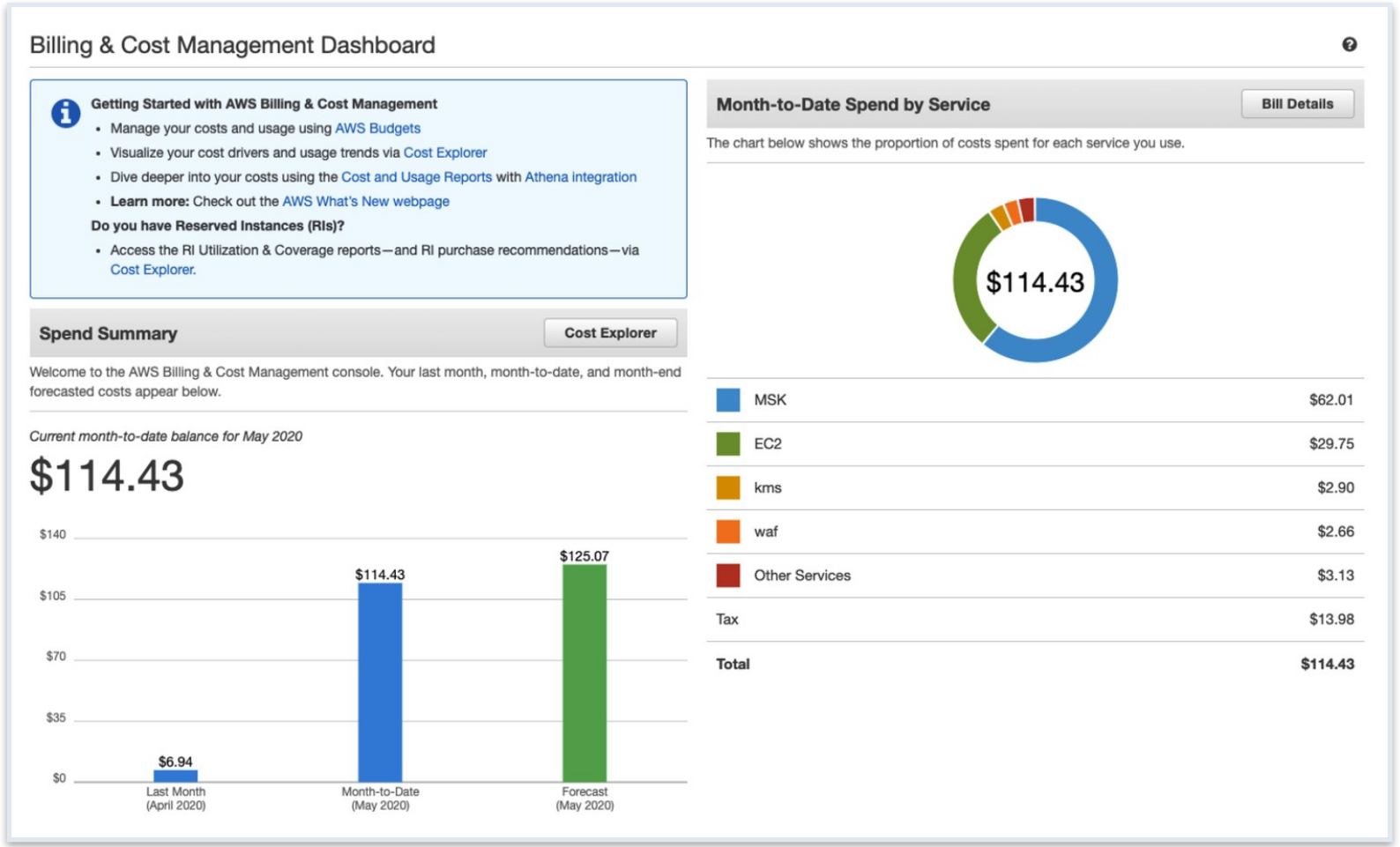
AWS Billing Dashboard



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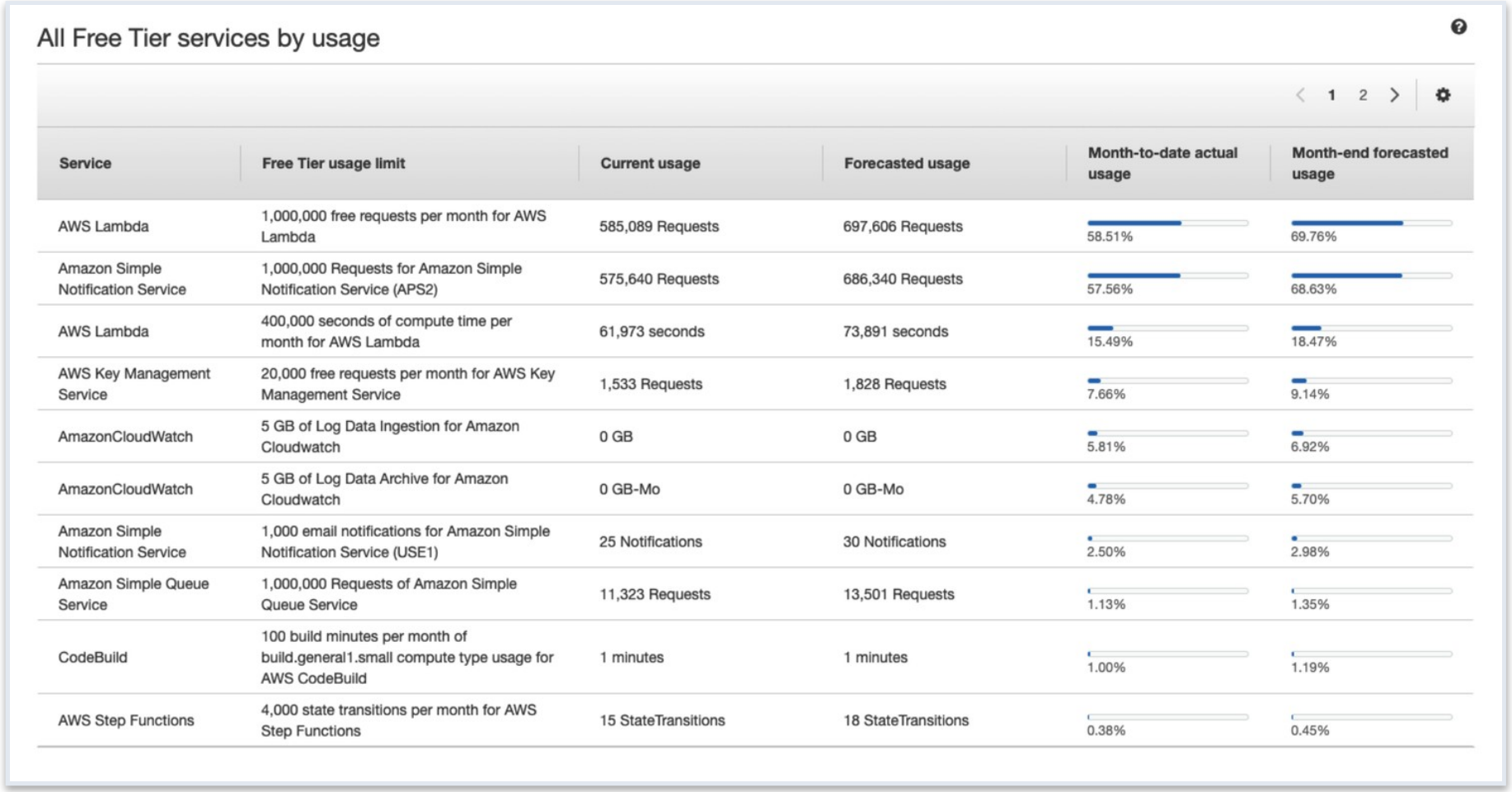
AWS Free Tier Dashboard



All Free Tier services by usage

1 2

Service	Free Tier usage limit	Current usage	Forecasted usage	Month-to-date actual usage	Month-end forecasted usage
AWS Lambda	1,000,000 free requests per month for AWS Lambda	585,089 Requests	697,606 Requests	58.51%	69.76%
Amazon Simple Notification Service	1,000,000 Requests for Amazon Simple Notification Service (APS2)	575,640 Requests	686,340 Requests	57.56%	68.63%
AWS Lambda	400,000 seconds of compute time per month for AWS Lambda	61,973 seconds	73,891 seconds	15.49%	18.47%
AWS Key Management Service	20,000 free requests per month for AWS Key Management Service	1,533 Requests	1,828 Requests	7.66%	9.14%
AmazonCloudWatch	5 GB of Log Data Ingestion for Amazon Cloudwatch	0 GB	0 GB	5.81%	6.92%
AmazonCloudWatch	5 GB of Log Data Archive for Amazon Cloudwatch	0 GB-Mo	0 GB-Mo	4.78%	5.70%
Amazon Simple Notification Service	1,000 email notifications for Amazon Simple Notification Service (USE1)	25 Notifications	30 Notifications	2.50%	2.98%
Amazon Simple Queue Service	1,000,000 Requests of Amazon Simple Queue Service	11,323 Requests	13,501 Requests	1.13%	1.35%
CodeBuild	100 build minutes per month of build.general1.small compute type usage for AWS CodeBuild	1 minutes	1 minutes	1.00%	1.19%
AWS Step Functions	4,000 state transitions per month for AWS Step Functions	15 StateTransitions	18 StateTransitions	0.38%	0.45%

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Cost Allocation Tags

Total Cost	user:Owner	user:Stack	user:Cost Center	user:Application
0.95	DbAdmin	Test	80432	Widget2
0.01	DbAdmin	Test	80432	Widget2
3.84	DbAdmin	Prod	80432	Widget2
6.00	DbAdmin	Test	78925	Widget1
234.63	SysEng	Prod	78925	Widget1
0.73	DbAdmin	Test	78925	Widget1
0.00	DbAdmin	Prod	80432	Portal
2.47	DbAdmin	Prod	78925	Portal

Use cost allocation tags to track your AWS costs on a detailed level

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Tagging and Resource Groups

- Tags are used for organizing resources:
- EC2: instances, images, load balancers, security groups…
- RDS, VPC resources, Route 53, IAM users, etc…
- Resources created by CloudFormation are all tagged the same way
- Free naming, common tags are: Name, Environment, Team …
- Tags can be used to create Resource Groups
- Create, maintain, and view a collection of resources that share common tags
- Manage these tags using the Tag Edit

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Cost and Usage Reports

- Dive deeper into your AWS costs and usage
- The AWS Cost & Usage Report contains the most comprehensive se of AWS cost and usage data available, including additional metadata about AWS services, pricing, and reservations (e.g., Amazon EC2 Reserved Instances (RIs)).
- The AWS Cost & Usage Report lists AWS usage for each service category used by an account and its IAM users in hourly or daily line items, as well as any tags that you have activated for cost allocation purposes.
- Can be integrated with Athena, Redshift or QuickSight

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Cost and Usage Reports

	M	N	O	P	R	S	T
1	lineItem/ProductCode	lineItem/UsageType	lineItem/Operation	lineItem/AvailabilityZone	lineItem/UsageAmount	lineItem/CurrencyCode	lineItem/LineItemDescription
2	AmazonEC2	CW:AlarmMonitorUsage	Unknown		0.00134409	USD	$0.00 per alarm-month - first 10 alarms
3	AmazonS3	Requests-Tier1	ListAllMyBuckets		2	USD	$0.00 per request - PUT, COPY, POST, or LIST requests under the monthly global free tier
4	AmazonEC2	CW:AlarmMonitorUsage	Unknown		0.00134409	USD	$0.00 per alarm-month - first 10 alarms
5	AmazonEC2	APS2-EBS:VolumeUsage.gp2	CreateVolume-Gp2		0.01344086	USD	$0.00 per GB-month of General Purpose (SSD) provisioned storage under monthly free tier
6	AmazonEC2	APS2-EBS:VolumeUsage.gp2	CreateVolume-Gp2		0.01344086	USD	$0.00 per GB-month of General Purpose (SSD) provisioned storage under monthly free tier
7	AmazonEC2	USW2-BoxUsage:t2.micro	RunInstances:0002	us-west-2a	1	USD	$0.00 per Windows t2.micro instance-hour (or partial hour) under monthly free tier
8	AmazonEC2	USW2-USE1-AWS-Out-Bytes	PublicIP-Out		0.00000174	USD	$0.000 per GB - data transfer out under the monthly global free tier
9	AmazonEC2	USW2-USE1-AWS-In-Bytes	PublicIP-In		0.00000138	USD	$0.00 per GB - US West (Oregon) data transfer from US East (Northern Virginia)
10	AmazonEC2	USW2-USW1-AWS-In-Bytes	PublicIP-In		0.00000149	USD	$0.00 per GB - US West (Oregon) data transfer from US West (Northern California)
11	AmazonS3	Requests-Tier1	ListAllMyBuckets		2	USD	$0.00 per request - PUT, COPY, POST, or LIST requests under the monthly global free tier
12	AmazonEC2	USW2-DataTransfer-Out-Bytes	RunInstances		0.00038144	USD	$0.000 per GB - data transfer out under the monthly global free tier
13	AmazonEC2	USW2-USW1-AWS-Out-Bytes	PublicIP-Out		0.00000174	USD	$0.000 per GB - data transfer out under the monthly global free tier
14	AmazonEC2	USW2-DataTransfer-In-Bytes	RunInstances		0.00030951	USD	$0.000 per GB - data transfer in per month
15	AmazonEC2	USW2-BoxUsage:t2.micro	RunInstances:0002	us-west-2a	1	USD	$0.00 per Windows t2.micro instance-hour (or partial hour) under monthly free tier
16	AmazonEC2	USW2-USW1-AWS-Out-Bytes	PublicIP-Out		0.00000349	USD	$0.000 per GB - data transfer out under the monthly global free tier
17	AmazonEC2	USW2-USW1-AWS-In-Bytes	PublicIP-In		0.00000276	USD	$0.00 per GB - US West (Oregon) data transfer from US West (Northern California)
18	AmazonEC2	APS2-EBS:VolumeUsage.gp2	CreateVolume-Gp2		0.01344086	USD	$0.00 per GB-month of General Purpose (SSD) provisioned storage under monthly free tier
19	AmazonEC2	CW:AlarmMonitorUsage	Unknown		0.00134409	USD	$0.00 per alarm-month - first 10 alarms
20	AmazonEC2	USW2-BoxUsage:t2.micro	RunInstances:0002	us-west-2a	1	USD	$0.00 per Windows t2.micro instance-hour (or partial hour) under monthly free tier
21	AmazonEC2	USW2-DataTransfer-Regional-Bytes	PublicIP-Out		0.00000349	USD	$0.000 per GB - regional data transfer under the monthly global free tier
22	AmazonEC2	USW2-DataTransfer-In-Bytes	RunInstances		0.00032071	USD	$0.000 per GB - data transfer in per month
23	AmazonEC2	USW2-DataTransfer-Regional-Bytes	PublicIP-In		0.00000302	USD	$0.000 per GB - regional data transfer under the monthly global free tier
24	AmazonEC2	USW2-USE1-AWS-Out-Bytes	PublicIP-Out		0.00000174	USD	$0.000 per GB - data transfer out under the monthly global free tier
25	AmazonEC2	USW2-DataTransfer-Out-Bytes	RunInstances		0.00045736	USD	$0.000 per GB - data transfer out under the monthly global free tier
26	AmazonEC2	USW2-DataTransfer-In-Bytes	RunInstances		0.00036737	USD	$0.000 per GB - data transfer in per month
27	AmazonEC2	USW2-APN2-AWS-In-Bytes	PublicIP-In		0.00000005	USD	$0.00 per GB - US West (Oregon) data transfer from Asia Pacific (Seoul)
28	AmazonEC2	USW2-APN2-AWS-Out-Bytes	PublicIP-Out		0.00000018	USD	$0.000 per GB - data transfer out under the monthly global free tier
29	AmazonEC2	USW2-USE1-AWS-In-Bytes	PublicIP-In		0.00000153	USD	$0.00 per GB - US West (Oregon) data transfer from US East (Northern Virginia)
30	AmazonEC2	USW2-DataTransfer-Out-Bytes	RunInstances		0.00039945	USD	$0.000 per GB - data transfer out under the monthly global free tier
31	AmazonEC2	CW:AlarmMonitorUsage	Unknown		0.00134409	USD	$0.00 per alarm-month - first 10 alarms

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Cost Explorer

- Visualize, understand, and manage your AWS costs and usage over time
- Create custom reports that analyze cost and usage data.
- Analyze your data at a high level: total costs and usage across all accounts
- Or Monthly, hourly, resource level granularity
- Choose an optimal Savings Plan (to lower prices on your bill)
- Forecast usage up to 12 months based on previous usage

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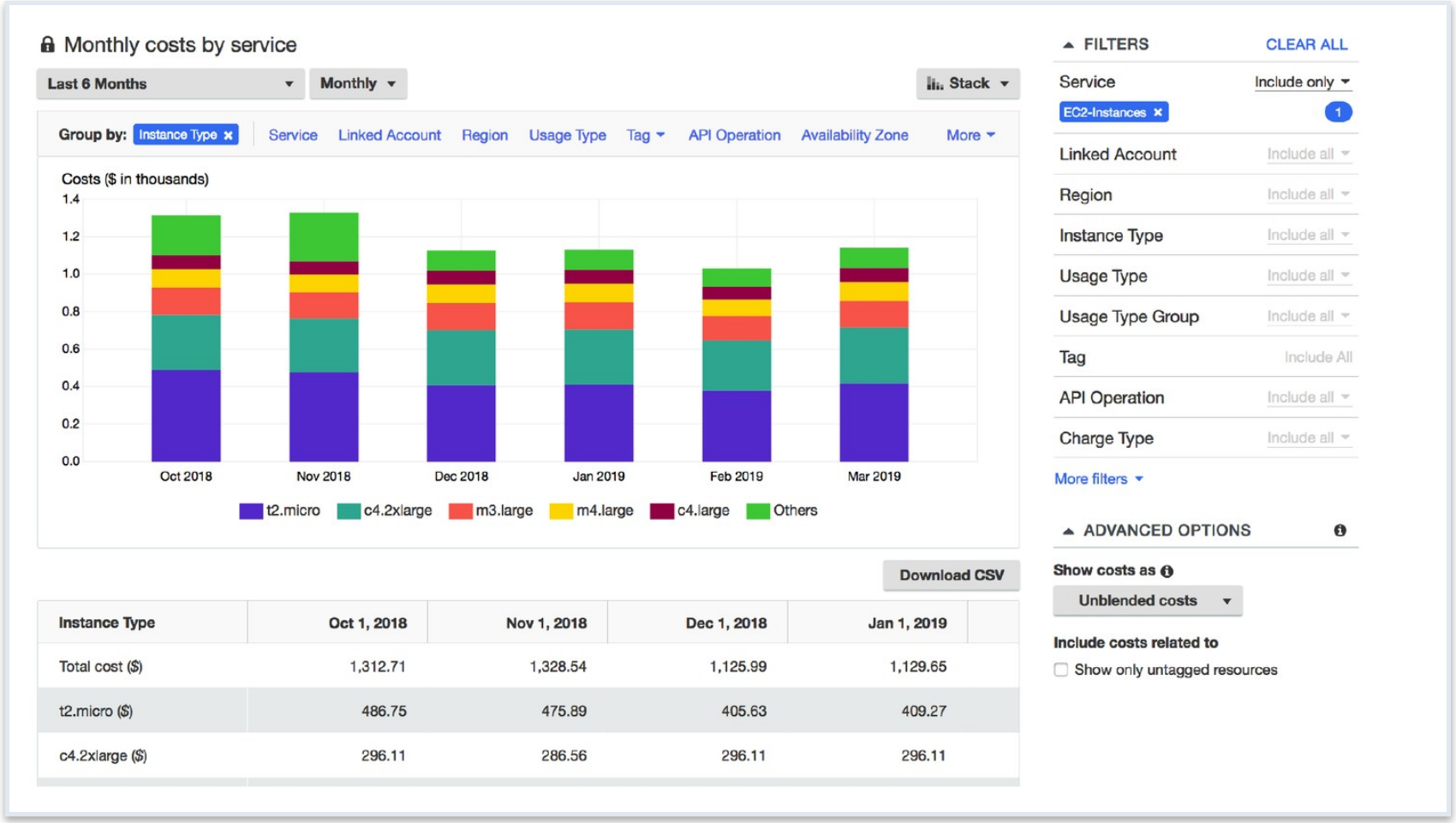
Cost Explorer – Monthly Cost by AWS Service



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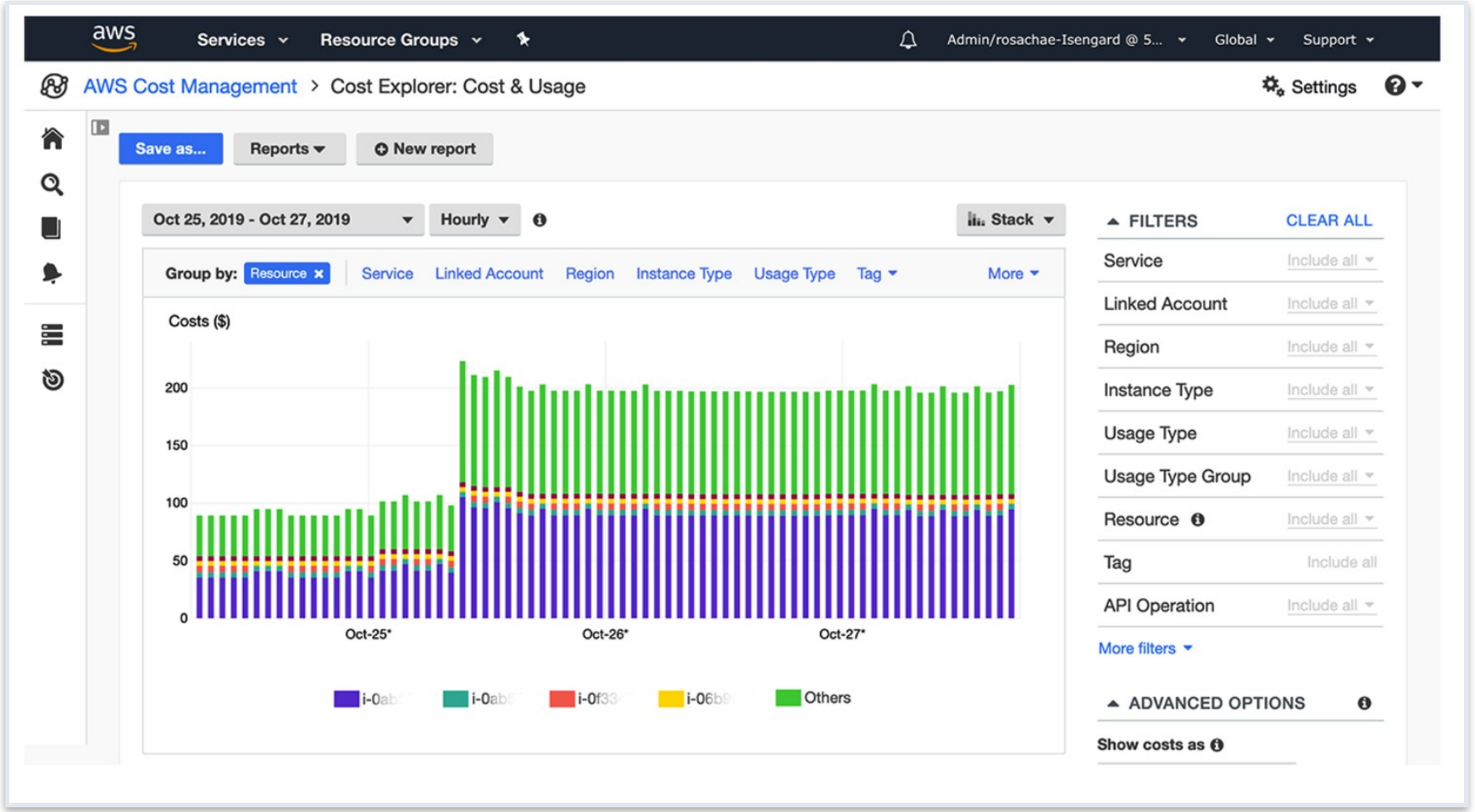
Cost Explorer– Hourly & Resource Level



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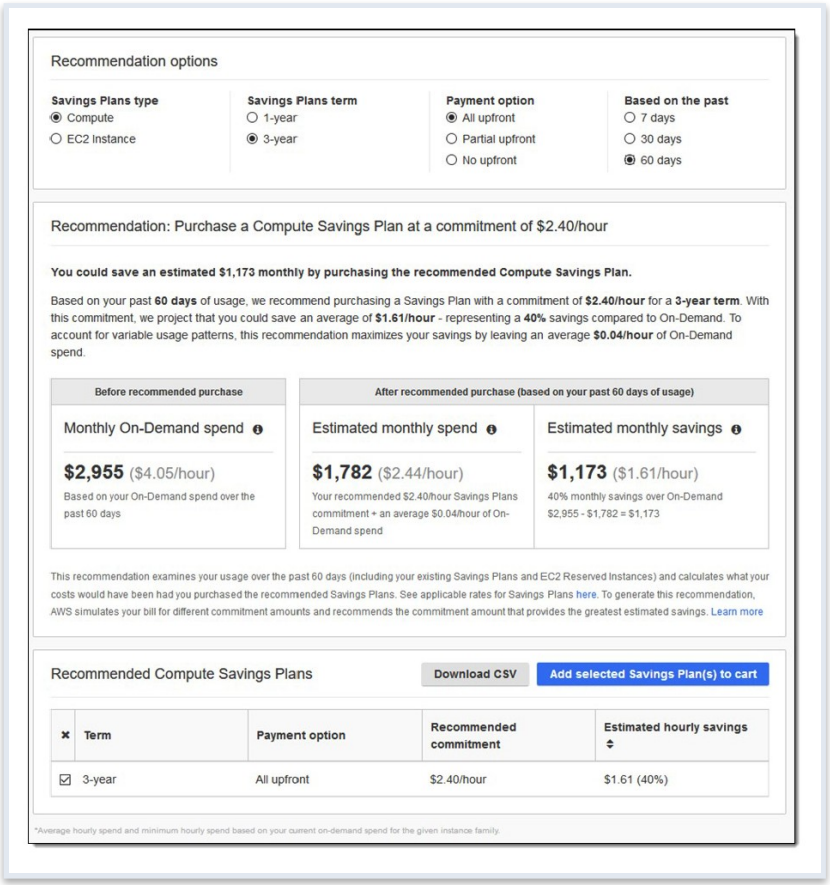
Cost Explorer – Savings Plan



Alternative to Reserved Instances

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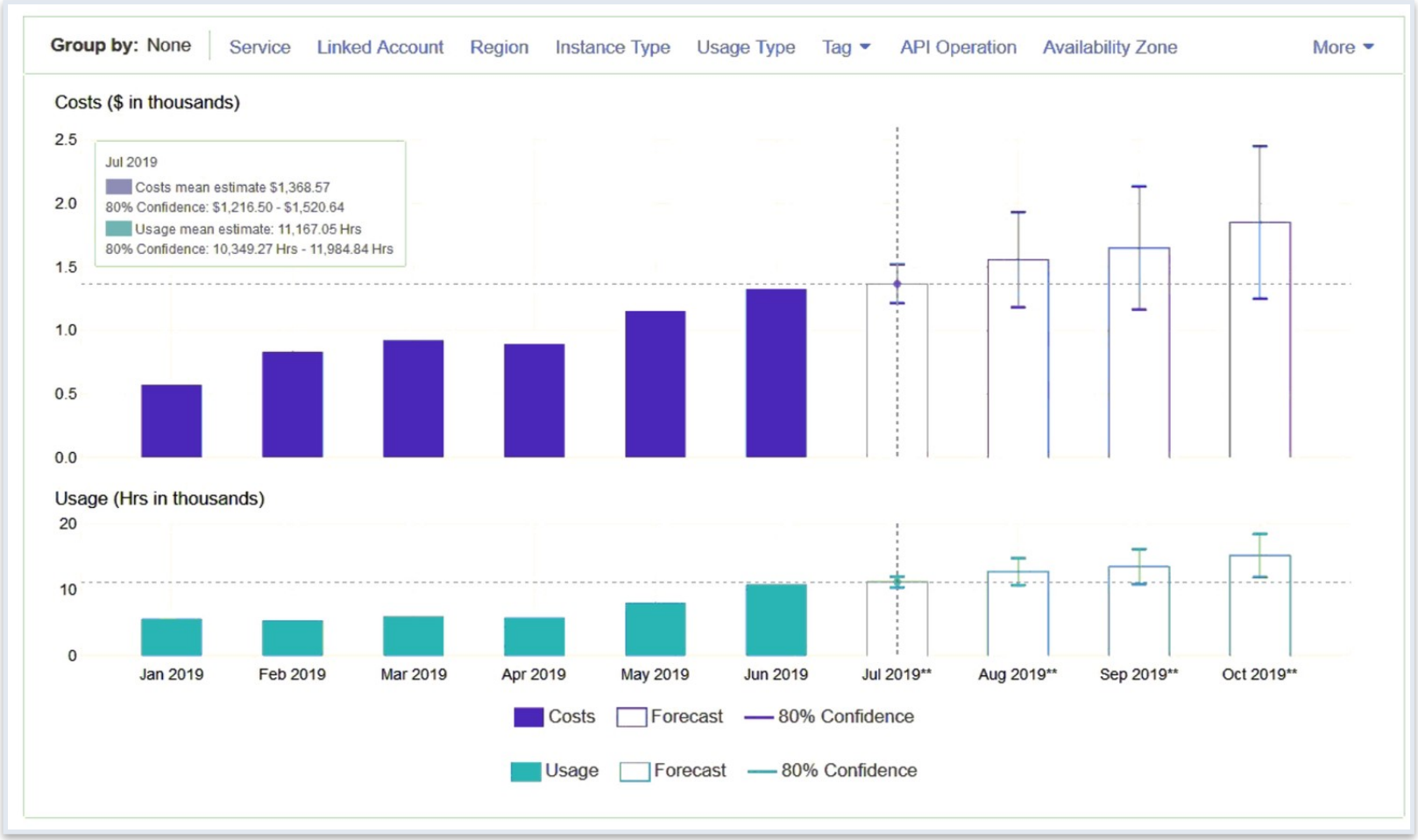
Cost Explorer – Forecast Usage



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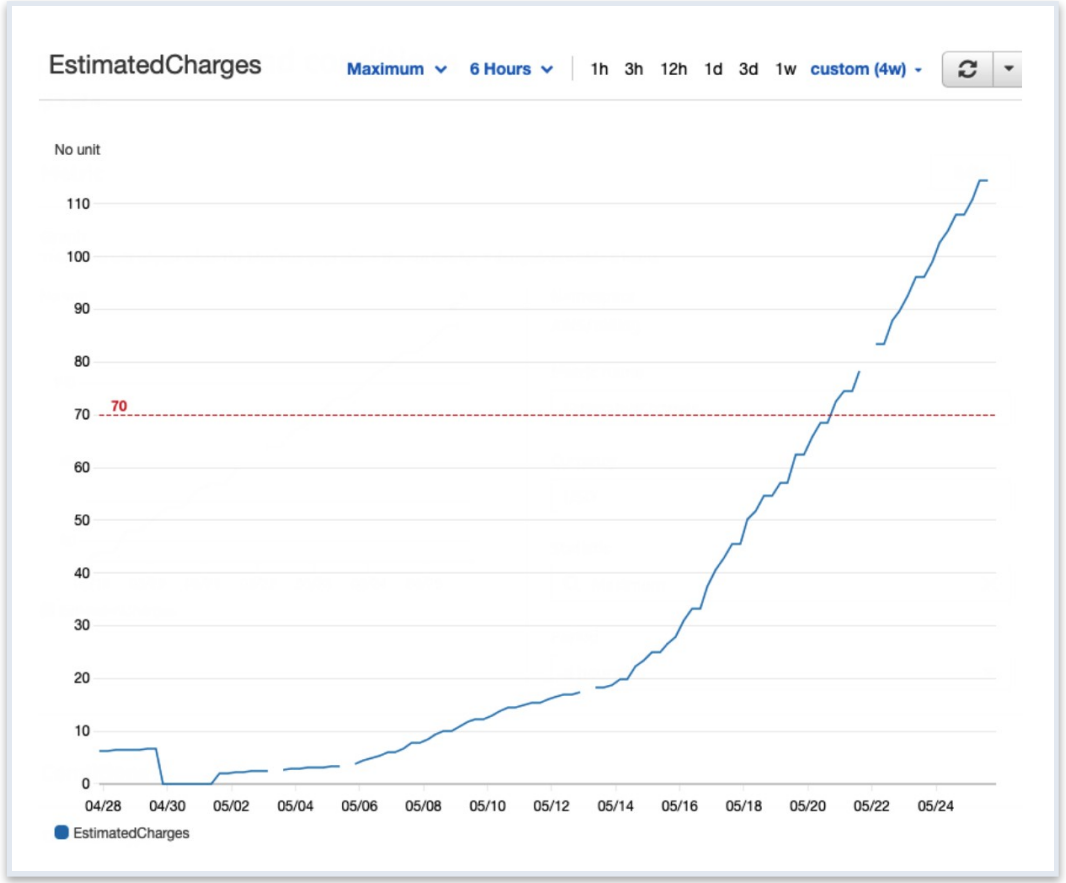
Billing Alarms in CloudWatch



Billing data metric is stored in CloudWatch us-east-1

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AWS Budgets

- Create budget and send alarms when costs exceeds the budget
- 4 types of budgets: Usage, Cost, Reservation, Savings Plans
- For Reserved Instances (RI)
- Track utilization
- Supports EC2, ElastiCache, RDS, Redshift
- Up to 5 SNS notifications per budget
- Can filter by: Service, Linked Account, Tag, Purchase Option, Instance Type, Region, Availability Zone, API Operation, etc…
- Same options as AWS Cost Explorer!
- •2 budgets are free, then $0.02/day/budget

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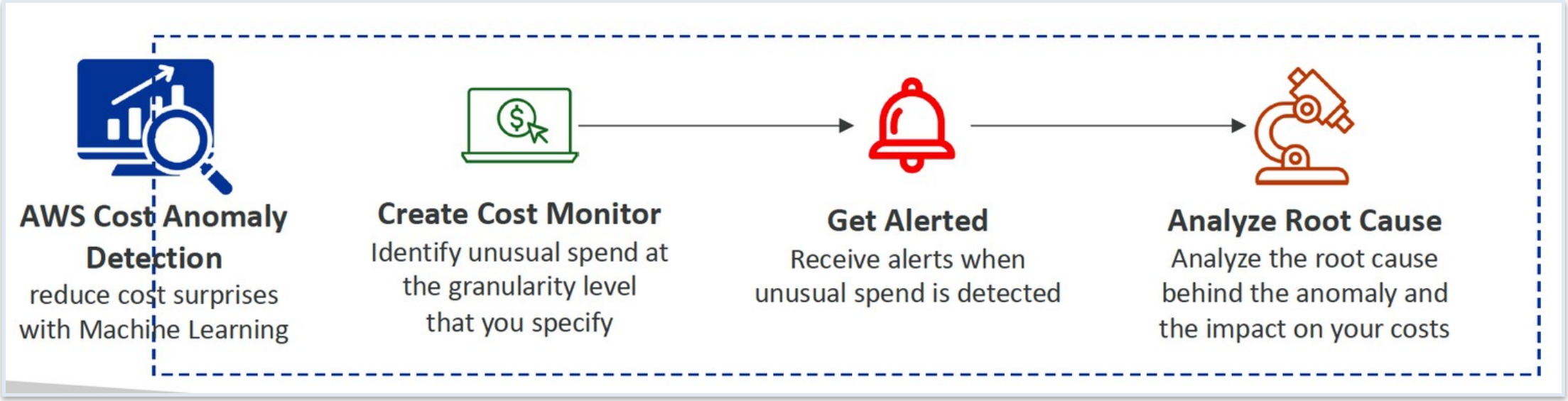
AWS Cost Anomaly Detection



Continuously monitor your cost and usage using ML to detect unusual spends

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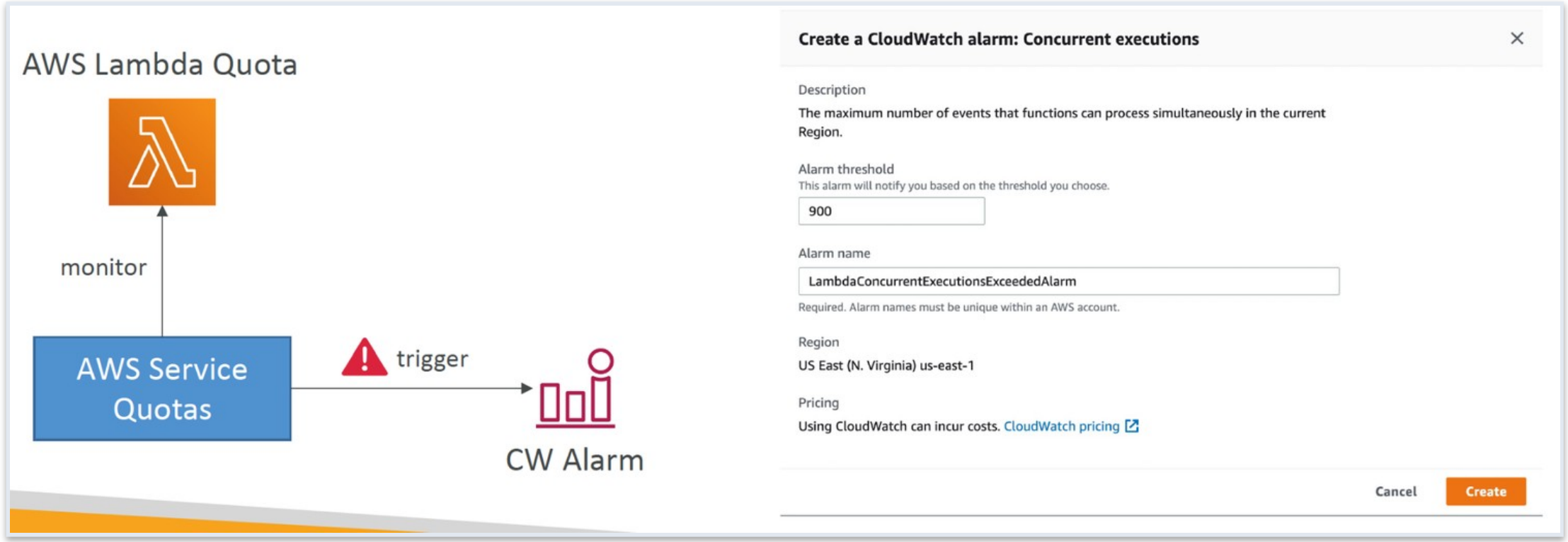
AWS Service Quotas



Notify you when you're close to a service quota value threshold

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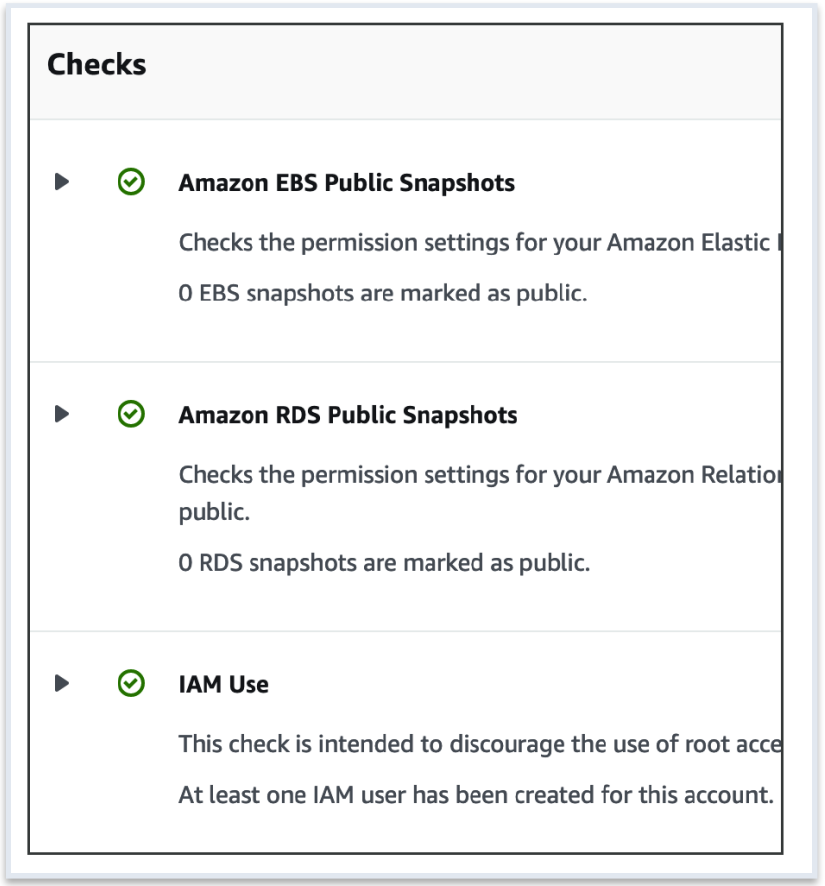
Trusted Advisor



No need to install anything – high level AWS account assessment

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AWS Support Plans Pricing

Developer	Business	Enterprise On-Ramp	Enterprise
Greater of $29.00 - or - 3% of monthly AWS charges	Greater of $100.00 - or - 10% of monthly AWS charges for the first $0--$10K 7% of monthly AWS charges from $10K--$80K 5% of monthly AWS charges from $80K--$250K 3% of monthly AWS charges over $250K	Greater of $5,500.00 - or - 10% of monthly AWS charges	Greater of $15,000.00 - or - 10% of monthly AWS charges for the first $0--$150K 7% of monthly AWS charges from $150K--$500K 5% of monthly AWS charges from $500K--$1M 3% of monthly AWS charges over $1M

Basic Support: free

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AWS Basic Support Plan

- Customer Service & Communities - 24x7 access to customer service, documentation, whitepapers, and support forums.
- AWS Trusted Advisor - Access to the 7 core Trusted Advisor checks and guidance to provision your resources following best practices to increase performance and improve security.
- AWS Personal Health Dashboard - A personalized view of the health of AWS services, and alerts when your resources are impacted.

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AWS Developer Support Plan

1. All Basic Support Plan +
2. Business hours email access to Cloud Support Associates
3. Unlimited cases / 1 primary contact
4. Case severity / response times:
5. General guidance: < 24 business hours
6. System impaired: < 12 business hours

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AWS Business Support Plan (24/7)

- Intended to be used if you have production workloads
- Trusted Advisor – Full set of checks + API access
- 24x7 phone, email, and chat access to Cloud Support Engineers
- Unlimited cases / unlimited contacts
- Access to Infrastructure Event Management for additional fee.
- Case severity / response times:
- General guidance: < 24 business hours
- System impaired: < 12 business hours
- Production system impaired: < 4 hours
- Production system down: < 1 hour

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AWS Enterprise On-Ramp Support Plan (24/7)

- Intended to be used if you have production or business critical workloads
- All of Business Support Plan +
- Access to a pool of Technical Account Managers (TAM)
- Concierge Support Team (for billing and account best practices)
- Infrastructure Event Management, Well-Architected & Operations Reviews
- Case severity / response times:
- …
- Production system impaired: < 4 hours
- Production system down: < 1 hour
- Business-critical system down: < 30 minutes

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AWS Enterprise Support Plan (24/7)

- Intended to be used if you have mission critical workloads
- All of Business Support Plan +
- Access to a designated Technical Account Manager (TAM)
- Concierge Support Team (for billing and account best practices)
- Infrastructure Event Management, Well-Architected & Operations Reviews
- Case severity / response times:
- …
- Production system impaired: < 4 hours
- Production system down: < 1 hour
- Business-critical system down: < 15 minutes

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Account Best Practices – Summary

- Operate multiple accounts using Organizations
- Use SCP (service control policies) to restrict account power
- Easily setup multiple accounts with best-practices with AWS Control Tower
- Use Tags & Cost Allocation Tags for easy management & billing
- IAM guidelines: MFA, least-privilege, password policy, password rotation
- Config to record all resources configurations & compliance over time
- CloudFormation to deploy stacks across accounts and regions
- Trusted Advisor to get insights, Support Plan adapted to your needs
- Send Service Logs and Access Logs to S3 or CloudWatch Logs
- CloudTrail to record API calls made within your account
- If your Account is compromised: change the root password, delete and rotate all passwords / keys, contact the AWS support
- Allow users to create pre-defined stacks defined by admins using AWS Service Catalog

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Billing and Costing Summary

- Compute Optimizer: recommends resources' configurations to reduce cost
- Pricing Calculator: cost of services on AWS
- Billing Dashboard: high level overview + free tier dashboard
- Cost Allocation Tags: tag resources to create detailed reports
- Cost and Usage Reports: most comprehensive billing dataset
- Cost Explorer: View current usage (detailed) and forecast usage
- Billing Alarms: in us-east-1 – track overall and per-service billing
- Budgets: more advanced – track usage, costs, RI, and get alerts
- Savings Plans: easy way to save based on long-term usage of AWS
- Cost Anomaly Detection: detect unusual spends using Machine Learning
- Service Quotas: notify you when you're close to service quota threshold

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AWS Architecting & Ecosystem

- AWS Architecting & Ecosystem

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Well Architected Framework

- General Guiding Principles
- Stop guessing your capacity needs
- Test systems at production scale
- Automate to make architectural experimentation easier
- Allow for evolutionary architectures
- Design based on changing requirements
- Drive architectures using data
- Improve through game days
- Simulate applications for flash sale days

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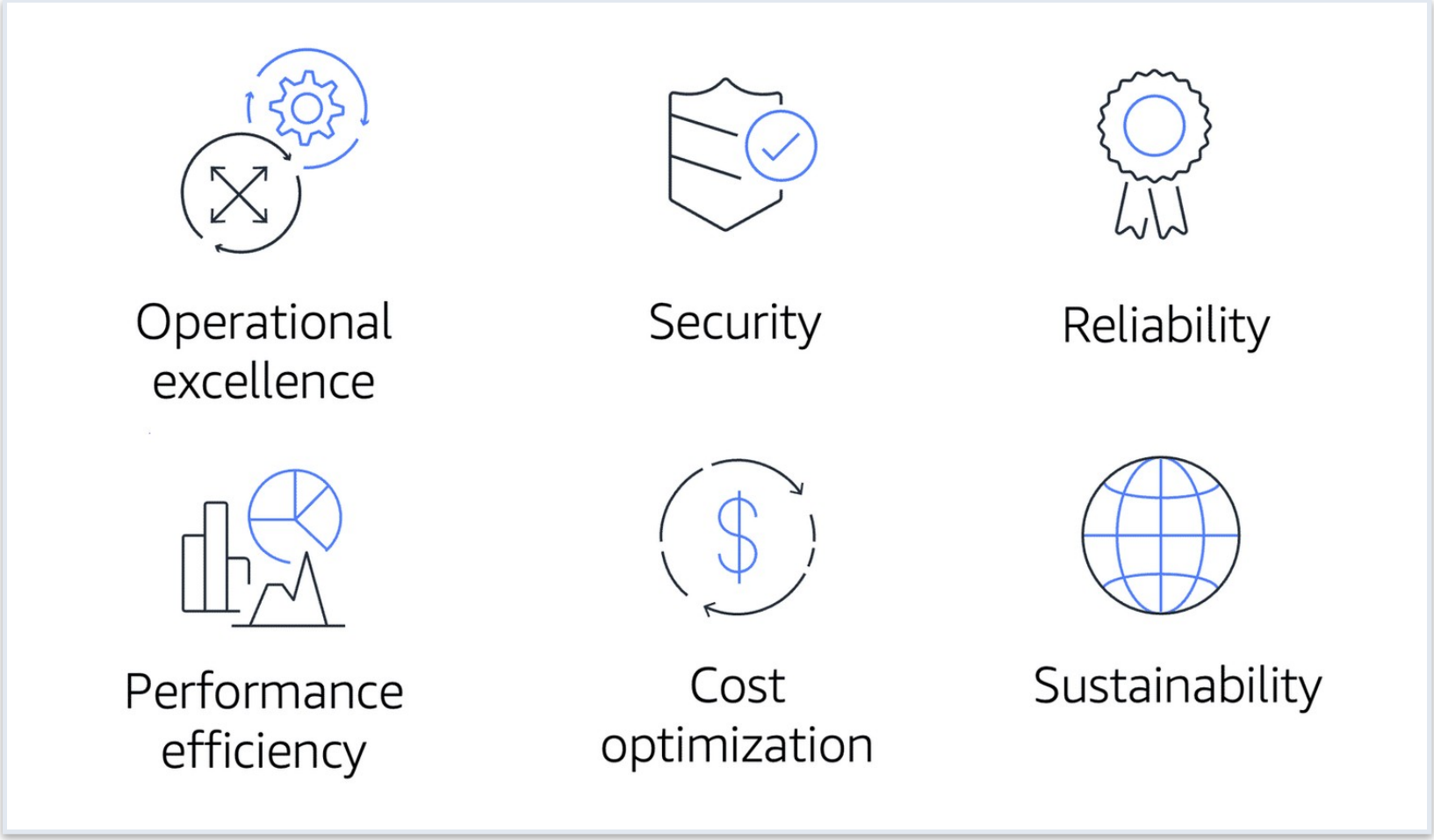
The AWS Well-Architected Framework



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AWS Cloud Best Practices – Design Principles

- Scalability: vertical & horizontal
- Disposable Resources: servers should be disposable & easily configured
- Automation: Serverless, Infrastructure as a Service, Auto Scaling…
- Loose Coupling:
- Monolith are applications that do more and more over time, become bigger
- Break it down into smaller, loosely coupled components
- A change or a failure in one component should not cascade to other components
- Services, not Servers:
- Don’t use just EC2
- Use managed services, databases, serverless, etc !

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Well Architected Framework 6 Pillars

- Operational Excellence
- Security
- Reliability
- Performance Efficiency
- Cost Optimization
- Sustainability
- They are not something to balance, or trade-offs, they’re a synergy

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Operational Excellence

- Includes the ability to run and monitor systems to deliver business value and to continually improve supporting processes and procedures
- Design Principles
- Perform operations as code - Infrastructure as code
- Make frequent, small, reversible changes - So that in case of any failure, you can reverse it
- Refine operations procedures frequently - And ensure that team members are familiar with it
- Anticipate failure
- Learn from all operational failures
- Use managed services - to reduce operational burden
- Implement observability for actionable insights - performance, reliability, cost

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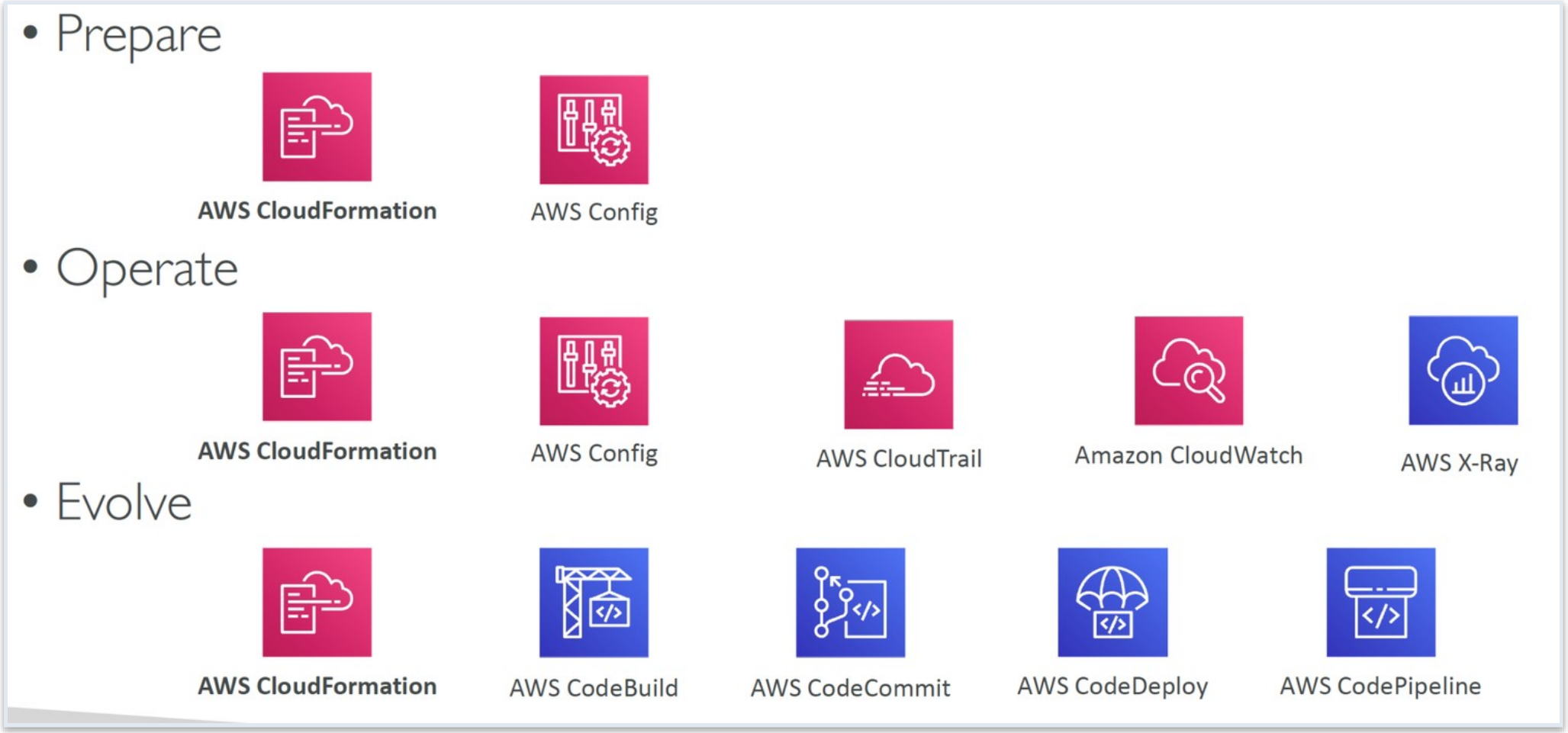
Operational Excellence AWS Services



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Security

- Includes the ability to protect information, systems, and assets while delivering business value through risk assessments and mitigation strategies
- Design Principles
- Implement a strong identity foundation - Centralize privilege management and reduce (or even
- eliminate) reliance on long-term credentials - Principle of least privilege - IAM
- Enable traceability - Integrate logs and metrics with systems to automatically respond and take
- action
- Apply security at all layers - Like edge network, VPC, subnet, load balancer, every instance,
- operating system, and application
- Automate security best practices
- Protect data in transit and at rest - Encryption, tokenization, and access control
- Keep people away from data - Reduce or eliminate the need for direct access or manual
- processing of data

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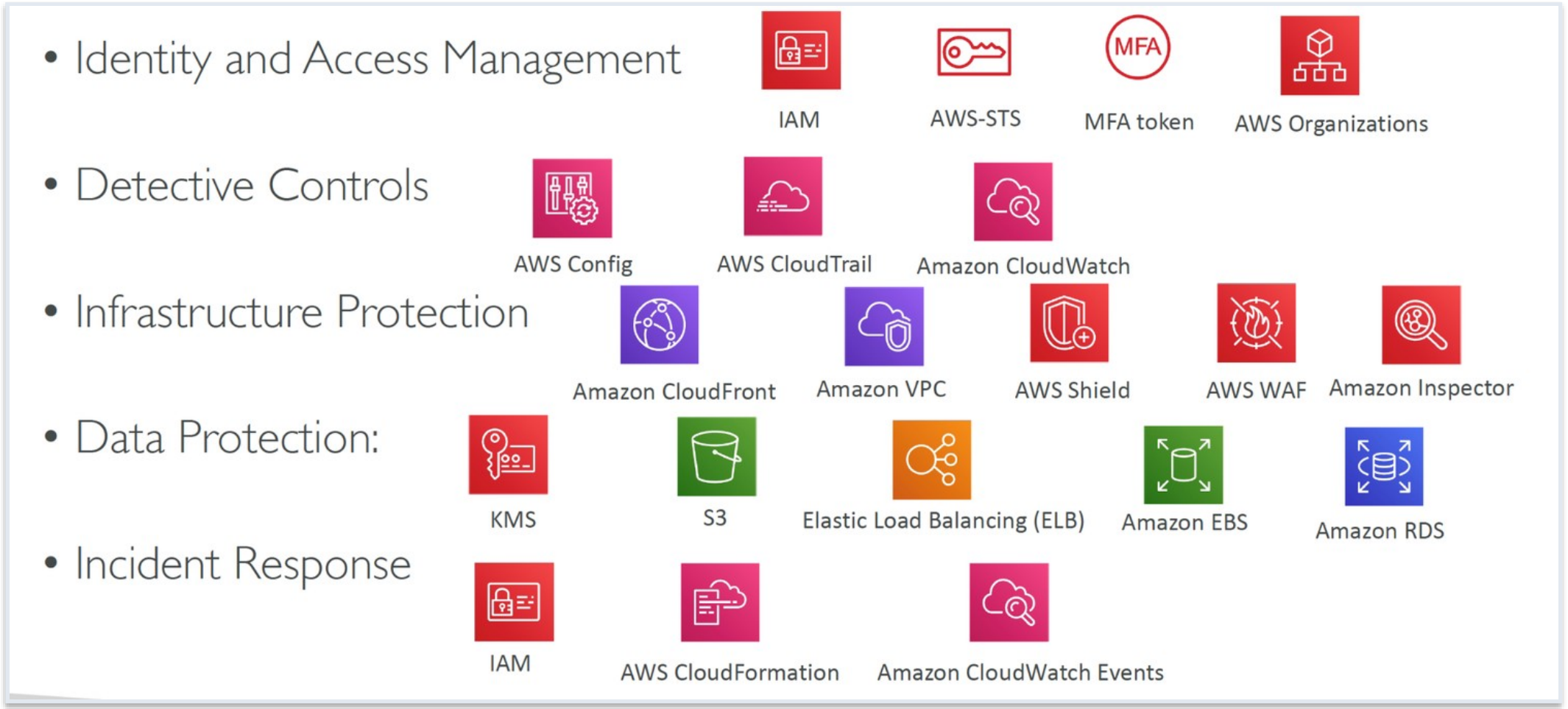
Security AWS Services



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Reliability

- Ability of a system to recover from infrastructure or service disruptions, dynamically acquire computing resources to meet demand, and mitigate disruptions such as misconfigurations or transient network issues
- Design Principles
- Test recovery procedures - Use automation to simulate different failures or to recreate scenarios that led to failures before
- Automatically recover from failure - Anticipate and remediate failures before the occur
- Scale horizontally to increase aggregate system availability - Distribute requests across multiple, smaller resources to ensure that they don't share a common point of failure
- Stop guessing capacity - Maintain the optimal level to satisfy demand without over or under provisioning - Use Auto Scaling
- Manage change in automation - Use automation to make changes to infrastructure

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Reliability AWS Services

- Foundations

IAM

Amazon VPC

Service Quotas

AWS Trusted Advisor

- Change Management

AWS Auto Scaling

Amazon CloudWatch

AWS CloudTrail

AWS Config

- Failure Management

Backups

AWS CloudFormation

Amazon S3

Amazon S3 Glacier

Amazon Route 53

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Performance Efficiency

- Includes the ability to use computing resources efficiently to meet system requirements, and to maintain that efficiency as demand changes and technologies evolve
- Design Principles
- Democratize advanced technologies - Advance technologies become services and hence you can focus more on product development
- Go global in minutes - Easy deployment in multiple regions
- Use serverless architectures - Avoid burden of managing servers
- Experiment more often - Easy to carry out comparative testing
- Mechanical sympathy - Be aware of all AWS services

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Performance Efficiency AWS Services

- Selection
- Review
- Monitoring
- Tradeoffs

AWS Auto Scaling

AWS Lambda

Amazon Elastic Block Store (EBS)

Amazon Simple Storage Service (S3)

Amazon RDS

AWS CloudFormation

AWS News Blog

Amazon CloudWatch

AWS Lambda

Amazon RDS

Amazon ElastiCache

AWS Snowball

Amazon CloudFront

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Cost Optimization

- Includes the ability to run systems to deliver business value at the lowest price point
- Design Principles
- Adopt a consumption mode - Pay only for what you use
- Measure overall efficiency - Use CloudWatch
- Stop spending money on data center operations - AWS does the infrastructure part and enables customer to focus on organization projects
- Analyze and attribute expenditure - Accurate identification of system usage and costs, helps measure return on investment (ROI) - Make sure to use tags
- Use managed and application level services to reduce cost of ownership - As managed services operate at cloud scale, they can offer a lower cost per transaction or service

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Cost Optimization AWS Services

- Expenditure Awareness

AWS Budgets

AWS Cost and Usage Report

AWS Cost Explorer

Reserved Instance Reporting

- Cost-Effective Resources

Spot instance

Reserved instance

Amazon S3 Glacier

- Matching supply and demand

AWS Auto Scaling

AWS Lambda

- Optimizing Over Time

AWS Trusted Advisor

AWS Cost and Usage Report

AWS News Blog

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Sustainability

- The sustainability pillar focuses on minimizing the environmental impacts of running cloud workloads.
- Design Principles
- Understand your impact – establish performance indicators, evaluate improvements
- Establish sustainability goals – Set long-term goals for each workload, model return on investment (ROI)
- Maximize utilization – Right size each workload to maximize the energy efficiency of the underlying hardware and minimize idle resources.
- Anticipate and adopt new, more efficient hardware and software offerings – and design for flexibility to adopt new technologies over time.
- Use managed services – Shared services reduce the amount of infrastructure; Managed services help automate sustainability best practices as moving infrequent accessed data to cold storage and adjusting compute capacity.
- Reduce the downstream impact of your cloud workloads – Reduce the amount of energy or resources required to use your services and reduce the need for your customers to upgrade their devices

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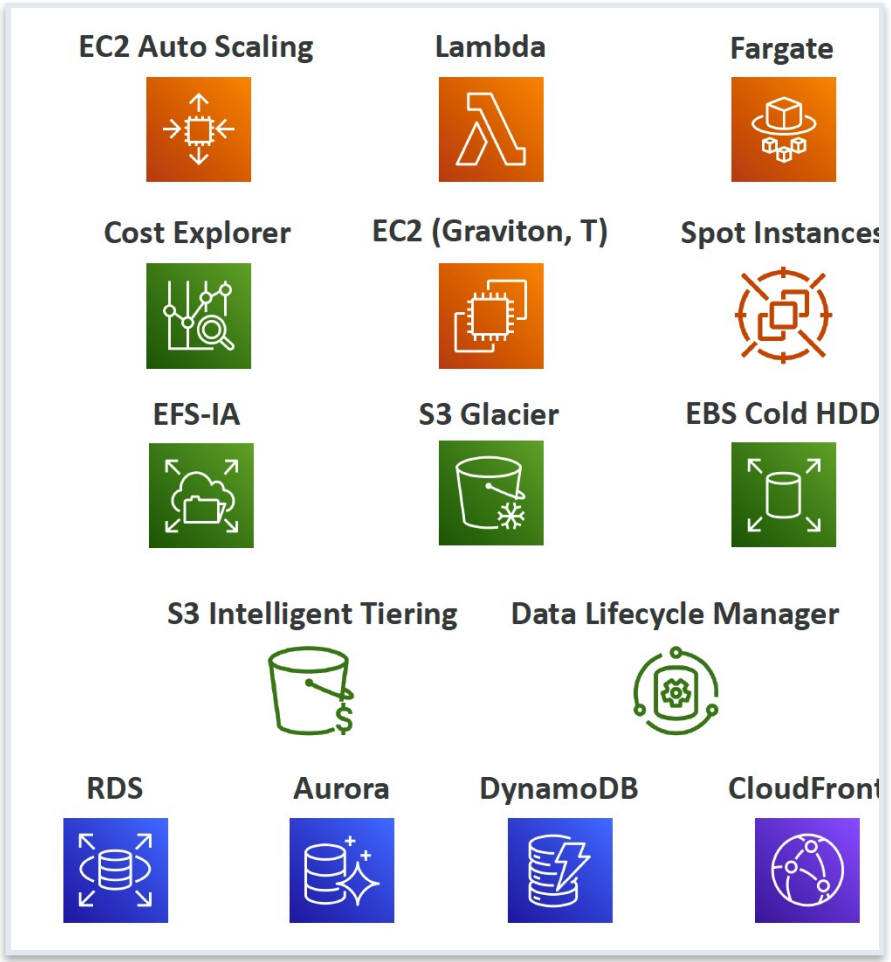
Sustainability AWS Services



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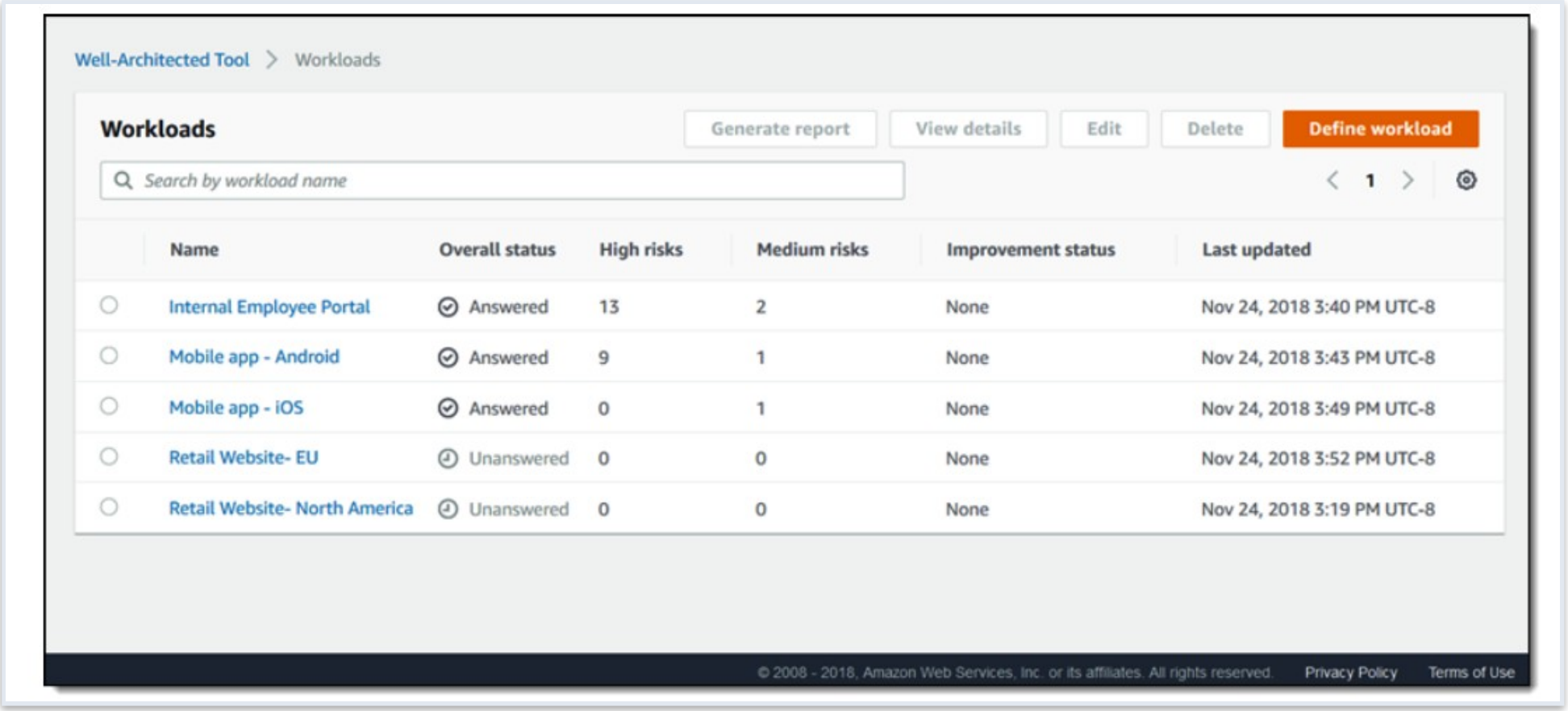
AWS Well-Architected Tool



Free tool to review your architectures against the 6 pillars Well-Architected

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AWS Cloud Adoption Framework

Helps you build and then execute a comprehensive plan for your digital transformation through innovative use of AWS

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CAF Perspectives and Foundational Capabilities Business Capabilities

- Business Perspective helps ensure that your cloud investments
- accelerate your digital transformation ambitions and business outcomes.
- People Perspective serves as a bridge between technology and
- business, accelerating the cloud journey to help organizations more
- rapidly evolve to a culture of continuous growth, learning, and where change becomes business-as-normal, with focus on culture,
- organizational structure, leadership, and workforce.
- Governance Perspective helps you orchestrate your cloud initiatives
- while maximizing organizational benefits and minimizing transformation related risks.

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CAF Perspectives and Foundational Capabilities Business Capabilities

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CAF Perspectives and Foundational Capabilities Technical Capabilities

- Platform Perspective helps you build an enterprise-grade, scalable, hybrid cloud platform; modernize existing workloads; and implement new cloud-native solutions.
- Security Perspective helps you achieve the confidentiality, integrity, and availability of your data and cloud workloads.
- Operations Perspective helps ensure that your cloud services are delivered at a level that meets the needs of your business.

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CAF Perspectives and Foundational

Technical Capabilities

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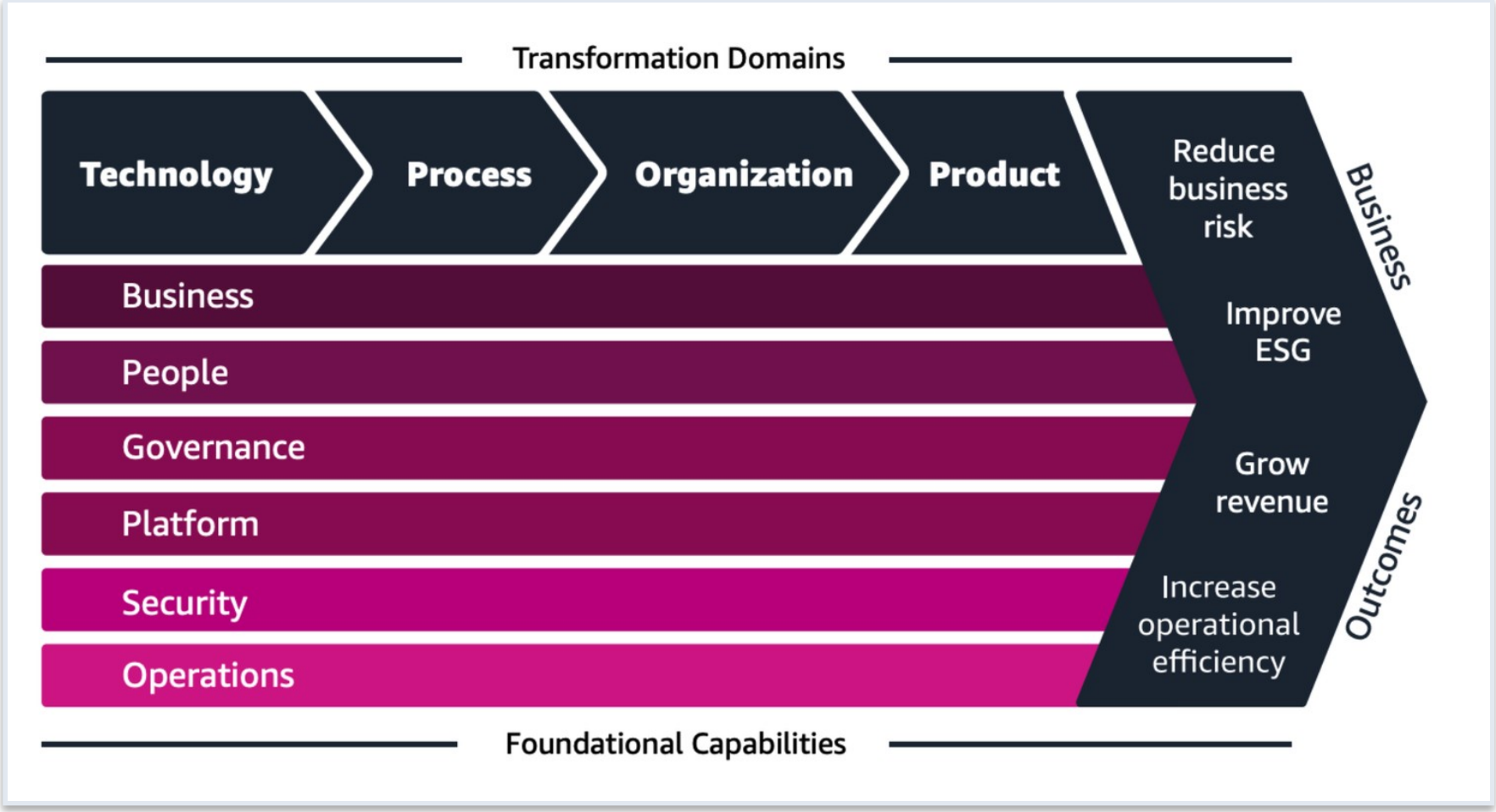
Cloud transformation value chain



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AWS CAF – Transformation Domains

- Technology - using the cloud to migrate and modernize legacy infrastructure, applications, data and analytics platforms…
- Process - digitizing, automating, and optimizing your business operations leveraging new data and analytics platforms to create actionable insights using machine learning (ML) to improve your customer service experience…
- Organization - Reimagining your operating model
- Organizing your teams around products and value streams
- Leveraging agile methods to rapidly iterate and evolve
- Product – reimagining your business model by creating new value propositions (products & services) and revenue models

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AWS CAF – Transformation Phases

- Envision – demonstrate how the Cloud will accelerate business outcomes by identifying transformation opportunities and create a
- foundation for your digital transformation
- Align – identify capability gaps across the 6 AWS CAF Perspectives which results in an Action Plan
- Launch – build and deliver pilot initiatives in production and demonstrate incremental business value
- Scale – expand pilot initiatives to the desired scale while realizing th desired business benefits

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AWS Right Sizings

- EC2 has many instance types, but choosing the most powerful instance type isn't the best choice, because the cloud is elastic
- Right sizing is the process of matching instance types and sizes to your workload performance and capacity requirements at the lowest possible cost
- Scaling up is easy so always start small
- It's also the process of looking at deployed instances and identifying opportunities to eliminate or downsize without compromising capacity or other requirements, which results in lower costs
- It's important to Right Size…
- before a Cloud Migration
- continuously after the cloud onboarding process (requirements change over time)
- CloudWatch, Cost Explorer, Trusted Advisor, 3rd party tools can help

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AWS Ecosystem – Free resources

- AWS Blogs: https://aws.amazon.com/blogs/aws/
- AWS Forums (community): https://forums.aws.amazon.com/index.jspa
- AWS Whitepapers & Guides: https://aws.amazon.com/whitepapers
- AWS Partner Solutions (formerly Quick Starts):
- https://aws.amazon.com/quickstart/
- Automated, gold-standard deployments in the AWS Cloud
- Build your production environment quickly with templates
- Example: WordPress on AWS https://fwd.aws/P3yyv?did=qs_card&trk=qs_card
- Leverages CloudFormation
- AWS Solutions: https://aws.amazon.com/solutions/
- Vetted Technology Solutions for the AWS Cloud
- Example - AWS Landing Zone: secure, multi-account AWS environment https://aws.amazon.com/solutions/implementations/aws-landing-zone/

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AWS Ecosystem - AWS Support

DEVELOPER	• Business hours email access to Cloud Support Associates • General guidance: < 24 business hours • System impaired: < 12 business hours
BUSINESS	• 24x7 phone, email, and chat access to Cloud Support Engineers • Production system impaired: < 4 hours • Production system down: < 1 hour
ENTERPRISE	• Access to a Technical Account Manager (TAM) • Concierge Support Team (for billing and account best practices) • Business-critical system down: < 15 minutes

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AWS Marketplace

- Digital catalog with thousands of software listings from independent software vendors (3rd party)
- Example:
- Custom AMI (custom OS, firewalls, technical solutions…)
- CloudFormation templates
- Software as a Service
- Containers
- If you buy through the AWS Marketplace, it goes into your AWS bill
- You can sell your own solutions on the AWS Marketplace

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AWS Professional Services & Partner Network

- The AWS Professional Services organization is a global team of experts
- They work alongside your team and a chosen member of the APN
- APN = AWS Partner Network
- APN Technology Partners: providing hardware, connectivity, and software
- APN Consulting Partners: professional services firm to help build on AWS
- APN Training Partners: find who can help you learn AWS
- AWS Competency Program: AWS Competencies are granted to APN
- Partners who have demonstrated technical proficiency and proven
- customer success in specialized solution areas.
- AWS Navigate Program: help Partners become better Partners

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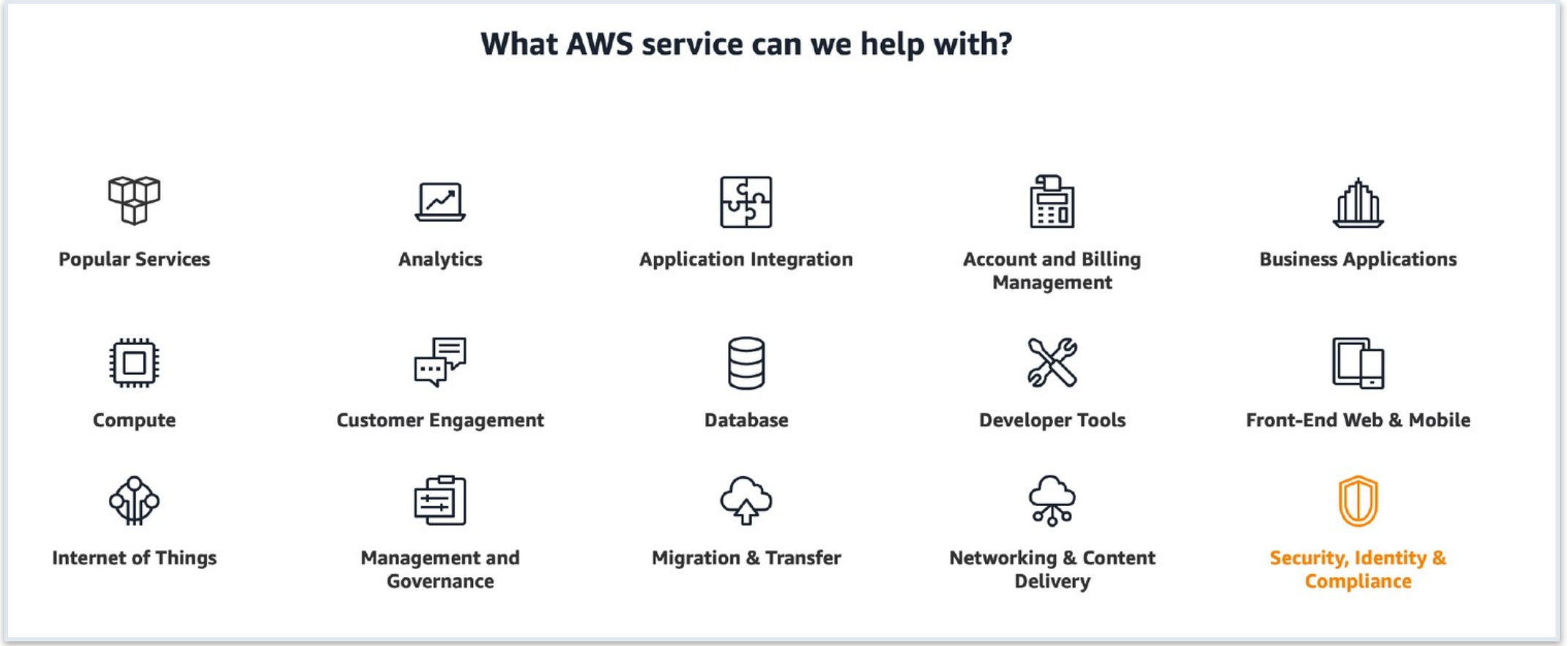
AWS Knowledge Center



https://repost.aws/knowledge-center

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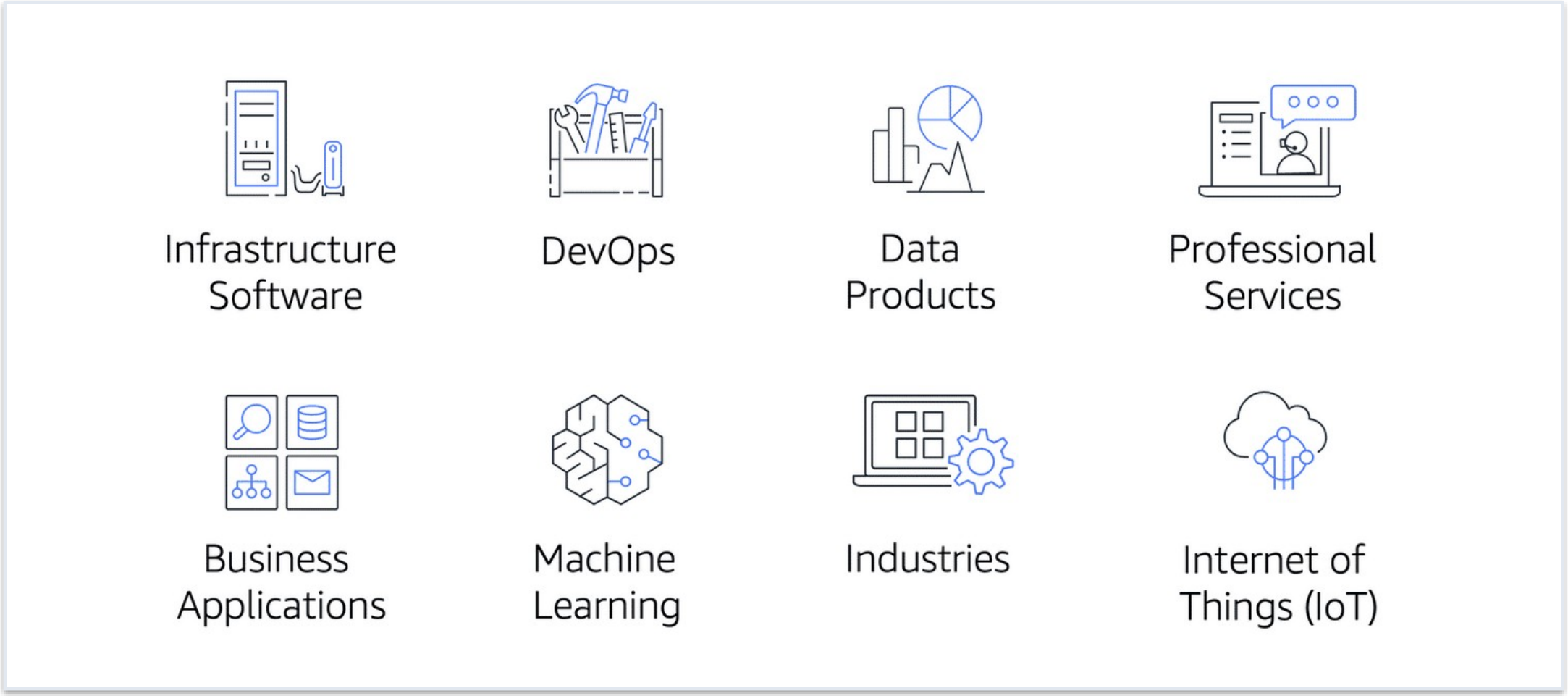
AWS Marketplace categories



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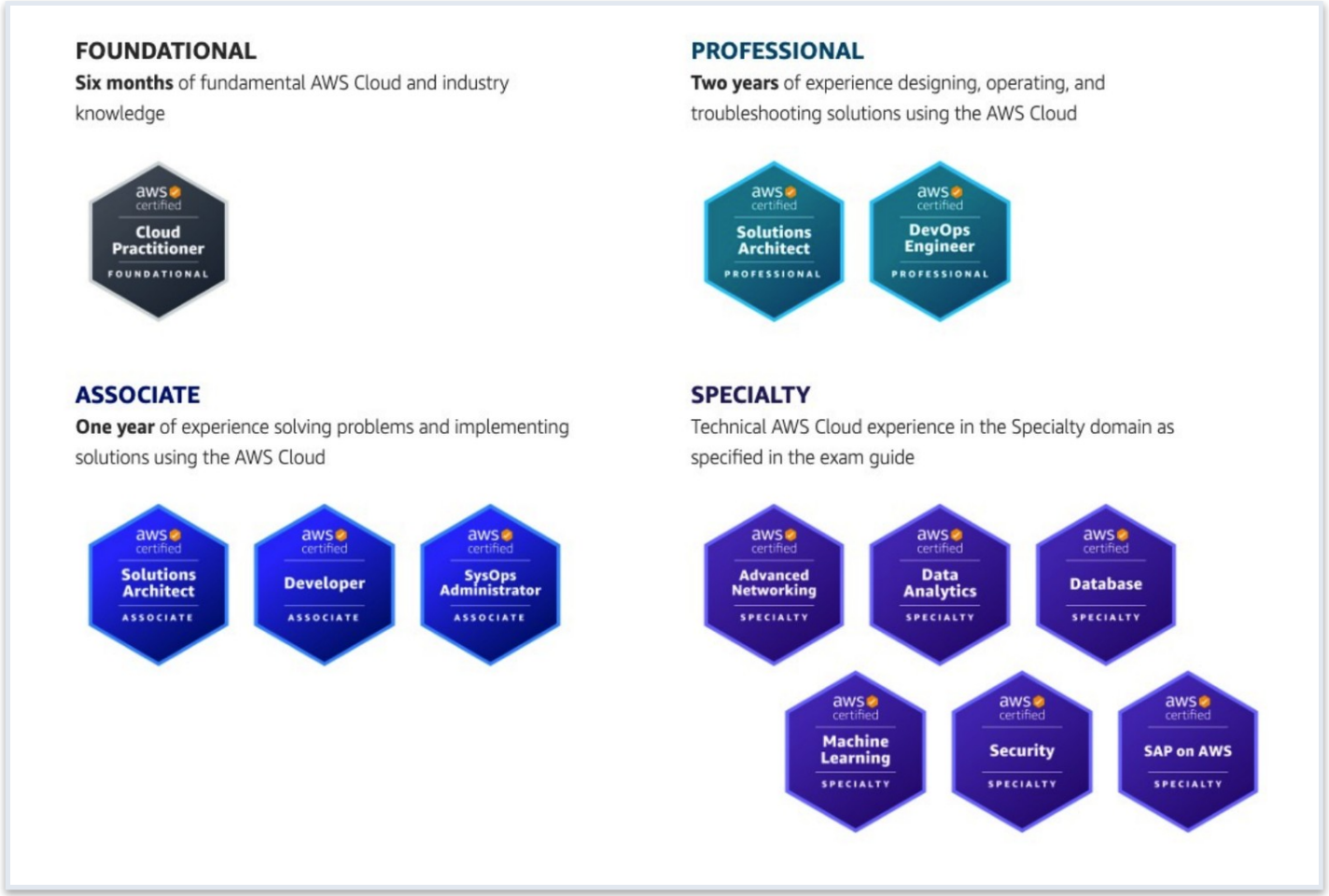
AWS Certification Journey



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AWS Certification Paths – Architecture

Architecture

Solutions Architect

Design, develop, and manage cloud infrastructure and assets, work with DevOps to migrate applications to the cloud

Dive Deep

Architecture

Application Architect

Design significant aspects of application architecture including user interface, middleware, and infrastructure, and ensure enterprise-wide scalable, reliable, and manageable systems

Dive Deep

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AWS Certification Paths – Operations

Operations

Systems Administrator

Install, upgrade, and maintain computer components and software, and integrate automation processes

Dive Deep

Operations

Cloud Engineer

Implement and operate an organization's networked computing infrastructure and Implement security systems to maintain data safety

Dive Deep

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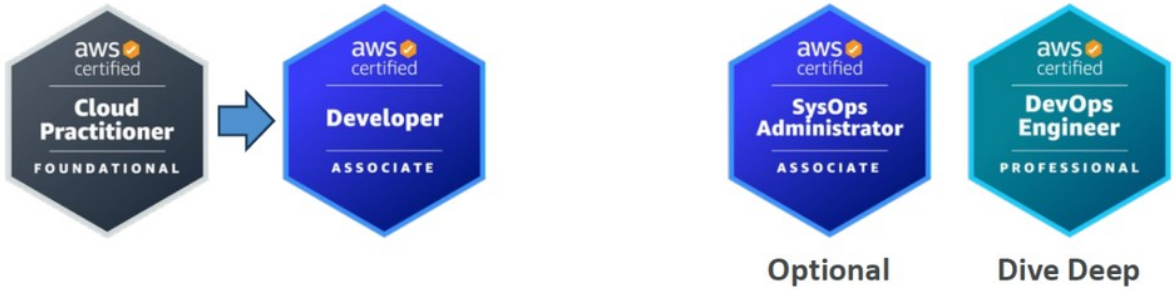
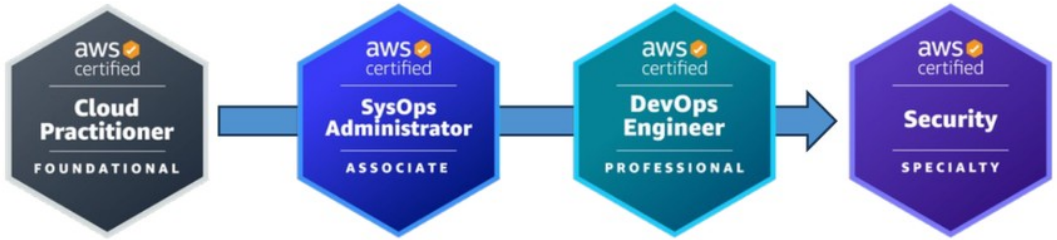
AWS Certification Paths – DevOps

DevOps
Test Engineer
Embed testing and quality best practices for software development from design to release, throughout the product life cycle

DevOps
Cloud DevOps Engineer
Design, deployment, and operations of large-scale global hybrid cloud computing environment, advocating for end-to-end automated CI/CD DevOps pipelines



DevOps
DevSecOps Engineer
Accelerate enterprise cloud adoption while enabling rapid and stable delivery of capabilities using CI/CD principles, methodologies, and technologies



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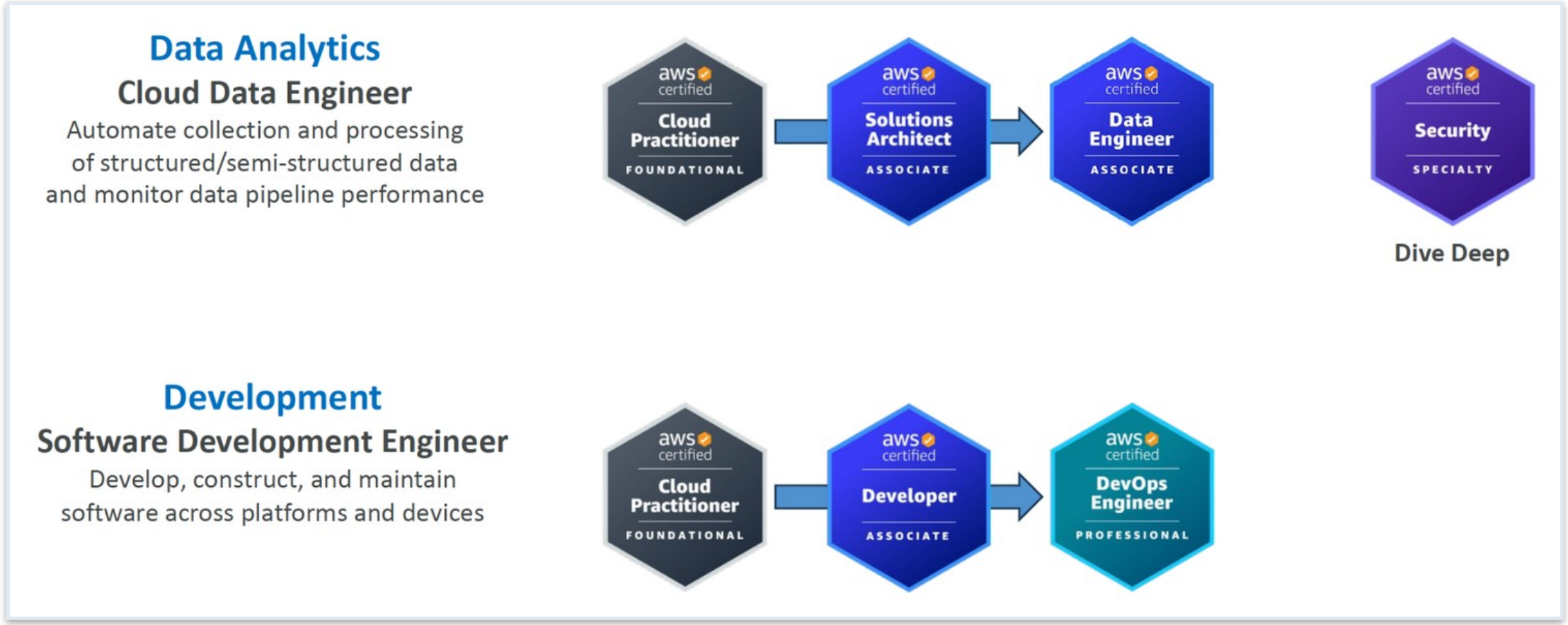
AWS Certification Paths – Data Analytics &



Development

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AWS Certification Paths – Networking & AI/ML

Networking

Network Engineer

Design and implement computer and information networks, such as local area networks (LAN), wide area networks (WAN), intranets, extranets, etc.

Dive Deep

AI/ML

Machine Learning Engineer

Research, build, and design artificial intelligence (AI) systems to automate predictive models, and design machine learning systems, models, and schemes

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Practice Exams

Access the Practice Exams from the Google Classroom

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SUMMARY

Q&A

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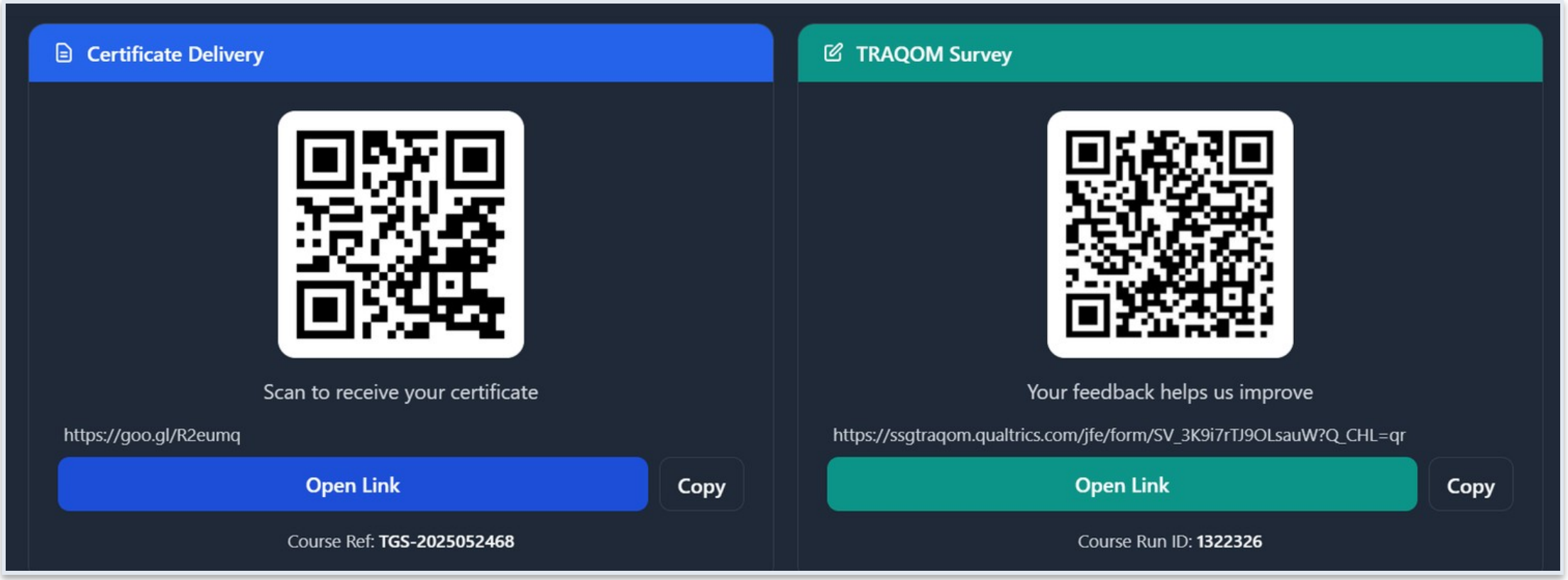
Cert & TRAQOM Survey (Mandatory)



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It is mandatory for you to take both AM, PM and Assessment digital attendance for WSQ funded courses.

- The trainer or administrator will show you the digital attendance QR code generated from SSG portal.
- Please scan the QR code from your mobile phone camera and submit your attendance.
- Digital Attendance (Mandatory)

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FINAL

Assessment

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Recommended Courses

- WSQ - AWS Certified DevOps Engineer Training (Synchronous e-Learning)
- WSQ - Microsoft Azure AI Fundamentals (AI-900)
- WSQ - Microsoft Azure Fundamentals (AZ-900)
- WSQ - Microsoft Azure Data Scientist Associate (DP-100)
- WSQ - Microsoft Certified Azure AI Engineer Associate (AI-102) Training

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Support

- If you have any enquiries during and after the class, you can contact us below
- Email: enquiry@tertiaryinfotech.com
- Tel: 61000613

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THANK YOU!

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